

LINUX ADMINISTRATION ASSIGNMENT

EDUV12936584

ITLIA0-44

Formative Assessment

EDUV12936584

ABSTRACT

Students will learn to plan, install, and manage an Ubuntu 24.04 LTS system, including user management, storage, packages, services, and logs. They will also connect it to an IPv4 network and integrate it with a Windows client.

Branny Aurthur Branfield

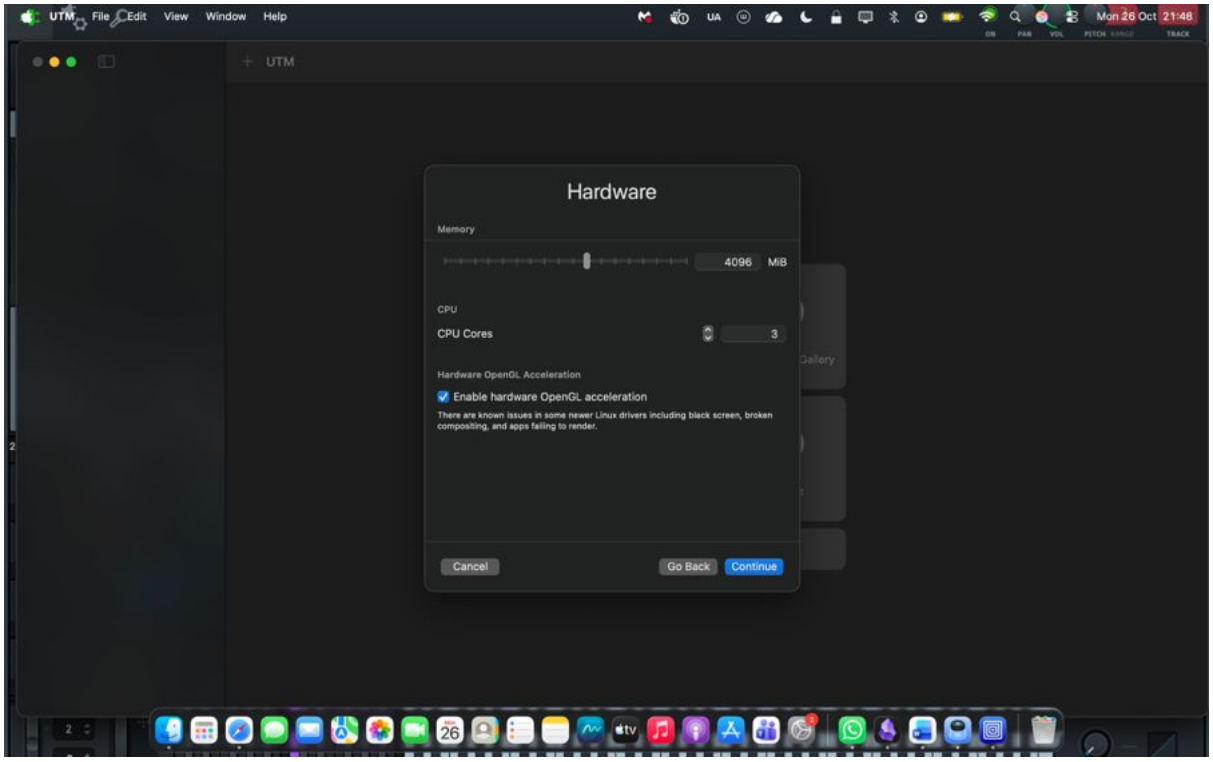
Administration

1.1 The recommended requirements for a small-form-factor server include an Intel Core i3 processor with a minimum speed of 4.70GHz, 16GB of RAM, and a disk storage capacity of 500GB or more, depending on the scale of the database and file size. A second disk should be purchased to store department data separately from the operating system, preventing disruptions caused by a full log partition. For Ubuntu, the minimum requirements are a 2GHz+ dual-core processor, 6GB of RAM, and 25GB of free hard drive space. The retired workstation with an Intel Core i3 at 1.90GHz, 4GB of RAM, and 500GB of disk storage may be insufficient due to its lower CPU speed and RAM, which could introduce lag. CUPS requires sufficient temporary disk space in /var/spool/cups for print jobs and lightweight RAM/CPU usage. PostgreSQL and Samba storage needs scale with data volume, with SSDs recommended. PostgreSQL demands more CPU and RAM for complex queries and high concurrent connections, while Samba is light on resources. The server should have two disks, with one dedicated to department data, and two VM network layouts Shared Network and Host-Only on

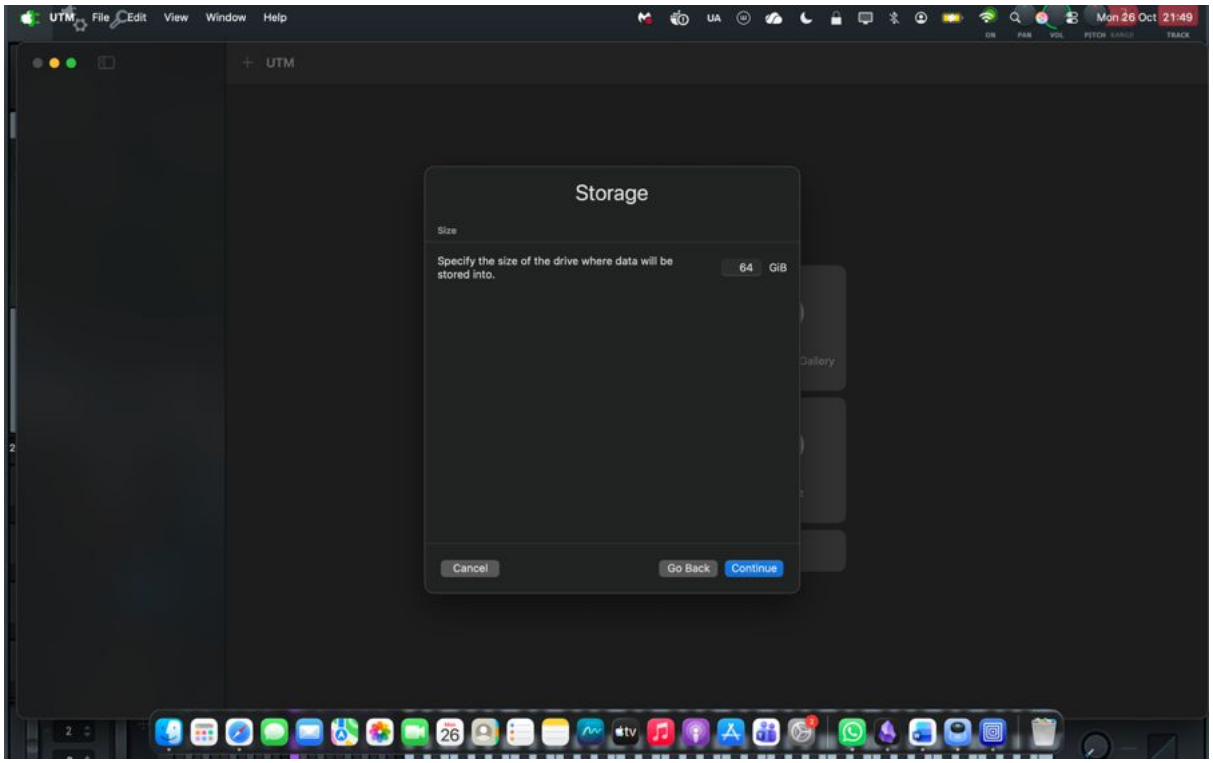
UTM VM for MacBook M Chip to enable internet connectivity and communication between VMs.

Note: The Recommended requirements will not be the same in the following VM configurations.

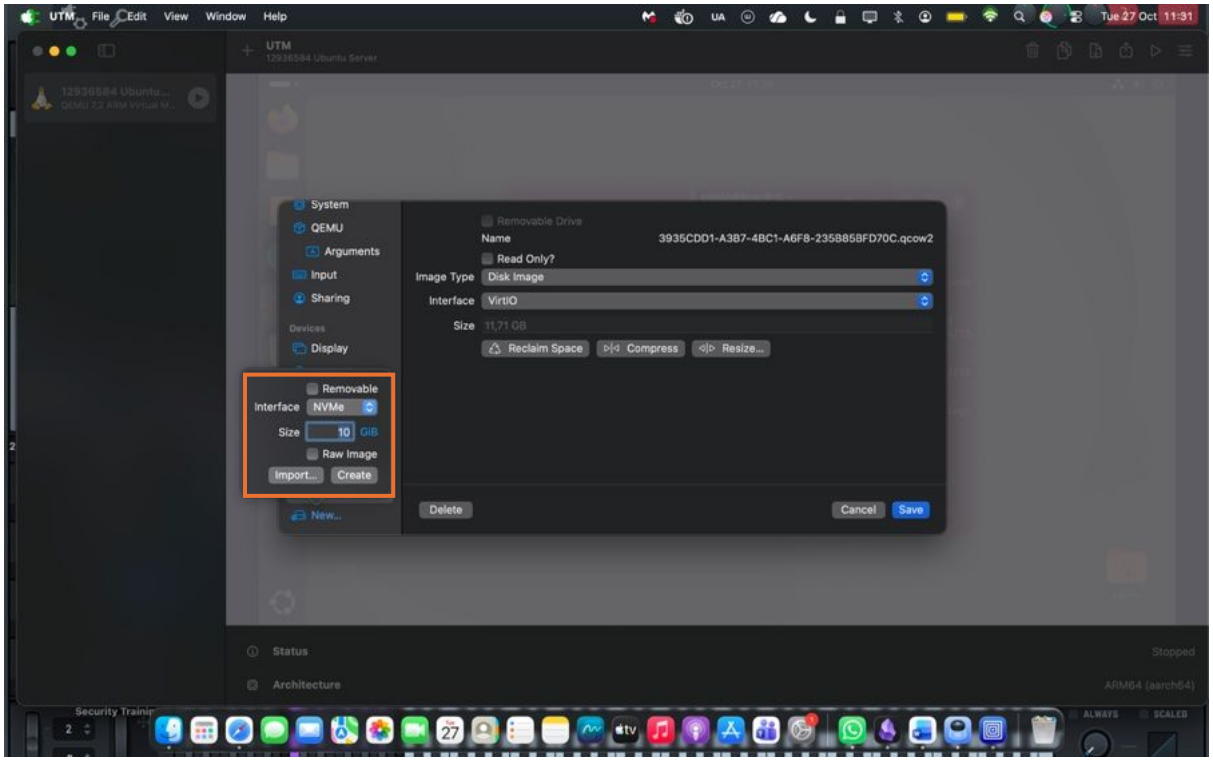
Note: I'll be using UTM as my VM which compatible with M Chip arm64 MacBook machines. Instead of VMware or VirtualBox.



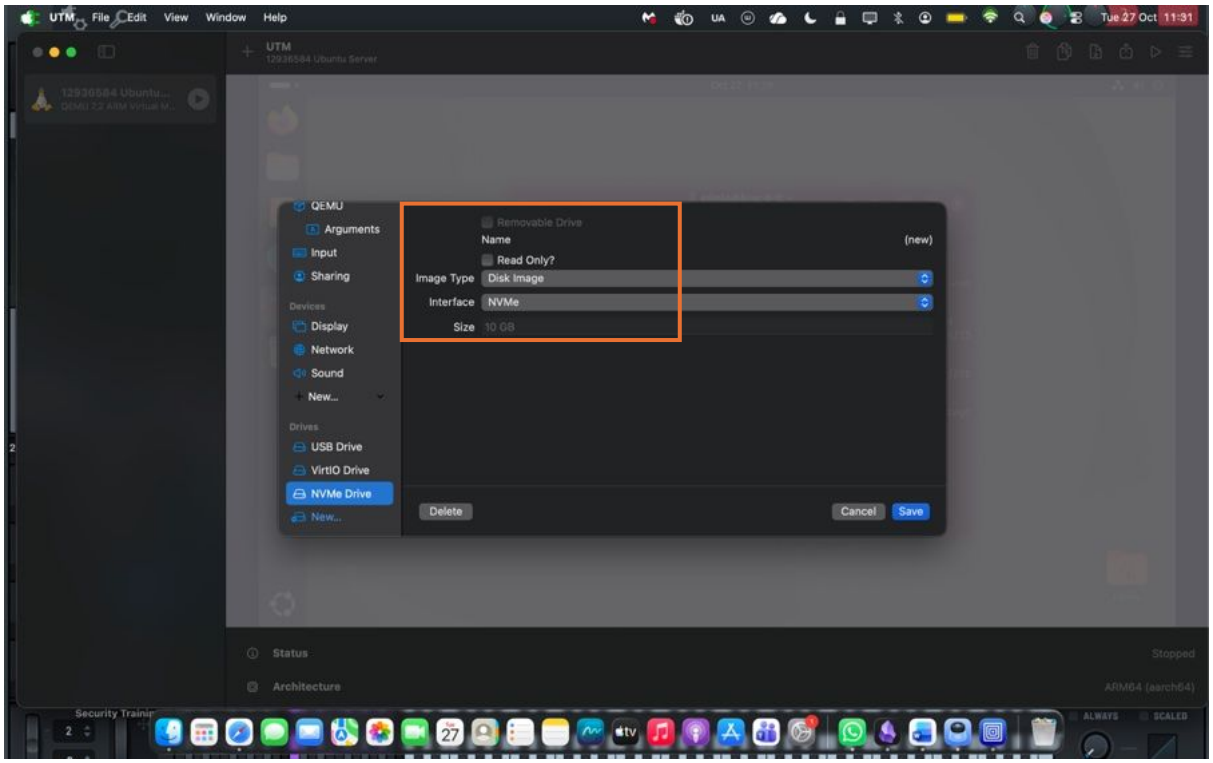
Setting CPU and Core.



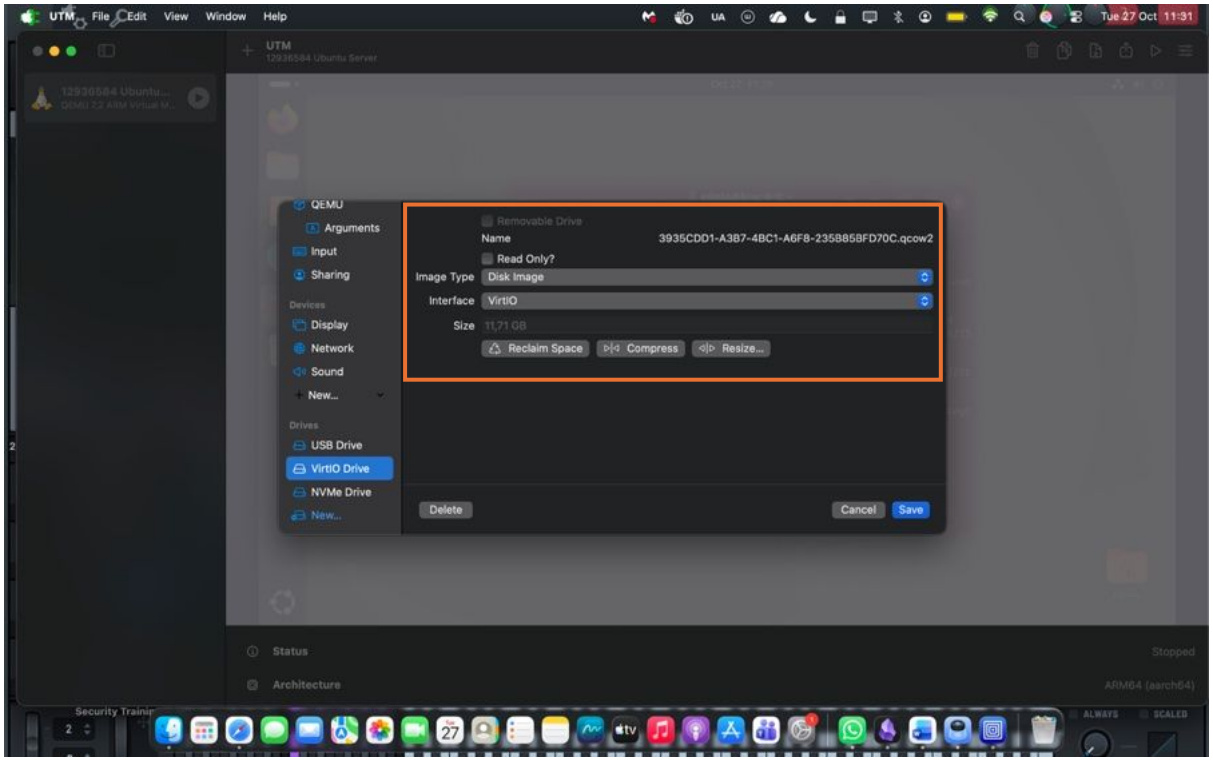
Setting Storage.



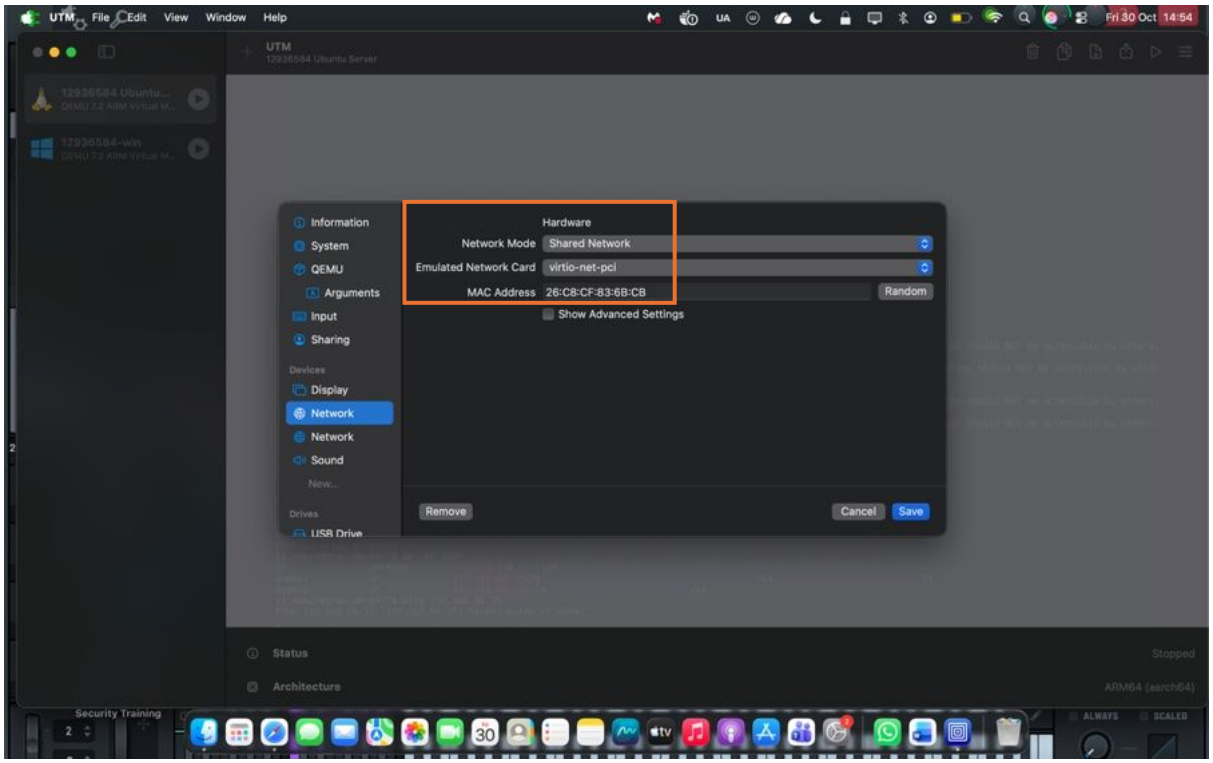
Setting up second disk storage for department data.



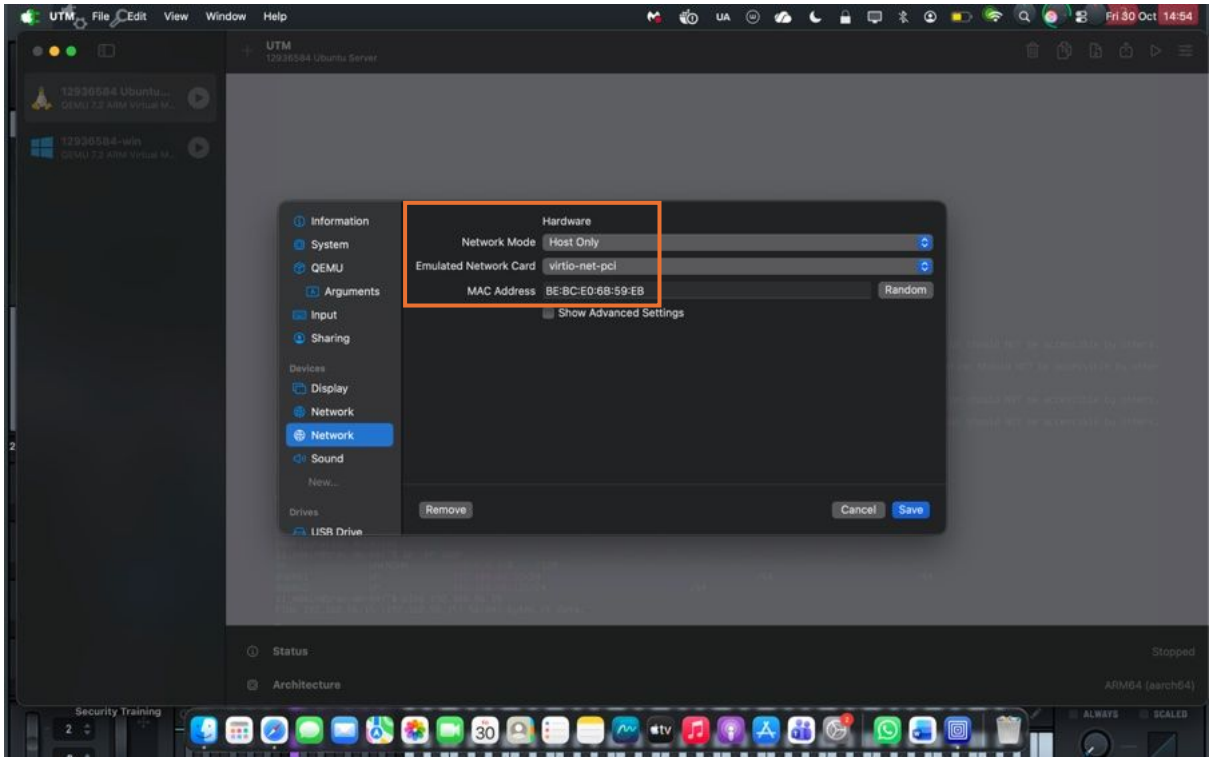
Department disk storage successfully set up.



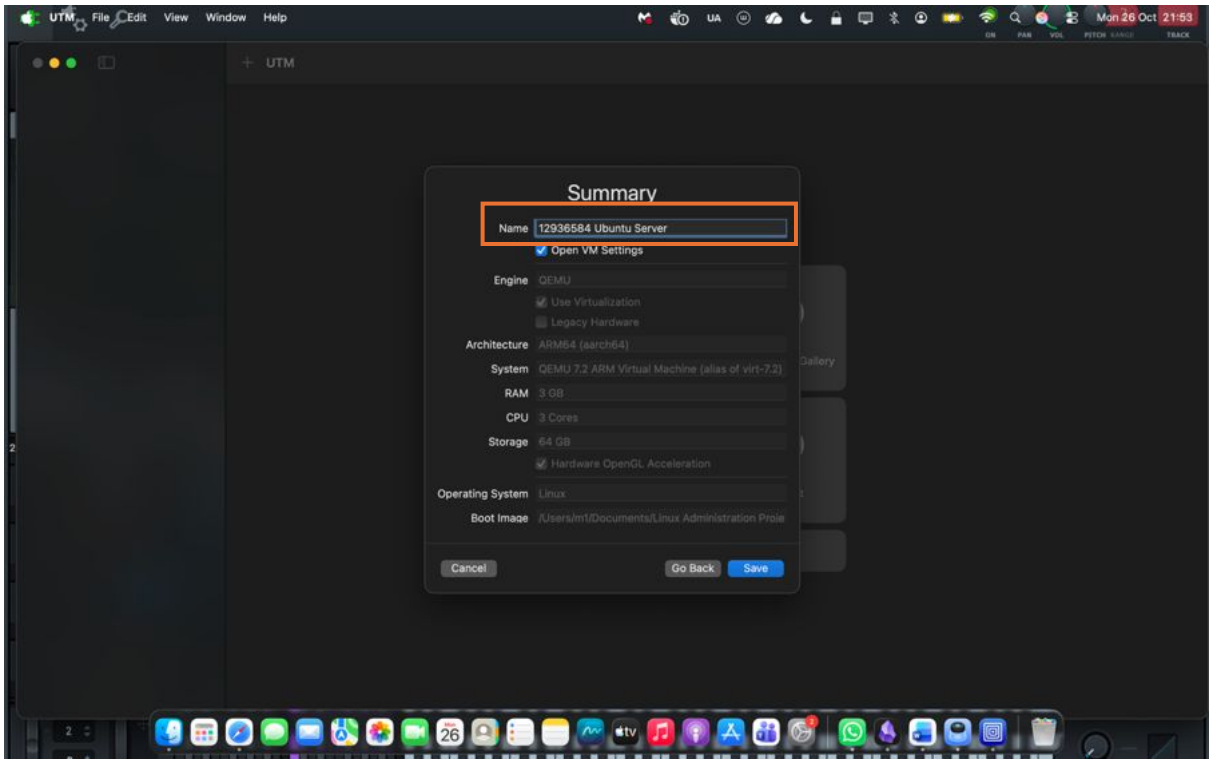
Main disk with storage used already by OS Ubuntu.



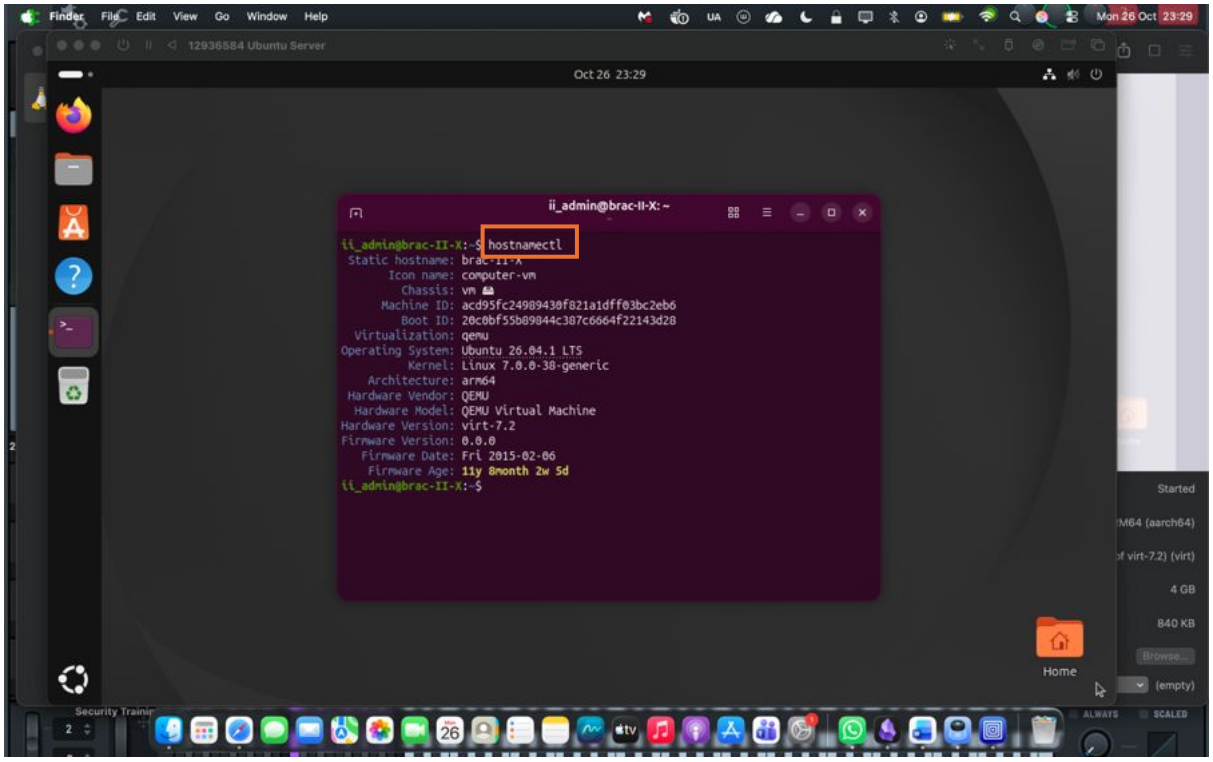
Shared Network Adapter for internet Access.



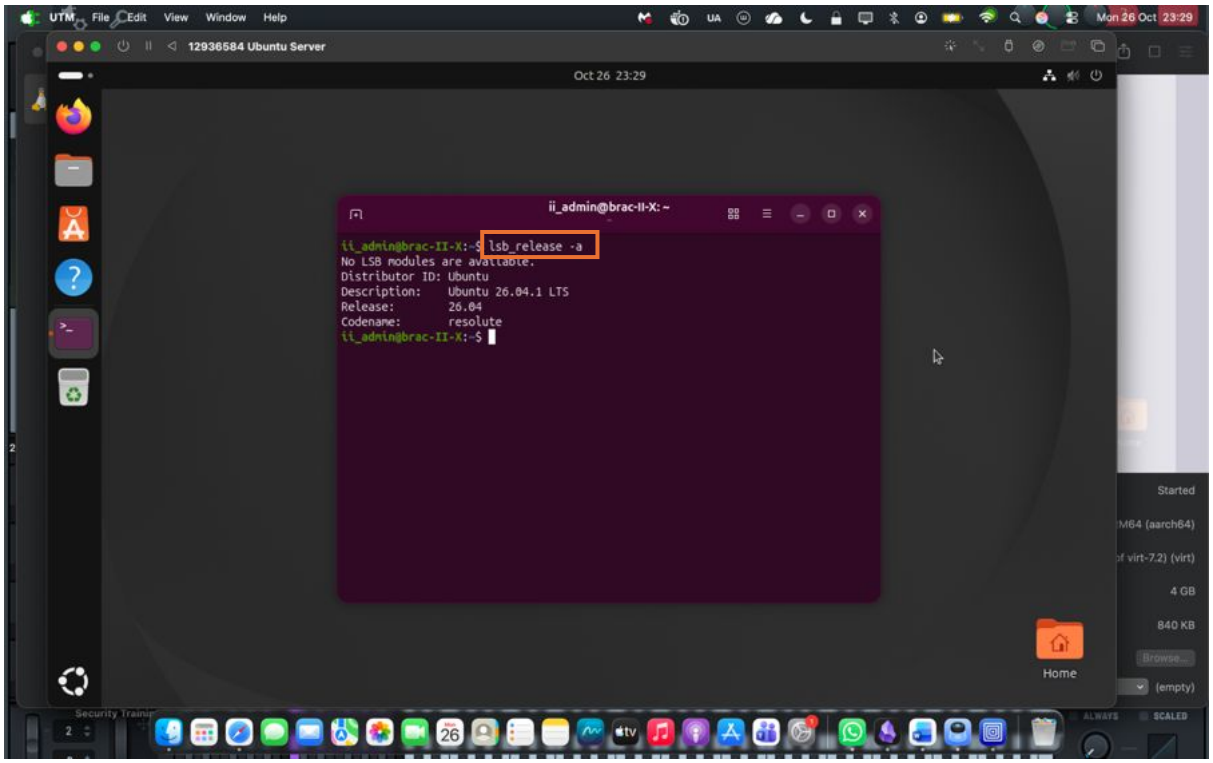
Host Only network adapter for communication between two VMs.



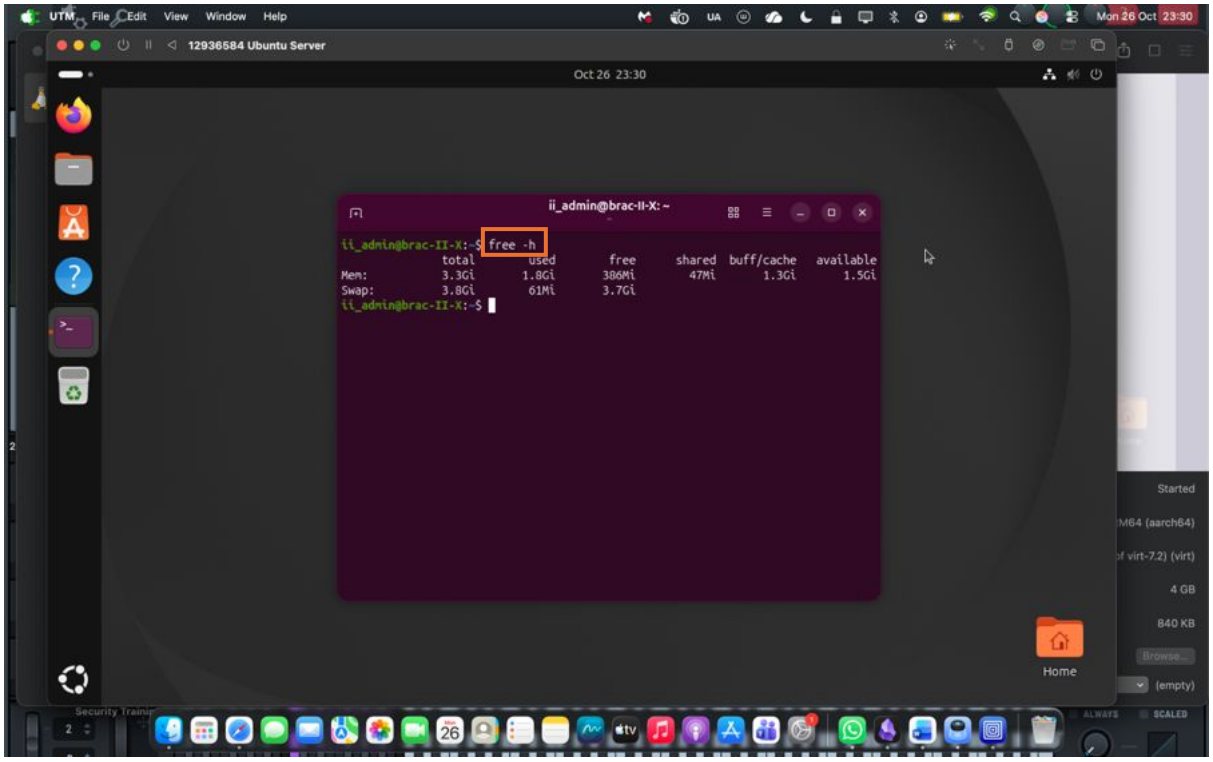
Naming Machine with my Student Number.



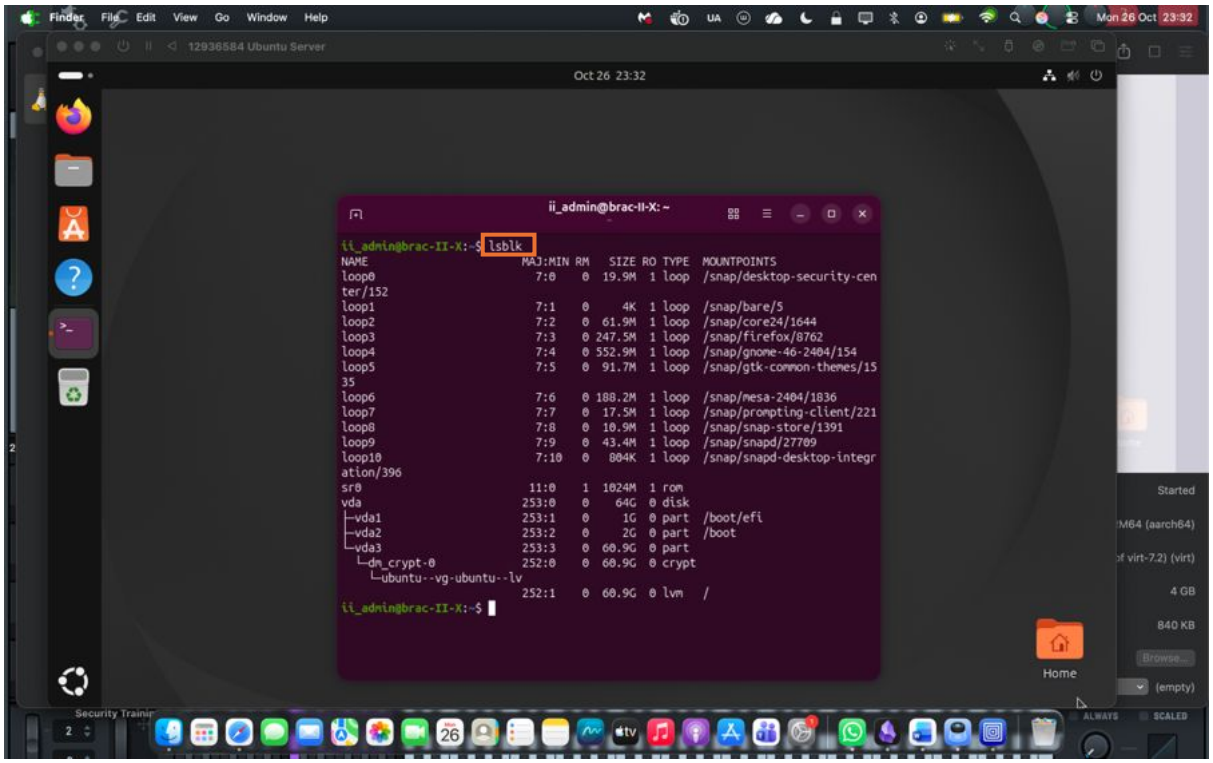
Post-installation verification with hostnamectl.



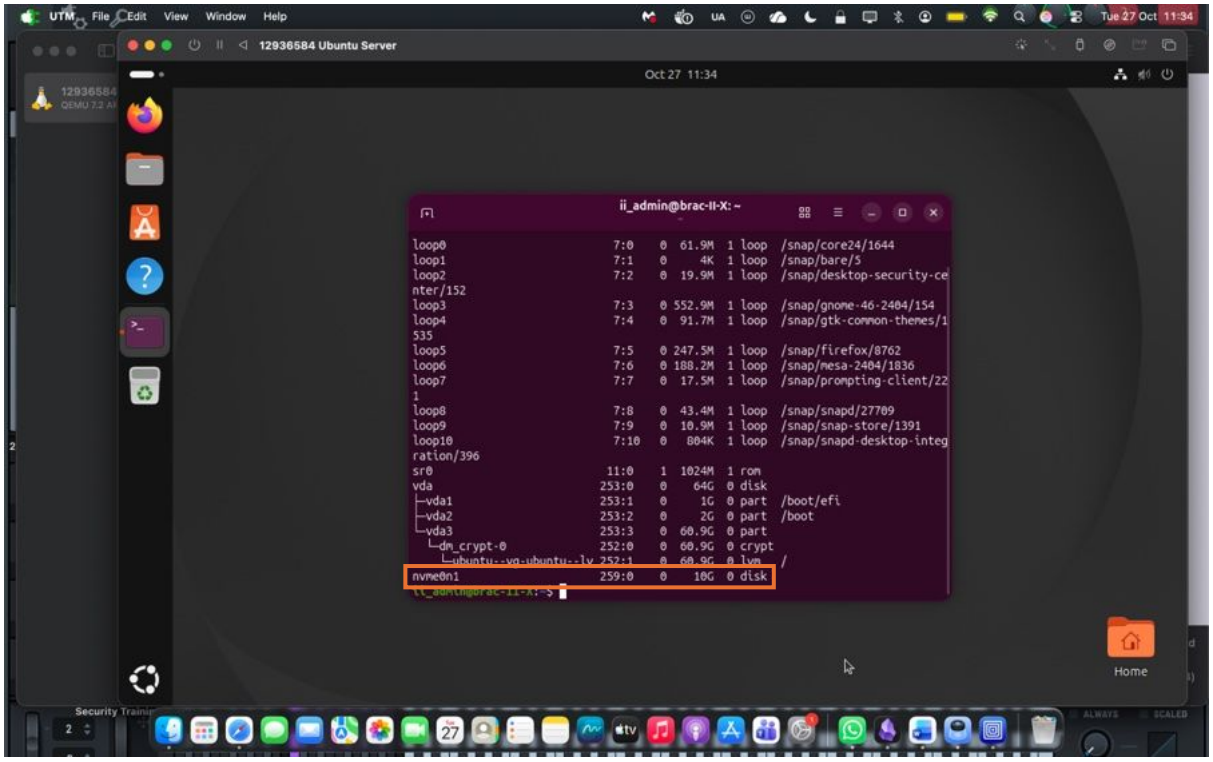
Post-installation verification with isb_release -a (OS Version Check)



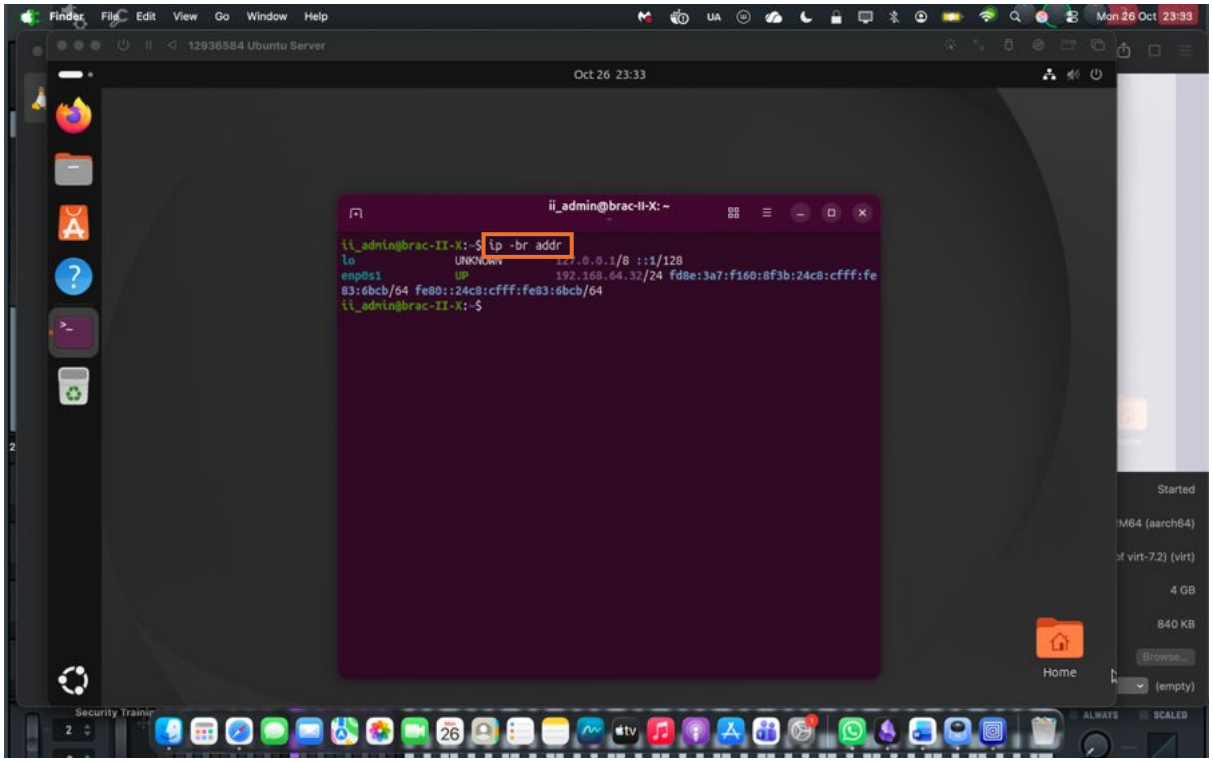
Post-installation verification with free -h (human readable free and used RAM Check).



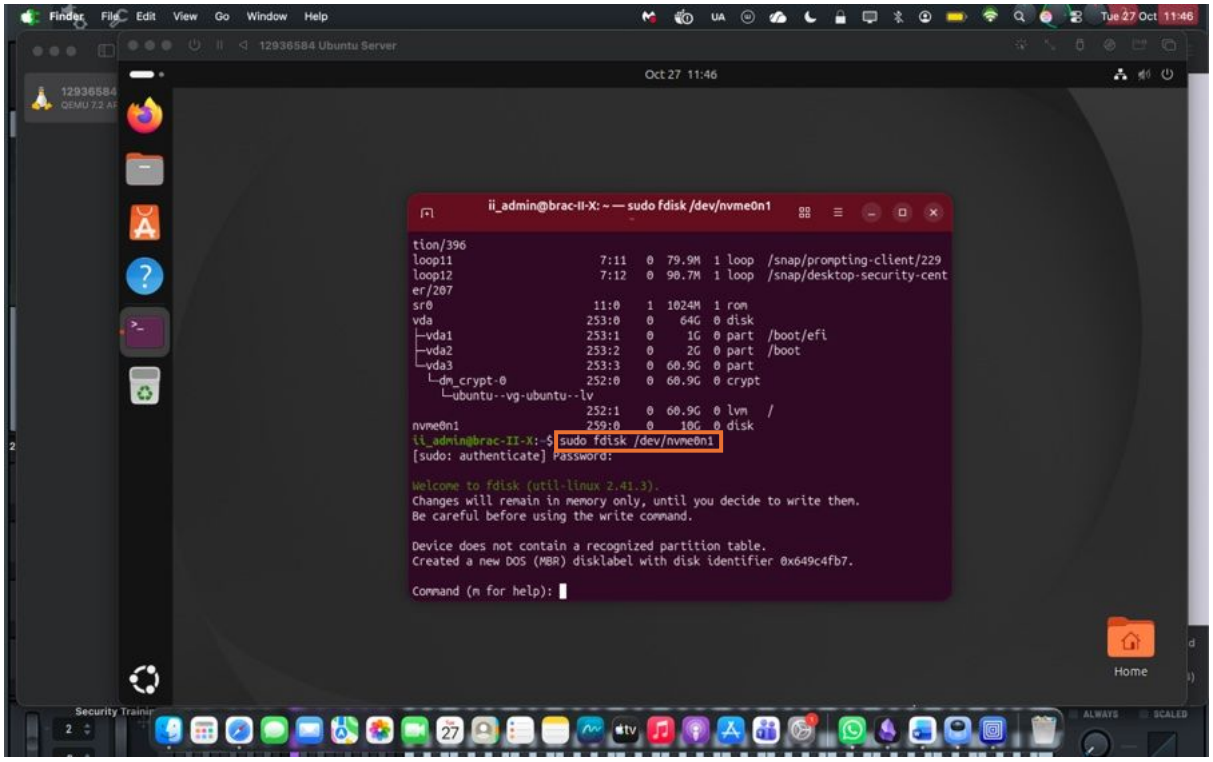
Post-installation verification with lsblk (storage device and partition check) before adding second disk.



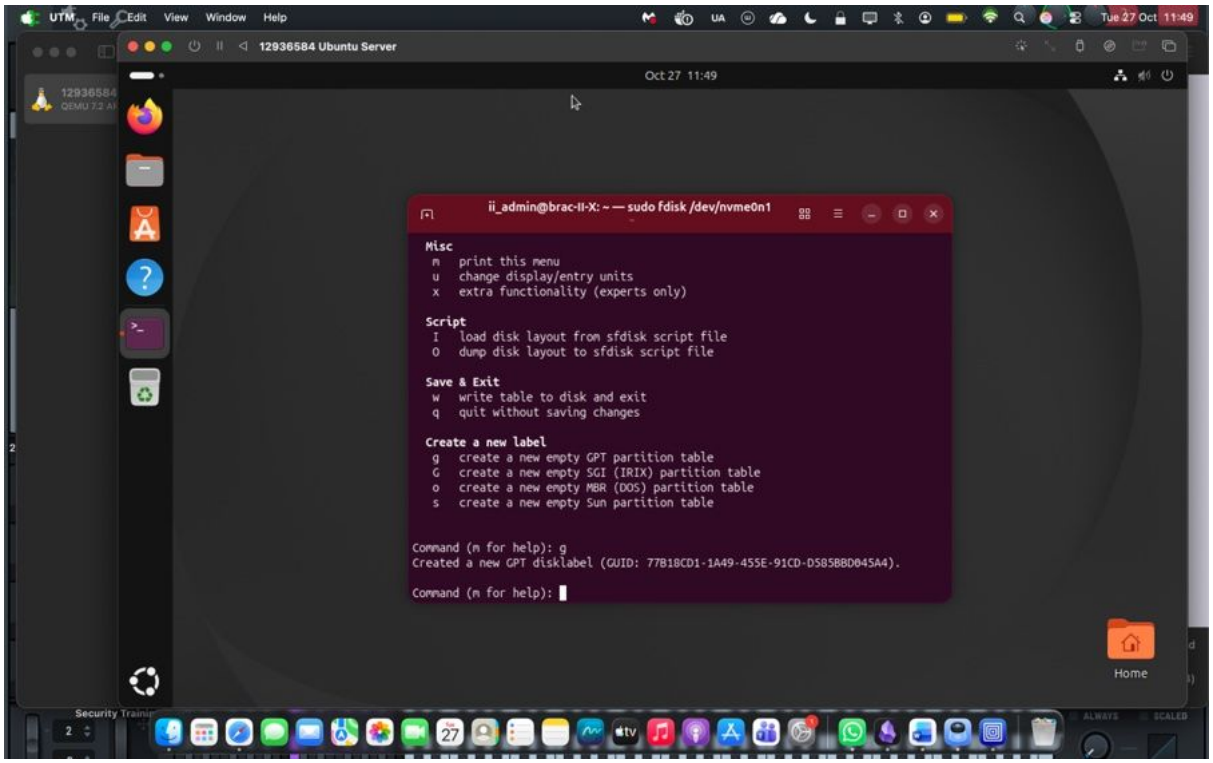
Post-installation verification with lsblk (storage device and partition check) with second disk.

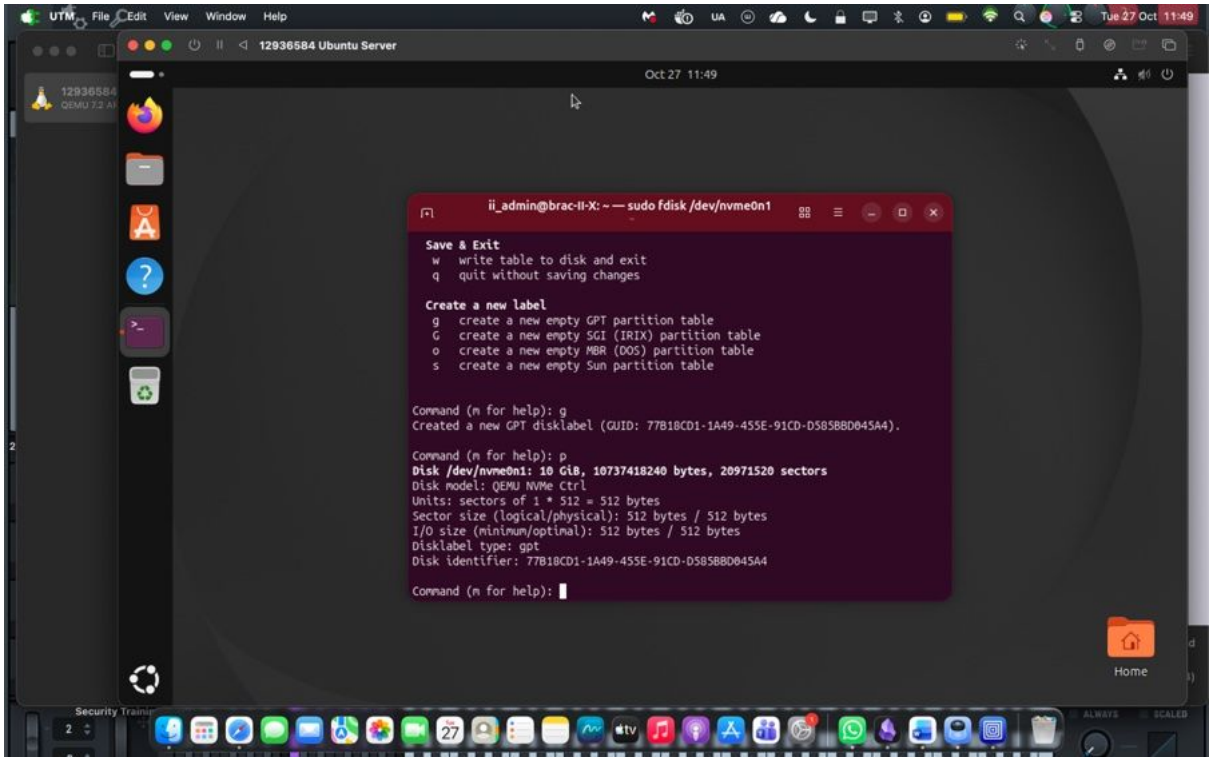


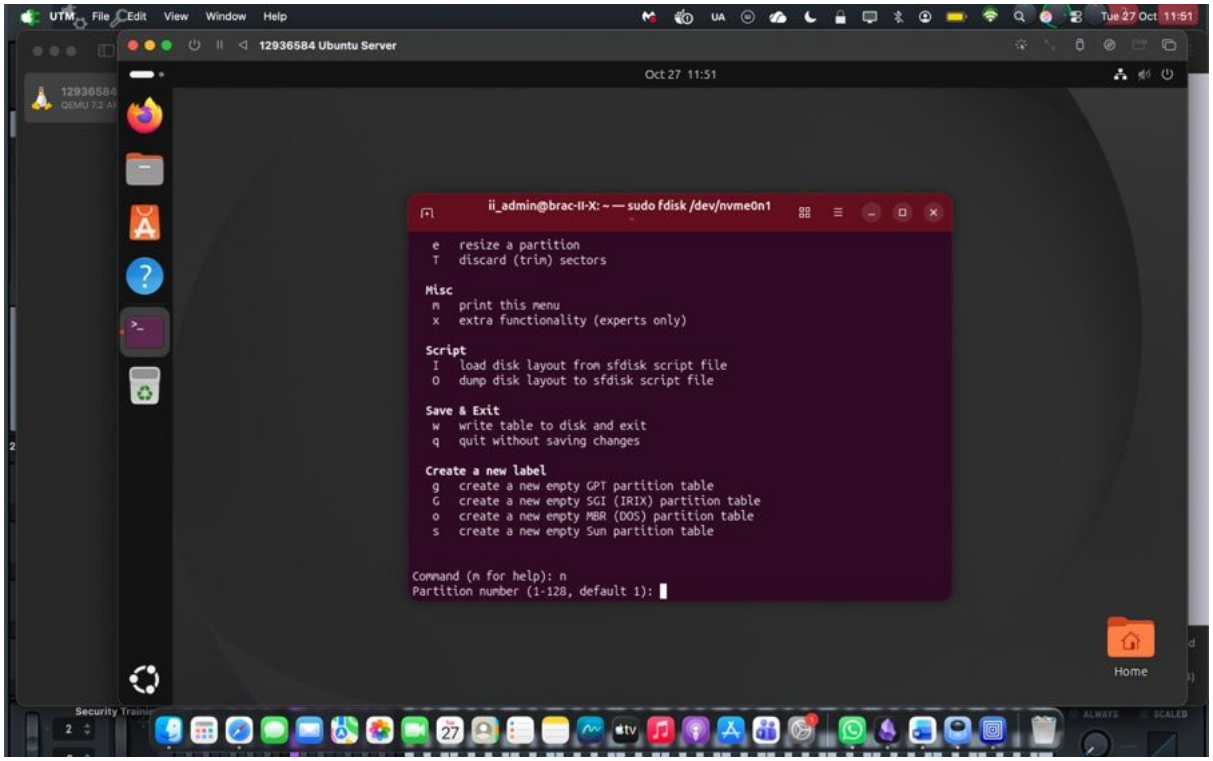
Post-installation verification with ip -br addr (IP Check) before adding second network adapter.

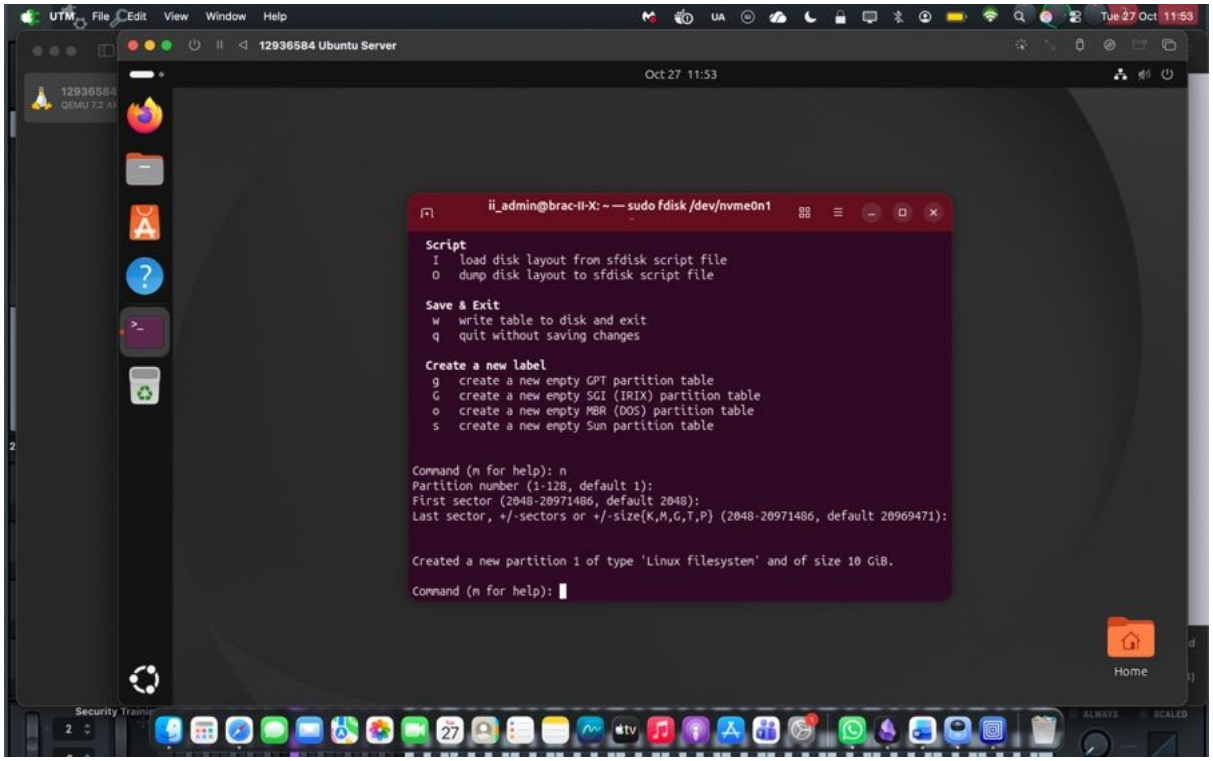


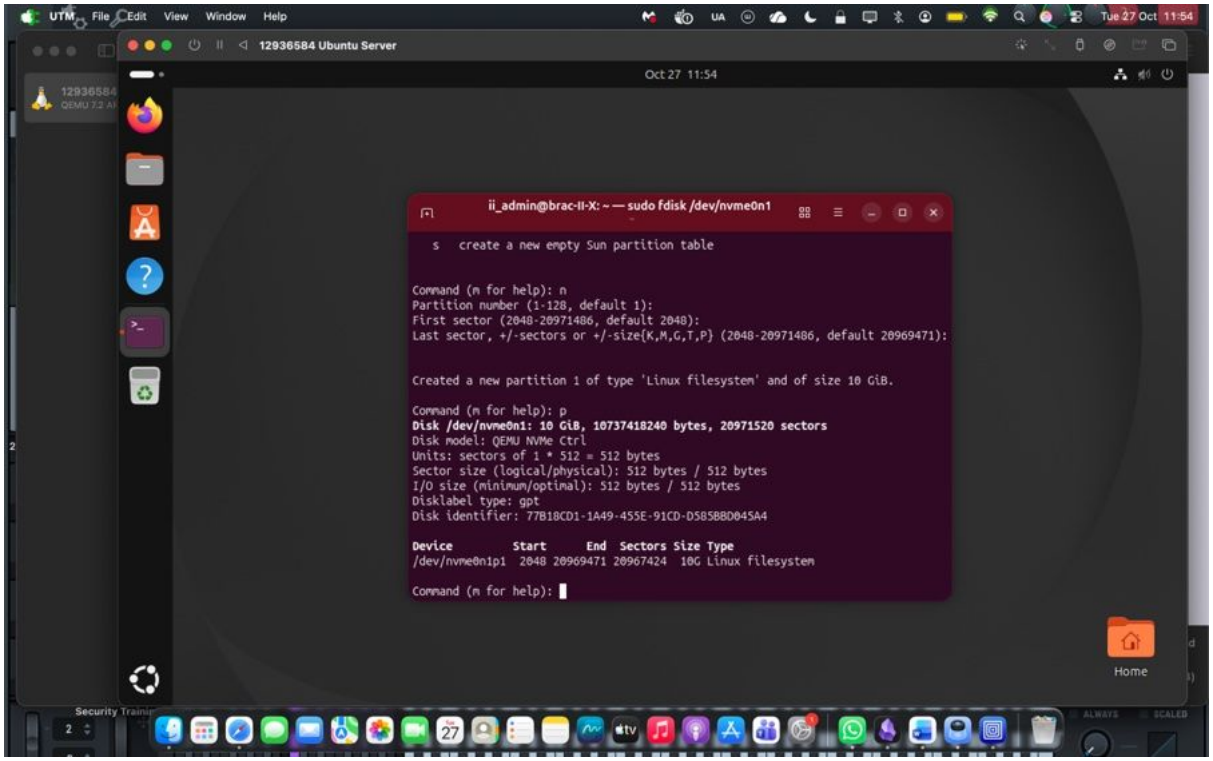
Checking disks with lsblk and fdisk to partition it.

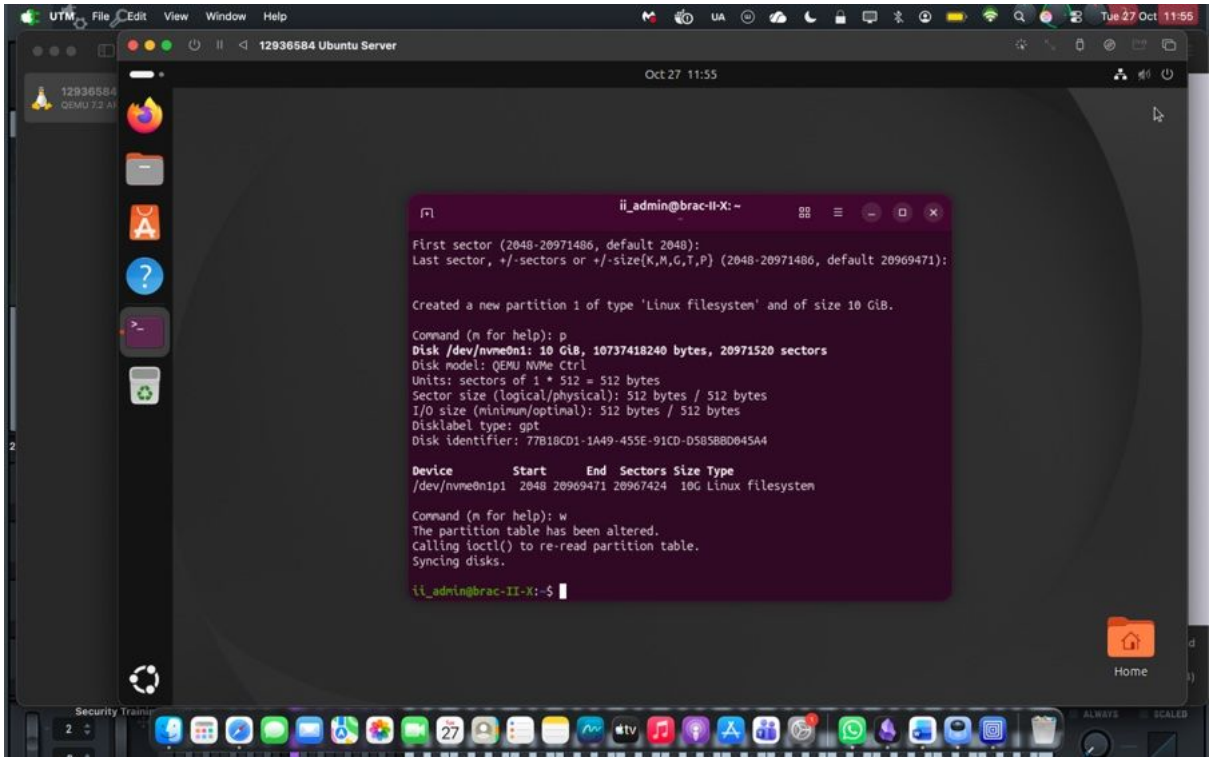


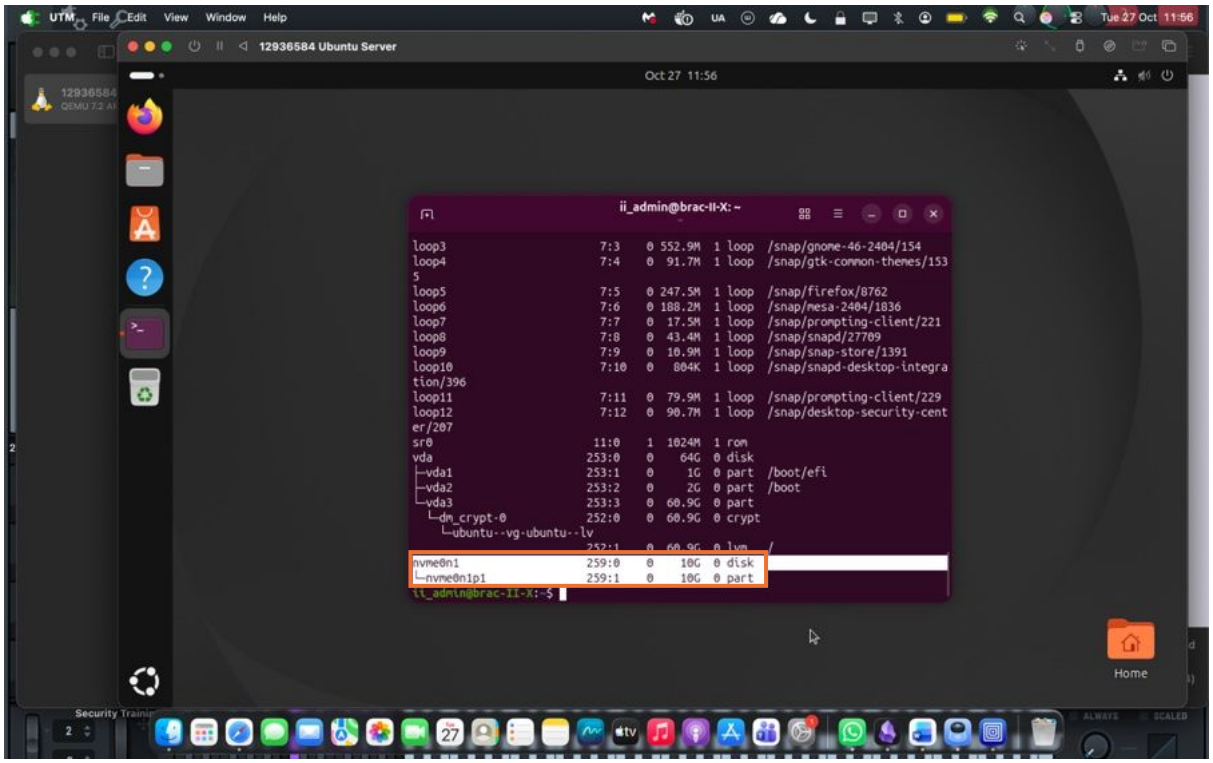

UTM File Edit View Window Help
Tue 27 Oct 11:49
12936584 Ubuntu Server
Oct 27 11:49
ii_admin@brac-II-X: ~ — sudo fdisk /dev/nvme0n1
Save & Exit
w write table to disk and exit
q quit without saving changes
Create a new label
g create a new empty GPT partition table
G create a new empty SGI (IRIX) partition table
o create a new empty MBR (DOS) partition table
s create a new empty Sun partition table
Command (m for help): g
Created a new GPT disklabel (GUID: 77B18CD1-1A49-455E-91CD-D585BBD045A4).
Command (m for help): p
Disk /dev/nvme0n1: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: QEMU NVMe Ctrl
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 77B18CD1-1A49-455E-91CD-D585BBD045A4
Command (m for help):
Home


UTM File Edit View Window Help
Tue 27 Oct 11:51
12936584 Ubuntu Server
Oct 27 11:51
ii_admin@brac-II-X: ~ — sudo fdisk /dev/nvme0n1
e resize a partition
T discard (trim) sectors
Misc
m print this menu
x extra functionality (experts only)
Script
I load disk layout from sfdisk script file
O dump disk layout to sfdisk script file
Save & Exit
w write table to disk and exit
q quit without saving changes
Create a new label
g create a new empty GPT partition table
G create a new empty SGI (IRIX) partition table
o create a new empty MBR (DOS) partition table
s create a new empty Sun partition table
Command (m for help): n
Partition number (1-128, default 1):
Home

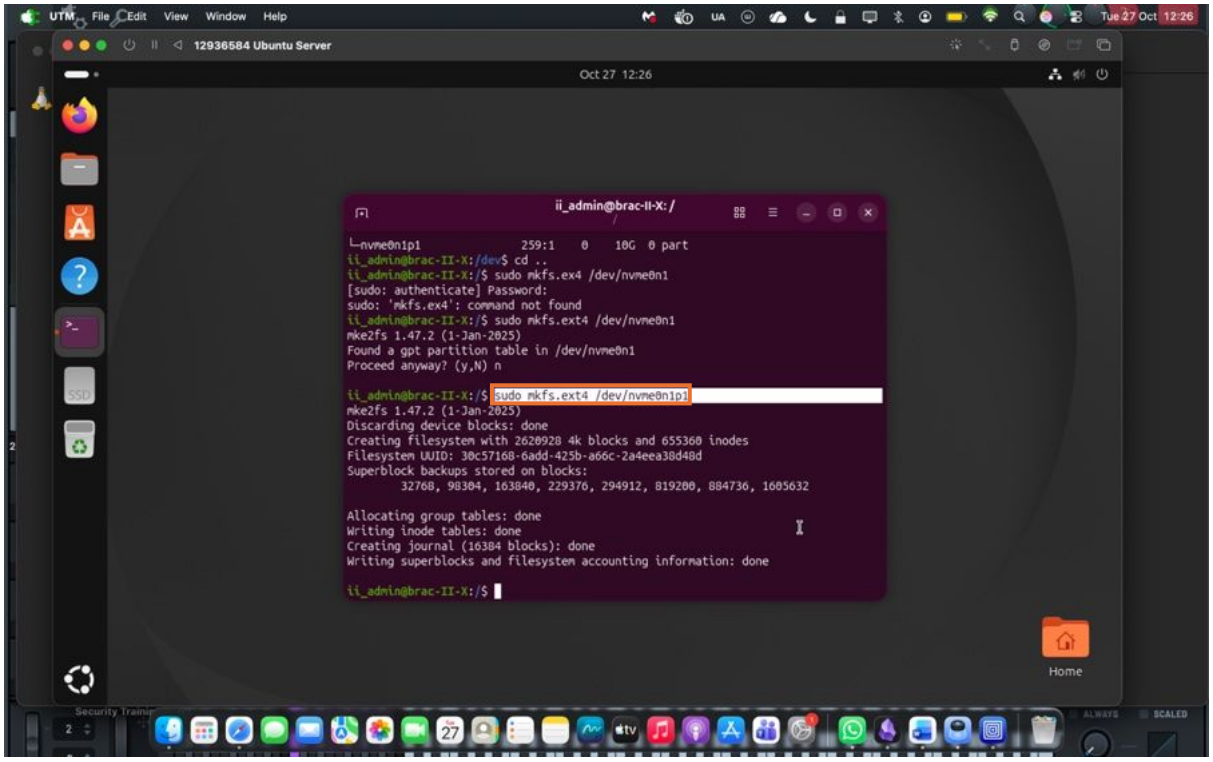

UTM File Edit View Window Help
Tue 27 Oct 11:53
12936584 Ubuntu Server
Oct 27 11:53
ii_admin@brac-II-X: ~ — sudo fdisk /dev/nvme0n1
Script
I load disk layout from sfdisk script file
O dump disk layout to sfdisk script file
Save & Exit
w write table to disk and exit
q quit without saving changes
Create a new label
g create a new empty GPT partition table
G create a new empty SGI (IRIX) partition table
o create a new empty MBR (DOS) partition table
s create a new empty Sun partition table
Command (m for help): n
Partition number (1-128, default 1):
First sector (2048-20971486, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-20971486, default 20969471):
Created a new partition 1 of type 'Linux filesystem' and of size 10 GiB.
Command (m for help):
Home


UTM File Edit View Window Help
Tue 27 Oct 11:54
12936584 Ubuntu Server
Oct 27 11:54
ii_admin@brac-II-X: ~ — sudo fdisk /dev/nvme0n1
s create a new empty Sun partition table
Command (m for help): n
Partition number (1-128, default 1):
First sector (2048-20971486, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-20971486, default 20969471):
Created a new partition 1 of type 'Linux filesystem' and of size 10 GiB.
Command (m for help): p
Disk /dev/nvme0n1: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: QEMU NVMe Ctrl
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 77B18CD1-1A49-455E-91CD-D585BBD045A4
Device Start End Sectors Size Type
/dev/nvme0n1p1 2048 20969471 20967424 10G Linux filesystem
Command (m for help):
Home

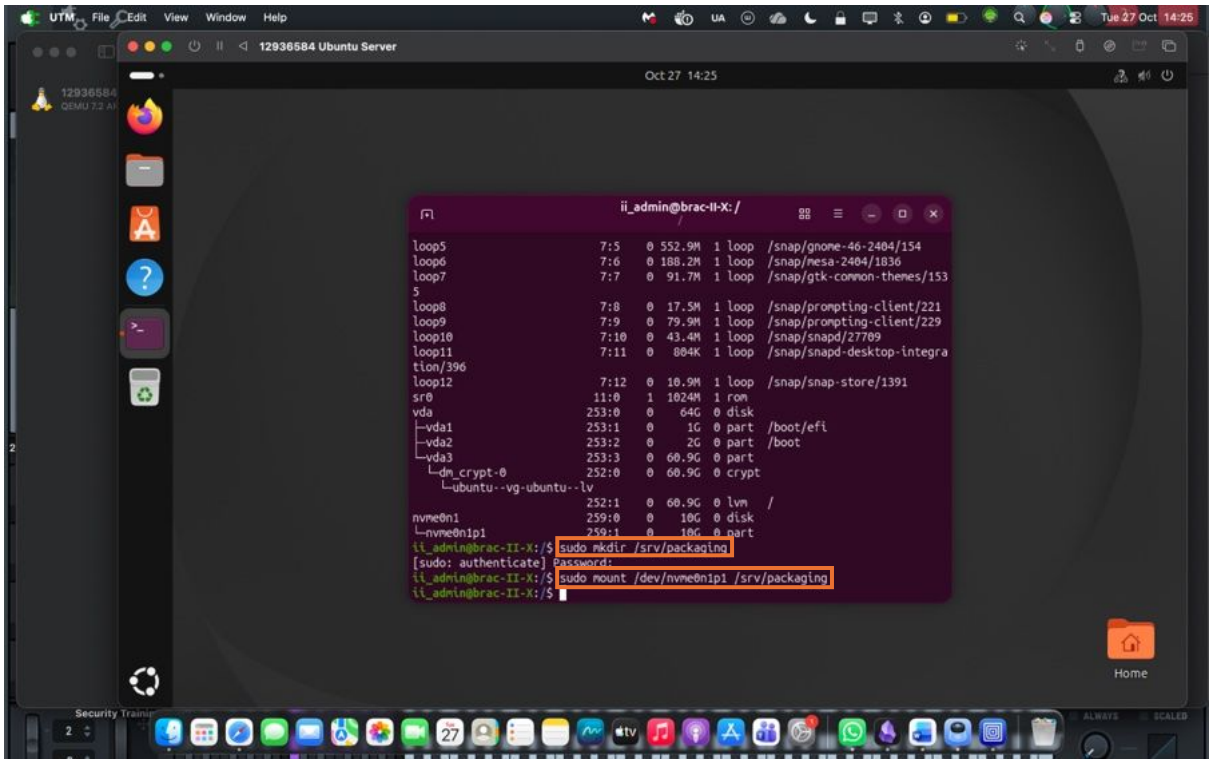




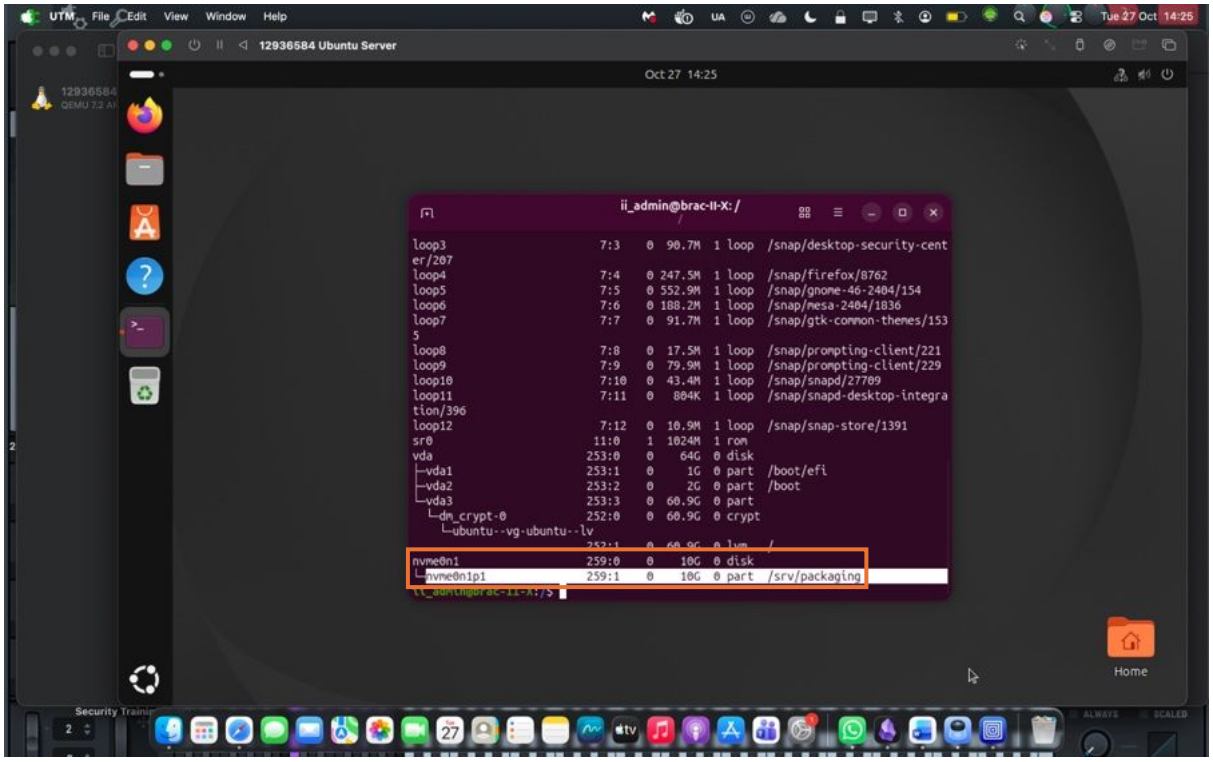
Successfully Partitioned disk nvme0n1.



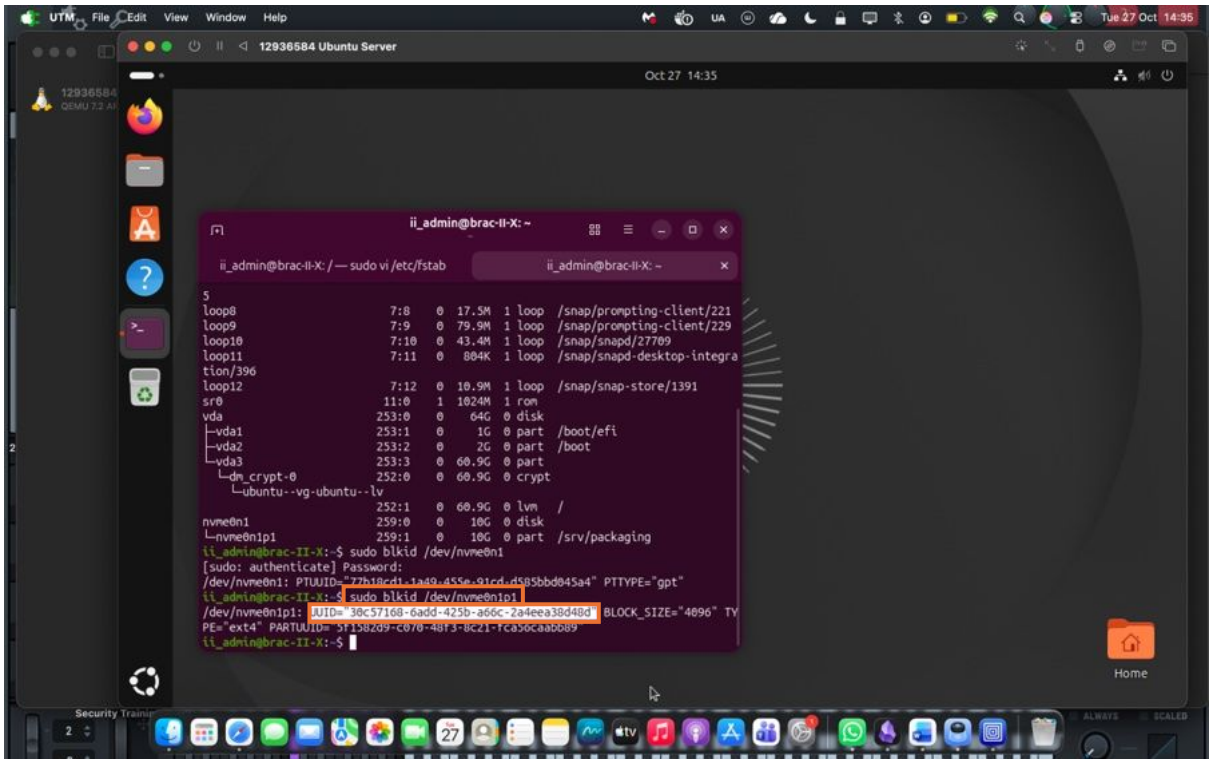
Formatting partition with mkfs.ext4 to ext4 format.



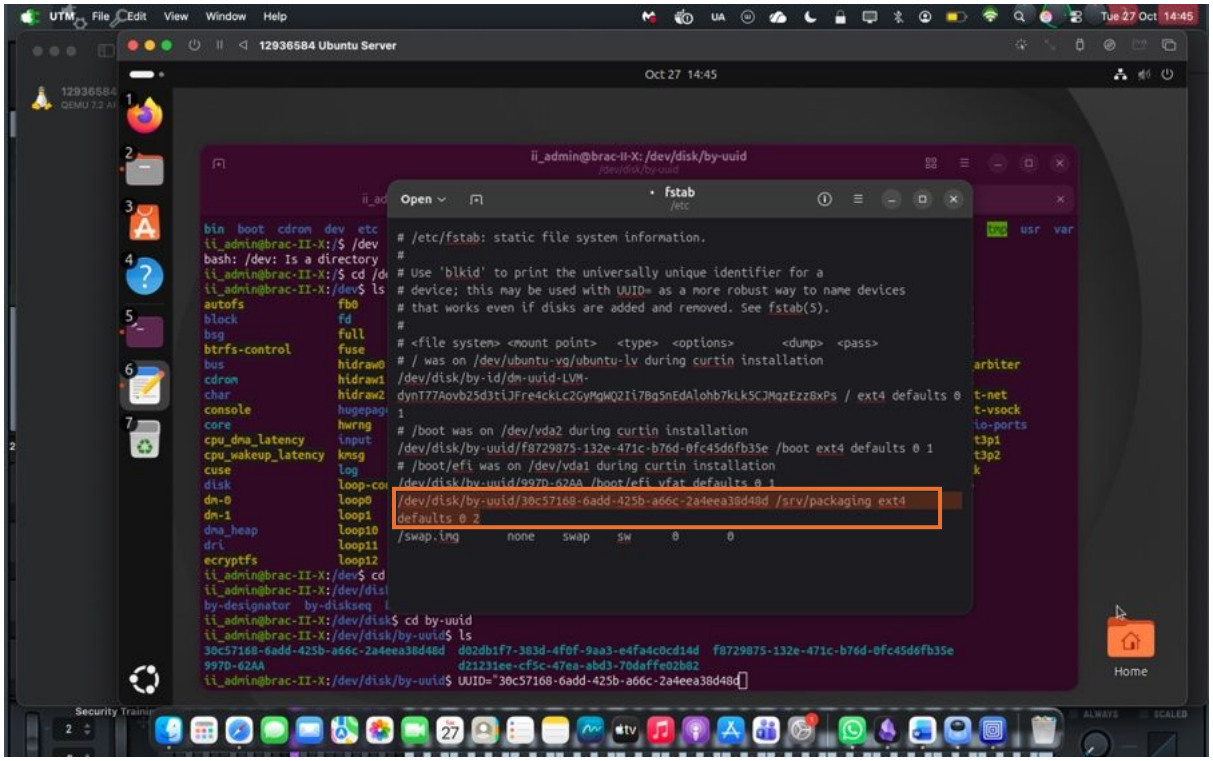
Making directory to Mount partition with mkdir and mounting with mount.



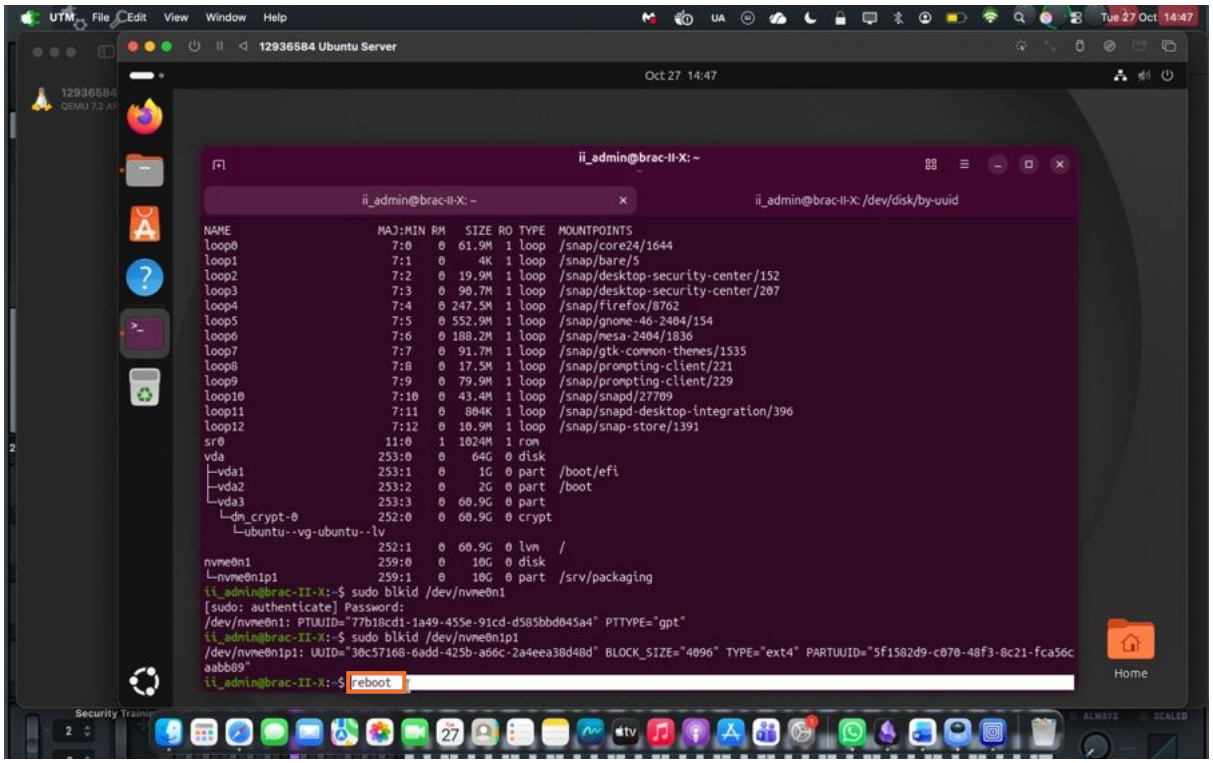
Partition successfully mounted.



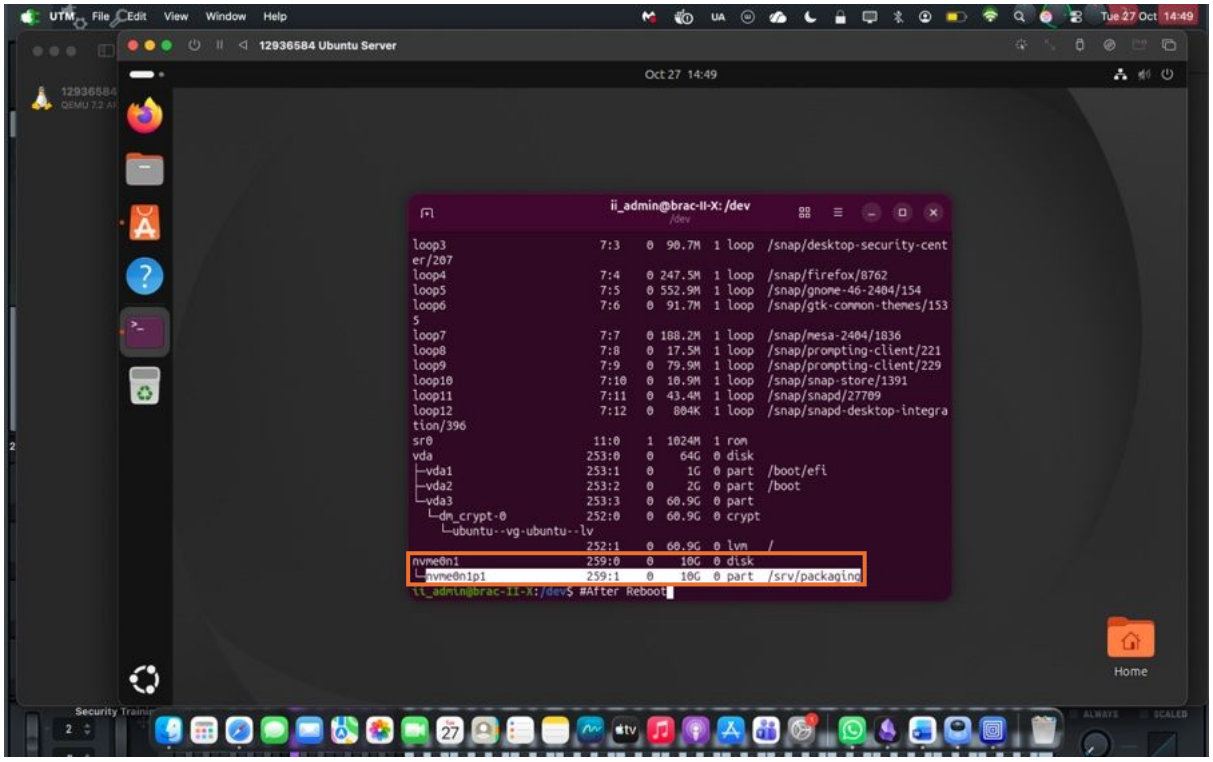
Finding UUID with blkid.



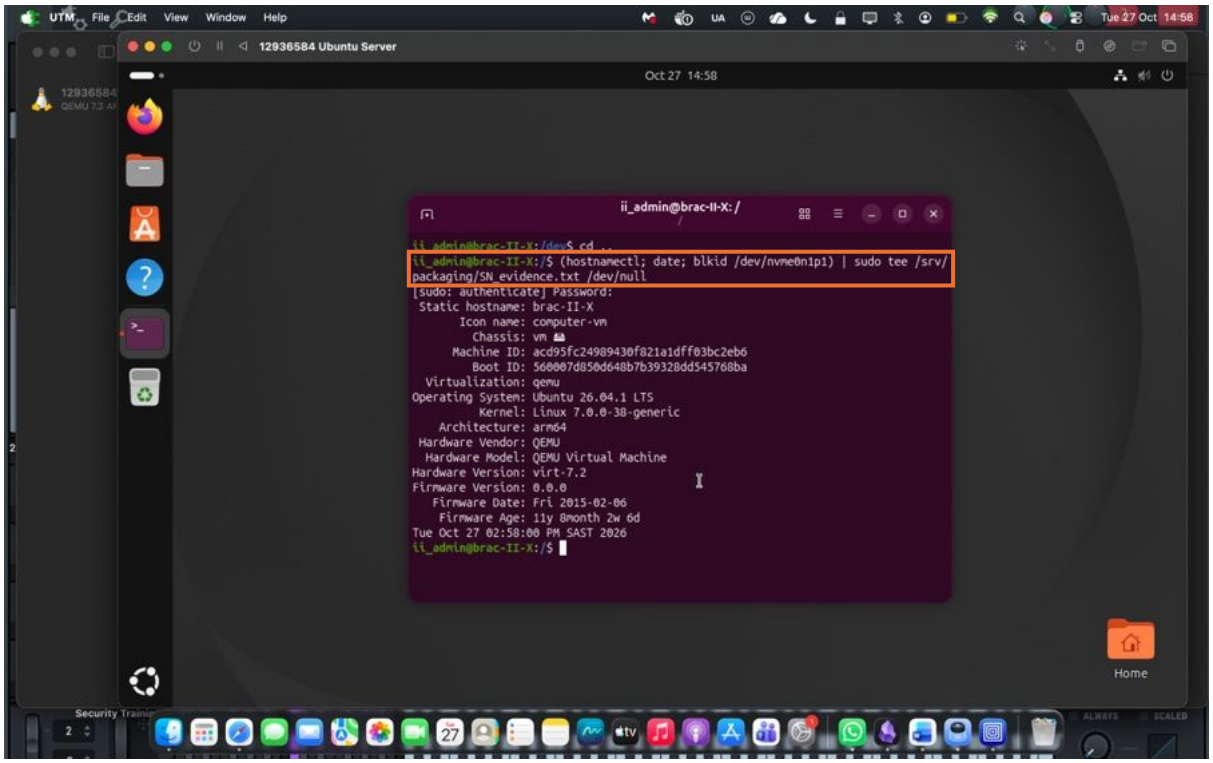
Making Mount persistent with /etc/fstab and text editor.



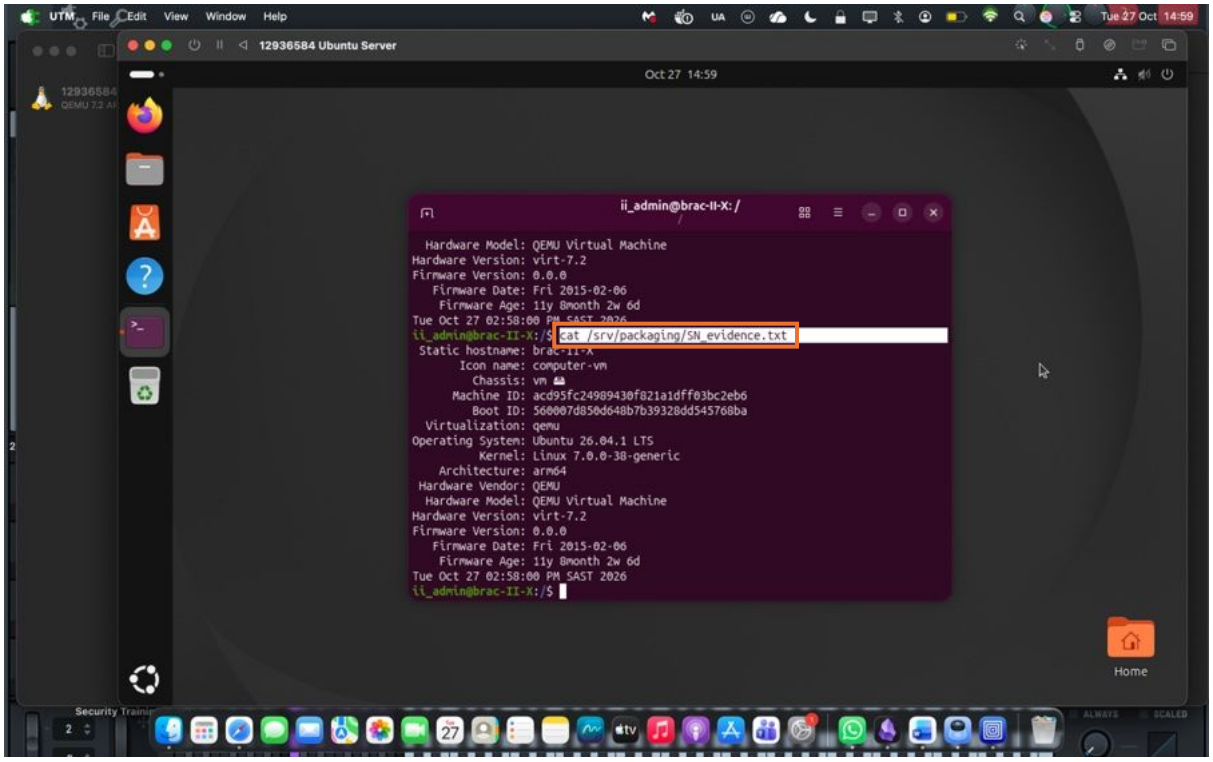
Rebooting Machine.



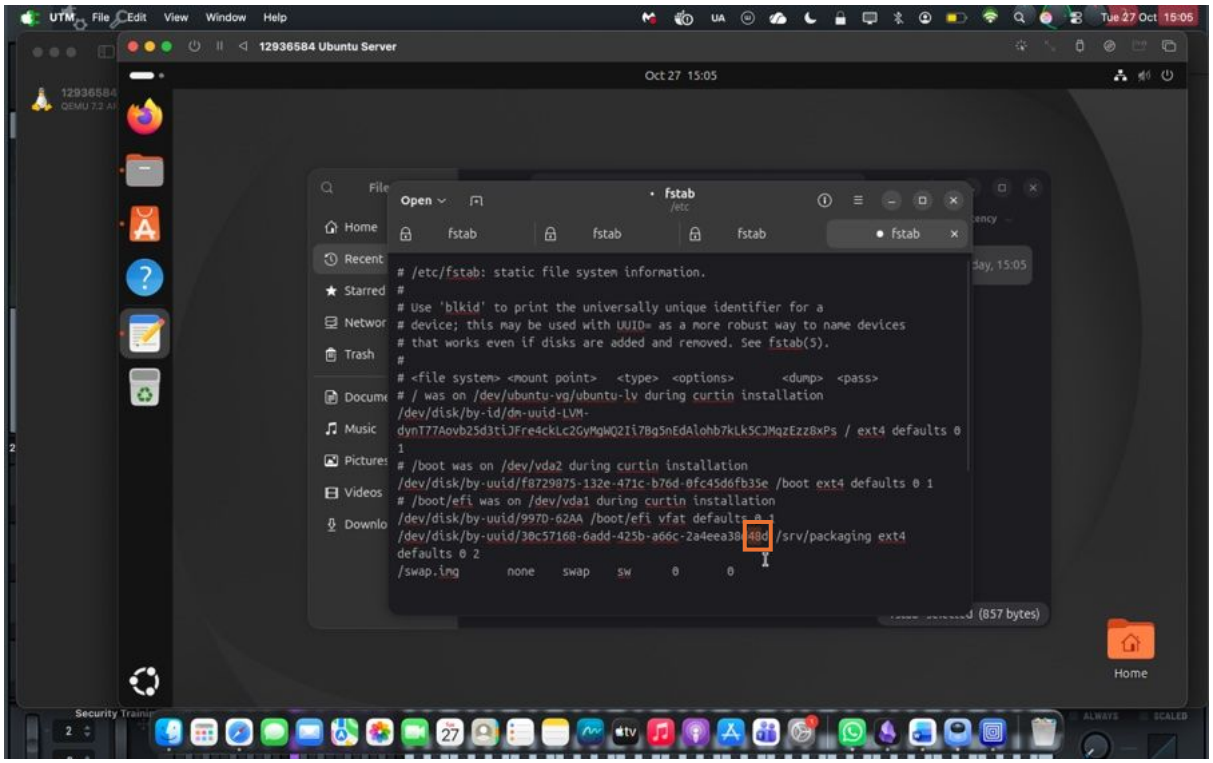
Mount Persistence Successful.



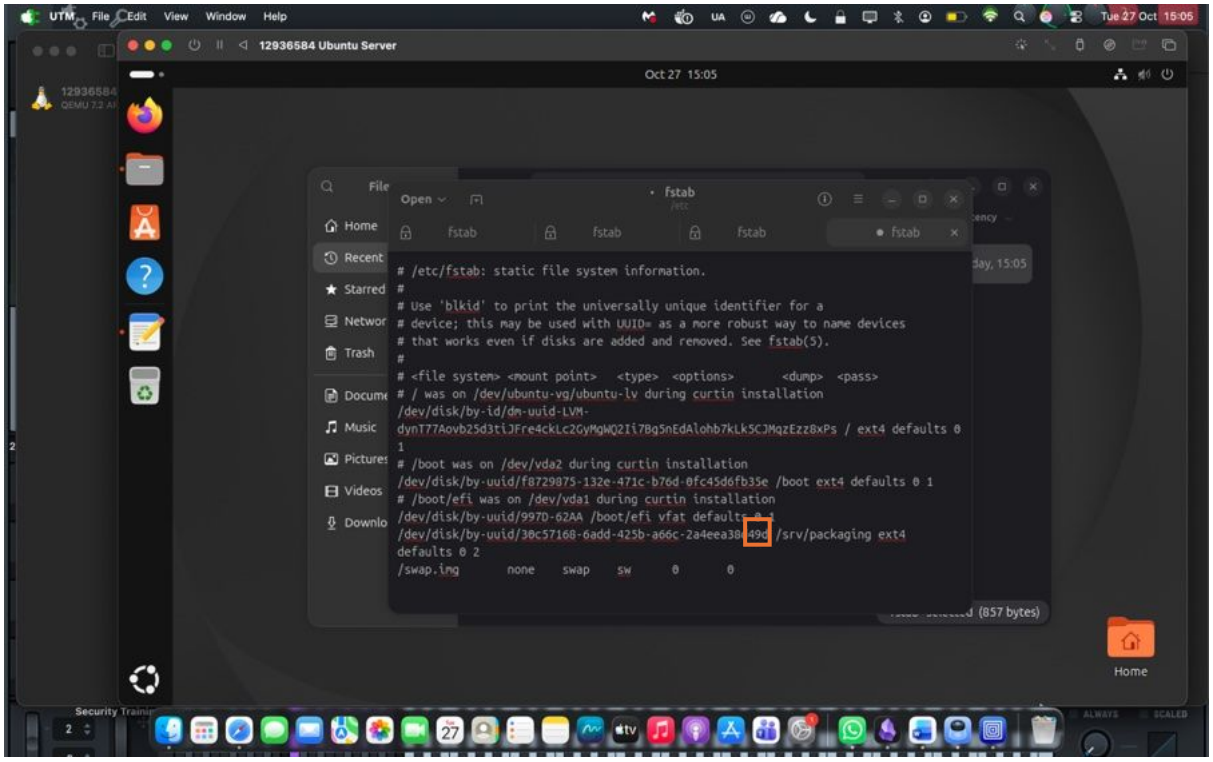
Creating File named SN_evidence.txt with hostnamectl; date and blkid using tee.



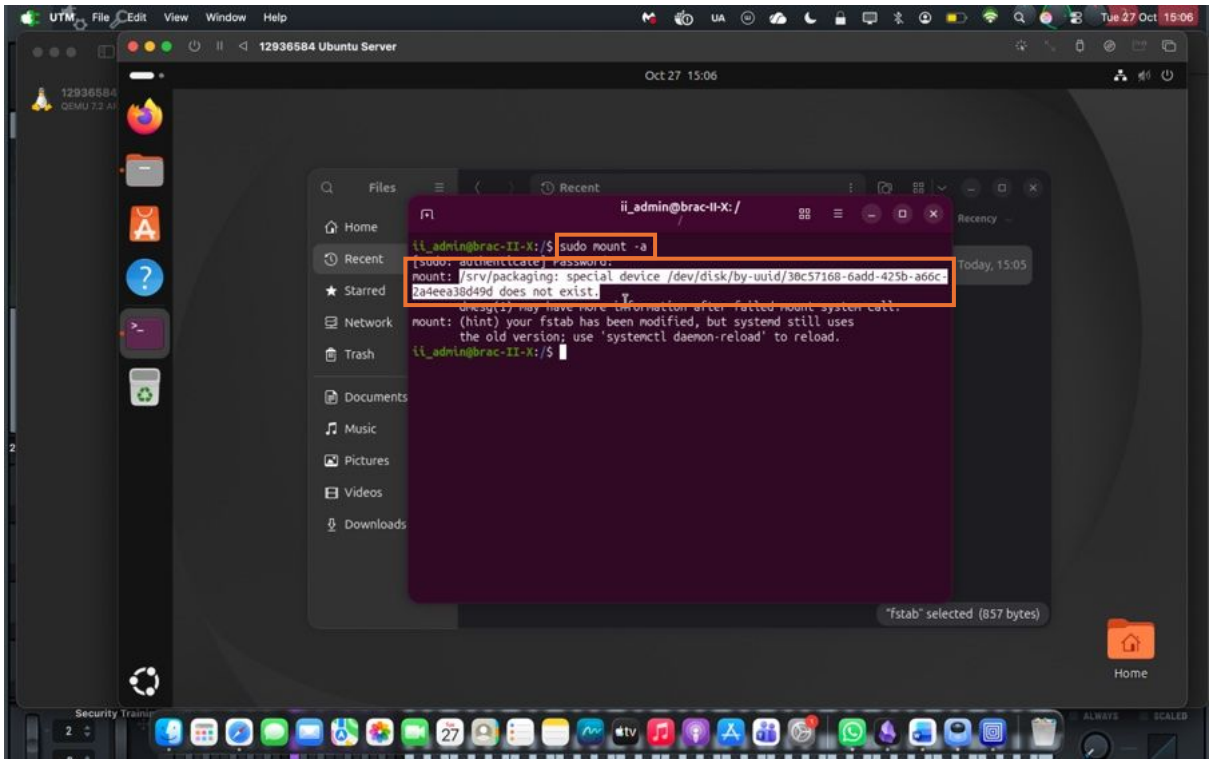
SN_evidence.txt successfully created and viewing content with cat.



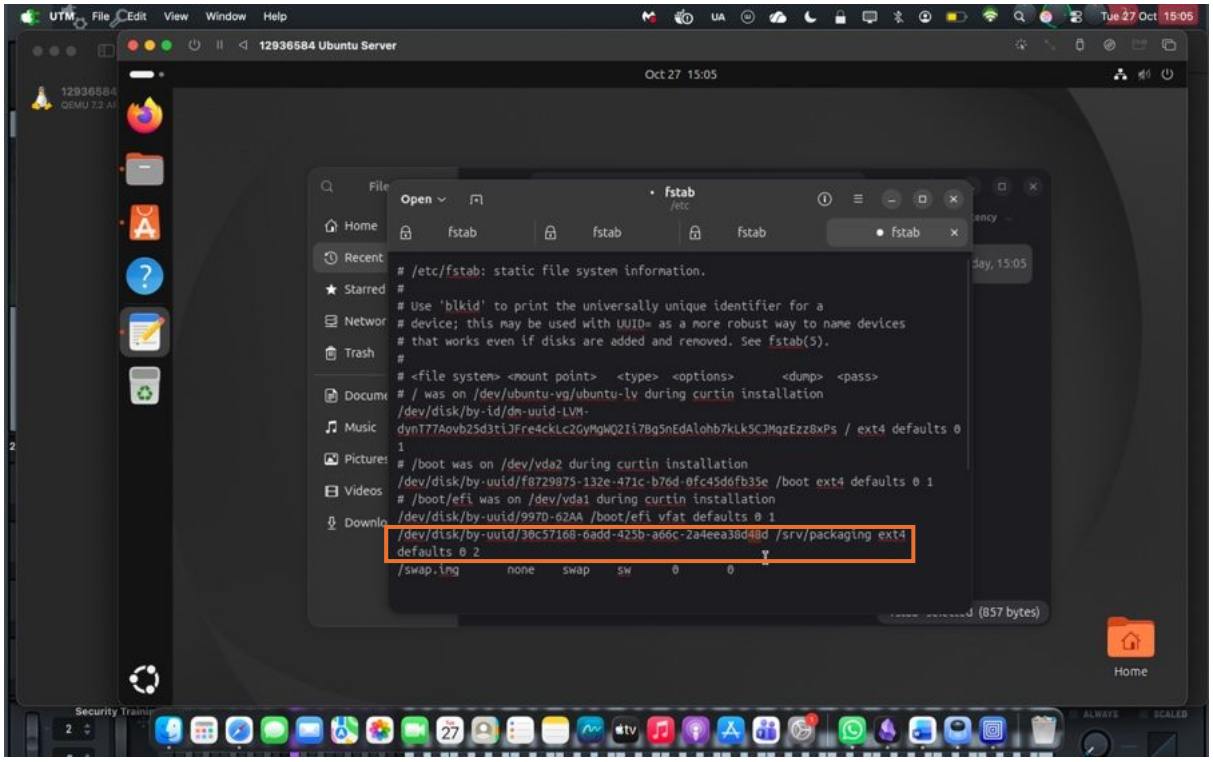
Creating a UUID error.



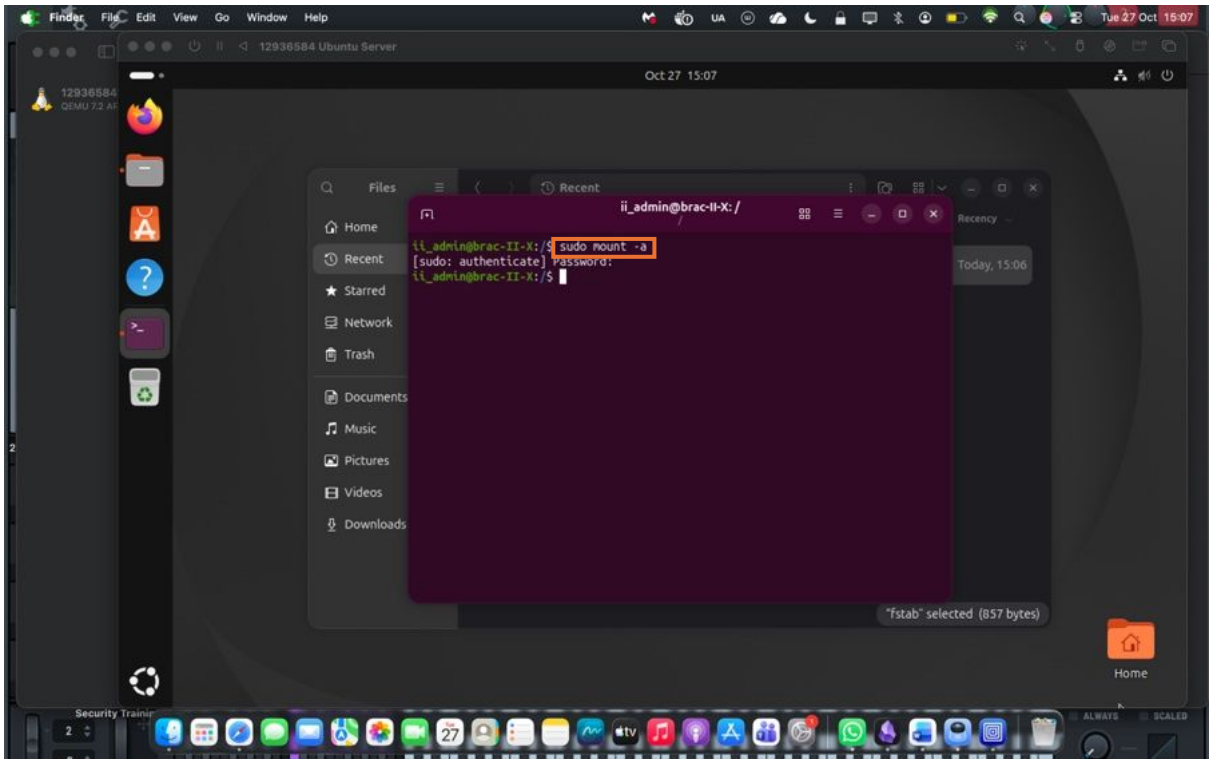
UUID error created.



Using sudo mount -a to view effects of created error.



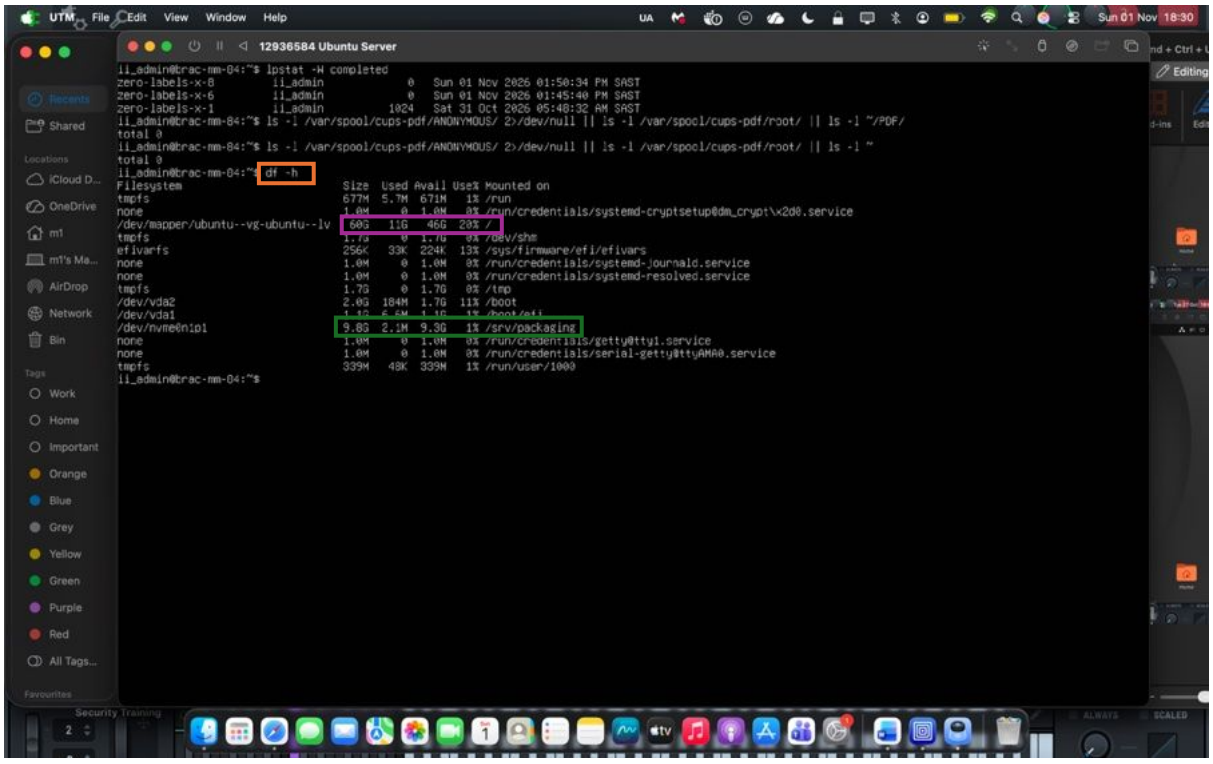
Repairing Error.



Error repaired no more error message.

1.4 Unvalidated fstab errors can lead to critical boot failures, potentially hanging the system or dropping it into an emergency shell. Using UUIDs is preferred as they consistently target the same data blocks, preventing dynamic drive snapping during reboots or hardware upgrades. UUIDs are persistent across reboots, avoiding disastrous overwrites and supporting safe multi-drive and hot-plug systems.

1.5

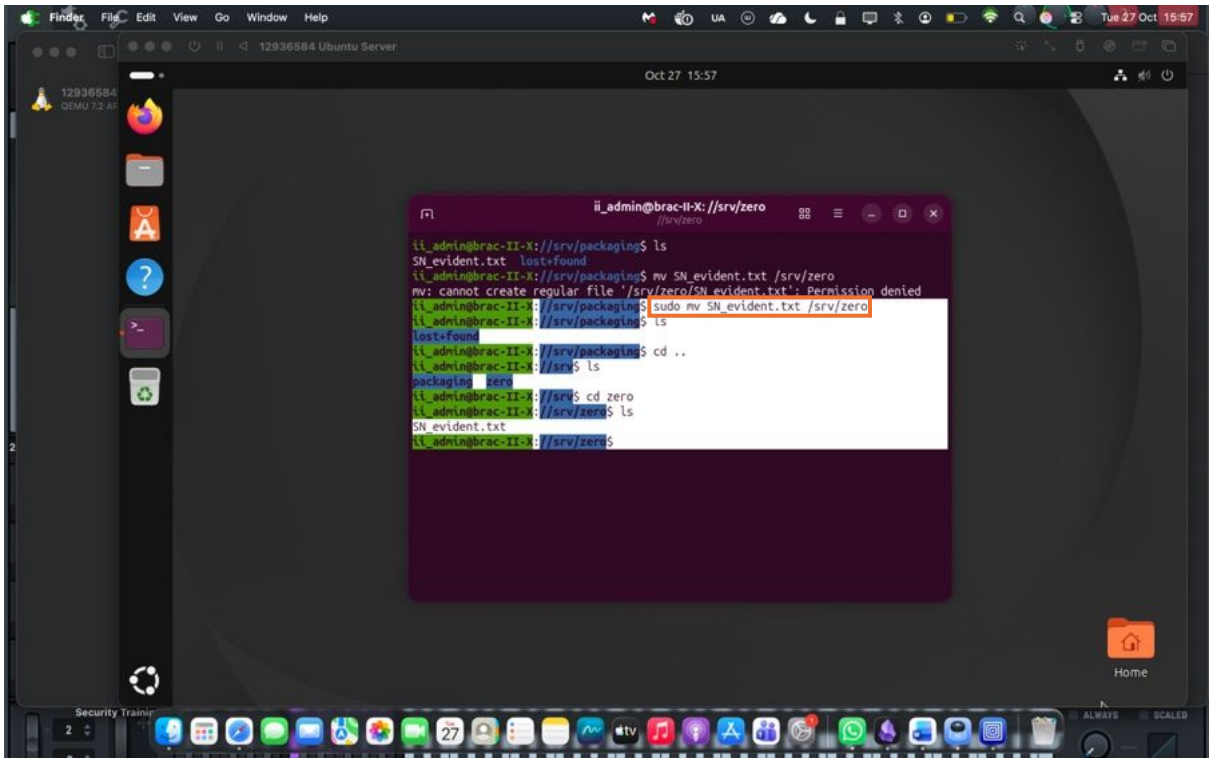


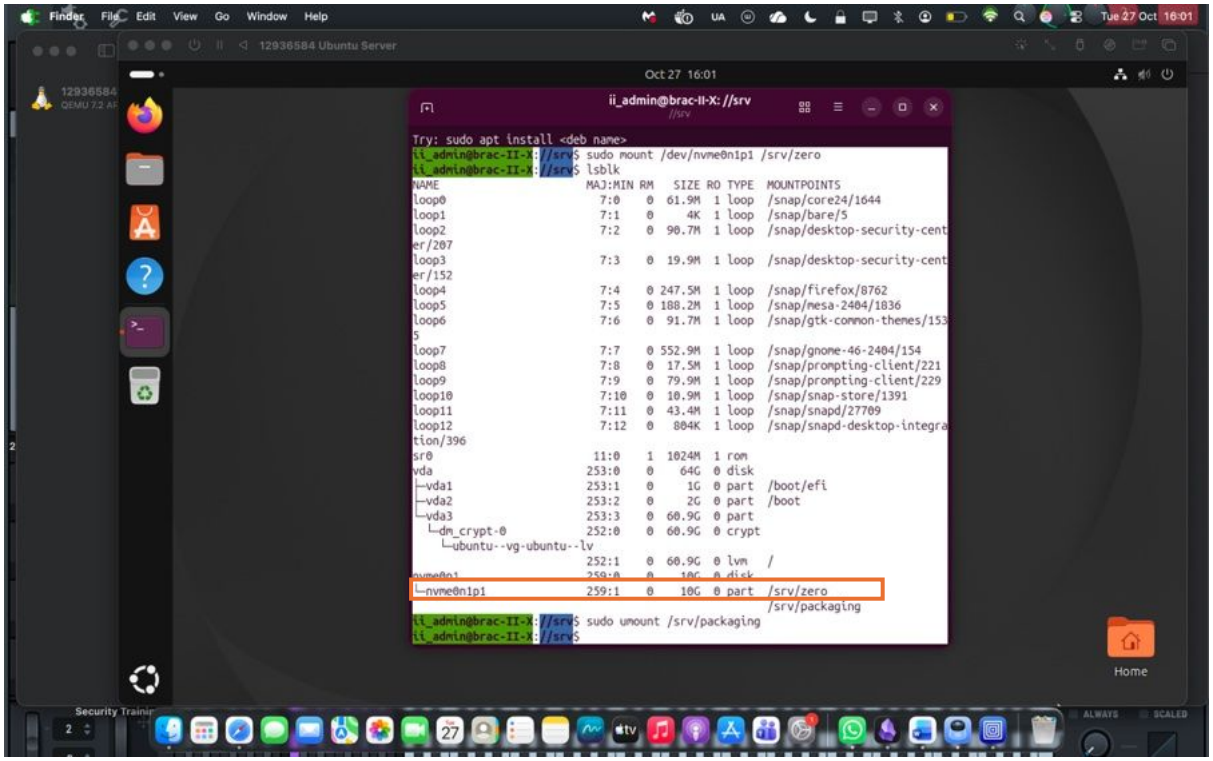
Above this I have performed some critical steps to ensure department data is separate from Operating System data using lsblk to identify the second disk, sudo fdisk /dev/nveme0n1 to partition the second

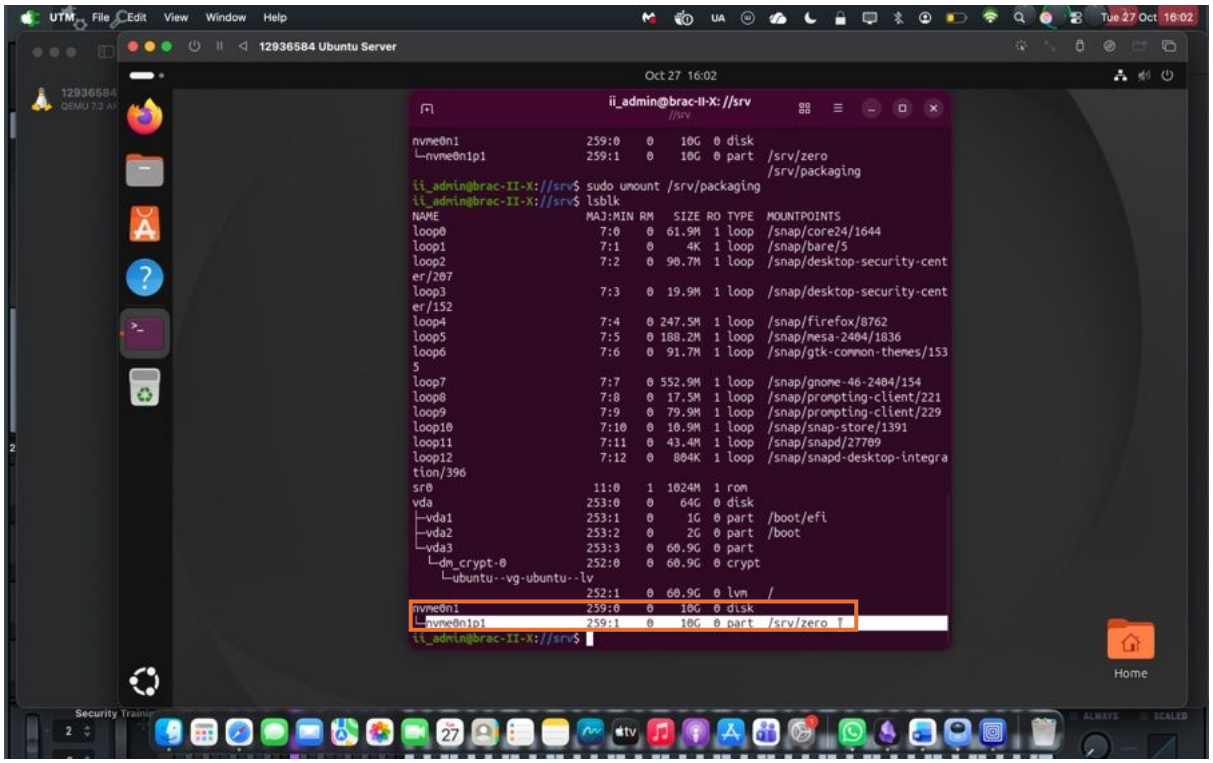
disk for department data, sudo mkfs.ex4 to format disk as ex4 format, sudo mkdir /srv/packaging and sudo mount /dev/nvme0n1 /srv/packaging to make directory and Mount to the created directory, sudo vi /etc/fstab to make Mount persistent adding to file system table fstab file /etc/fstab, sudo blkid /dev/nvme0n1 to find the UUIDs of the drive and finally (hostnamectl; date; blkid /dev/nvme0n1) sudo tee /srv/packaging/SN_evidence.txt > /dev/null and cat /srv/packaging /SN_evident.txt to create file with blkid, hostname, date and view it with cat.

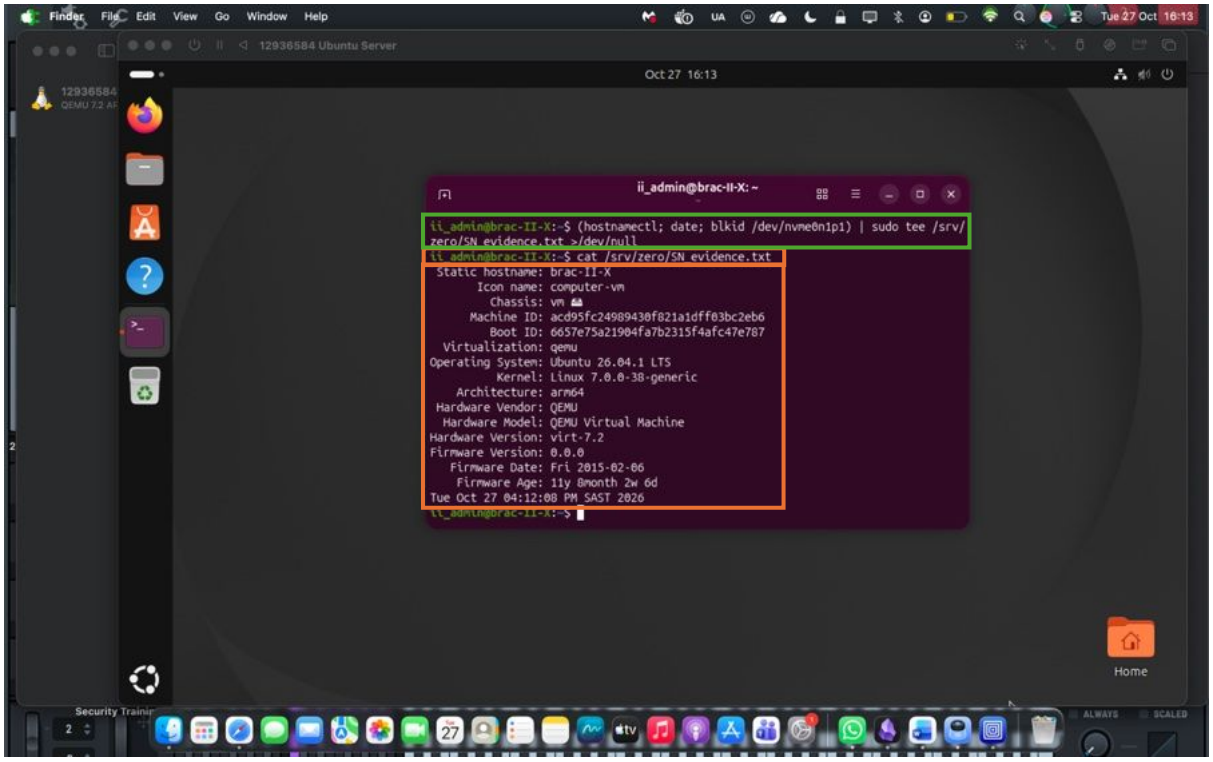
Separate file systems isolate disk space, preventing log file storms from crashing systems. Here, 9.3GB is isolated from the 46GB main disk, ensuring stability.

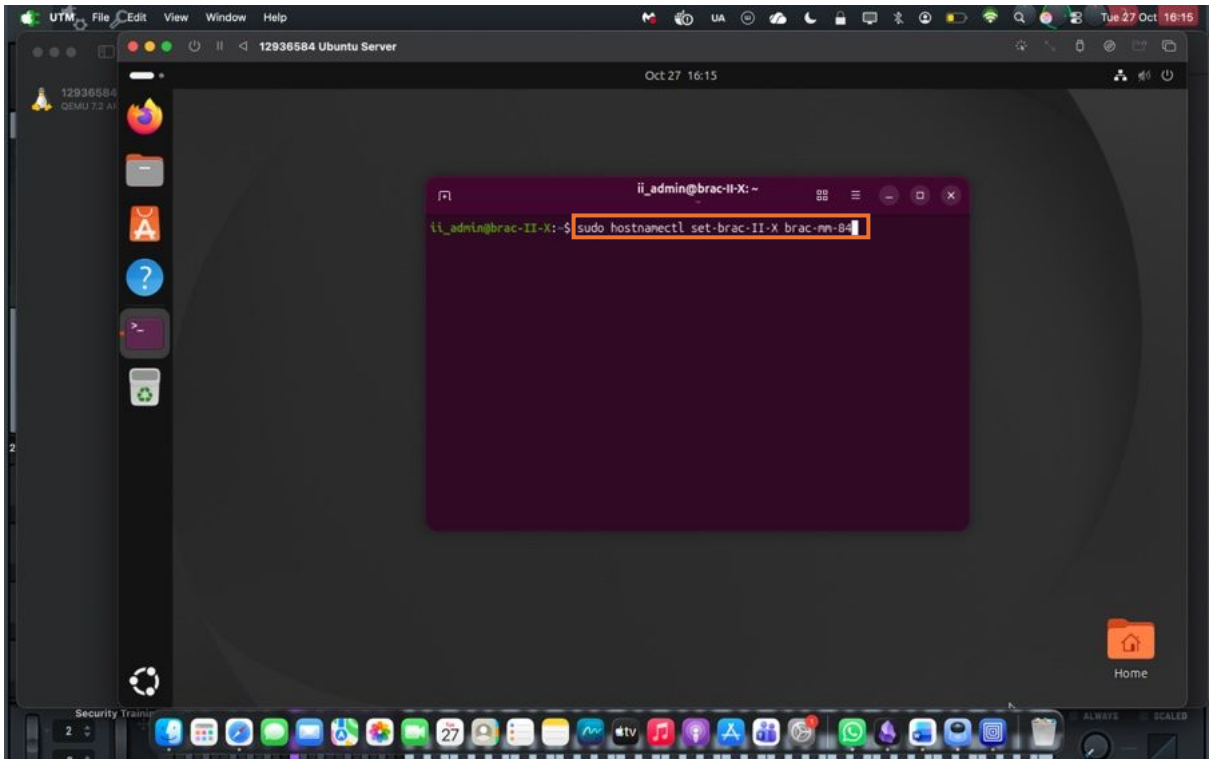
NOTE: In following I will be changing “packaging” directory to “Zero” and hostname from “brac-II-xx” to “brac-mm-84” applying PCVs (Personal Configuration Values).

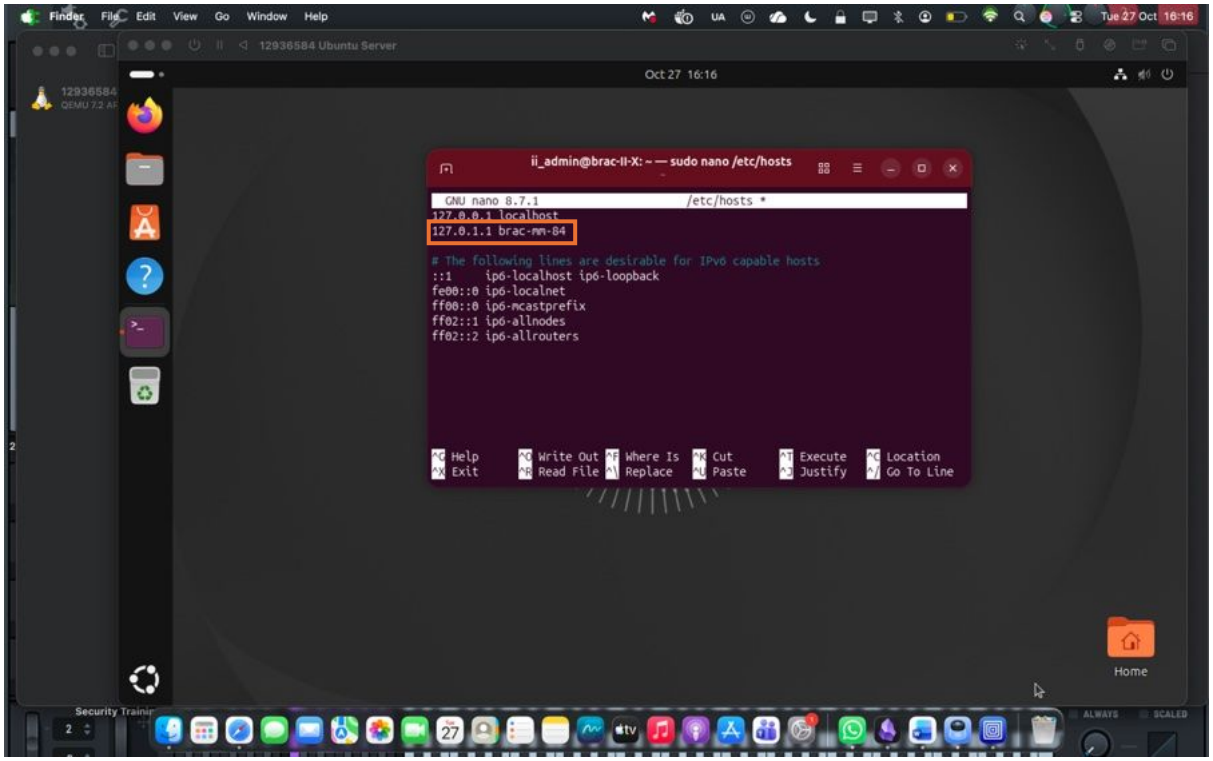



Finder File Edit View Go Window Help
Tue 27 Oct 16:01
12936584 Ubuntu Server
Oct 27 16:01
ii_admin@brac-II-X: //srv
Try: sudo apt install <deb name>
ii_admin@brac-II-X:/srv$ sudo mount /dev/nvme0n1p1 /srv/zero
ii_admin@brac-II-X:/srv$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
loop0 7:0 0 61.9M 1 loop /snap/core24/1644
loop1 7:1 0 4K 1 loop /snap/bare/5
loop2 7:2 0 90.7M 1 loop /snap/desktop-security-cent
er/207
loop3 7:3 0 19.9M 1 loop /snap/desktop-security-cent
er/152
loop4 7:4 0 247.5M 1 loop /snap/firefox/8762
loop5 7:5 0 188.2M 1 loop /snap/mesa-2404/1836
loop6 7:6 0 91.7M 1 loop /snap/gtk-common-themes/153
5
loop7 7:7 0 552.9M 1 loop /snap/gnome-46-2404/154
loop8 7:8 0 17.5M 1 loop /snap/prompting-client/221
loop9 7:9 0 79.9M 1 loop /snap/prompting-client/229
loop10 7:10 0 10.9M 1 loop /snap/snap-store/1391
loop11 7:11 0 43.4M 1 loop /snap/snapd/27709
loop12 7:12 0 804K 1 loop /snap/snapd-desktop-integra
tion/396
sr0 11:0 1 1024M 1 rom
vda 253:0 0 64G 0 disk
├─vda1 253:1 0 1G 0 part /boot/efi
├─vda2 253:2 0 2G 0 part /boot
└─vda3 253:3 0 60.9G 0 part
└─dm_crypt-0 252:0 0 60.9G 0 crypt
└─ubuntu--vg-ubuntu--lv
252:1 0 60.9G 0 lvm /
nvme0n1 259:0 0 10G 0 disk
└─nvme0n1p1 259:1 0 10G 0 part /srv/zero
/srv/packaging
ii_admin@brac-II-X:/srv$ sudo umount /srv/packaging
ii_admin@brac-II-X:/srv$
Home
Security Trainir

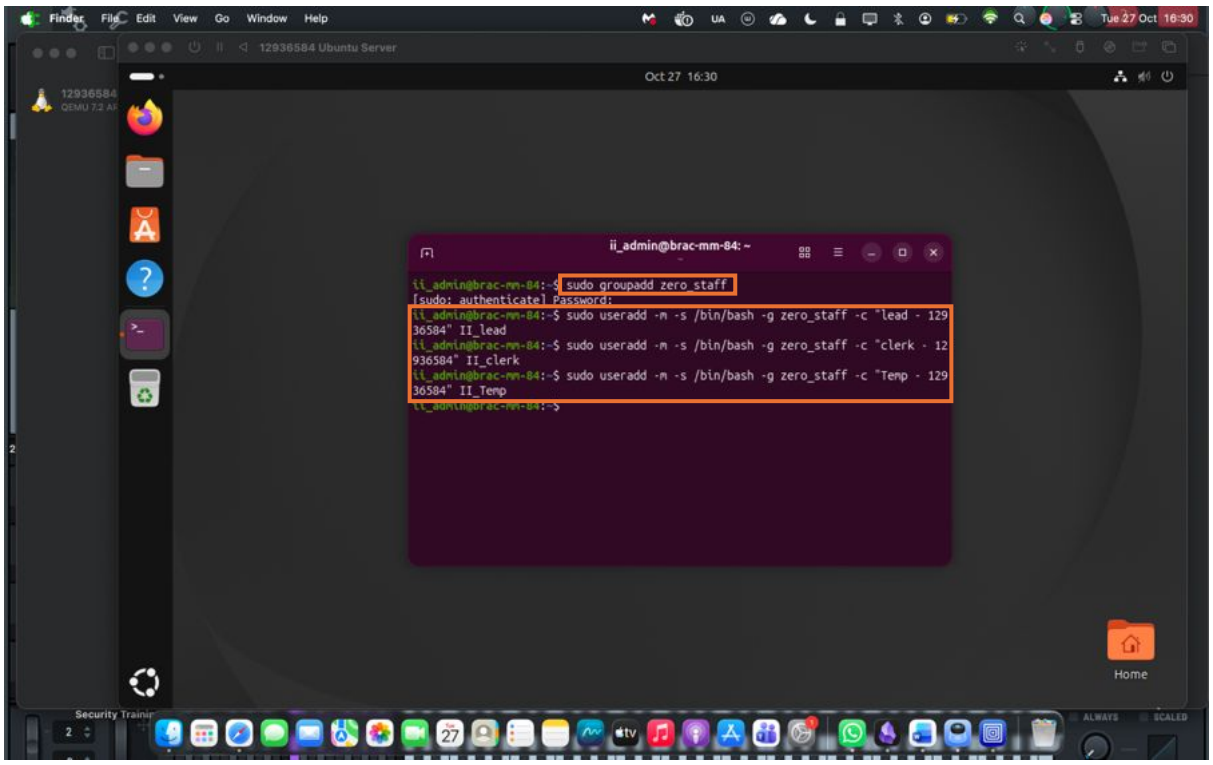

UTM File Edit View Window Help
Tue 27 Oct 16:02
12936584 Ubuntu Server
Oct 27 16:02
ii_admin@brac-II-X: //srv
nvme0n1 259:0 0 10G 0 disk
└─nvme0n1p1 259:1 0 10G 0 part /srv/zero
/srv/packaging
ii_admin@brac-II-X:/srv$ sudo umount /srv/packaging
ii_admin@brac-II-X:/srv$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
loop0 7:0 0 61.9M 1 loop /snap/core24/1644
loop1 7:1 0 4K 1 loop /snap/bare/5
loop2 7:2 0 90.7M 1 loop /snap/desktop-security-cent
er/207
loop3 7:3 0 19.9M 1 loop /snap/desktop-security-cent
er/152
loop4 7:4 0 247.5M 1 loop /snap/firefox/8762
loop5 7:5 0 188.2M 1 loop /snap/mesa-2404/1836
loop6 7:6 0 91.7M 1 loop /snap/gtk-common-themes/153
5
loop7 7:7 0 552.9M 1 loop /snap/gnome-46-2404/154
loop8 7:8 0 17.5M 1 loop /snap/prompting-client/221
loop9 7:9 0 79.9M 1 loop /snap/prompting-client/229
loop10 7:10 0 10.9M 1 loop /snap/snap-store/1391
loop11 7:11 0 43.4M 1 loop /snap/snapd/27709
loop12 7:12 0 804K 1 loop /snap/snapd-desktop-integra
tion/396
sr0 11:0 1 1024M 1 rom
vda 253:0 0 64G 0 disk
├─vda1 253:1 0 1G 0 part /boot/efi
├─vda2 253:2 0 2G 0 part /boot
└─vda3 253:3 0 60.9G 0 part
└─dm_crypt-0 252:0 0 60.9G 0 crypt
└─ubuntu--vg-ubuntu--lv
252:1 0 60.9G 0 lvm /
nvme0n1 259:0 0 10G 0 disk
└─nvme0n1p1 259:1 0 10G 0 part /srv/zero
ii_admin@brac-II-X:/srv$
Home
Security Trainir


Finder File Edit View Go Window Help
Tue 27 Oct 16:13
12936584 Ubuntu Server
Oct 27 16:13
ii_admin@brac-II-X: ~
ii_admin@brac-II-X:~$ (hostnamectl; date; blkid /dev/nvme0n1p1) | sudo tee /srv/zero/SN_evidence.txt >/dev/null
ii_admin@brac-II-X:~$ cat /srv/zero/SN_evidence.txt
Static hostname: brac-II-X
Icon name: computer-vm
Chassis: vm
Machine ID: acd95fc24989430f821a1dff03bc2eb6
Boot ID: 6657e75a21904fa7b2315f4afc47e787
Virtualization: qemu
Operating System: Ubuntu 26.04.1 LTS
Kernel: Linux 7.0.0-38-generic
Architecture: arm64
Hardware Vendor: QEMU
Hardware Model: QEMU Virtual Machine
Hardware Version: virt-7.2
Firmware Version: 0.0.0
Firmware Date: Fri 2015-02-06
Firmware Age: 11y 8month 2w 6d
Tue Oct 27 04:12:08 PM SAST 2026
Home

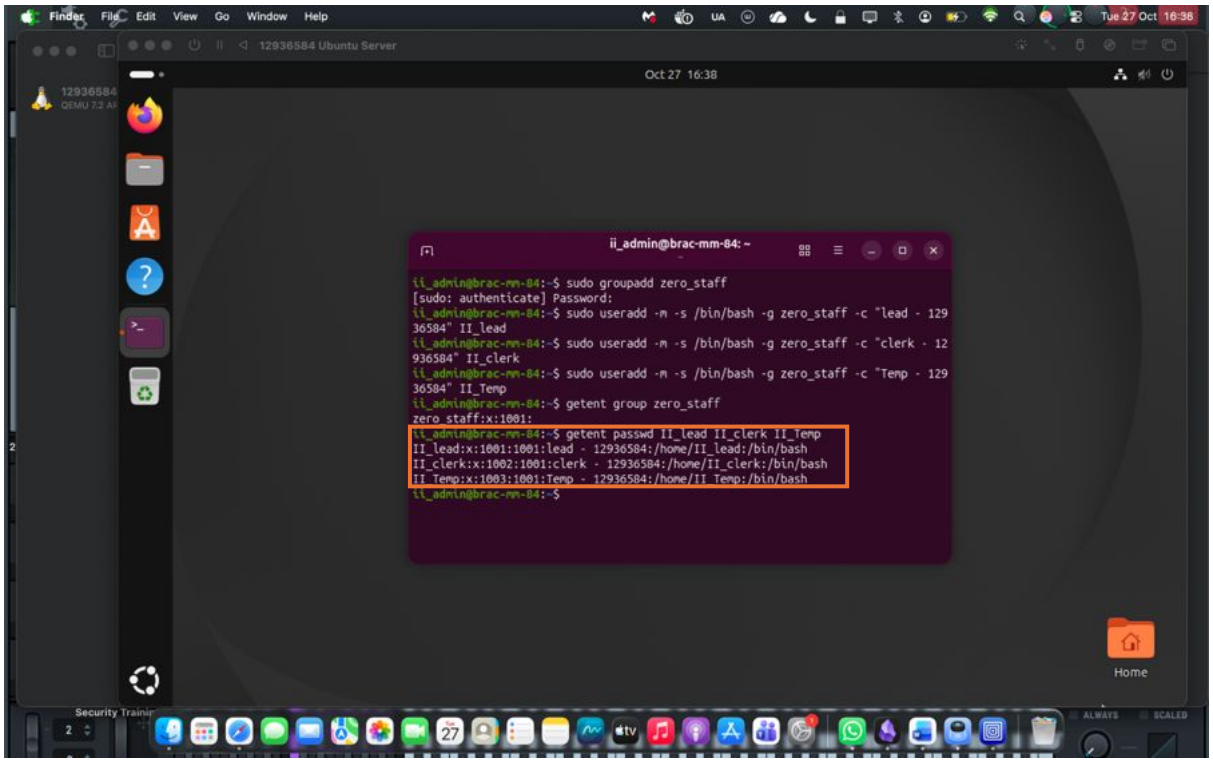

UTM File Edit View Window Help
Tue 27 Oct 16:15
12936584 Ubuntu Server
Oct 27 16:15
ii_admin@brac-II-X: ~
ii_admin@brac-II-X:~$ sudo hostnamectl set-brac-II-X brac-mm-84
Home


Finder File Edit View Go Window Help
Tue 27 Oct 16:16
12936584 Ubuntu Server
Oct 27 16:16
ii_admin@brac-II-X: ~ — sudo nano /etc/hosts
GNU nano 8.7.1 /etc/hosts *
127.0.0.1 localhost
127.0.1.1 brac-mm-84
The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line
Home
Security Trainir

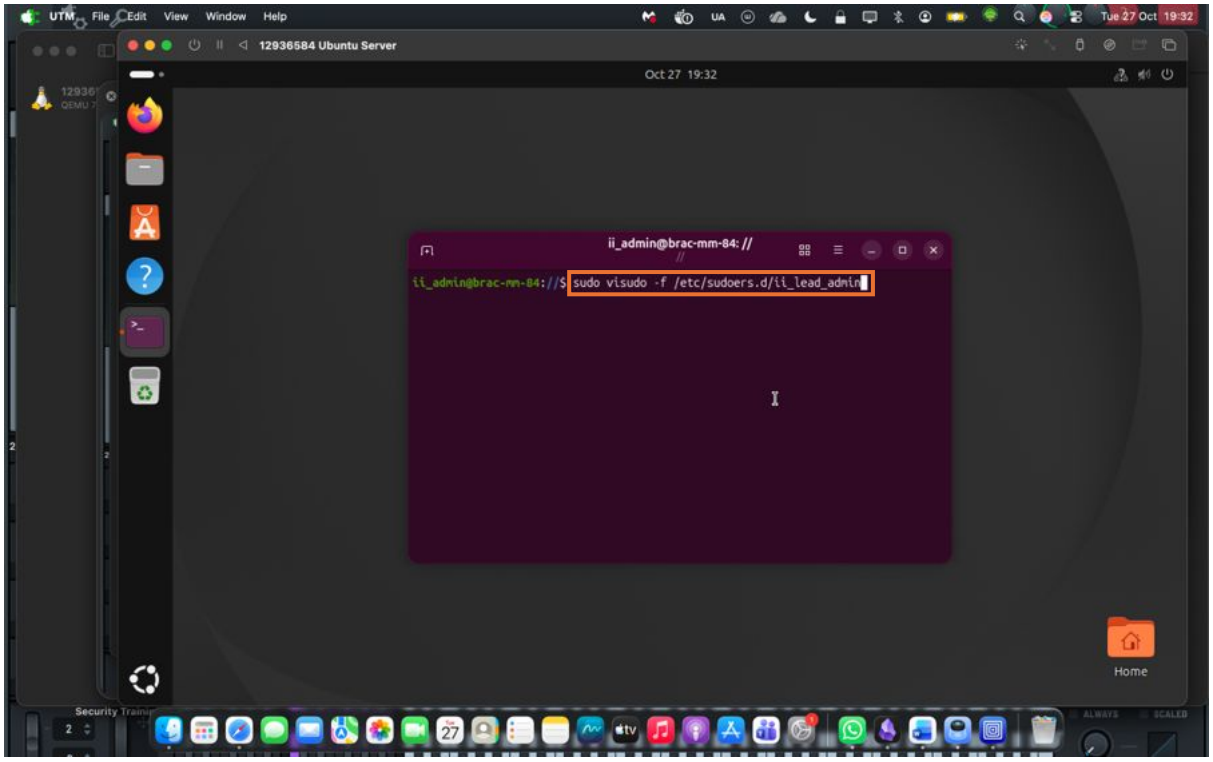
Question 2



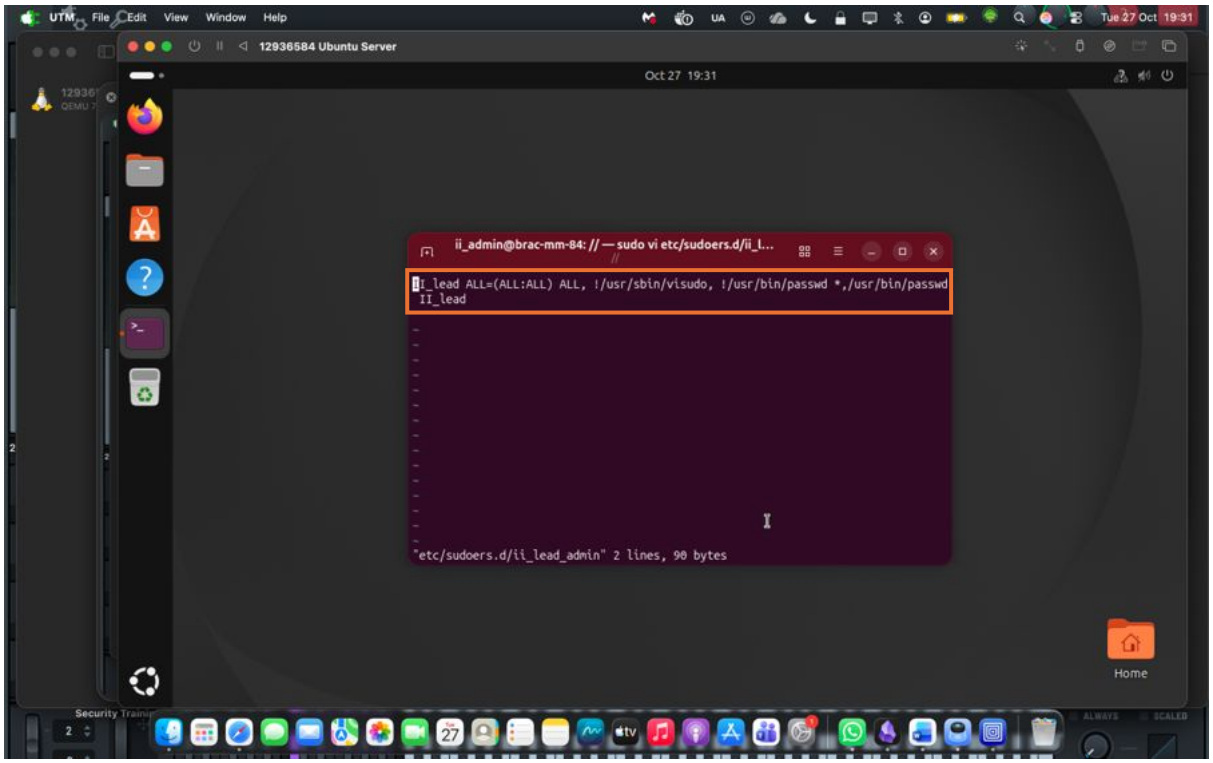
Creating Staff group and user accounts to staff group(-g). Using groupadd and useradd, bash as the login shell(-s) and comment field (-c) with persons role and my Student Number. (-m) force creation of the users home directory.



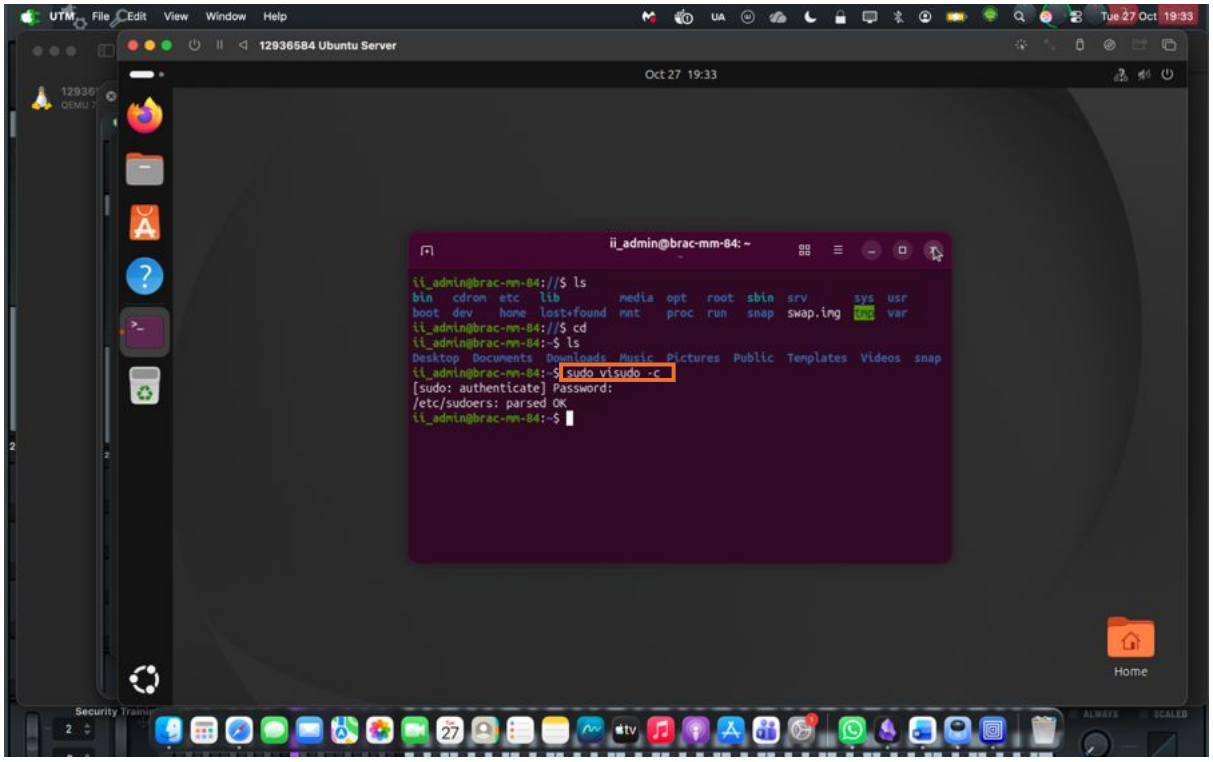
Proving results with getent group(for group) and getent passwd(for users).



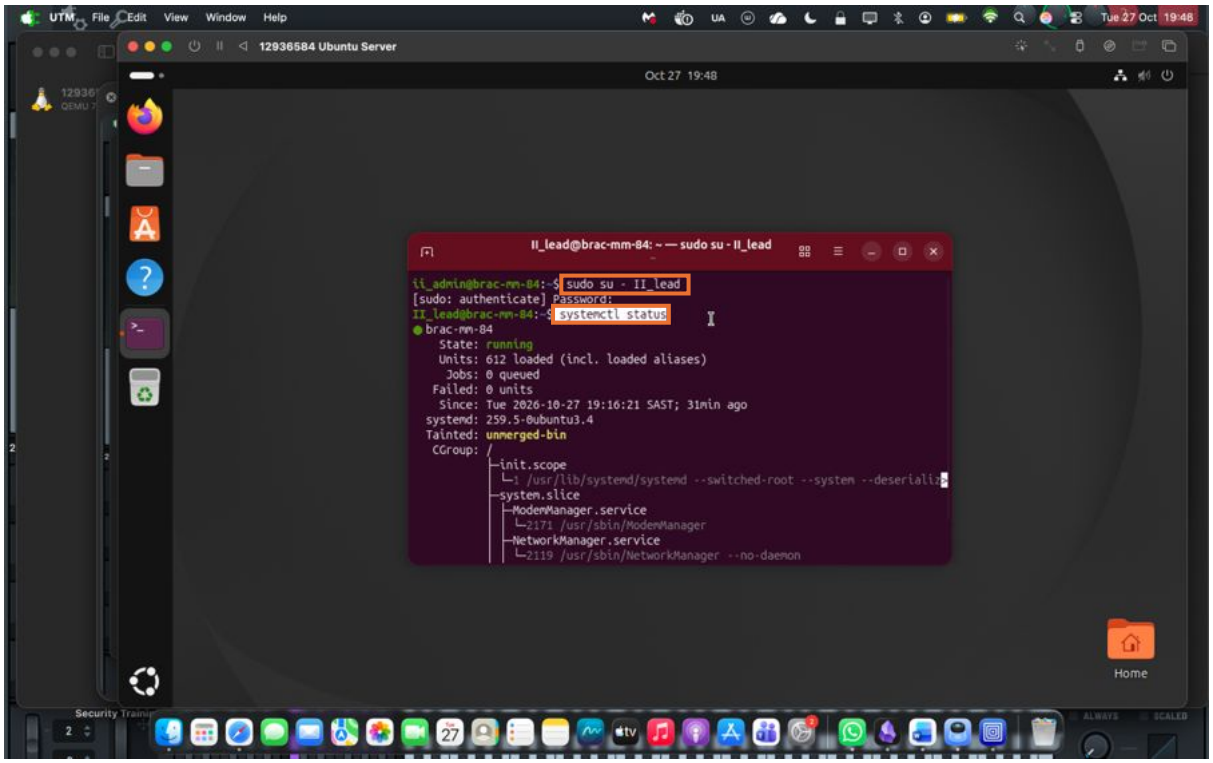
Creating Drop-in file with visudo -f under etc/sudoers.d for ii_lead.



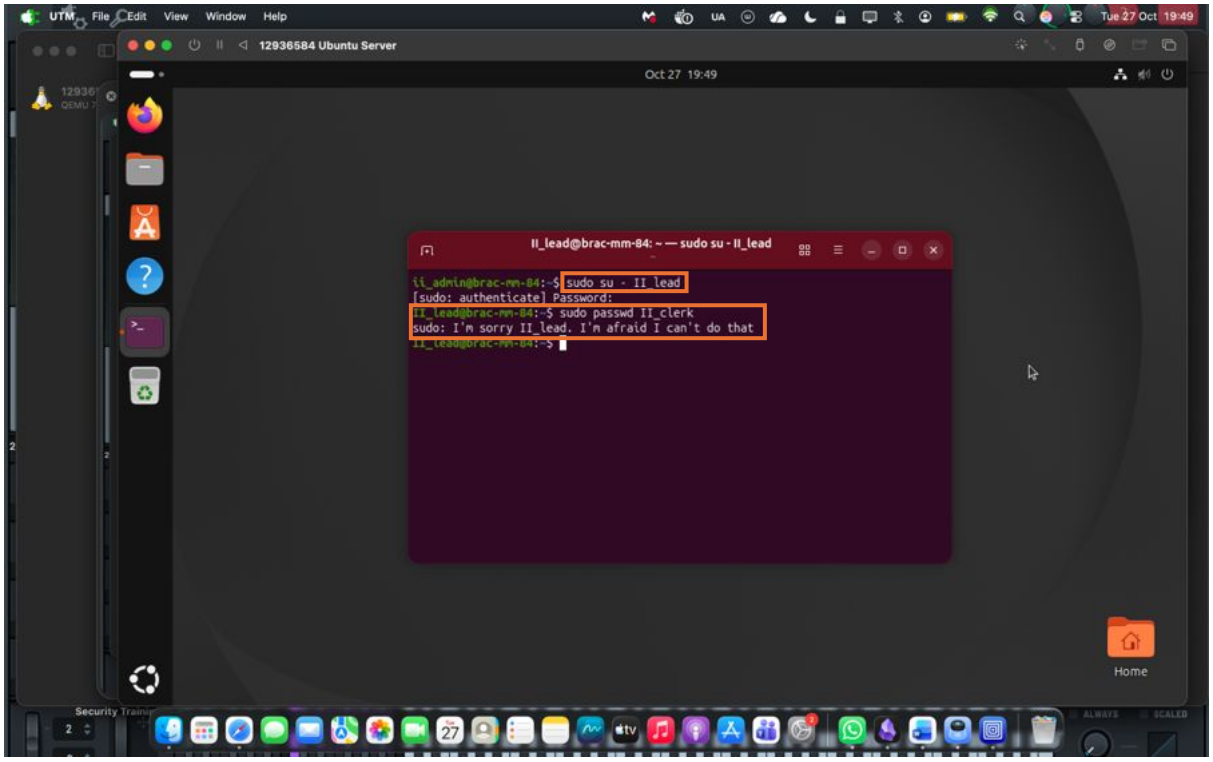
Adding Configuration Rule.



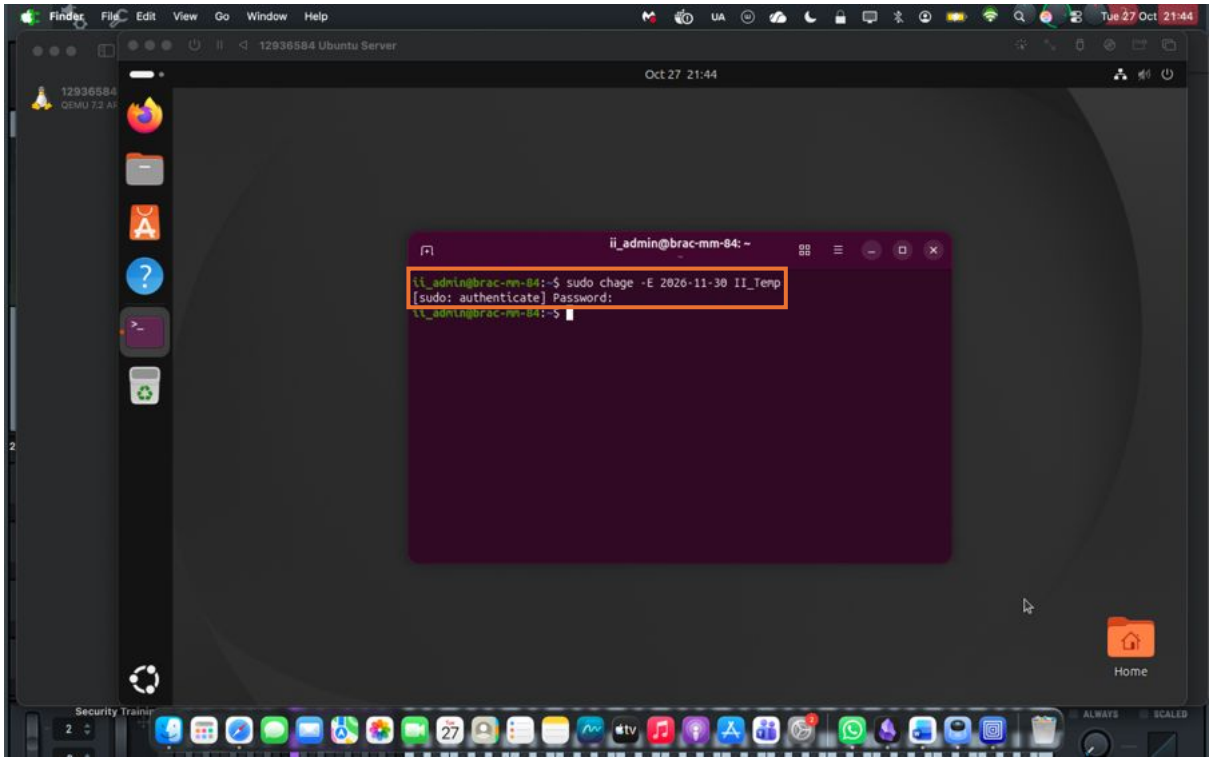
Verifying Configuration with visudo -c.



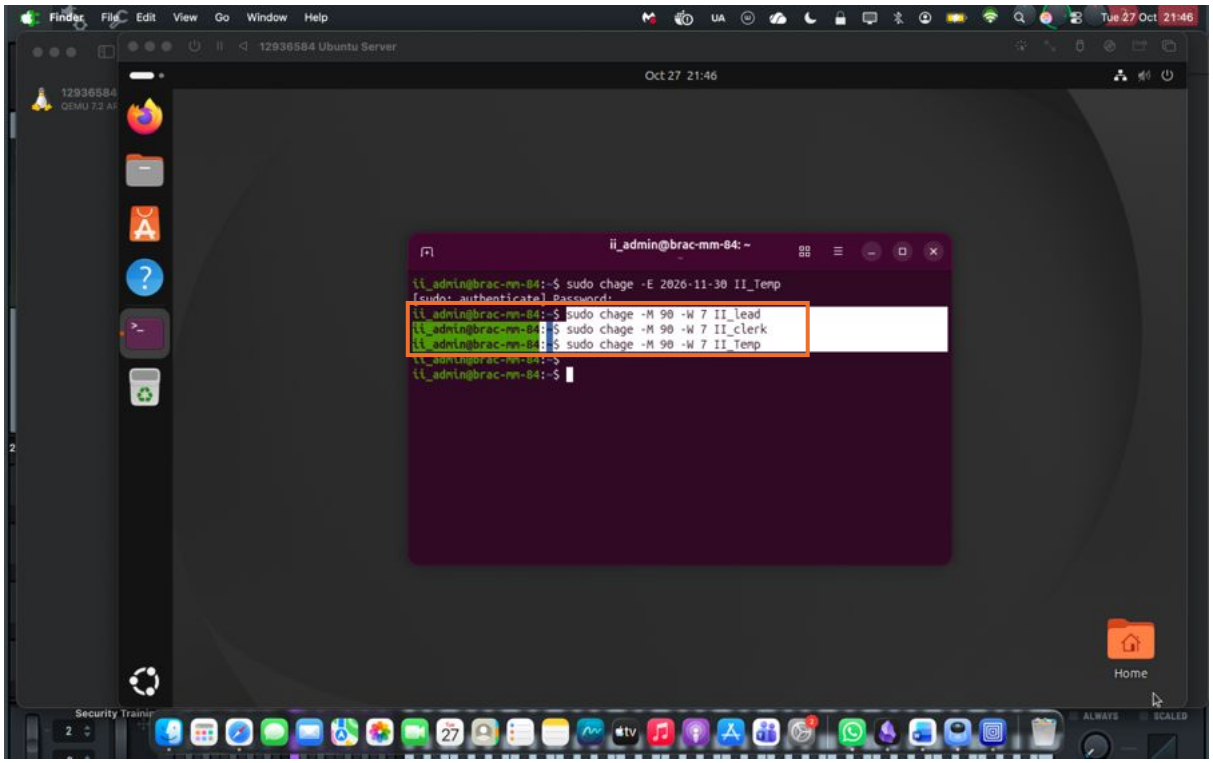
Demonstrating succeeding permitted sudo command as II_lead.



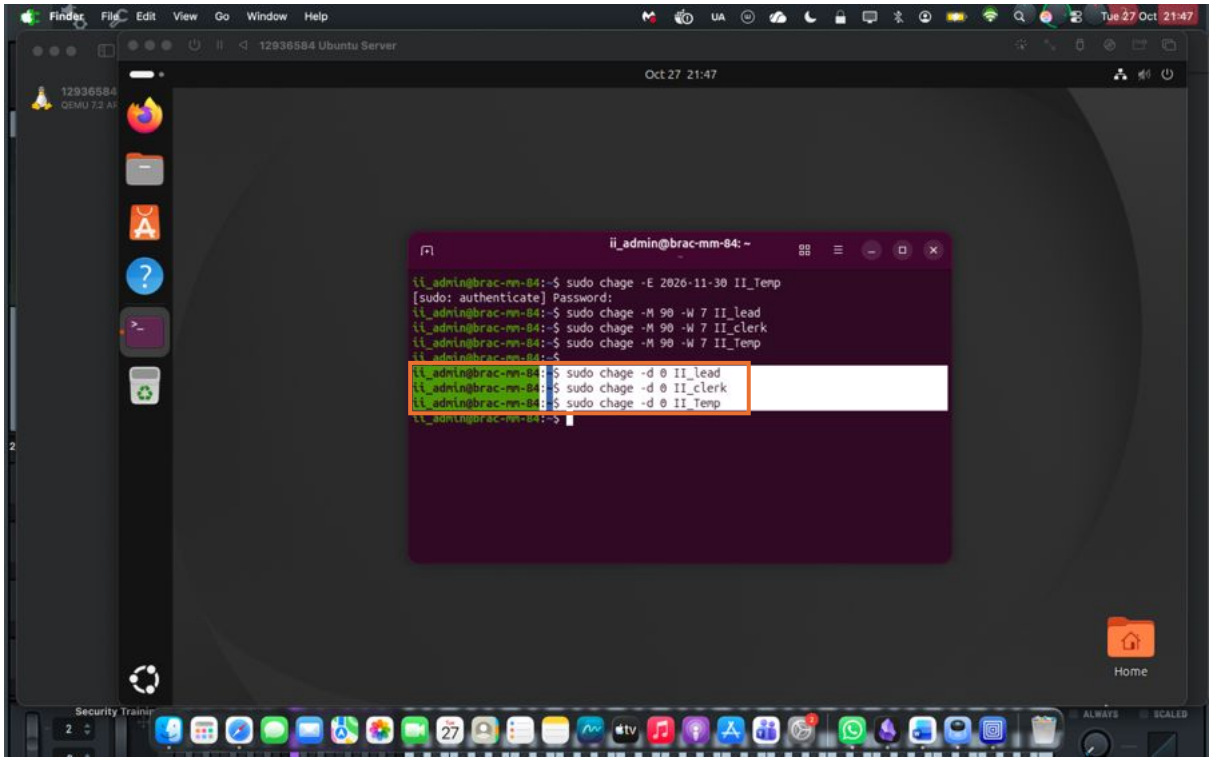
Demonstrating refusing forbidden sudo command as II_lead.



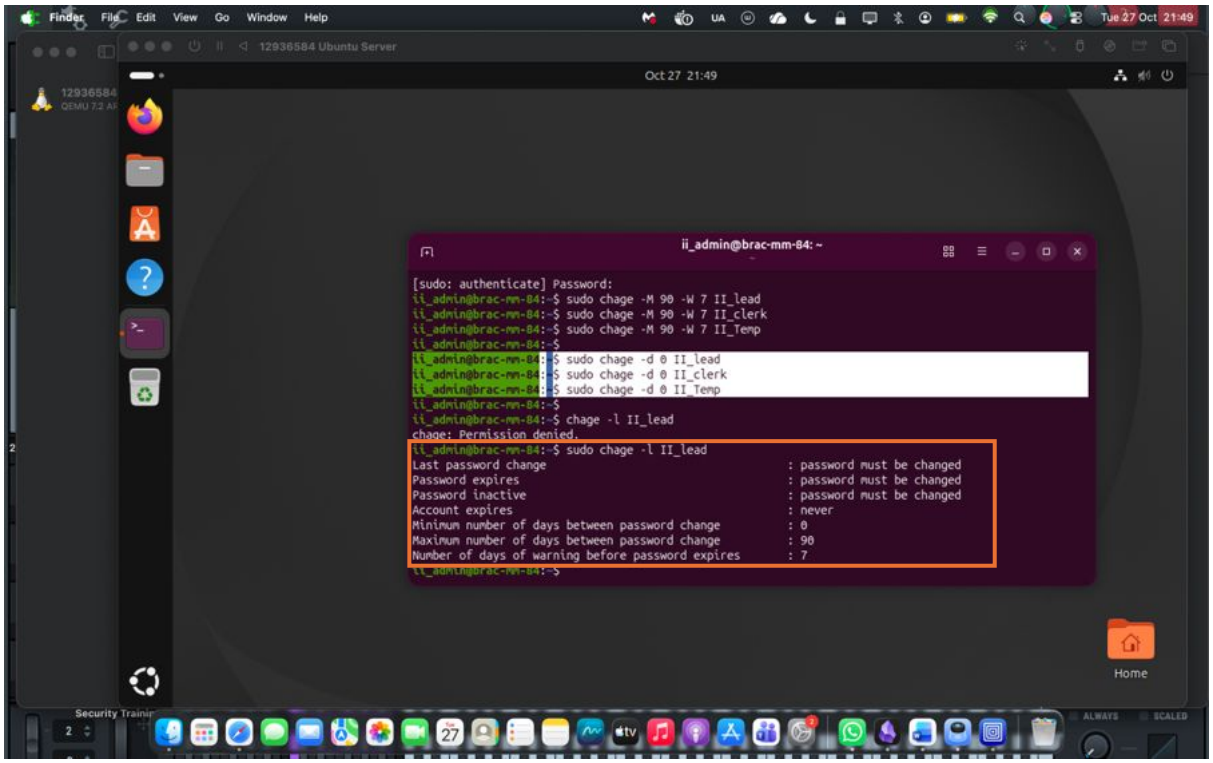
Setting account expiration date for II_temp.



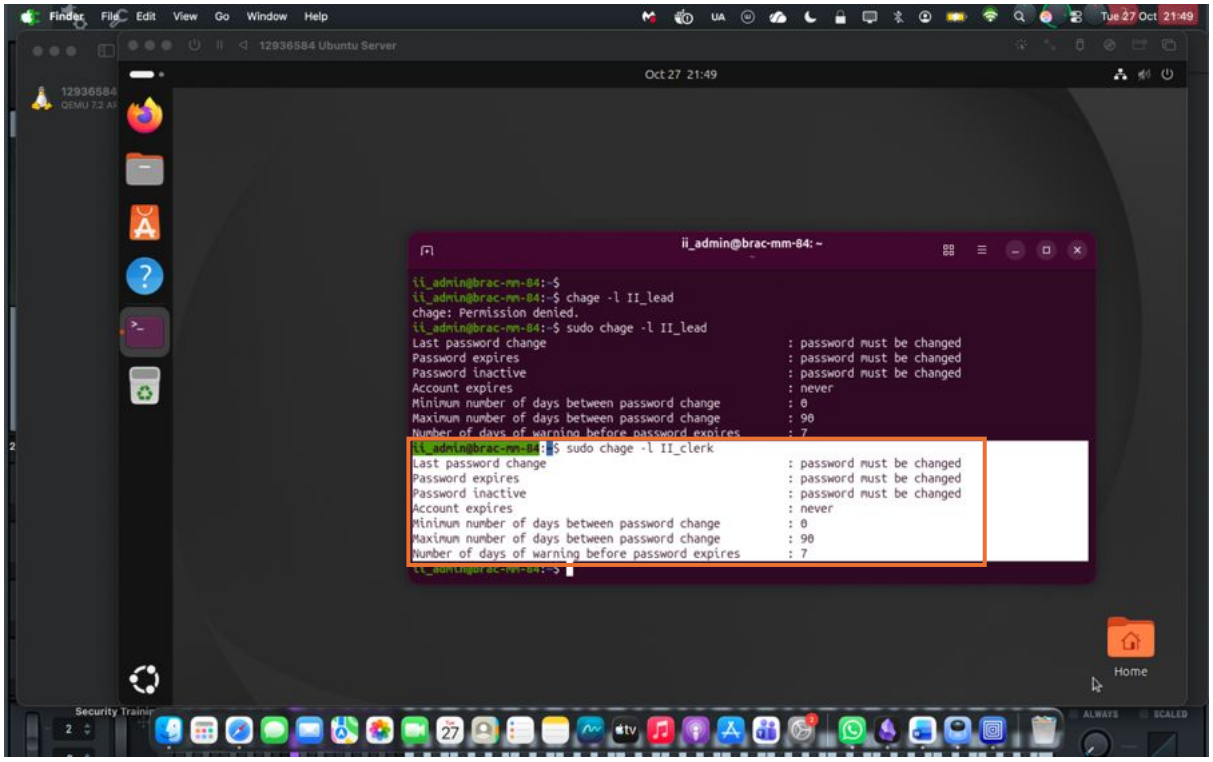
Configuring 90-day maximum password age and 7-day warning period for all three accounts.



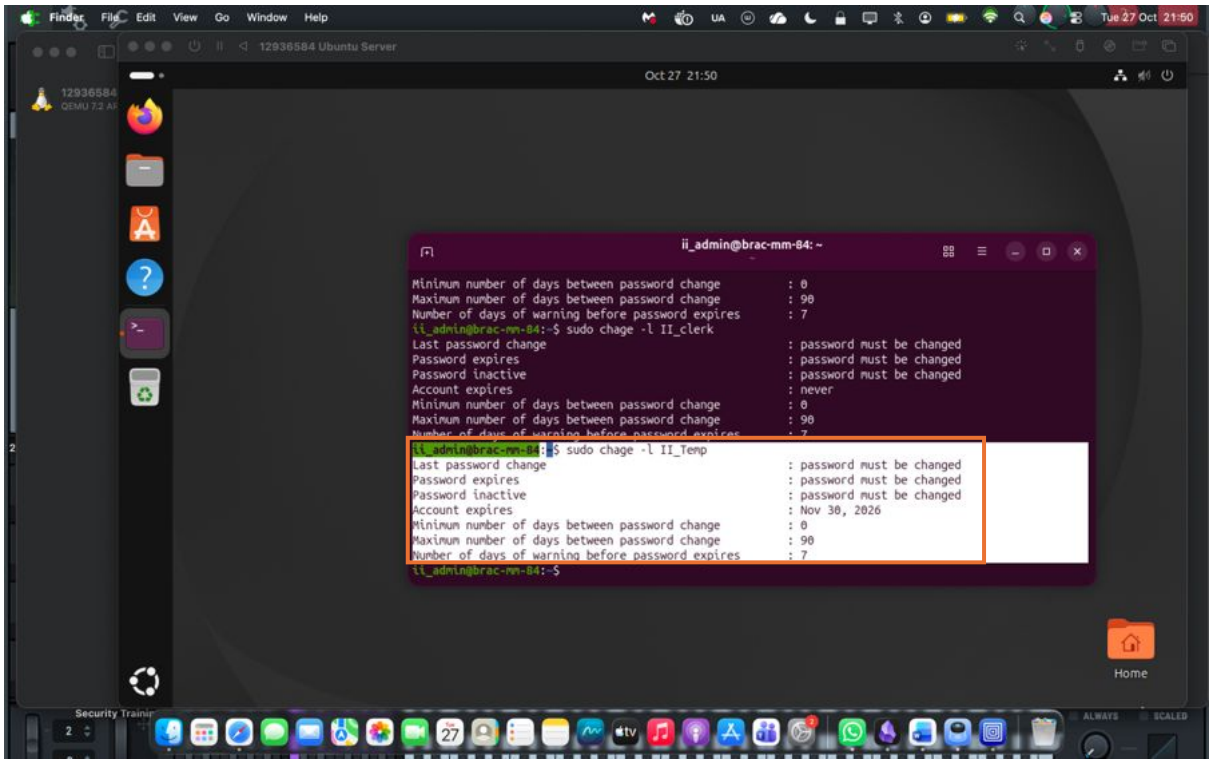
Forcing Password change at first login (set last password change to 0).



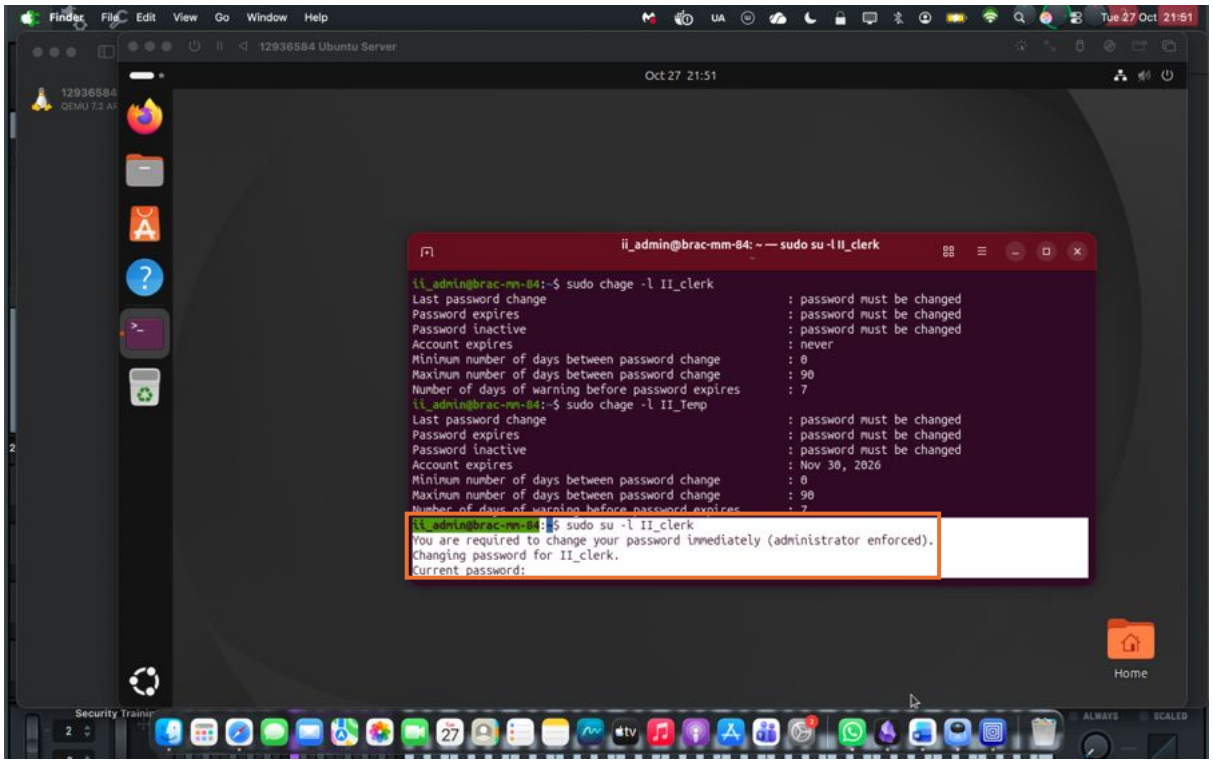
Proving the policy with chage -l for II_lead.



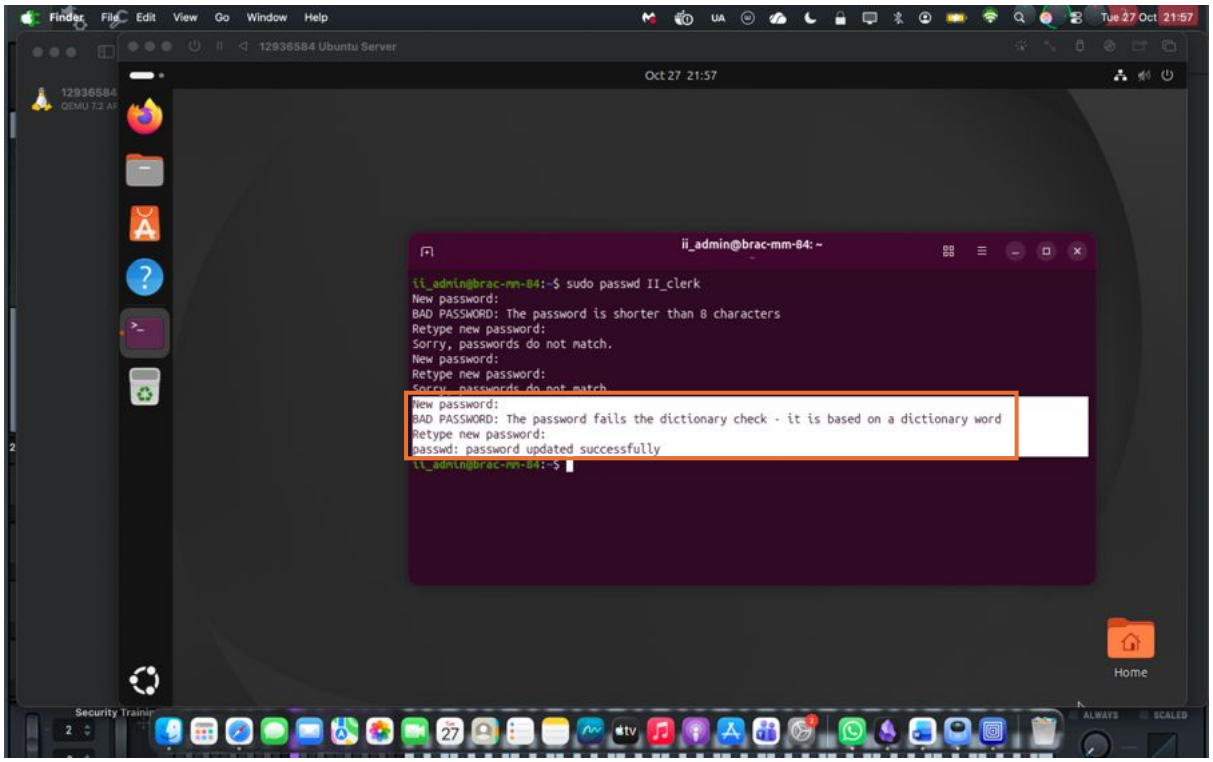
Proving the policy with chage -l for II_clerk.



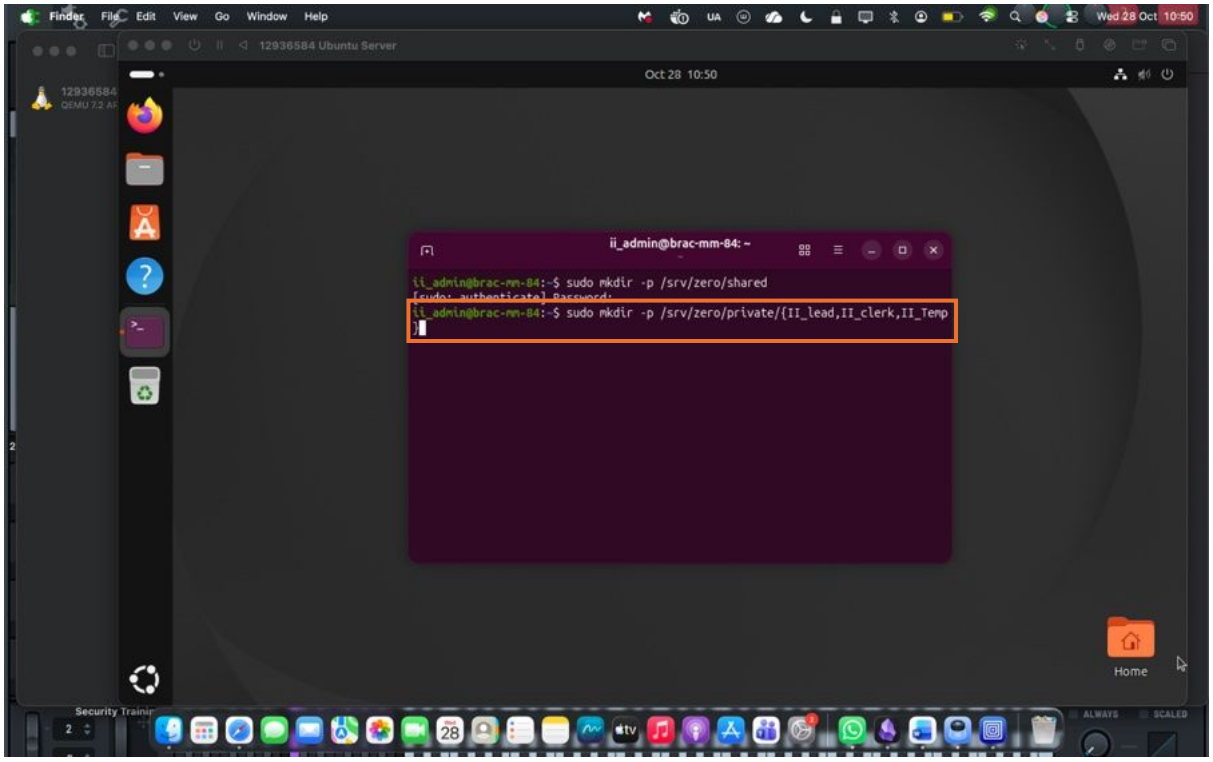
Proving the policy with chage -l for II_Temp.



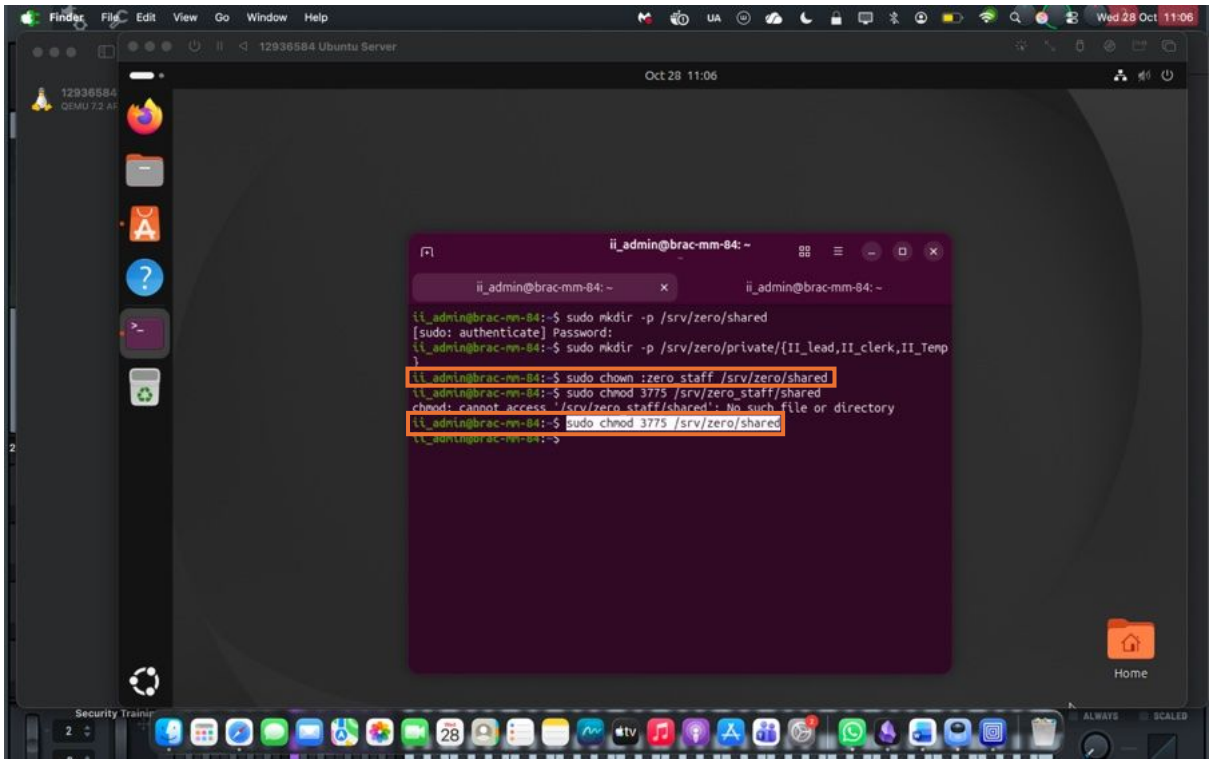
Demonstrating forced password change for II_clerk.



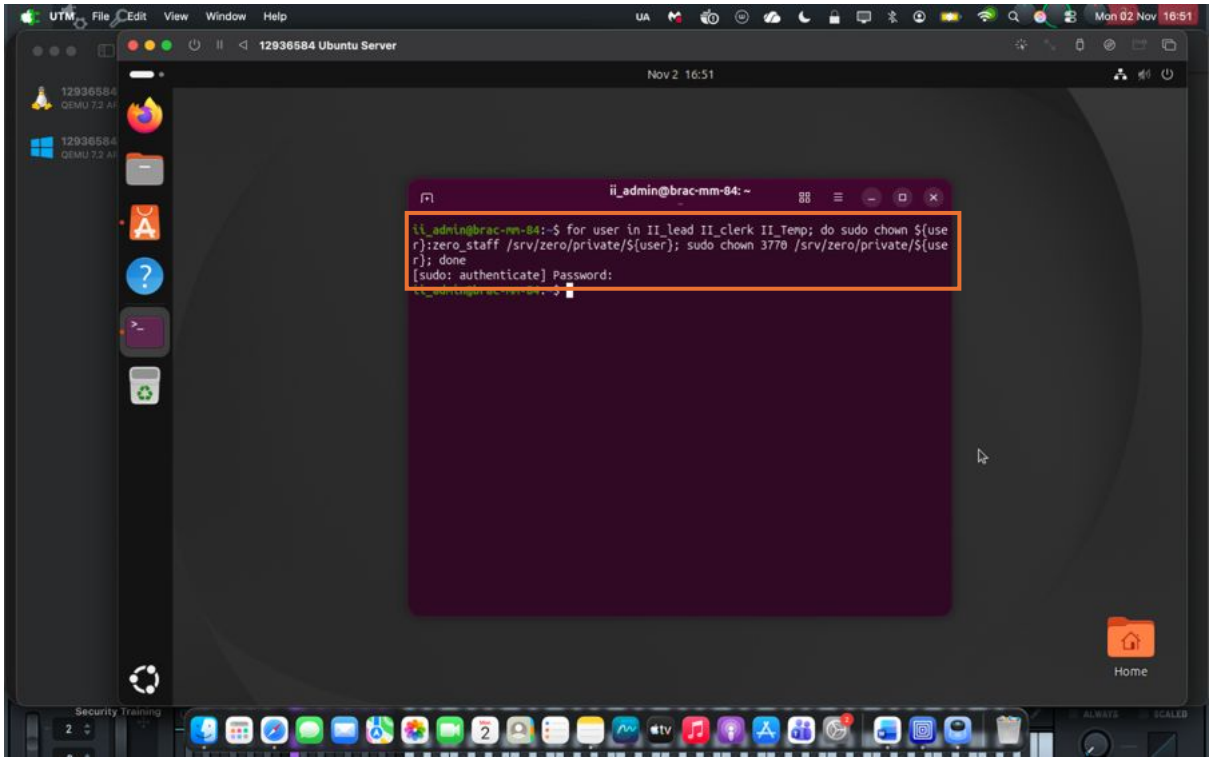
Setting initial placeholder password for II_clerk.



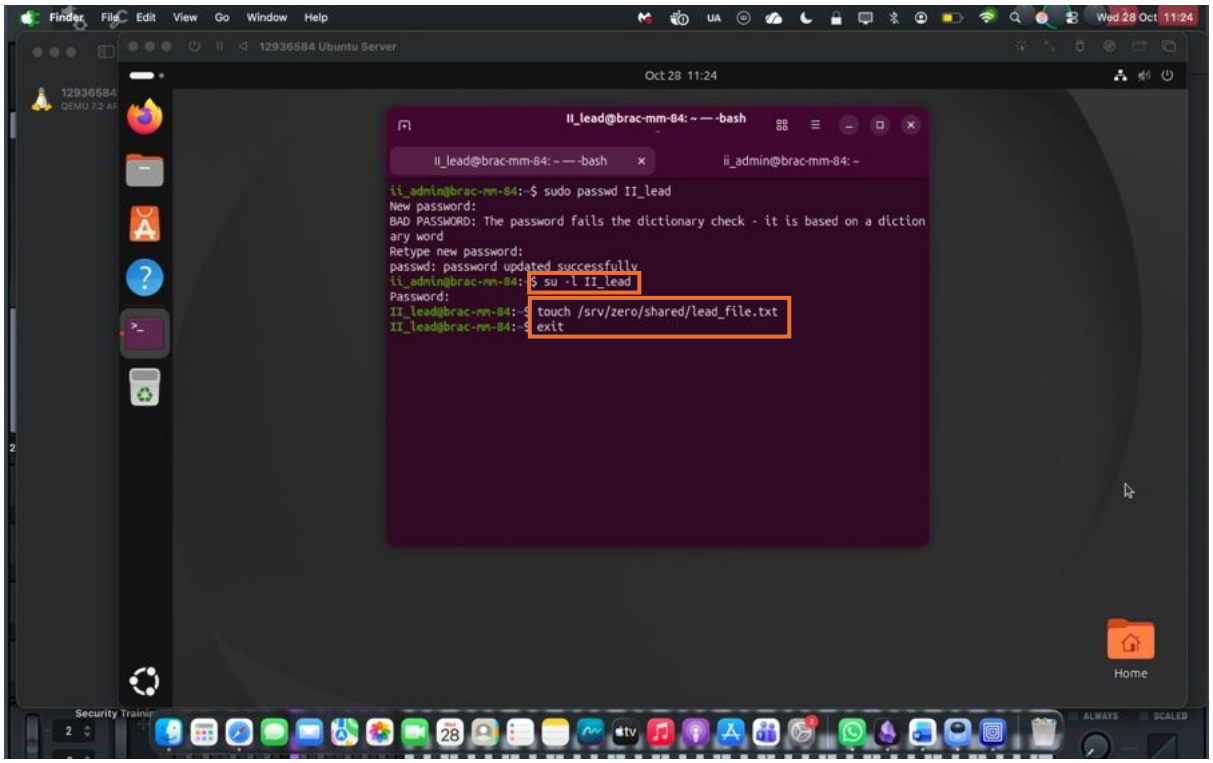
Creating a Directory Structure for folder requirements.



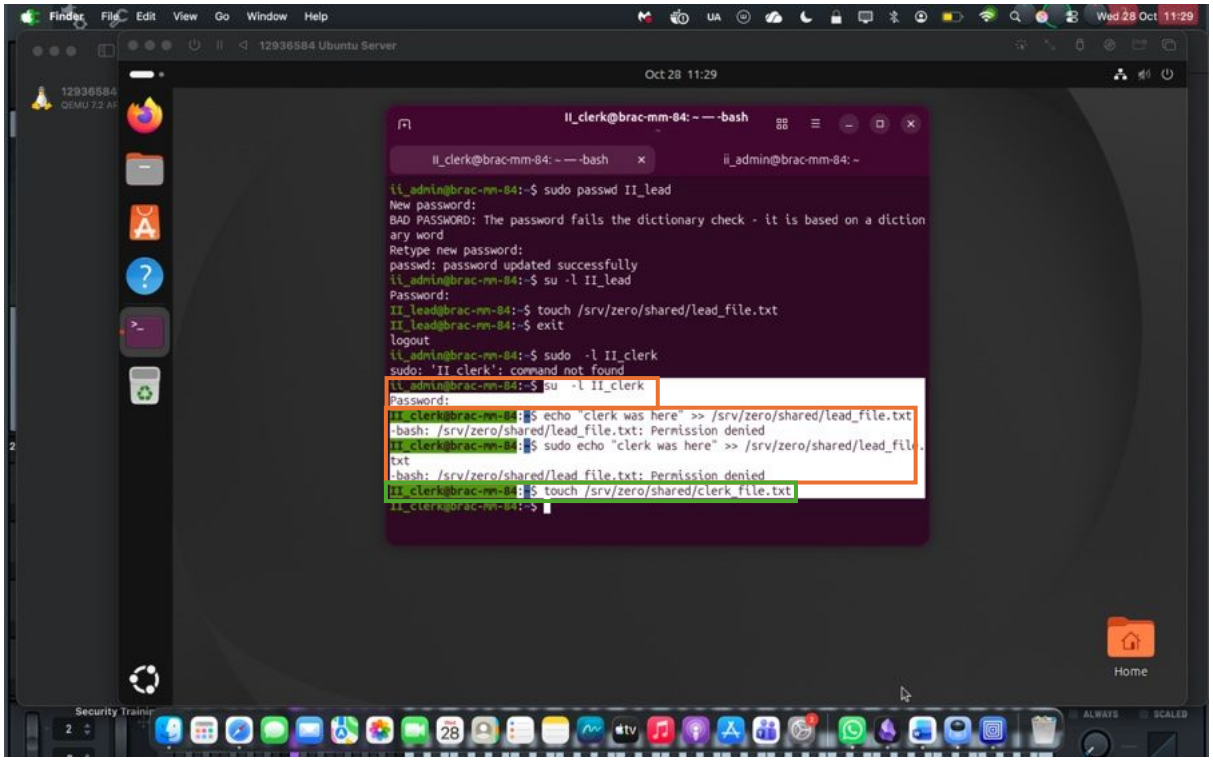
Configuring Shared Folder, setting group ownership to my department staff group and applying permissions with chmod preventing file delectation of each other files.



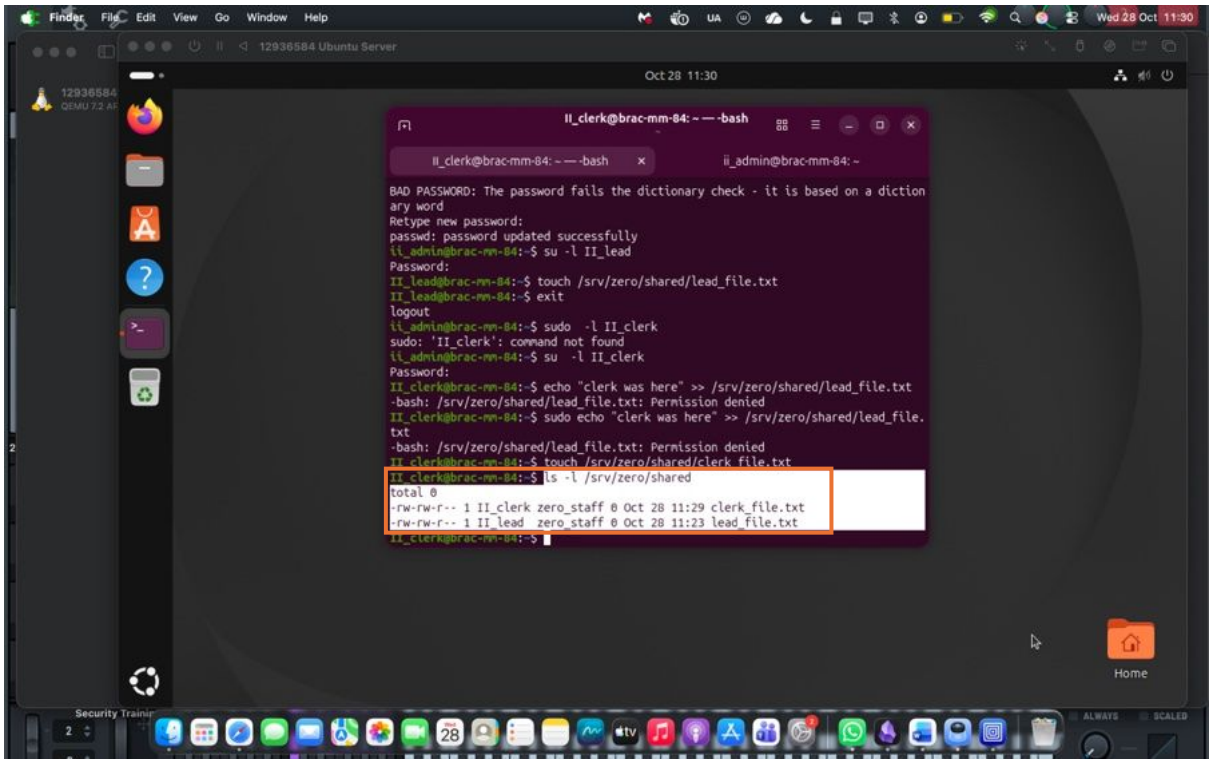
Configuring Private Folder for each account and permissions.



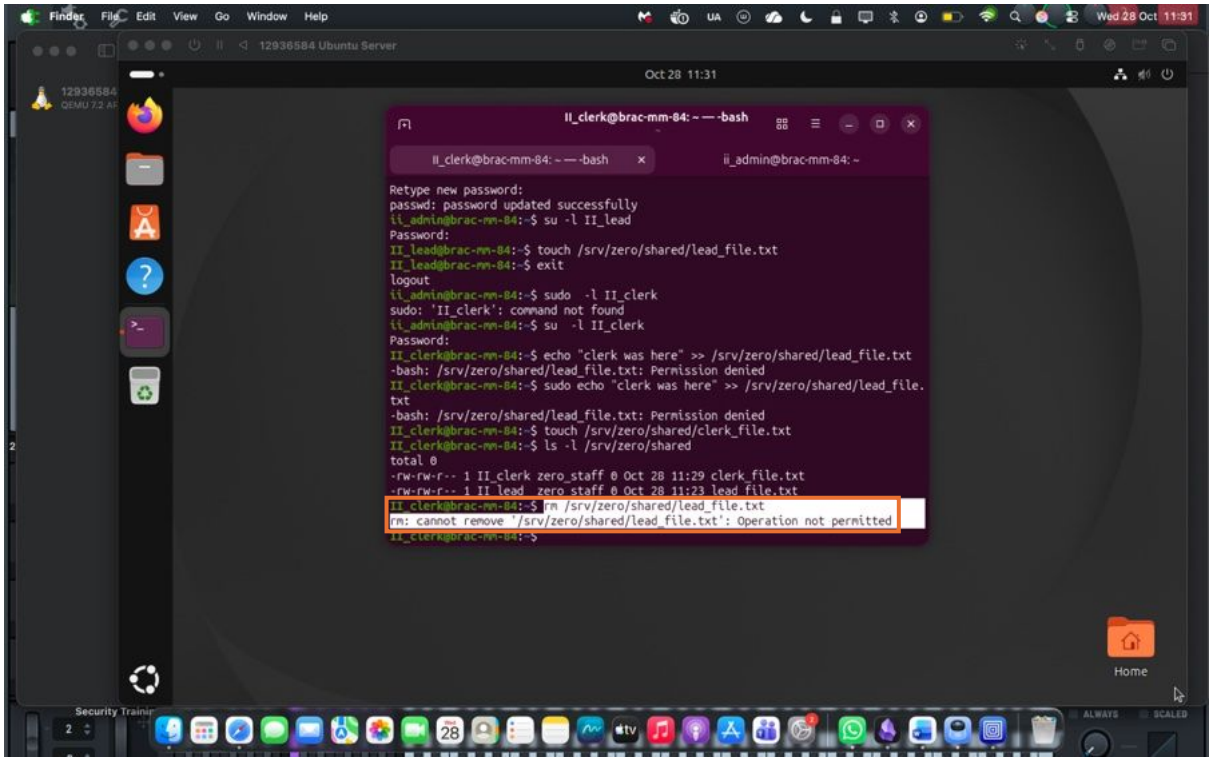
Proving requirement by logging in as II_lead and successfully accessing II-lead's private folder/file.



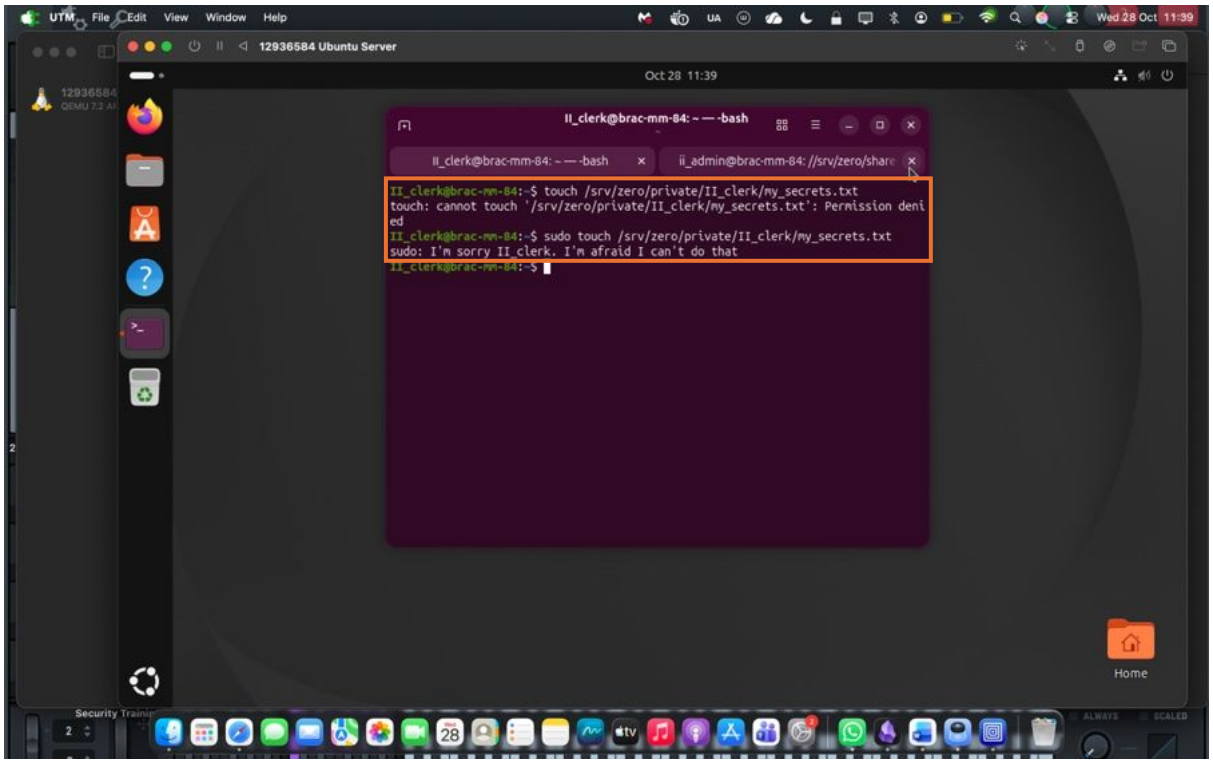
Proving requirement by logging in as II_clerk and failing to access II_lead's private folder/file. While accessing II_clerk's private folder/file.



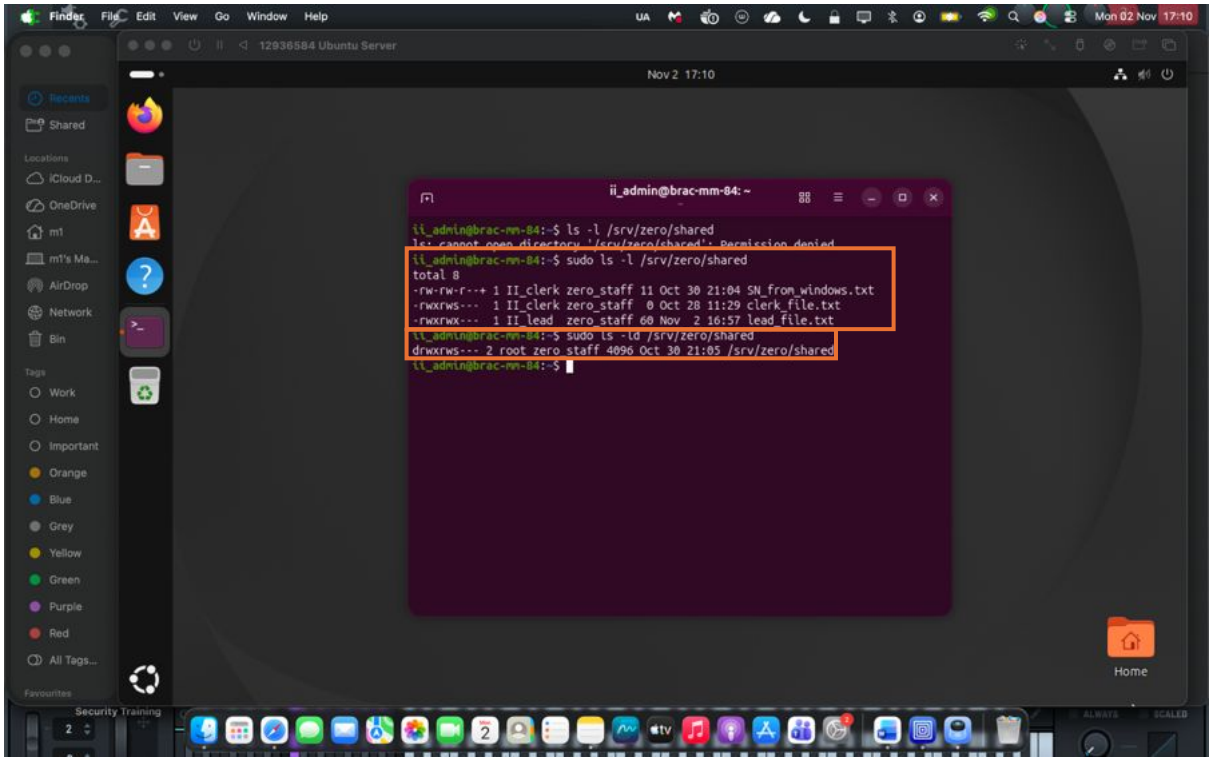
Viewing files users have access to. To verify.



Failing to remove other user (II_lead) file as II_clerk.

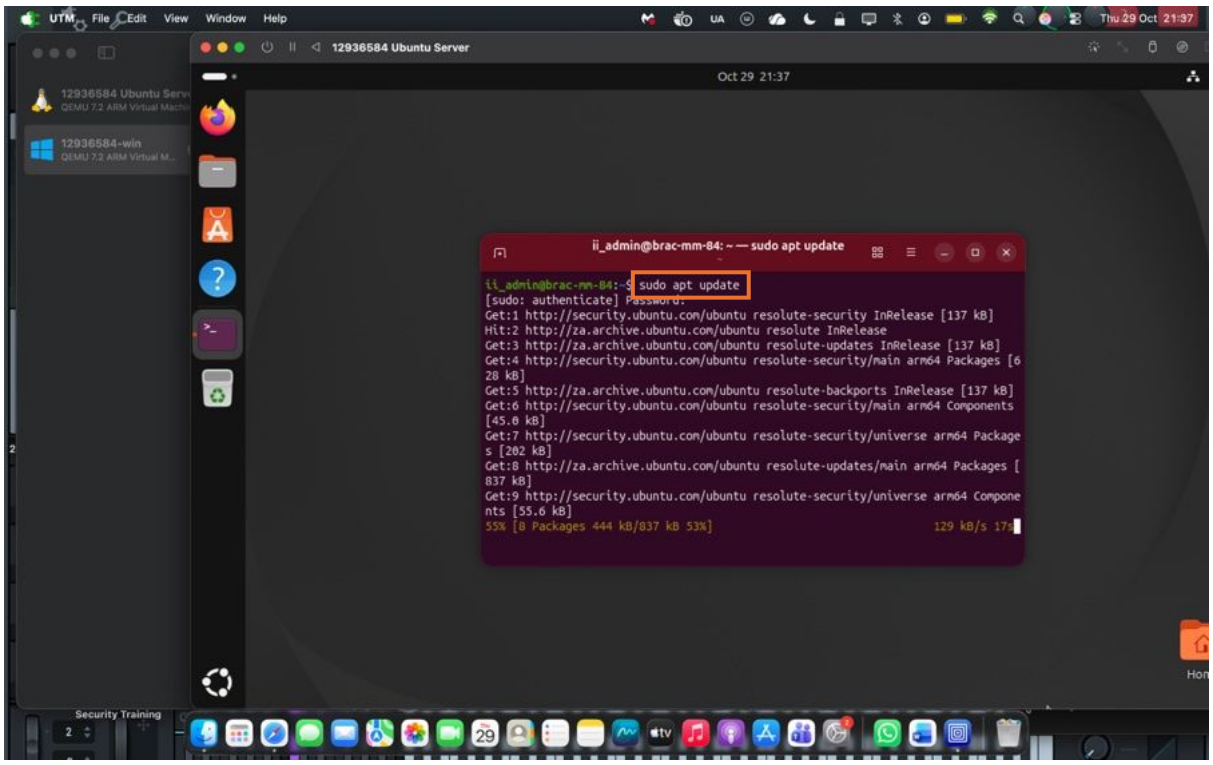


User failing to access secret file.

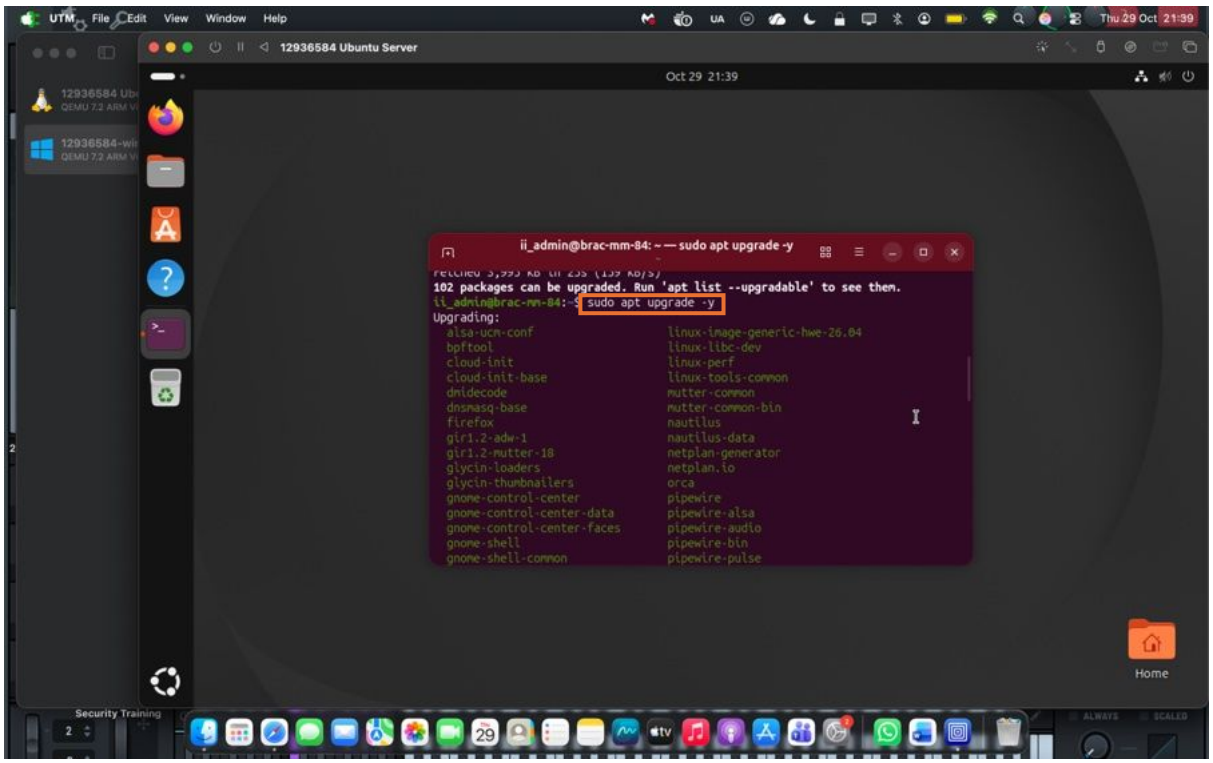


2.5 The two requirement cannot be met with chmod 770 alone as 770 alone does not include prevention which stops users from deleting each other's files which is where we need a Sticky Bit(1) e.g. 1770 to achieve this. 770 alone also does not ensure continuity collaboration which is where we need a SETGID(2) e.g. 2770 to achieve this. We can combine the SETGID and Stickly Bit value 2+1=3 e.g. 3770. with s in rws representing execute for SETGID, the second number representing user permissions, third number group permission and fourth number other.

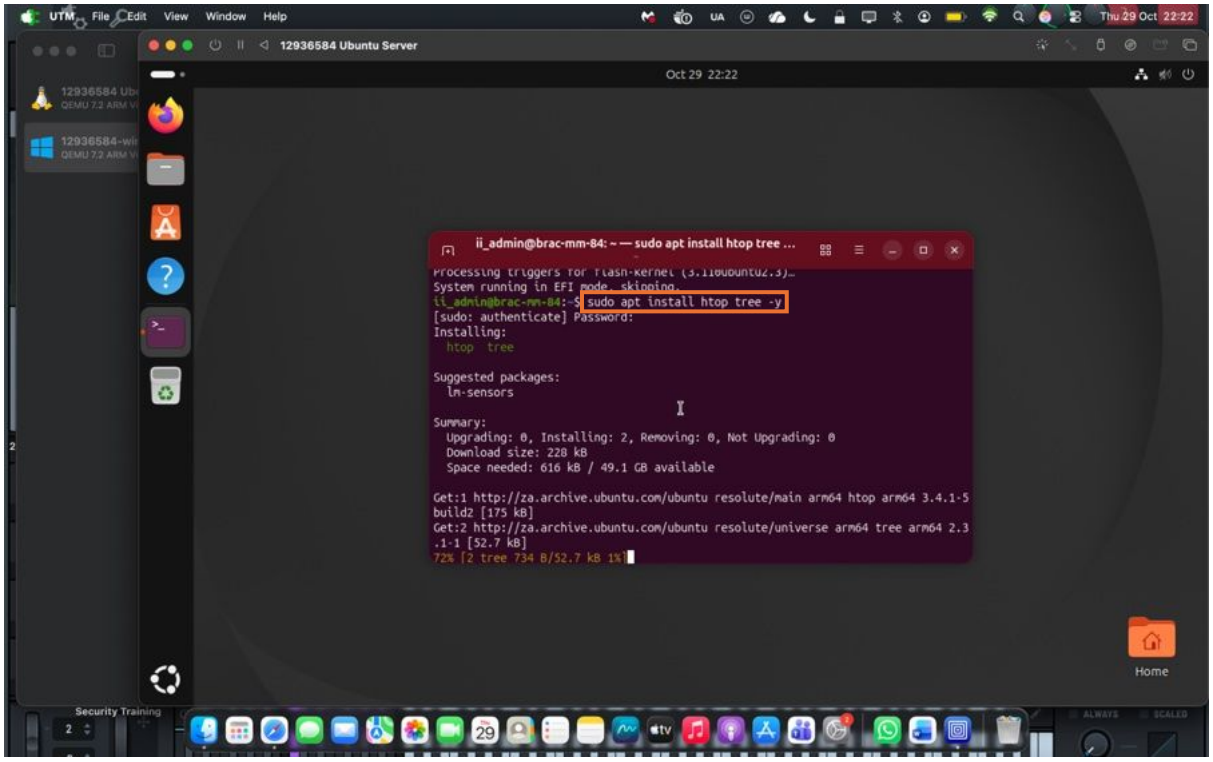
Question 3



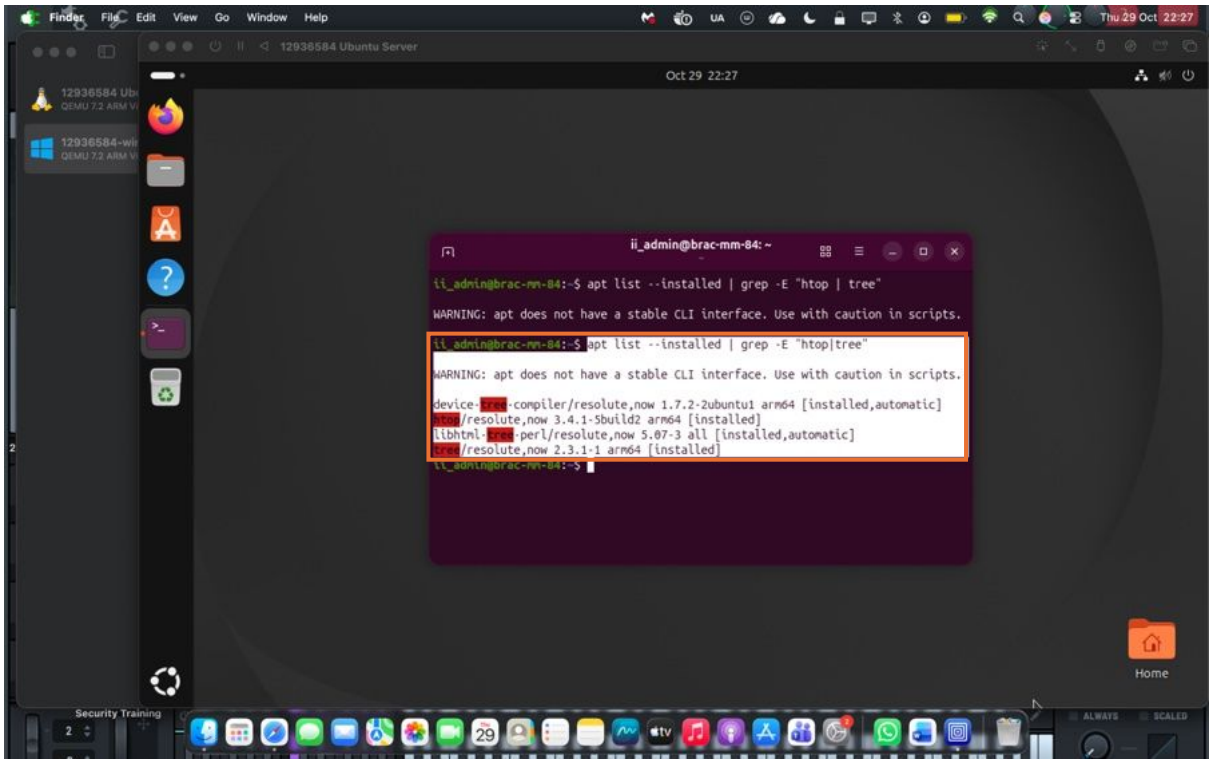
Updating the package index.



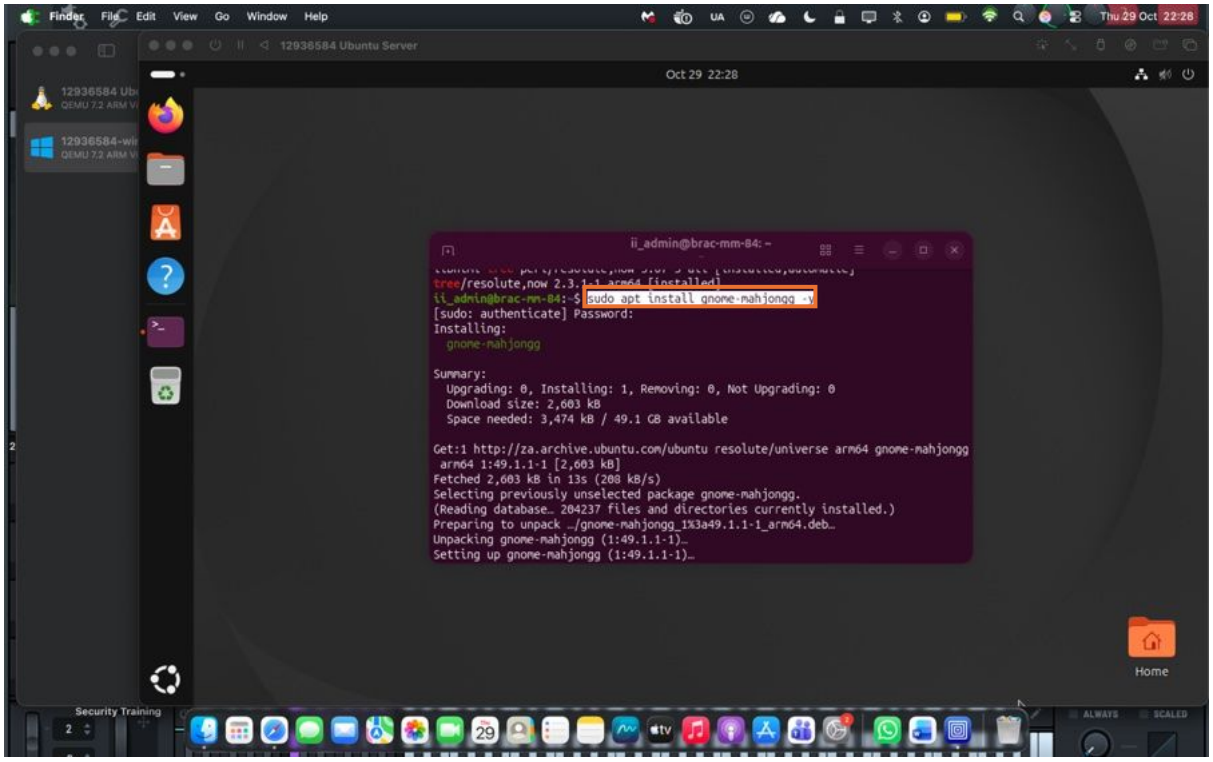
Upgrading the System.



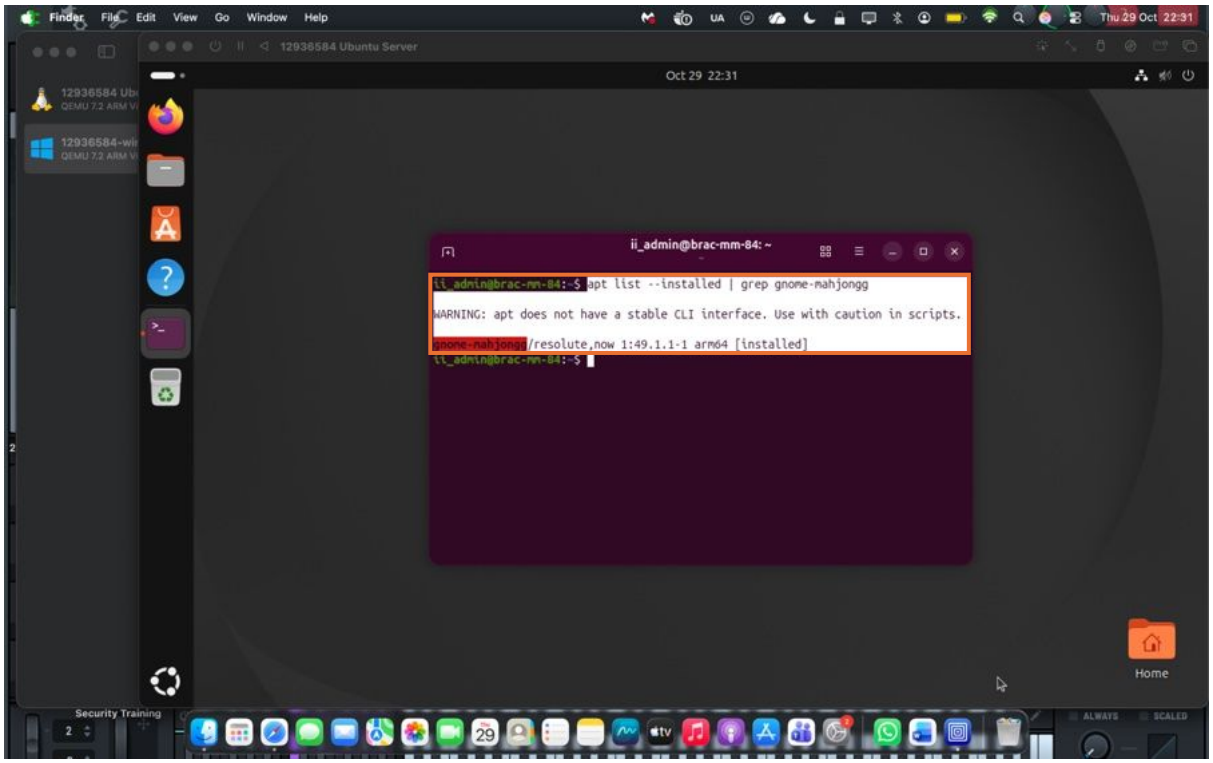
Installing htop tree -y.



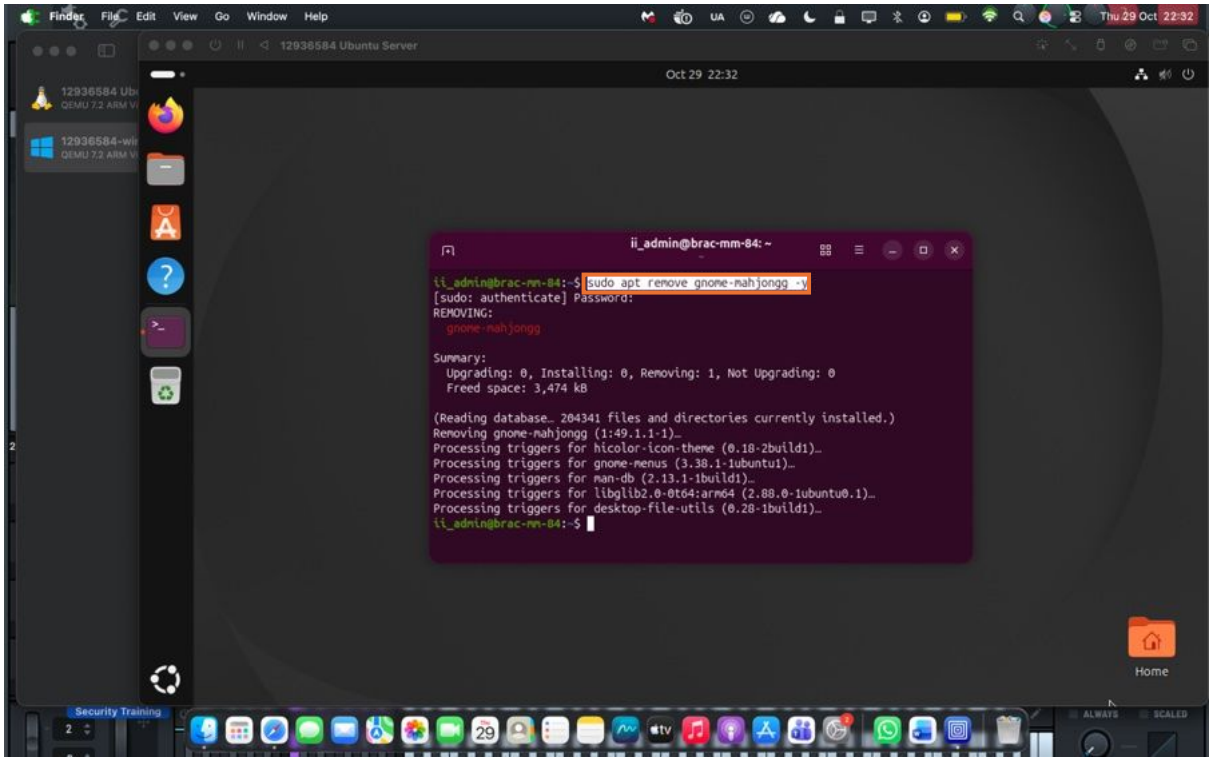
Listing proof of installation.



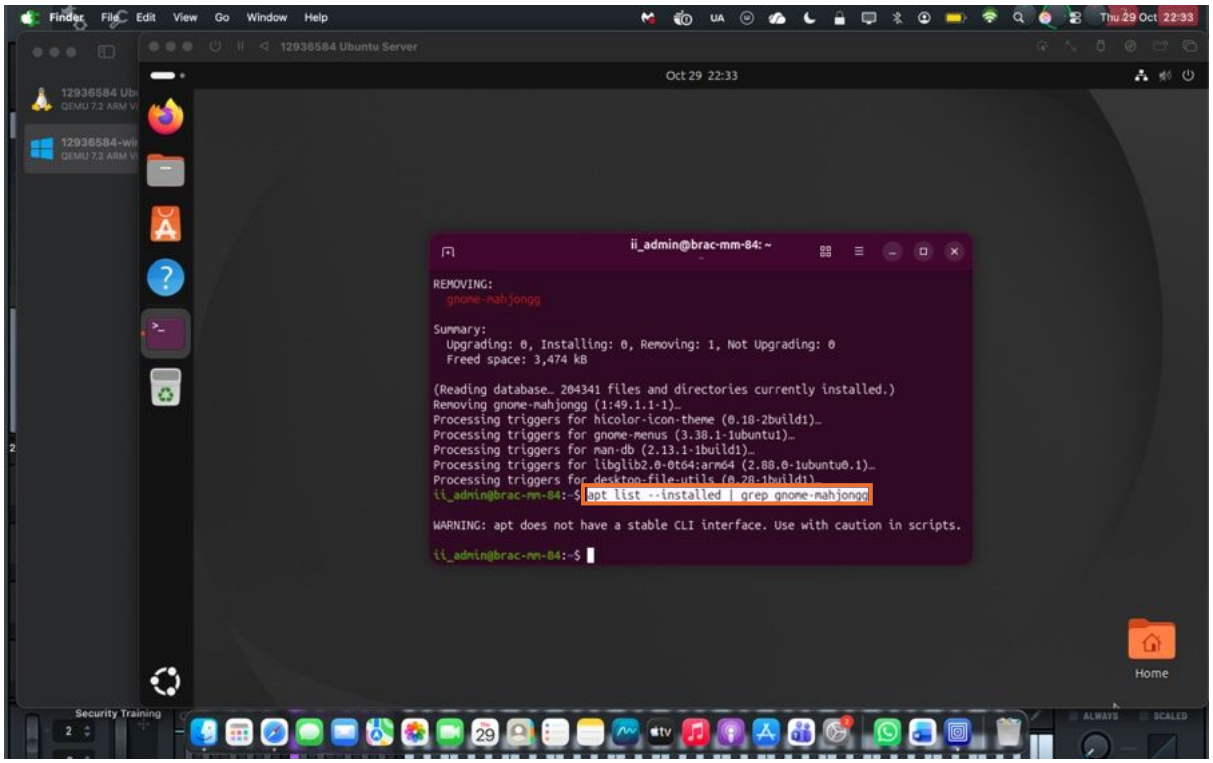
Installing gnome-mahjongg (Department Note Prep).



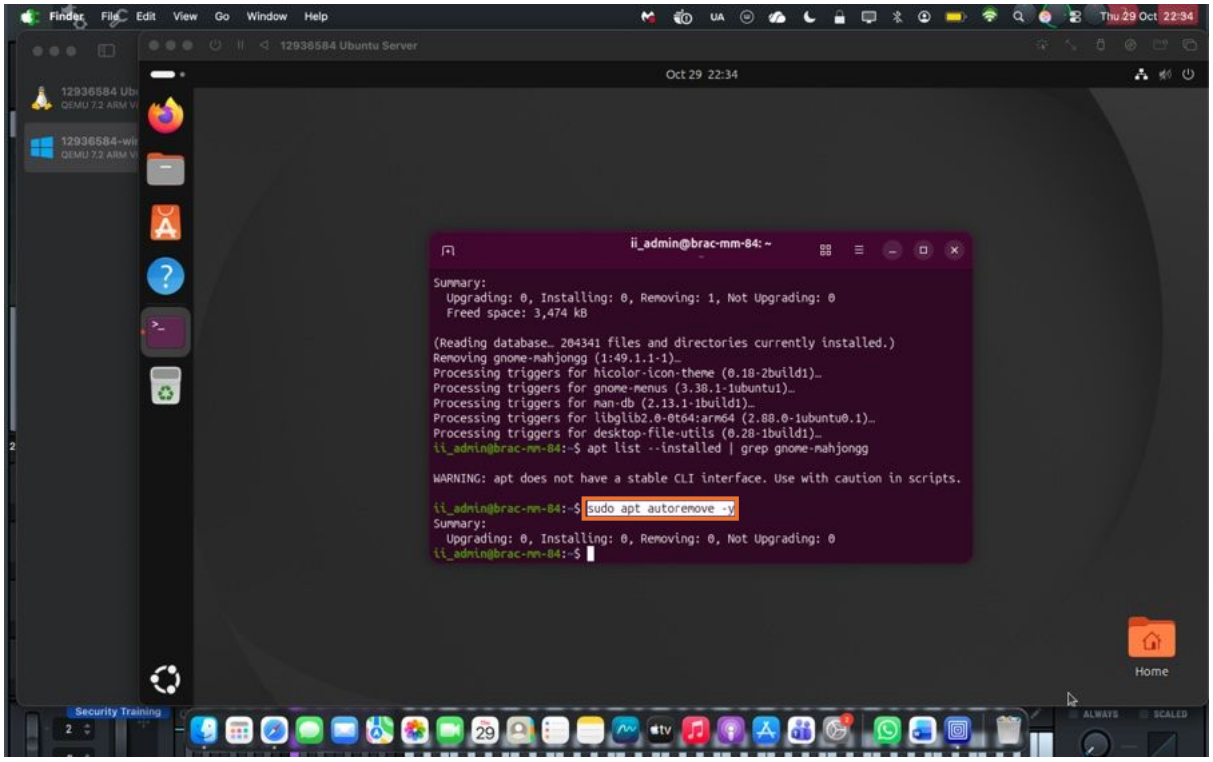
Listing proof of installation.



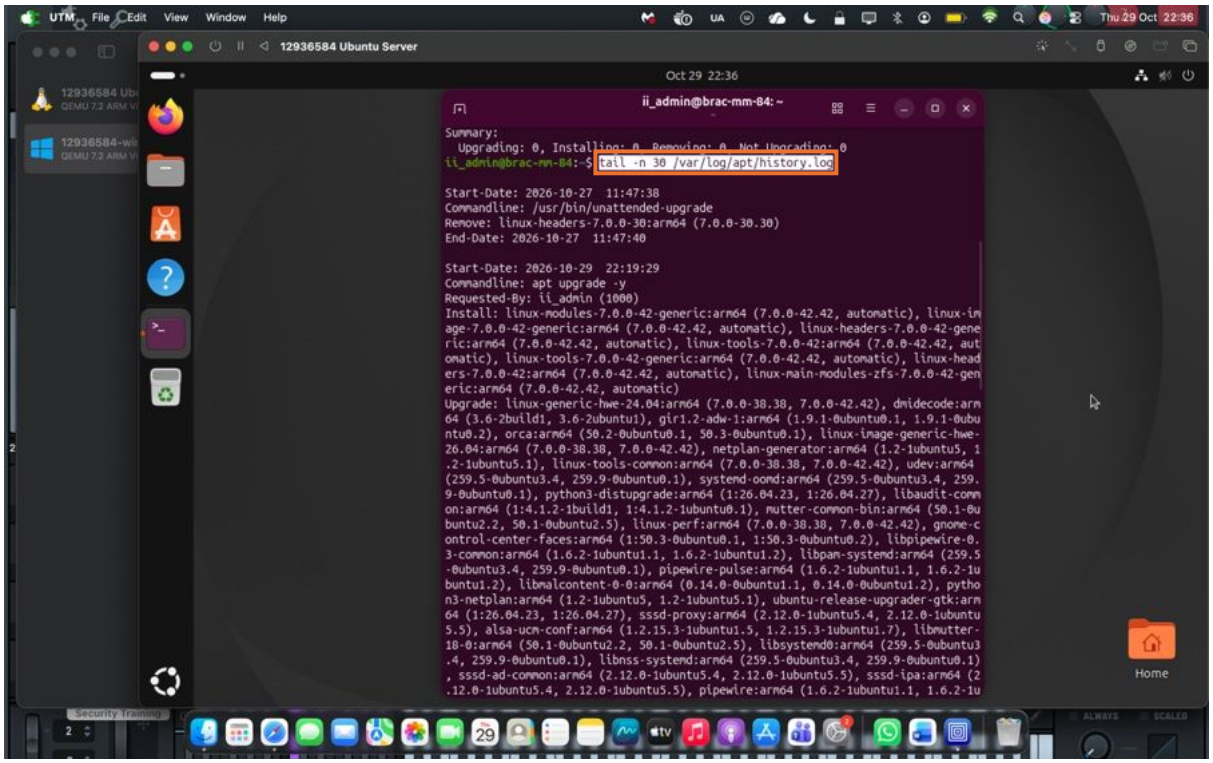
Removing the Department Package (gnome-mahjongg).



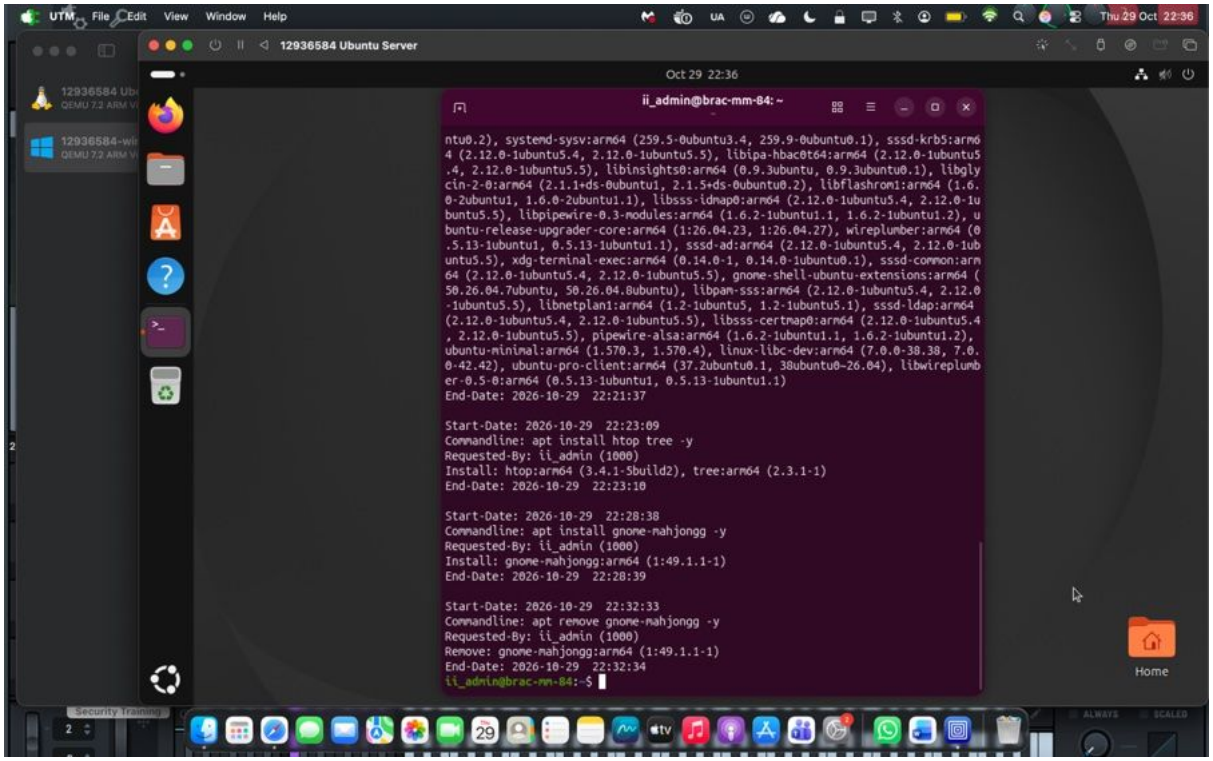
Listing proof of removal.

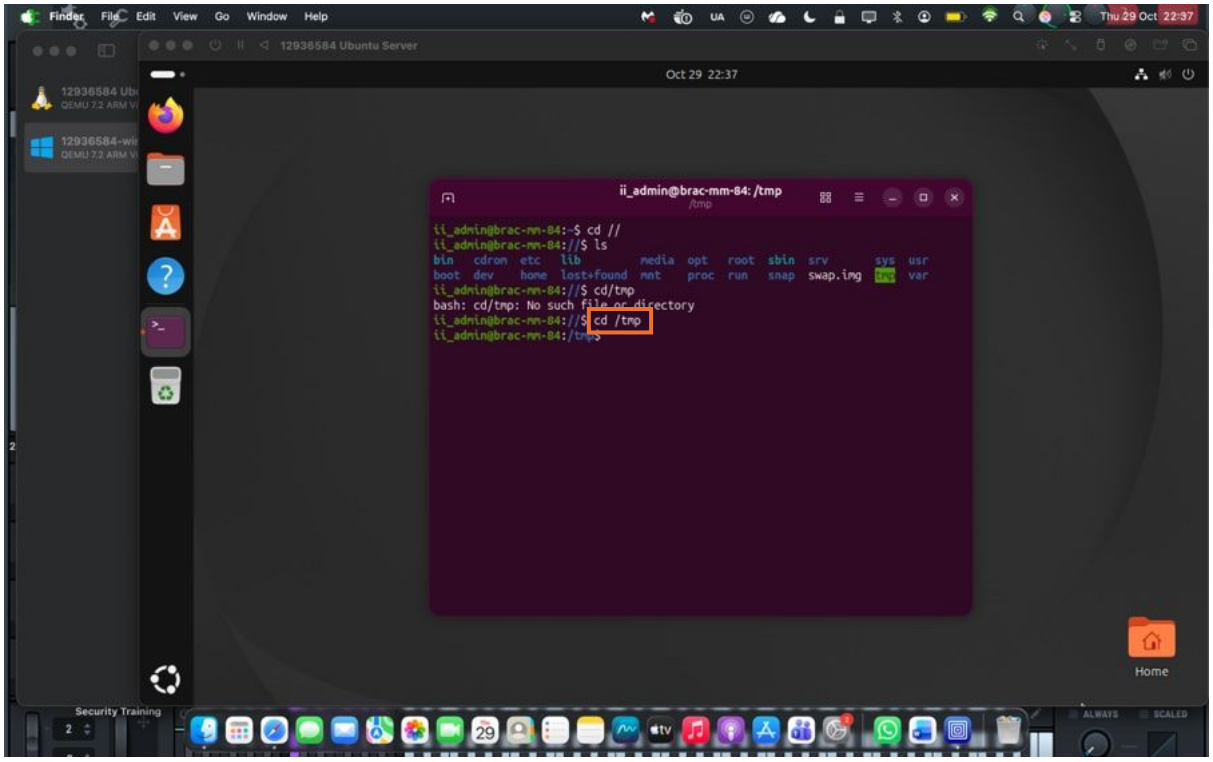


Running apt autoremove.

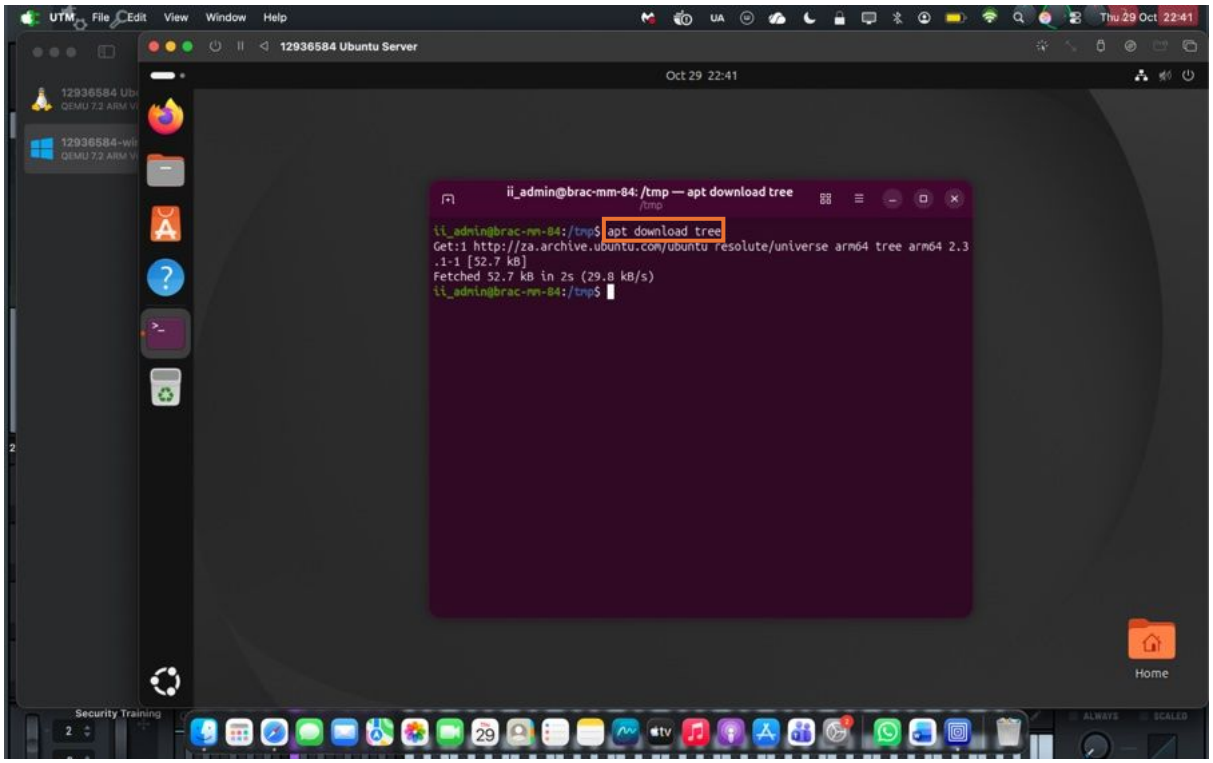


Checking APT History Log Entries with tail.

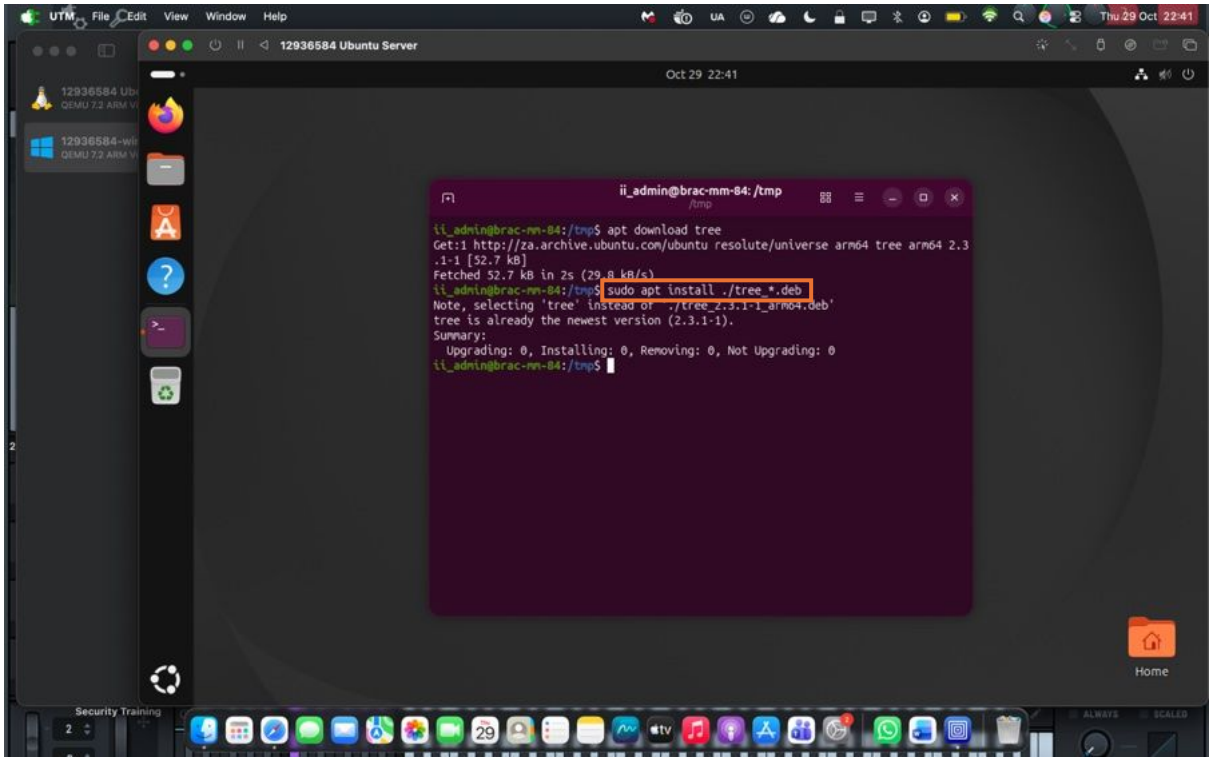

UTM File Edit View Window Help
Thu 29 Oct 22:36
12936584 Ubuntu Server
Oct 29 22:36
ii_admin@brac-mm-84: ~
ntu0.2), systemd-sysv:arm64 (259.5-0ubuntu3.4, 259.9-0ubuntu0.1), sssd-krb5:arm64 (2.12.0-1ubuntu5.4, 2.12.0-1ubuntu5.5), libipa-hbac0t64:arm64 (2.12.0-1ubuntu5.4, 2.12.0-1ubuntu5.5), libinsights0:arm64 (0.9.3ubuntu, 0.9.3ubuntu0.1), libglycin-2-0:arm64 (2.1.1+ds-0ubuntu1, 2.1.5+ds-0ubuntu0.2), libflashrom1:arm64 (1.6.0-2ubuntu1, 1.6.0-2ubuntu1.1), libsss-idmap0:arm64 (2.12.0-1ubuntu5.4, 2.12.0-1ubuntu5.5), libpipewire-0.3-modules:arm64 (1.6.2-1ubuntu1.1, 1.6.2-1ubuntu1.2), ubuntu-release-upgrader-core:arm64 (1:26.04.23, 1:26.04.27), wireplumber:arm64 (0.5.13-1ubuntu1, 0.5.13-1ubuntu1.1), sssd-ad:arm64 (2.12.0-1ubuntu5.4, 2.12.0-1ubuntu5.5), xdg-terminal-exec:arm64 (0.14.0-1, 0.14.0-1ubuntu0.1), sssd-common:arm64 (2.12.0-1ubuntu5.4, 2.12.0-1ubuntu5.5), gnome-shell-ubuntu-extensions:arm64 (50.26.04.7ubuntu, 50.26.04.8ubuntu), libpam-sss:arm64 (2.12.0-1ubuntu5.4, 2.12.0-1ubuntu5.5), libnetplan1:arm64 (1.2-1ubuntu5, 1.2-1ubuntu5.1), sssd-ldap:arm64 (2.12.0-1ubuntu5.4, 2.12.0-1ubuntu5.5), libsss-certmap0:arm64 (2.12.0-1ubuntu5.4, 2.12.0-1ubuntu5.5), pipewire-alsa:arm64 (1.6.2-1ubuntu1.1, 1.6.2-1ubuntu1.2), ubuntu-minimal:arm64 (1.570.3, 1.570.4), linux-libc-dev:arm64 (7.0.0-38.38, 7.0.0-42.42), ubuntu-pro-client:arm64 (37.2ubuntu0.1, 38ubuntu0~26.04), libwireplumber-0.5-0:arm64 (0.5.13-1ubuntu1, 0.5.13-1ubuntu1.1)
End-Date: 2026-10-29 22:21:37
Start-Date: 2026-10-29 22:23:09
Commandline: apt install htop tree -y
Requested-By: ii_admin (1000)
Install: htop:arm64 (3.4.1-5build2), tree:arm64 (2.3.1-1)
End-Date: 2026-10-29 22:23:10
Start-Date: 2026-10-29 22:28:38
Commandline: apt install gnome-mahjongg -y
Requested-By: ii_admin (1000)
Install: gnome-mahjongg:arm64 (1:49.1.1-1)
End-Date: 2026-10-29 22:28:39
Start-Date: 2026-10-29 22:32:33
Commandline: apt remove gnome-mahjongg -y
Requested-By: ii_admin (1000)
Remove: gnome-mahjongg:arm64 (1:49.1.1-1)
End-Date: 2026-10-29 22:32:34
ii_admin@brac-mm-84:~$
Home



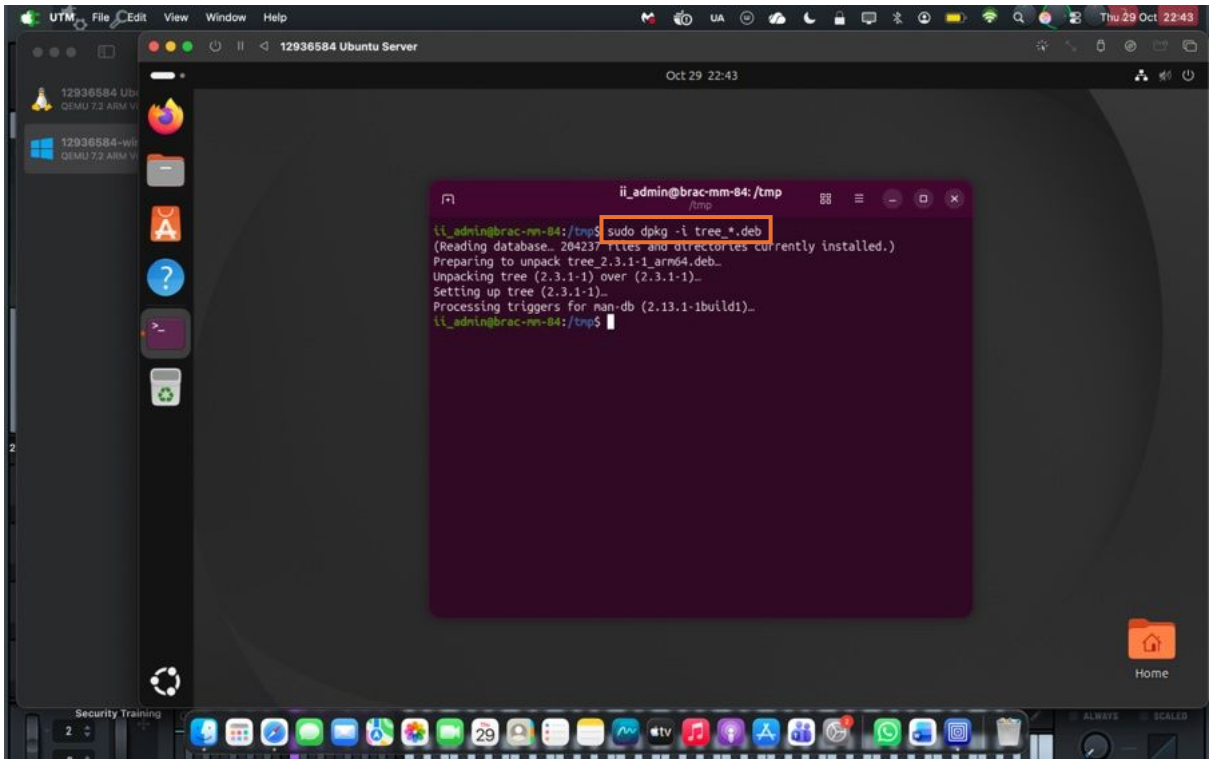
Navigating to /tmp.



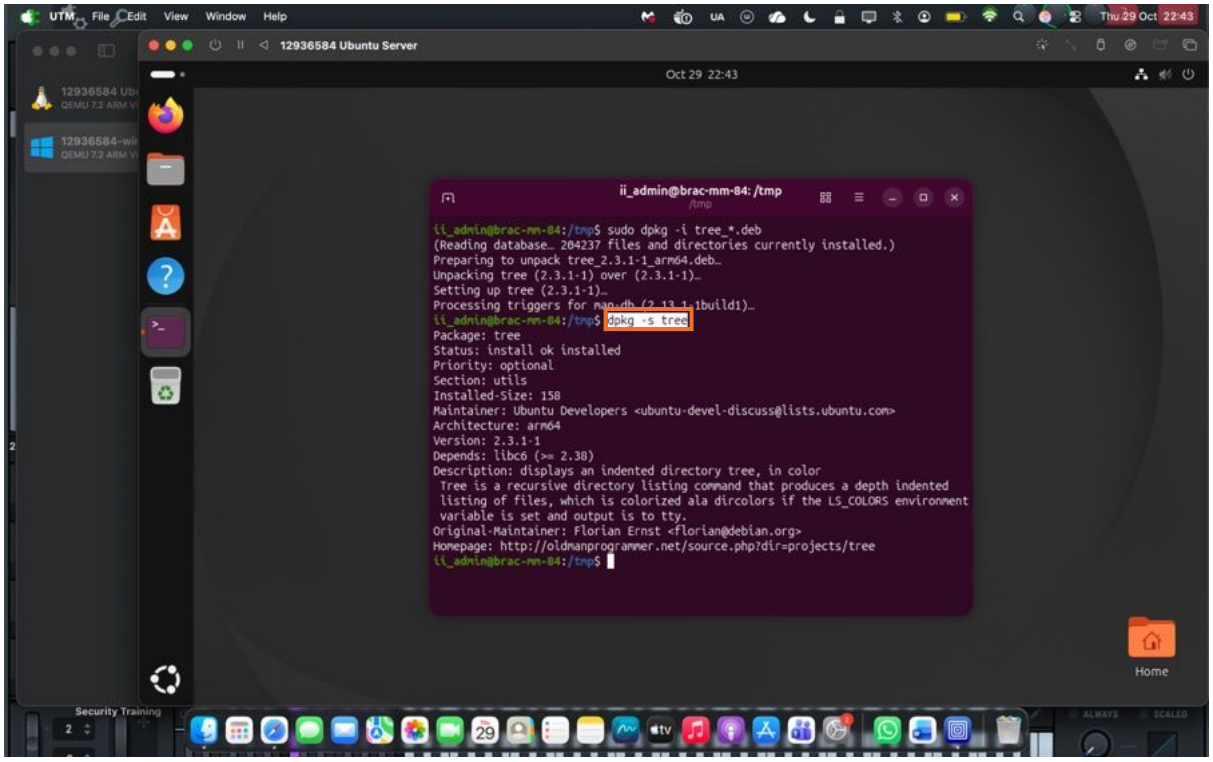
Downloading a sample .deb package.



Installing using apt.

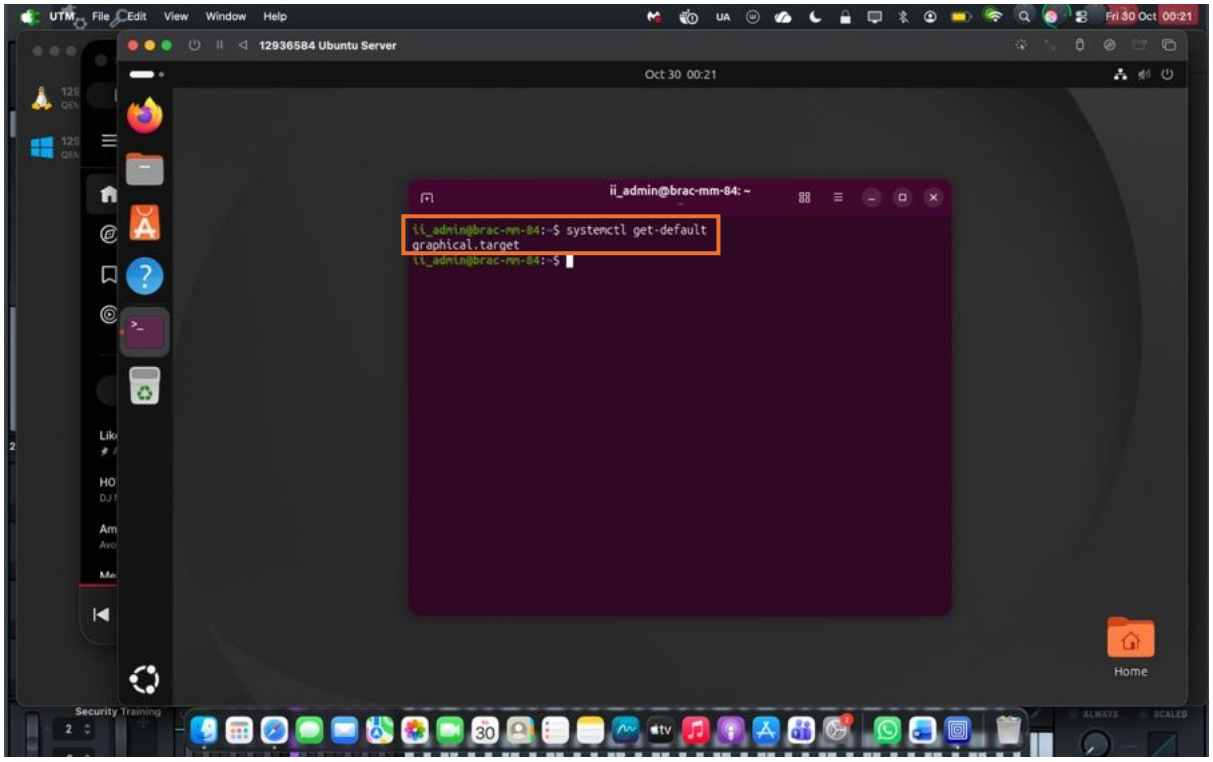


Installing using dpkg.

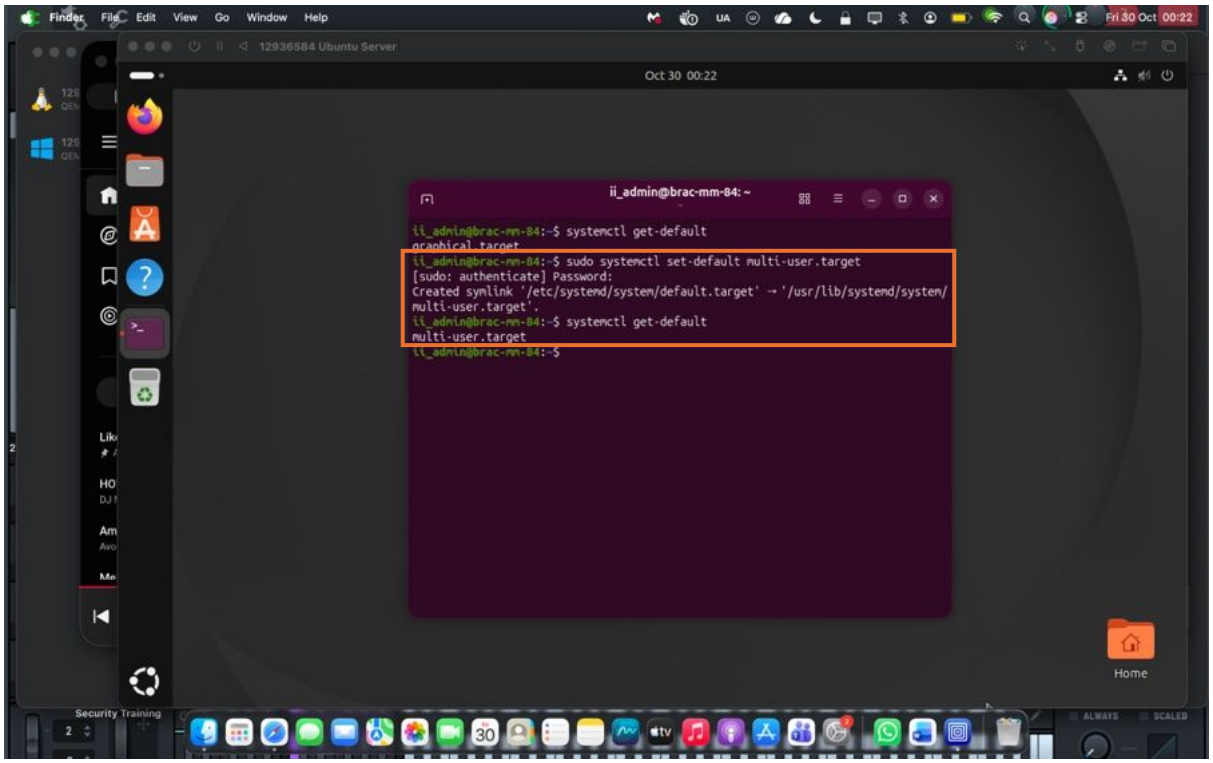


Proving the Installation.

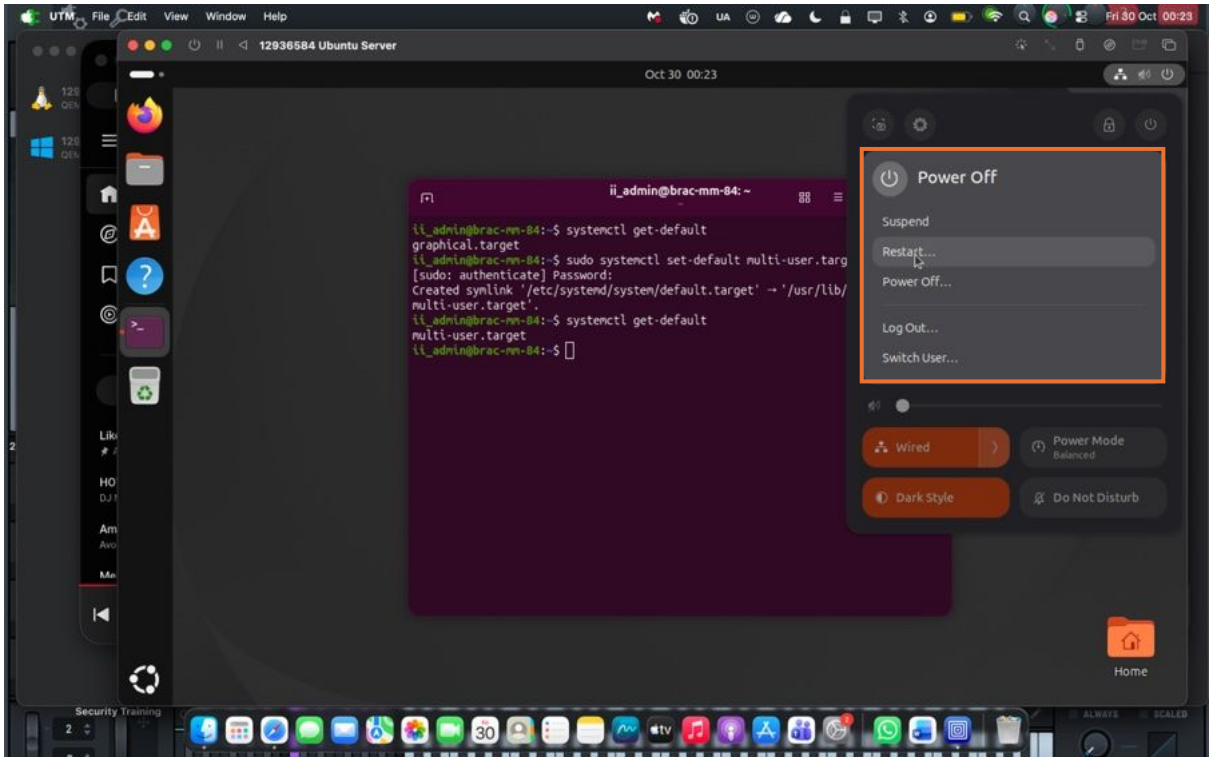
3.2 The key difference between sudo apt install <package> and sudo apt install ./file.deb or sudo dpkg -i file.deb lies in the package source and dependency handling. apt installs from remote repositories, automatically resolving dependencies and managing future updates. dpkg and ./file.deb installs local .deb files directly, dpkg failing if dependencies are missing and requiring manual fixes. When using local .deb files, updates must be managed manually, whereas apt ensures ongoing update support and smoother installation processes.



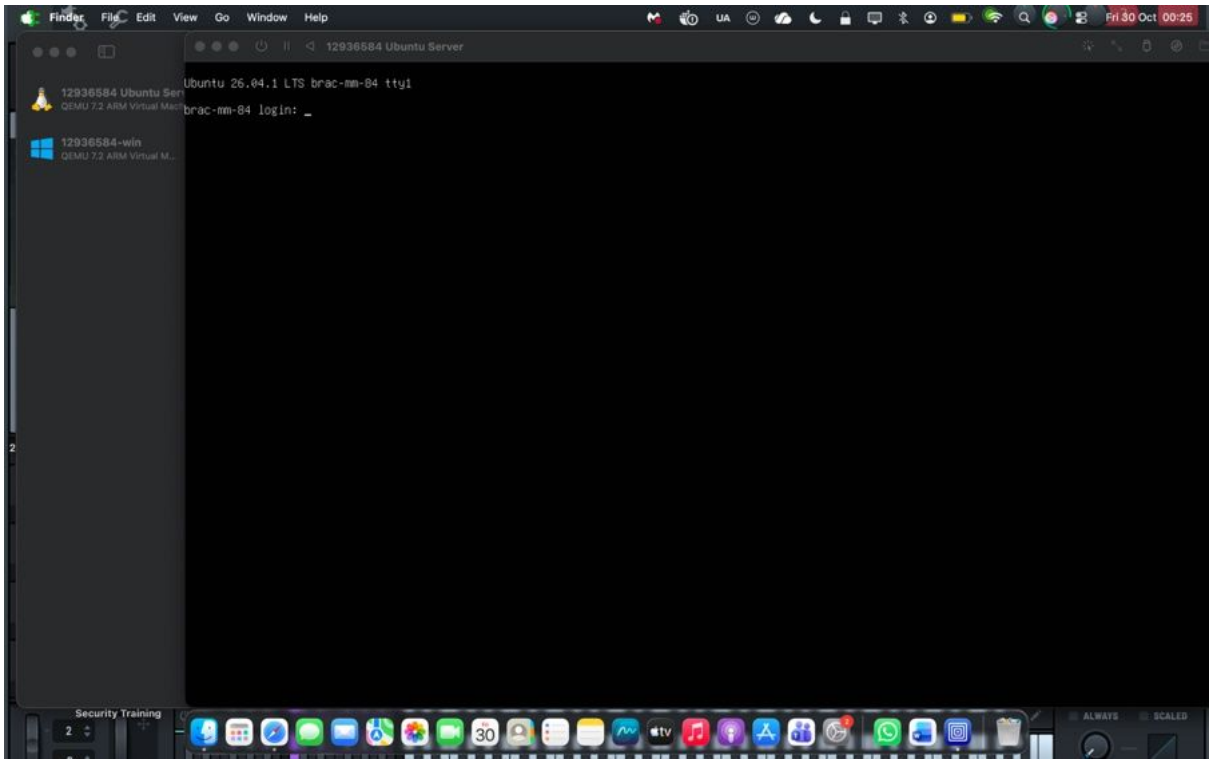
Checking the current default target.



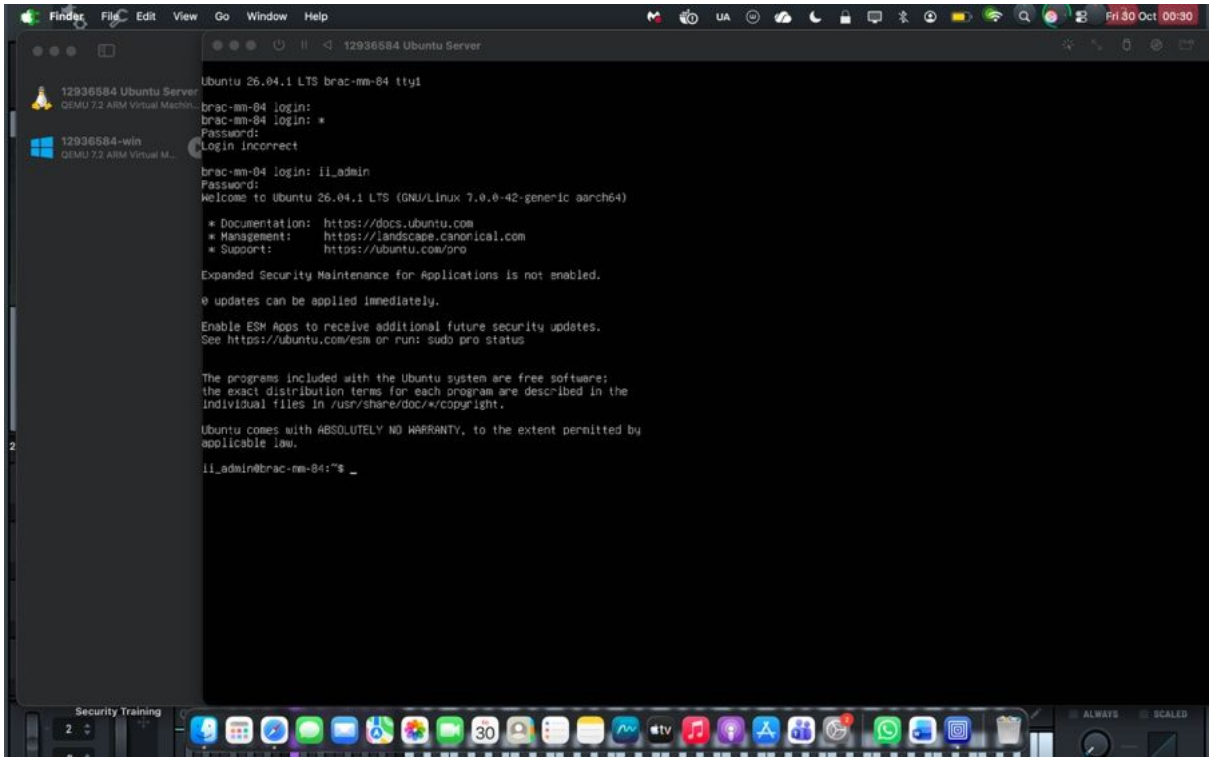
Setting the default to text mode.



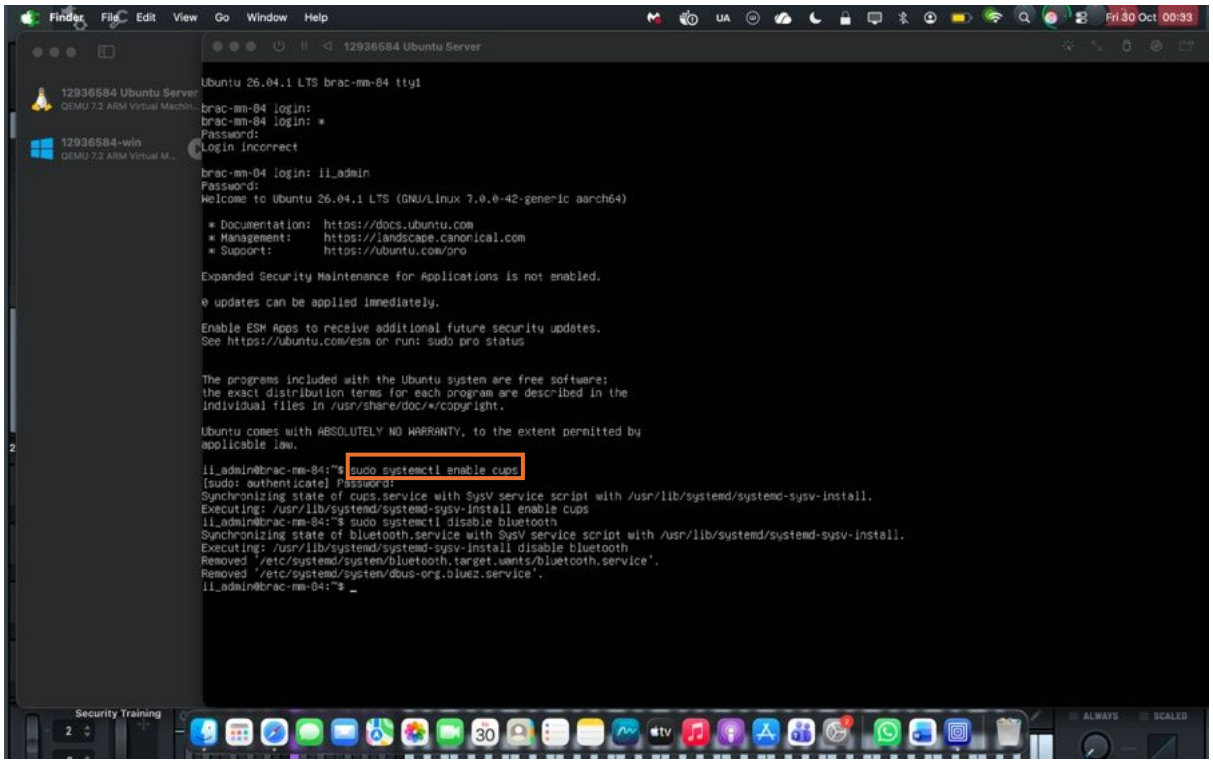
Verifying the change before rebooting.



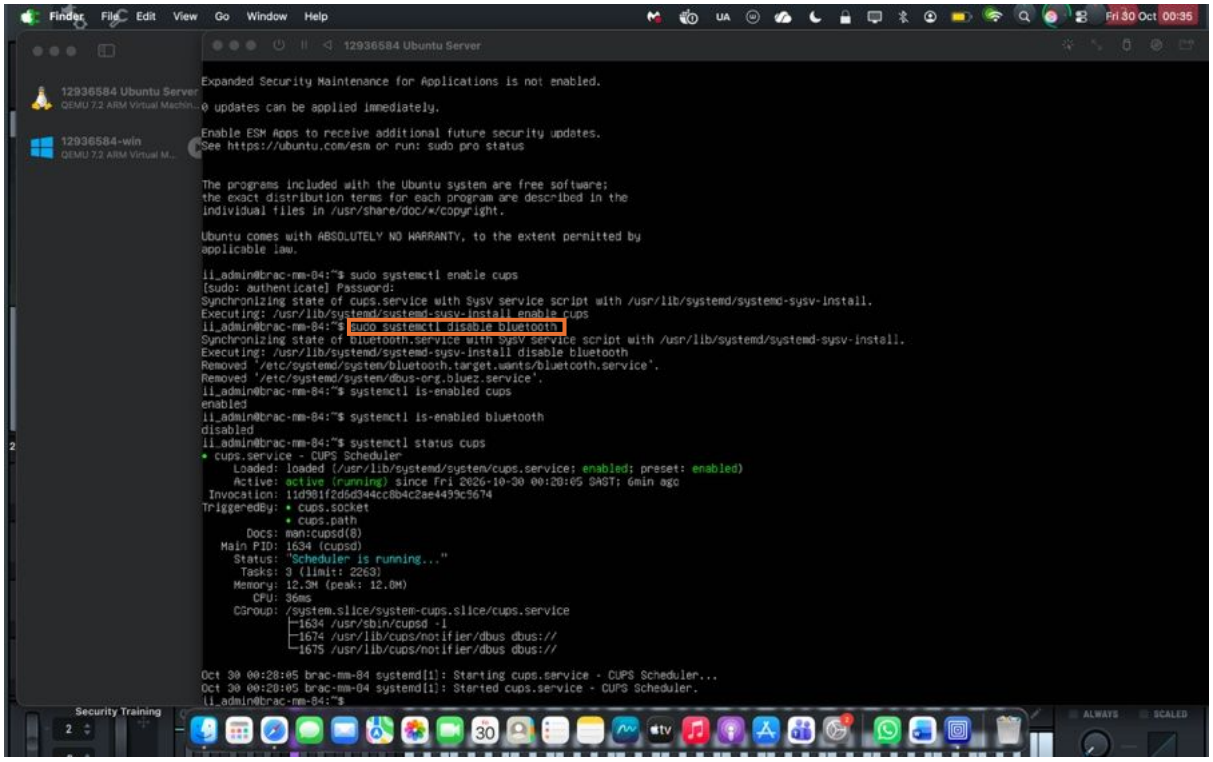
Successful changed to multi-user.target.



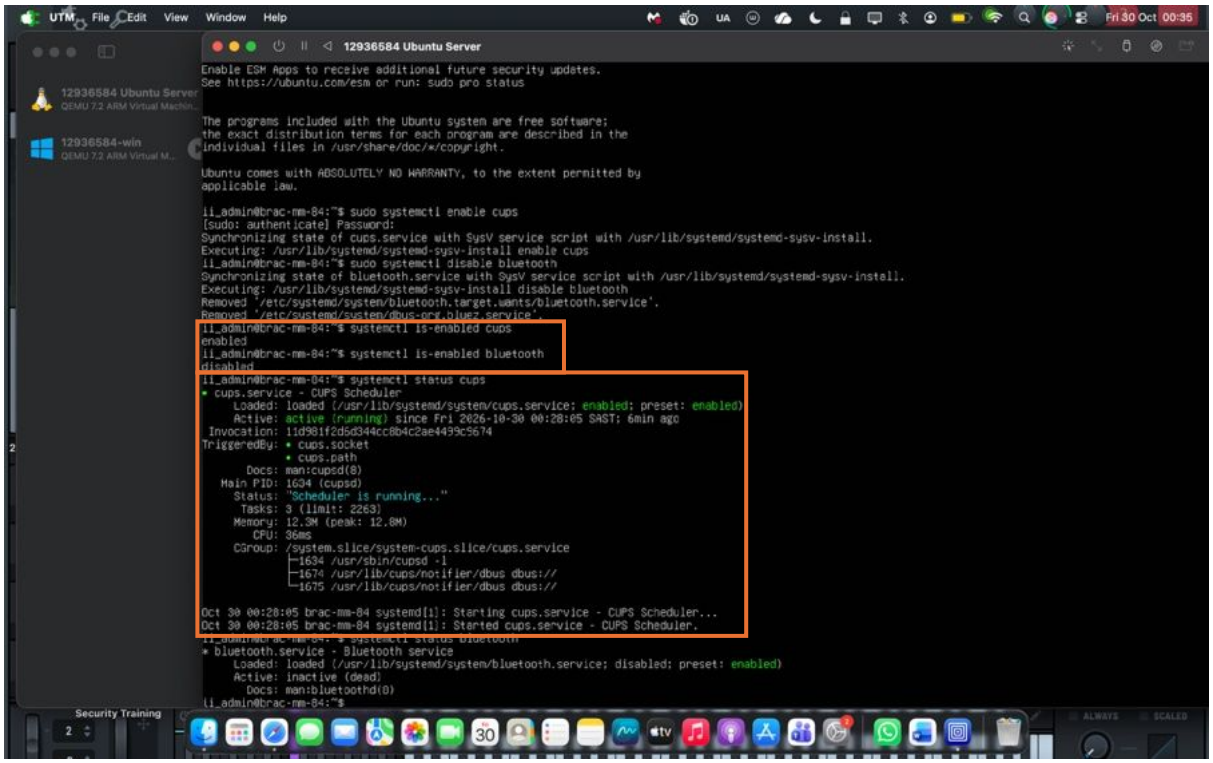
Successful changed to multi-user.target.



Enabling the CUPS service to start at boot.



Disabling an unneeded service (Bluetooth).

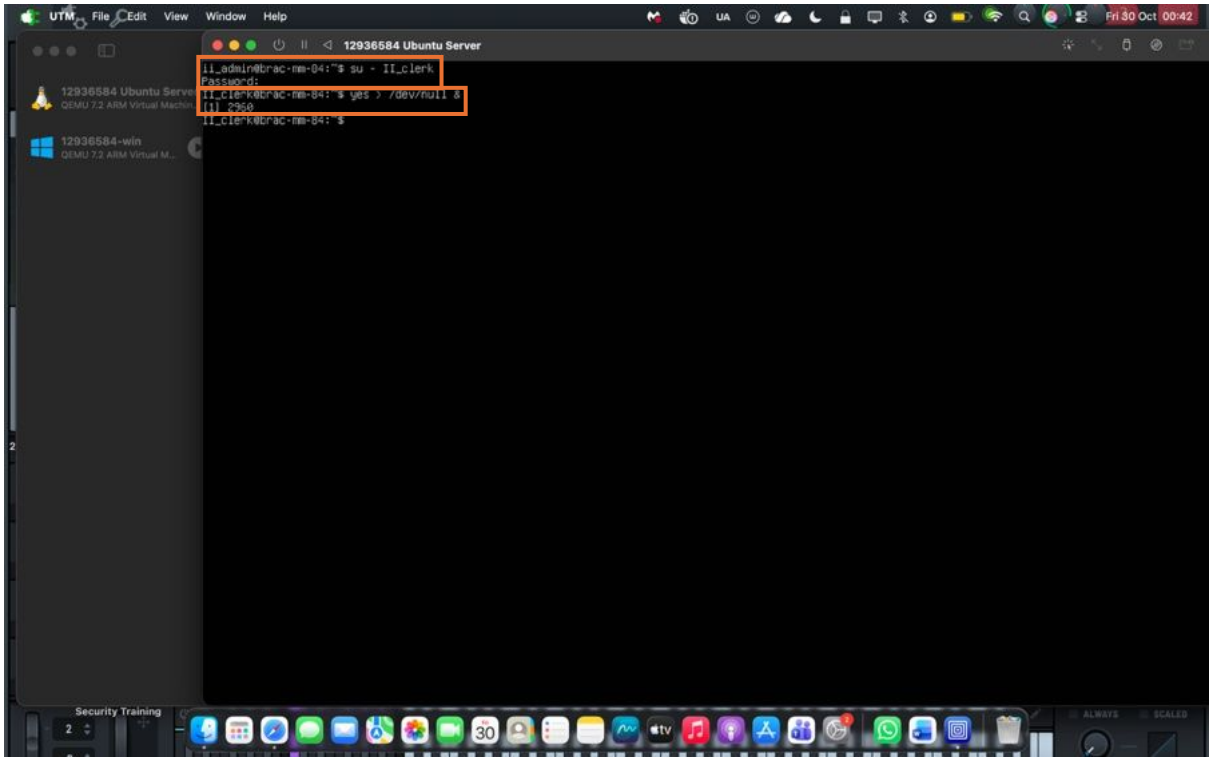


Proving the enabled/disabled status and active running status.

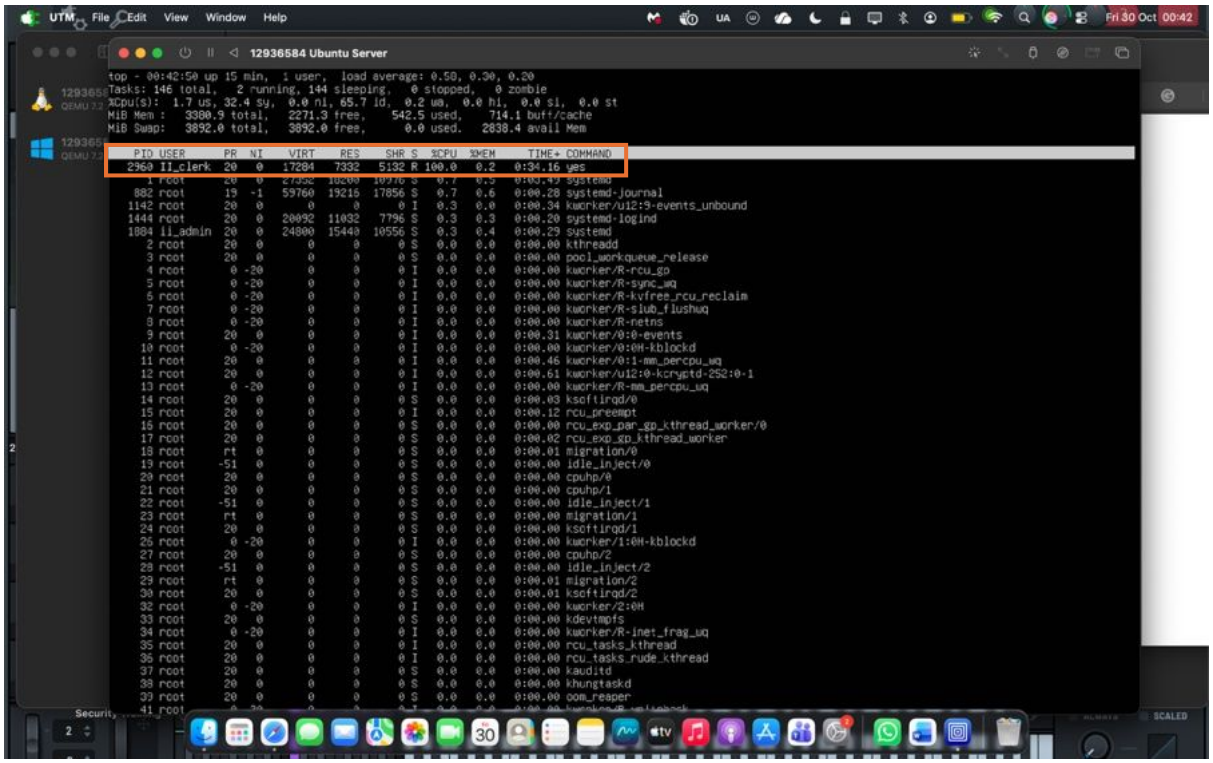
3.4

Multiuser.target maps directly to traditional Run Level 3. Representing a multi-user text console environment with networking enabled but without a graphical user interface.

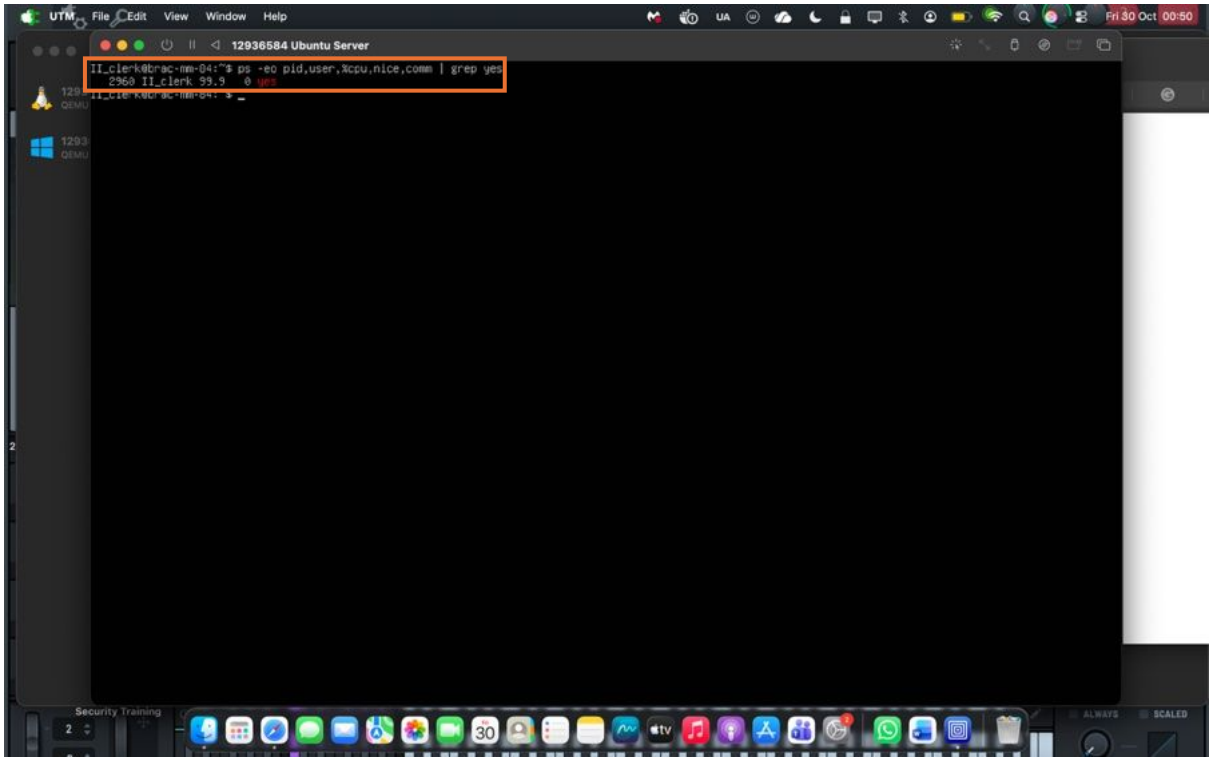
Graphical.target maps directly to traditional Run Level 5. Sets up a multiuser environment complete with networking and a graphical display manager.



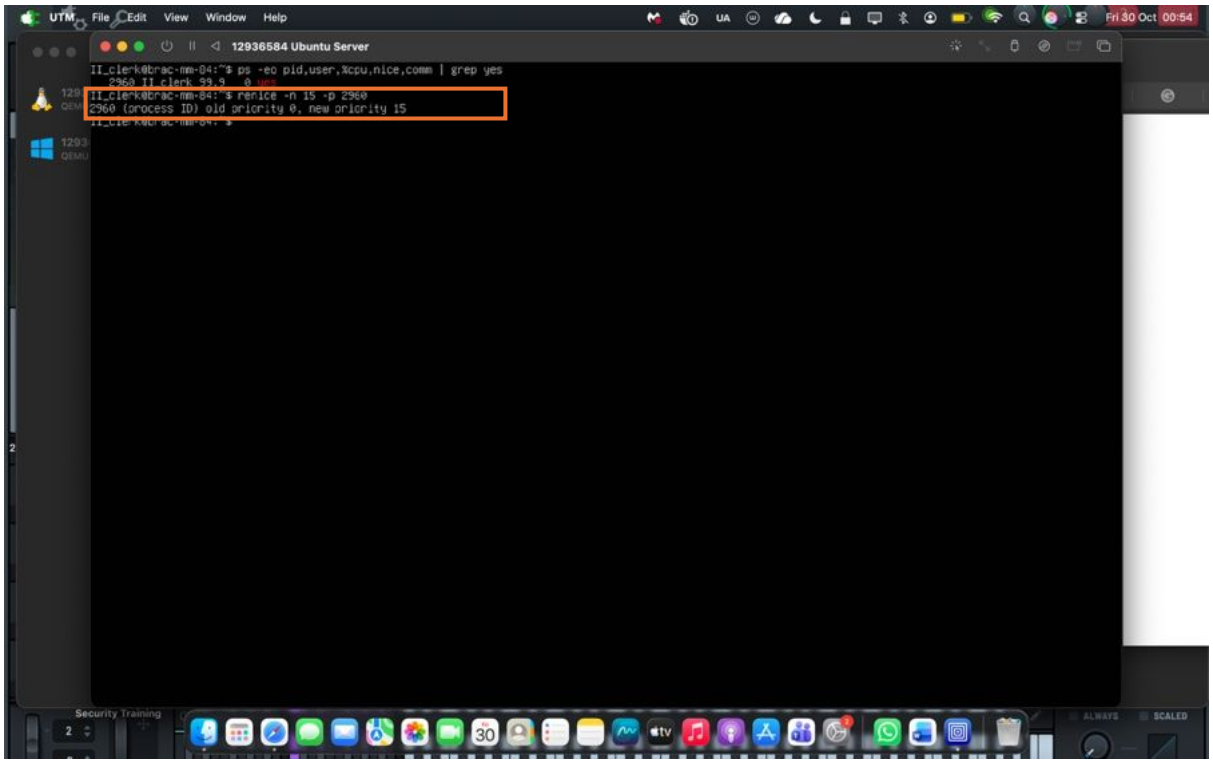
Creating the CPU Hog.



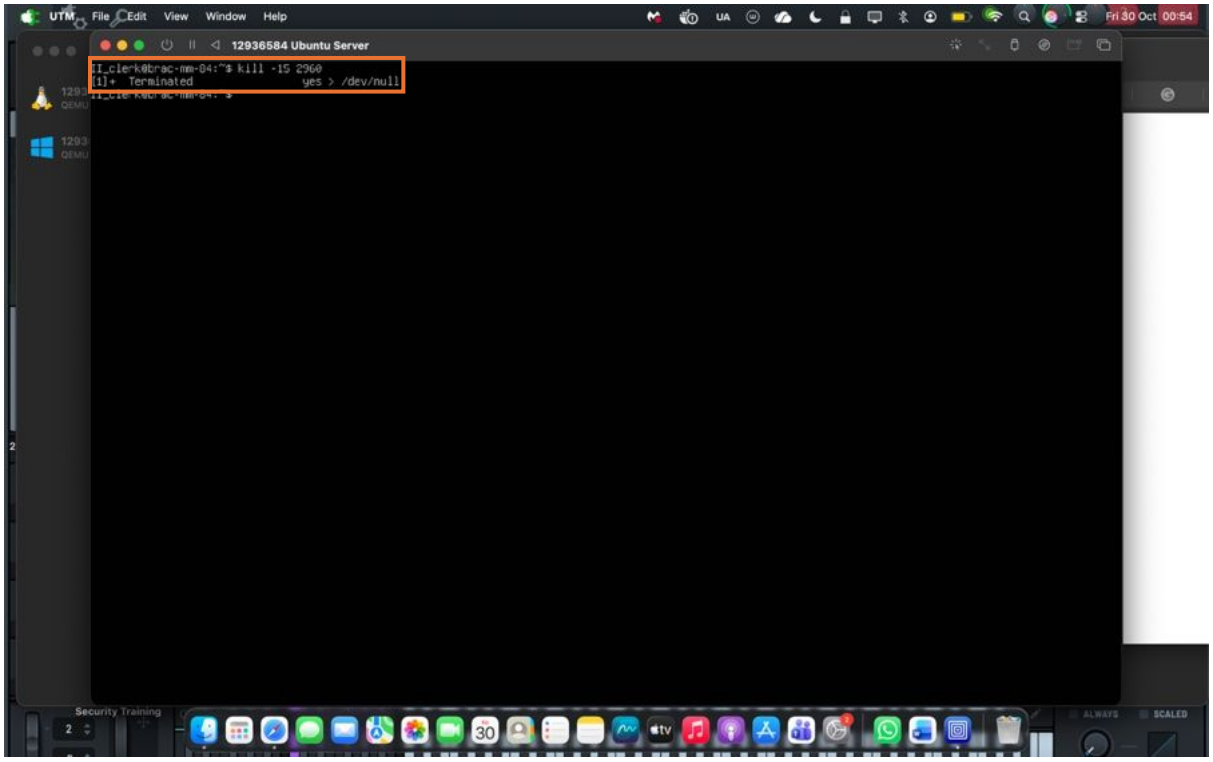
Identifying the process using top.



Identifying the process using ps.



Lowering the priority (renice).

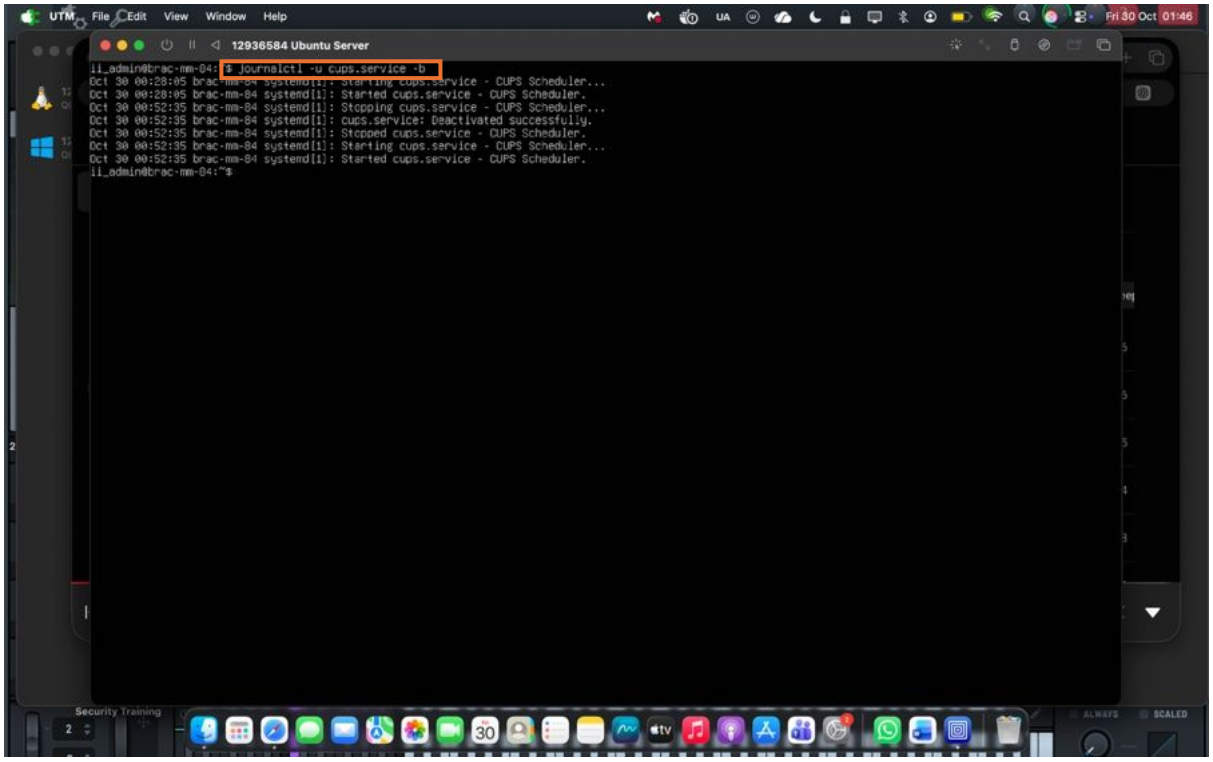


Terminating and confirming

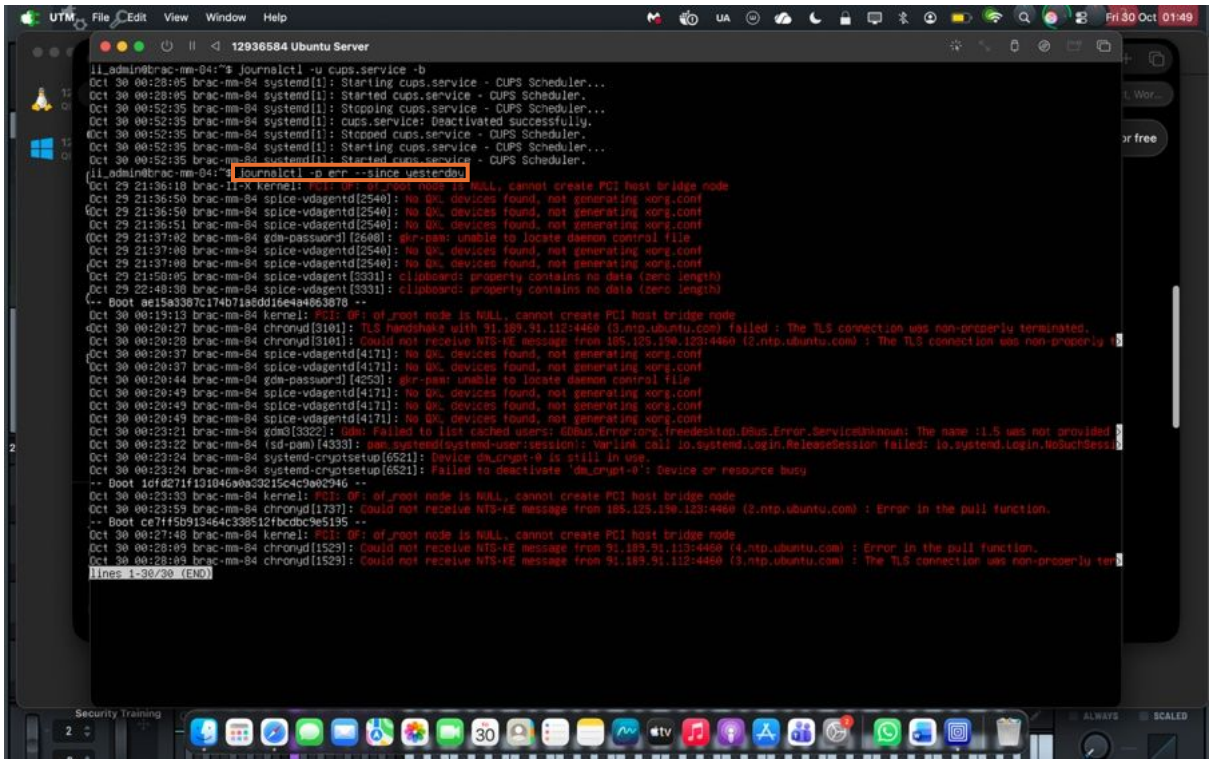
3.4

Kill -15, or SIGTERM, is a graphical termination signal. It politely asks a process to stop. It allows the process to catch the kill signal, save its current state, close files and clean up temporary data safely, preventing data corruption.

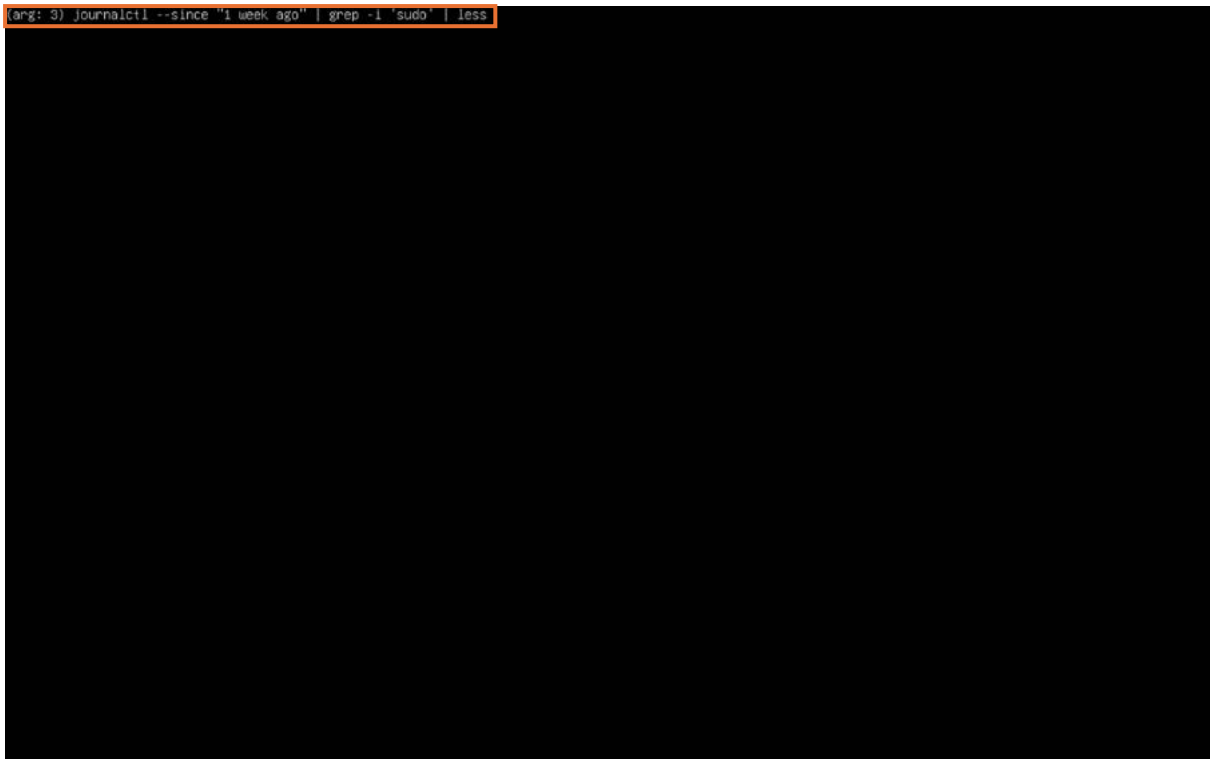
Kill -9, or SIGKILL, is a forced immediate kill. It abruptly halts the process without allowing it to catch ignore or handle the signal. This is useful for frozen unresponsive processes that ignore SIGTERM (kill -15).



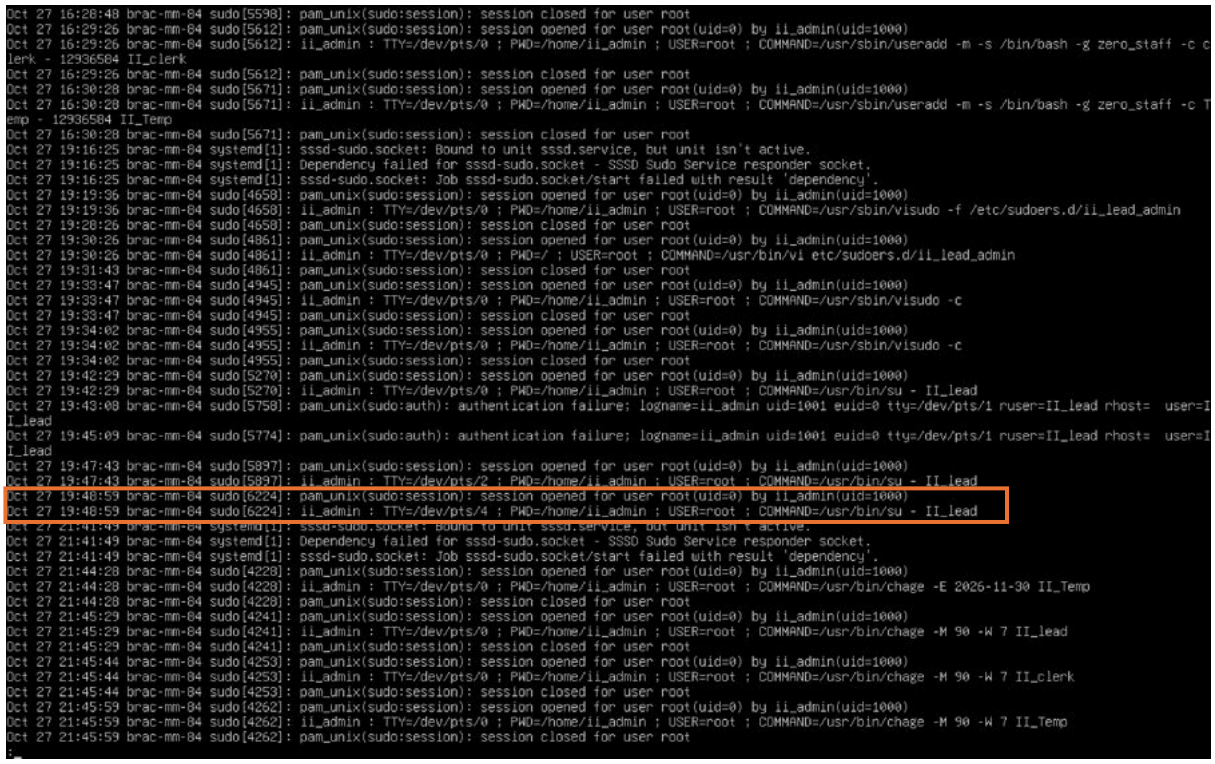
CUPS service logs since the last boot.



All entries of priority err or worse since yesterday.

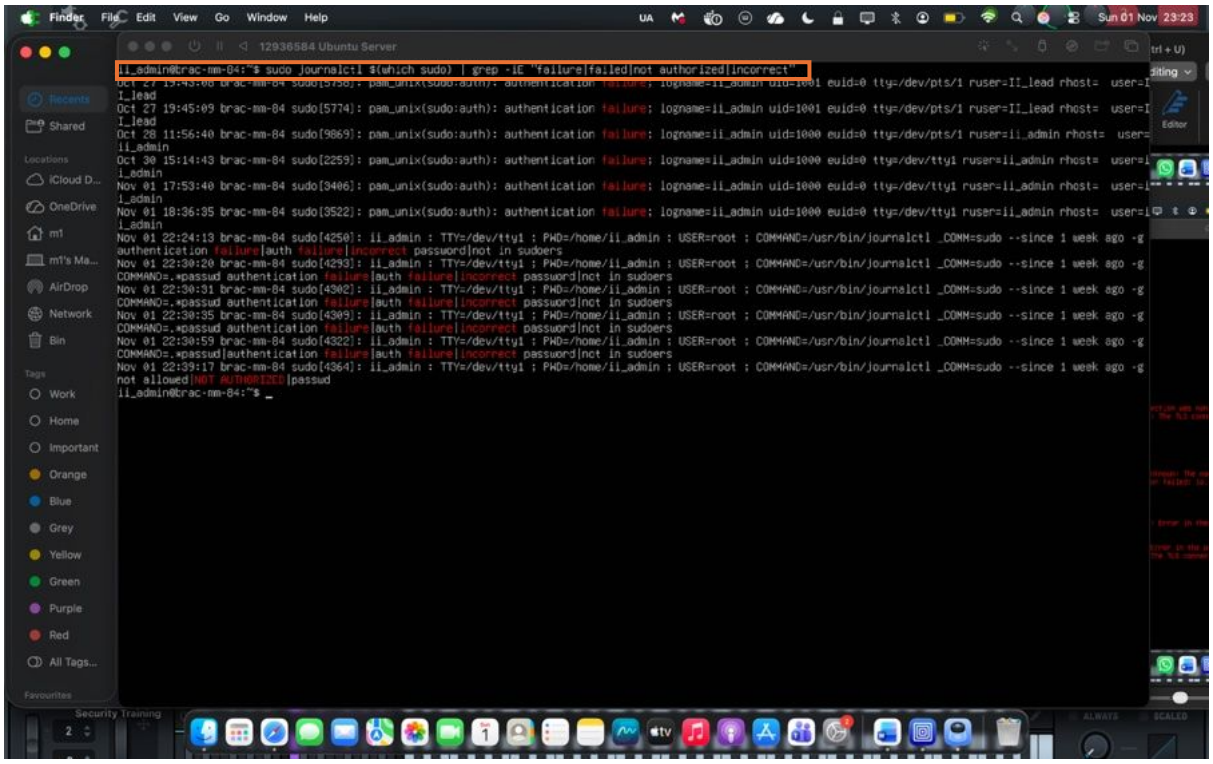


Entries generated by me for failed sudo attempts since last week.



At 19:49 user II_lead failed (was restricted) a sudo passwd command for II_clerk.

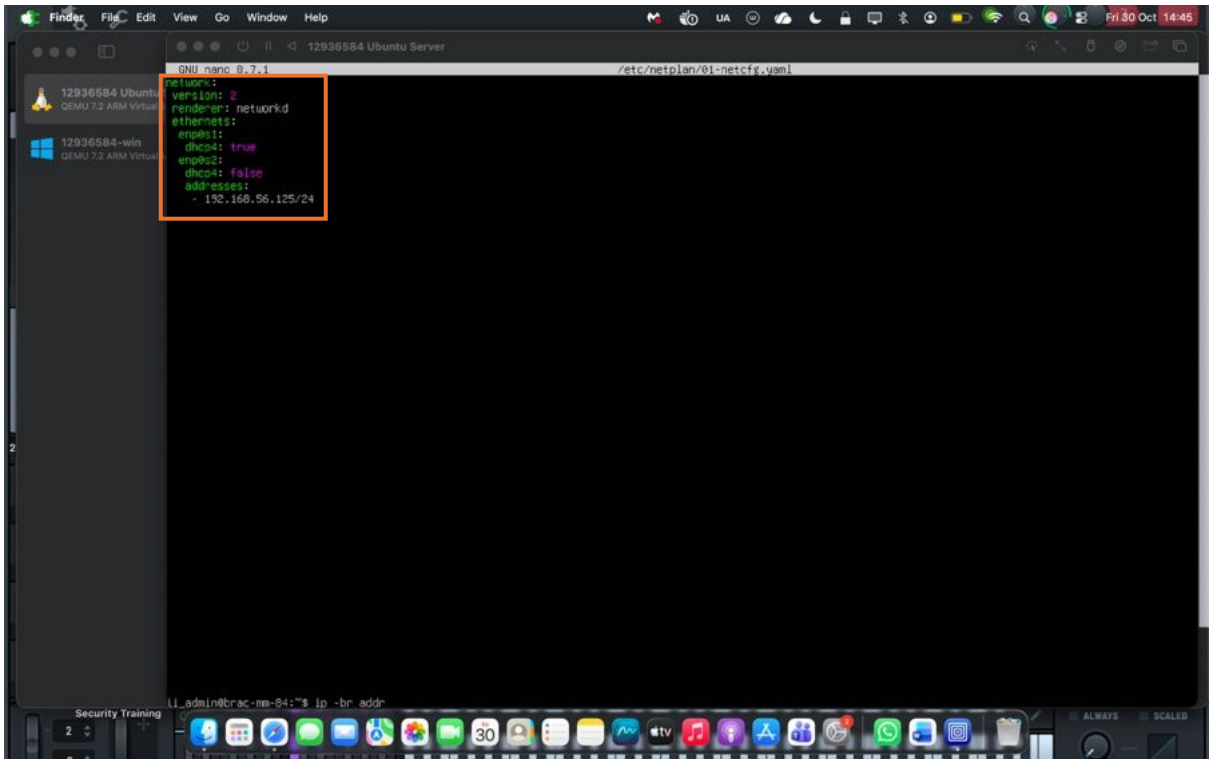
3.5



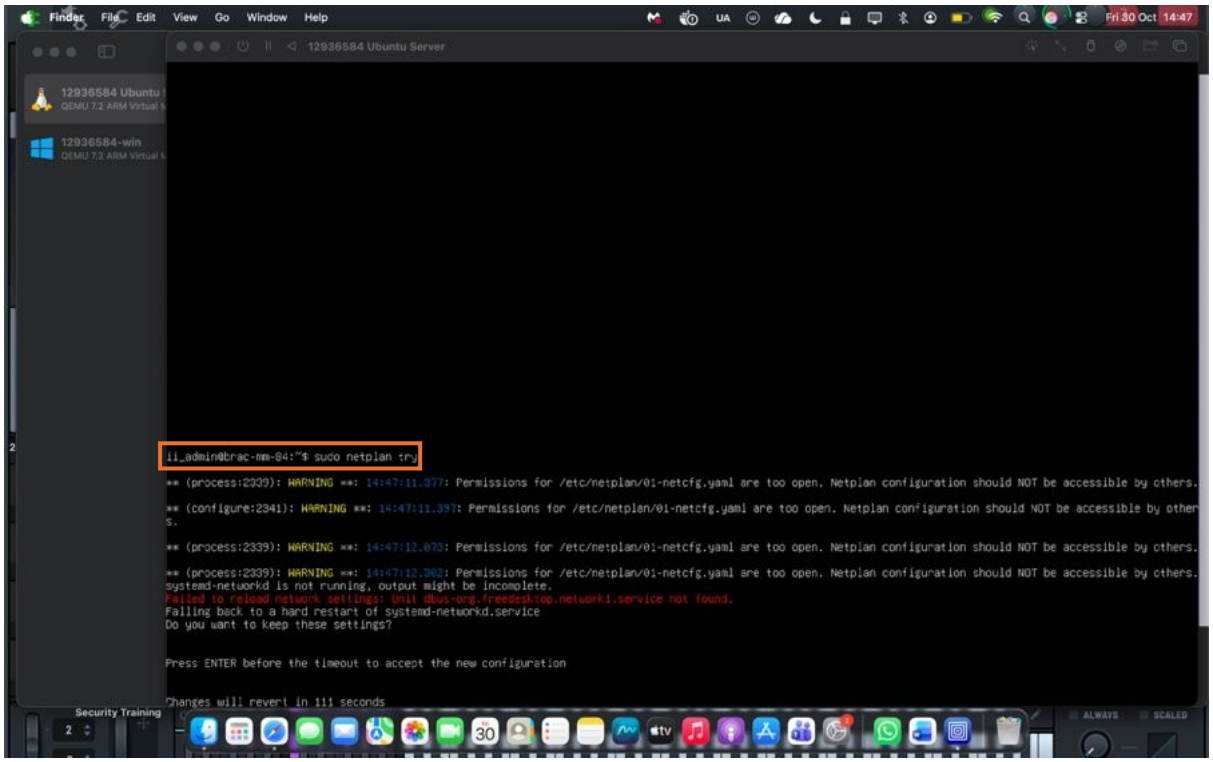
Between October 27th and November 1st, multiple failed sudo command attempts were made by user “II_lead” **and** “ii_admin” on brac-mm-04. On October 27th, at 19:43:00 and 19:45:09, “II_lead” attempted sudo commands, both failing authentications. The attempts were made by “ii_admin” with user ID 1001 and effective user ID 0, using terminal /dev/pts/1, with the remote user identified as “II_lead”.

On October 28th at 11:56:40, “ii_admin” attempted another sudo command, failing authentication with user ID 1000 and terminal /dev/tty1, with the remote user as “ii_admin”.

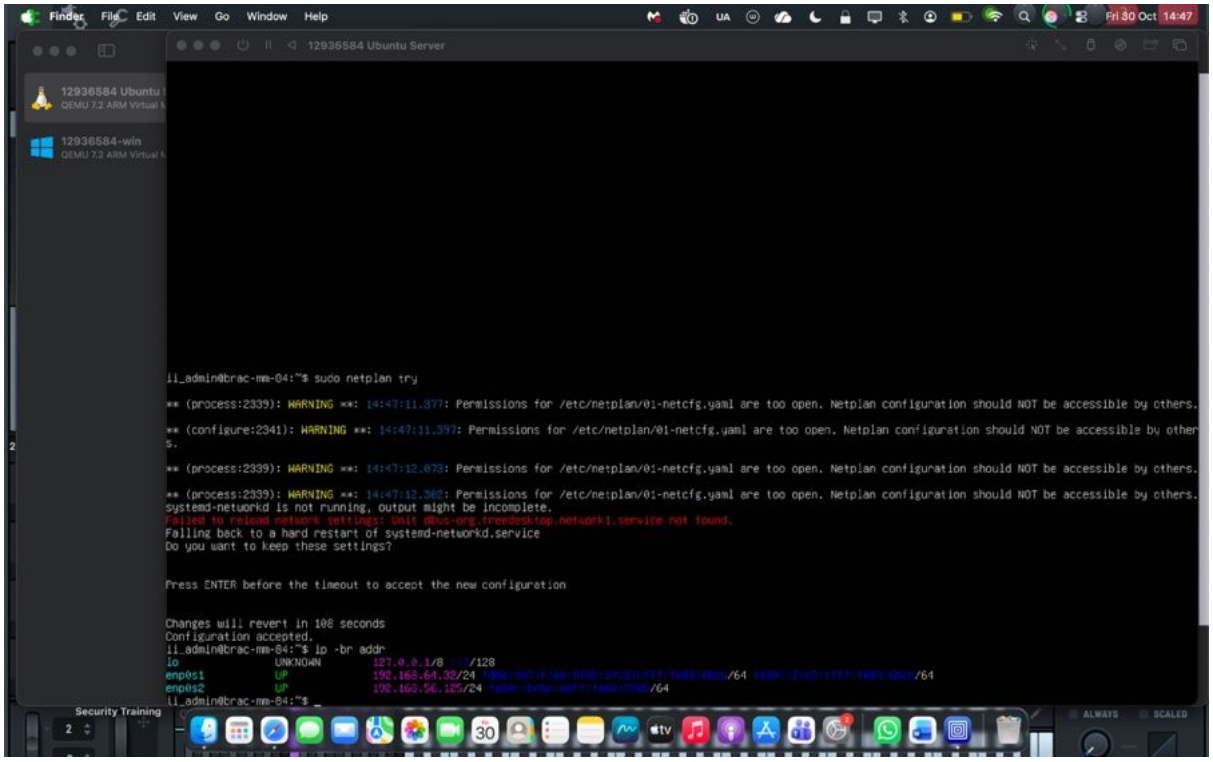
On October 30th **at** 15:14:43, another failed attempt was made by **“**ii_admin**”,** with remote user as “ii_admin”, Two more failed attempts occurred on November 1st at 17:50:40 and 18:30:35**,** with consistent user ID, terminal, and remote user details.

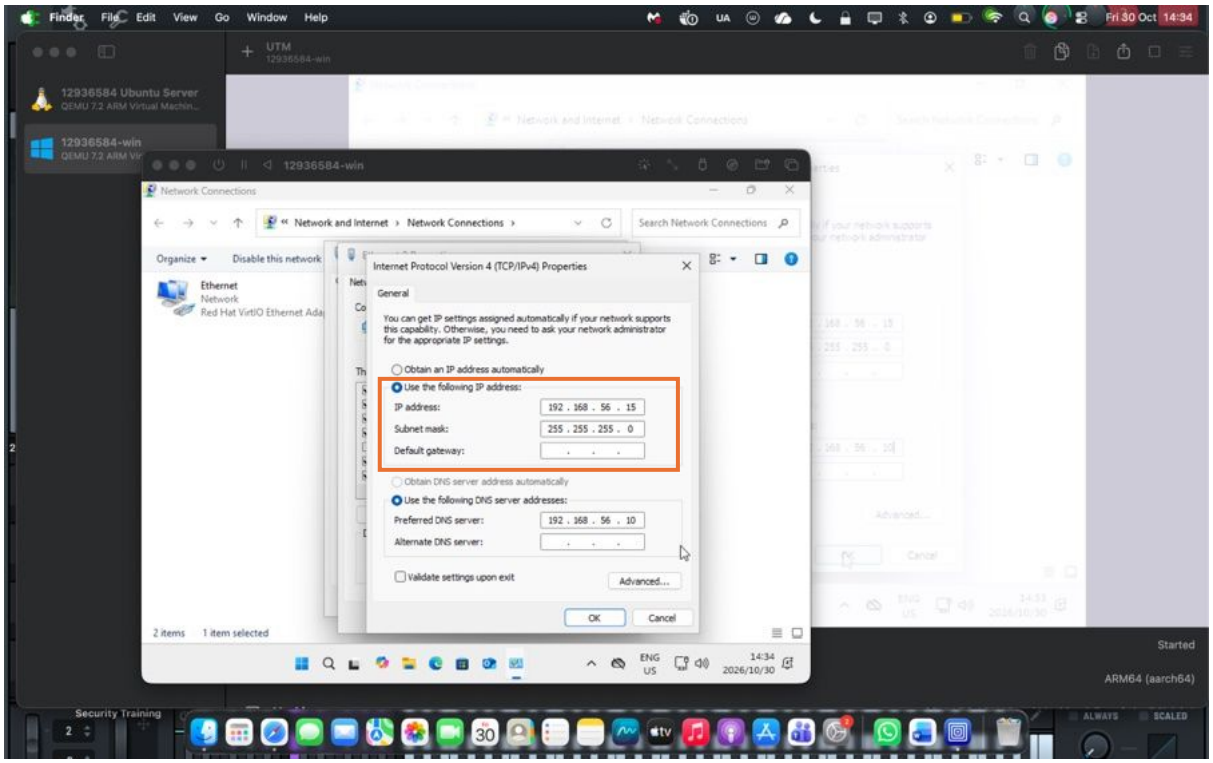


Configuring Netplan (Ubuntu Server).

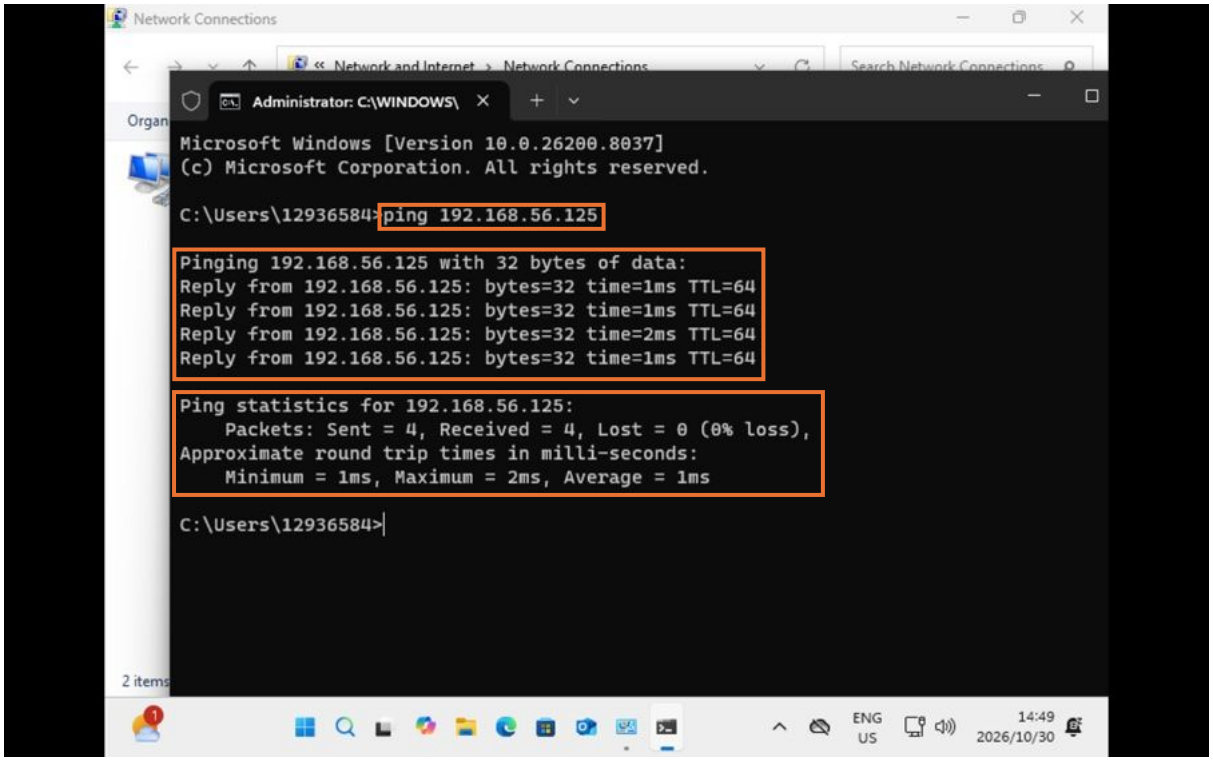


Testing the Configuration safely before applying with sudo netplan try.

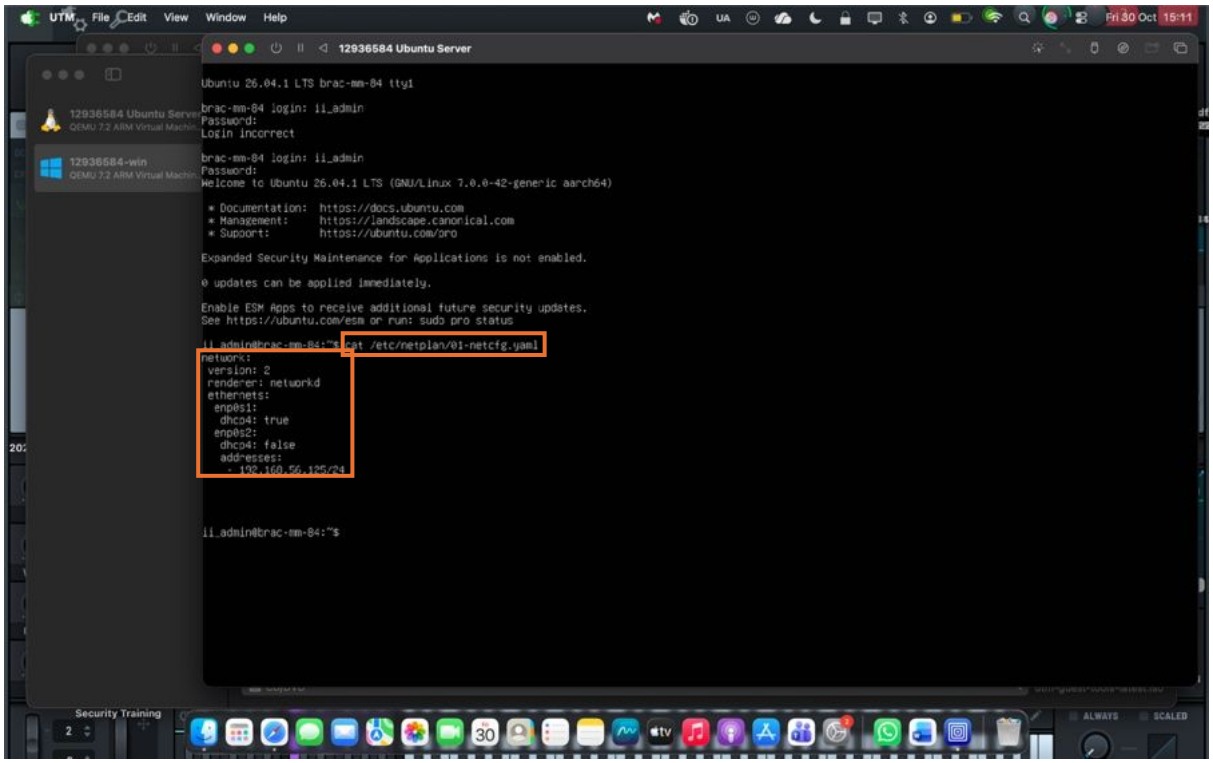




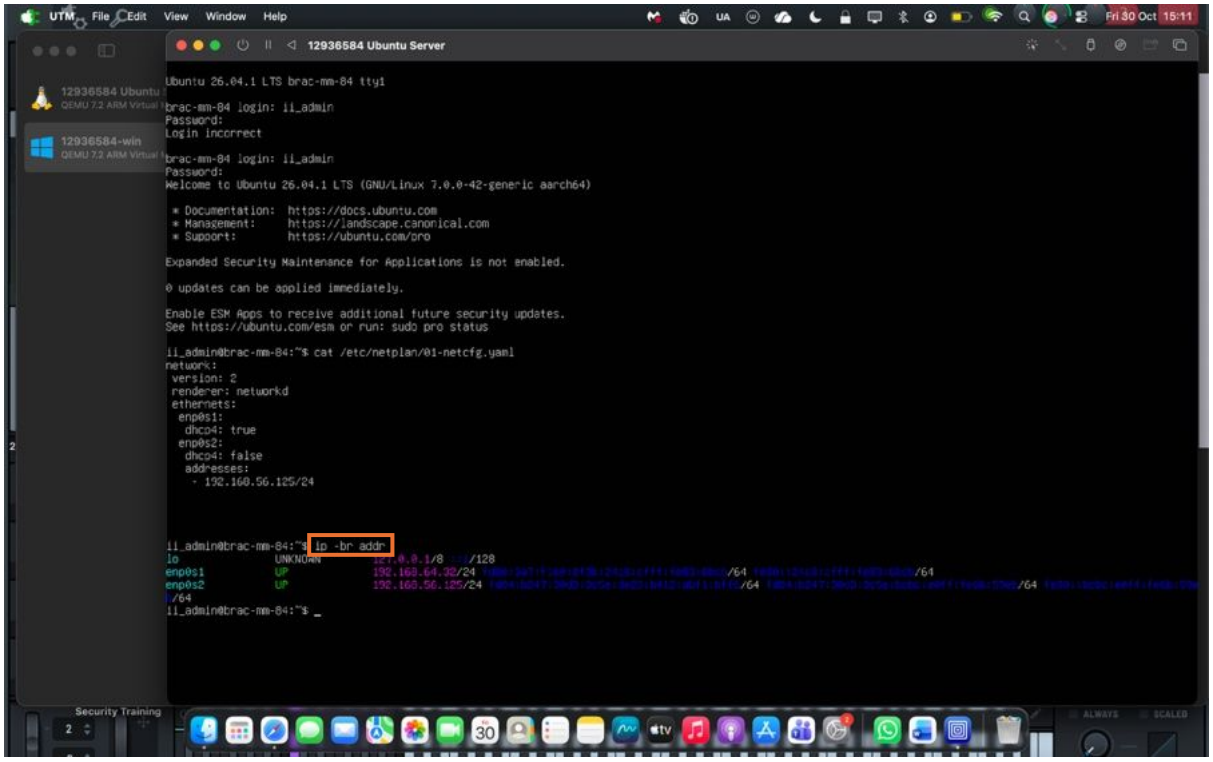
Setting IP Address and Subnet Mask on Host-Only network.



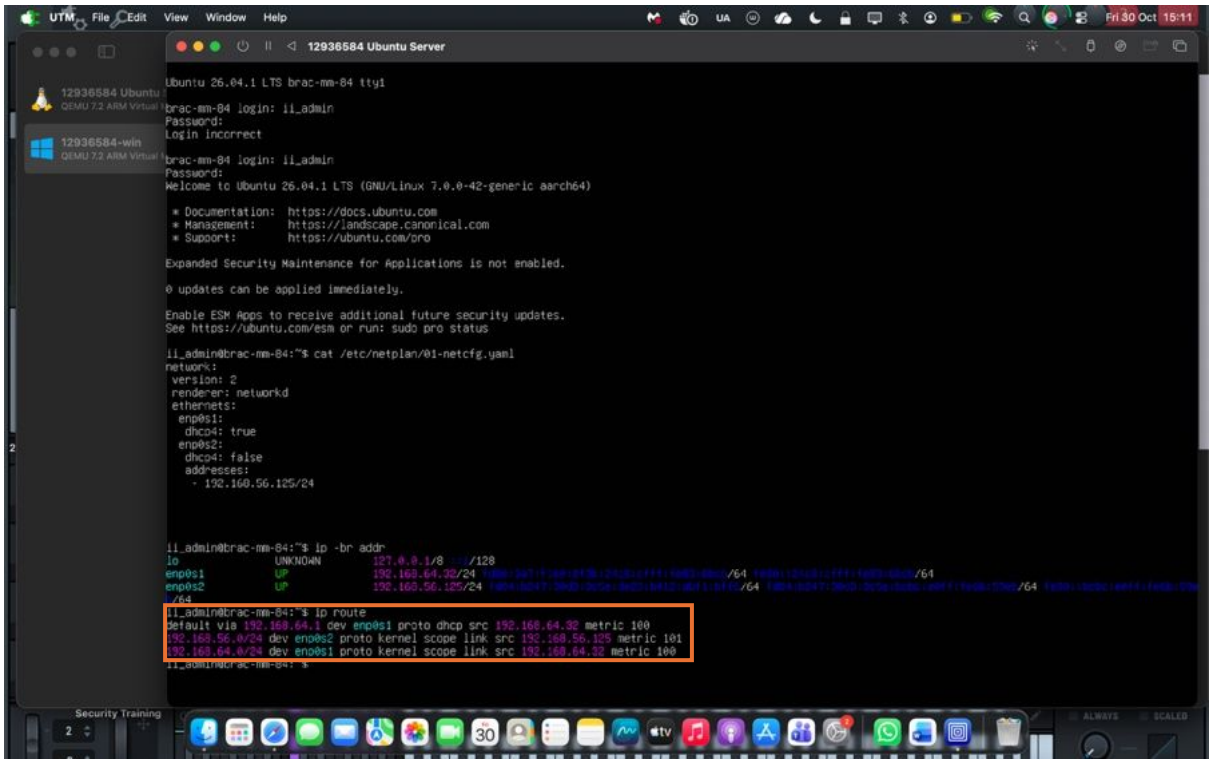
Checking if Pingable.



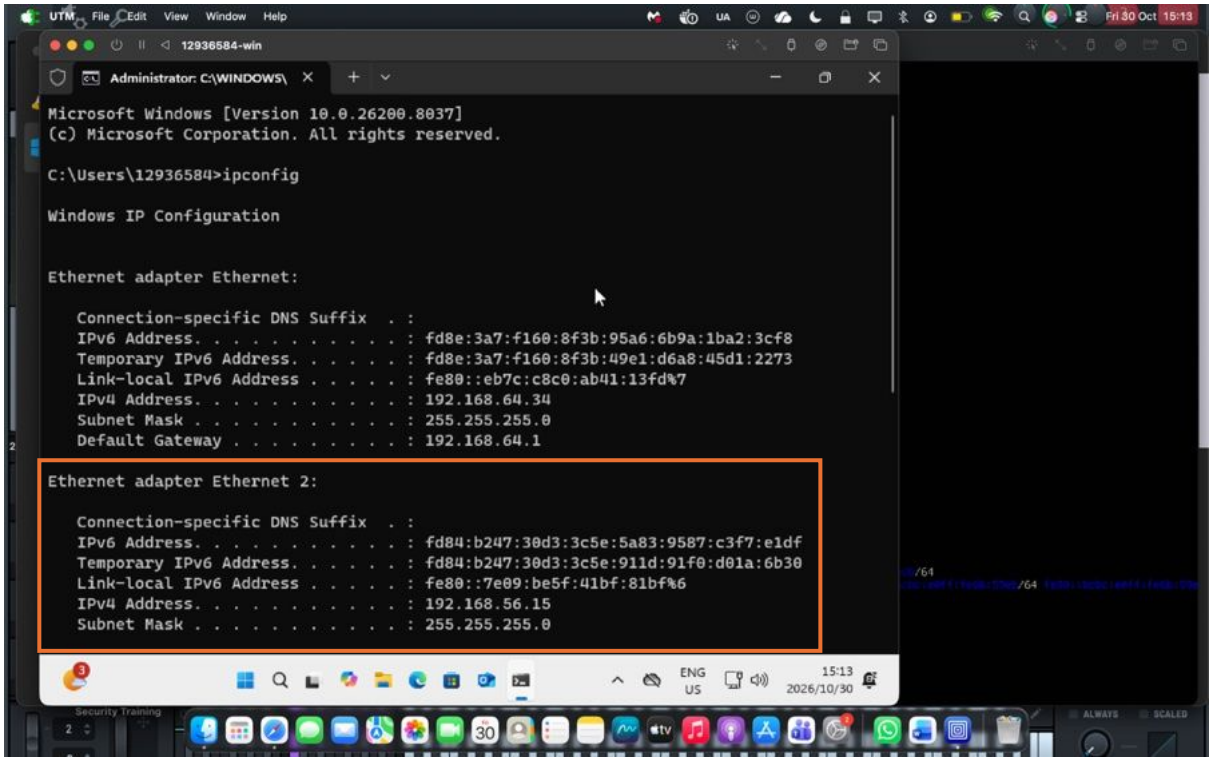
Verifying Netplan Ubuntu configuration file. (Server).



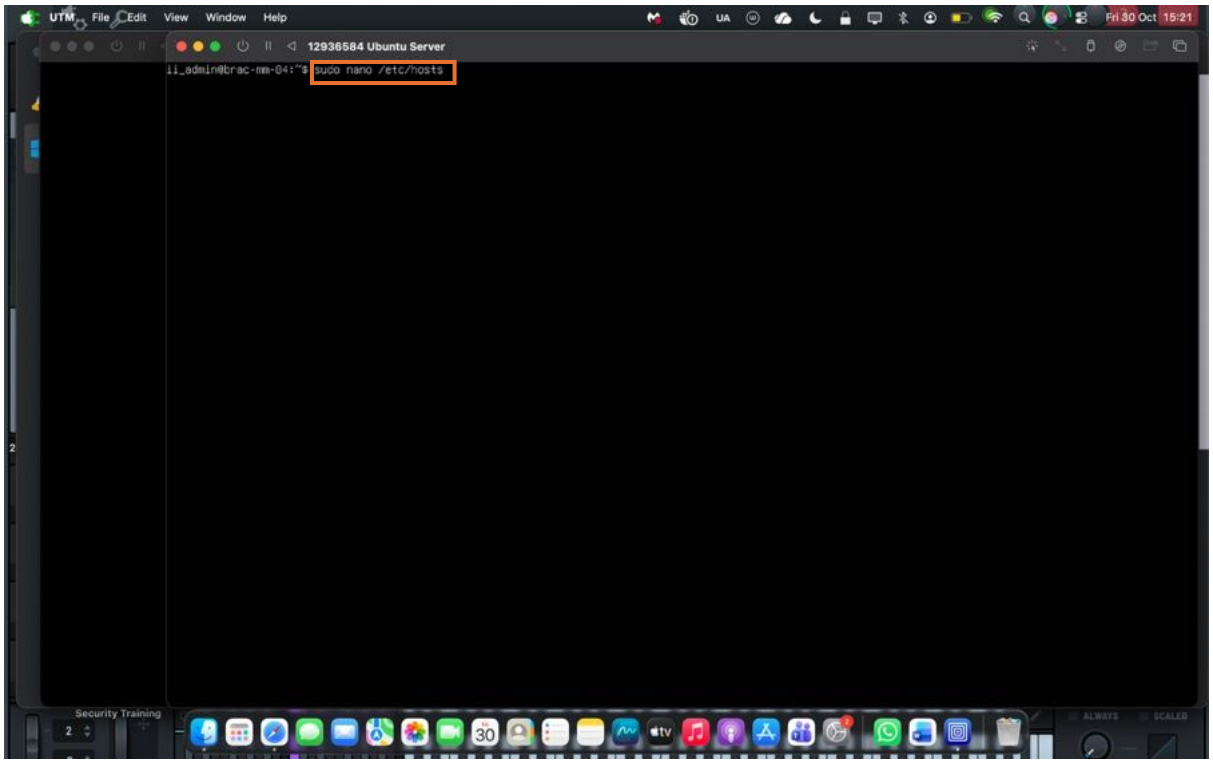
Verifying Ubuntu IP with ip -br addr



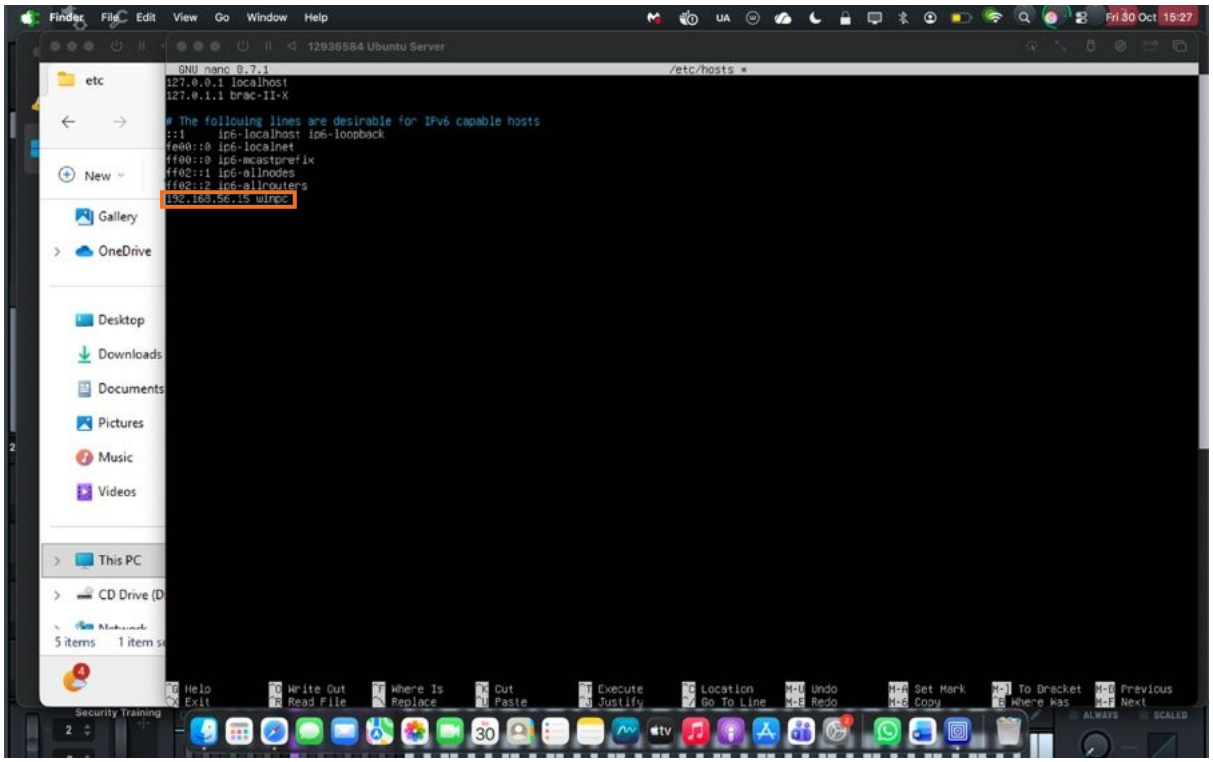
Verifying Ubuntu IP with ip route.



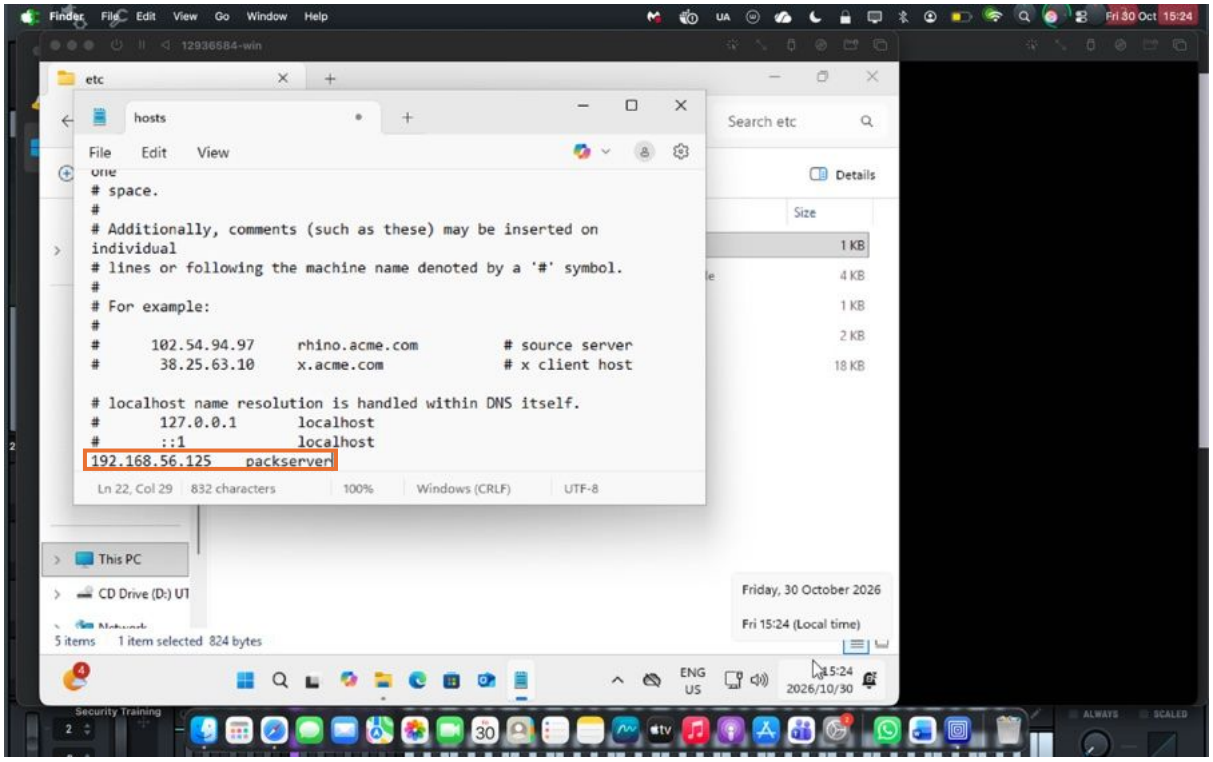
Verifying Windows IP.



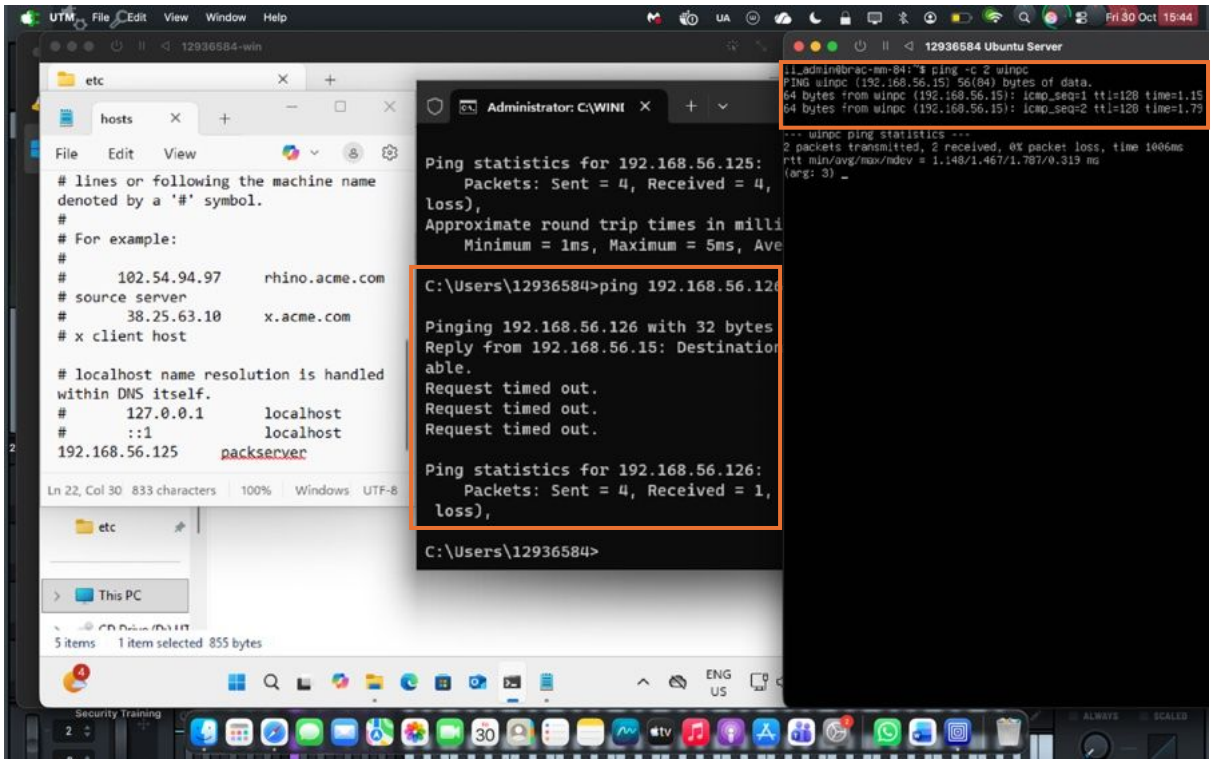
Opening /etc/hosts to edit/configure.



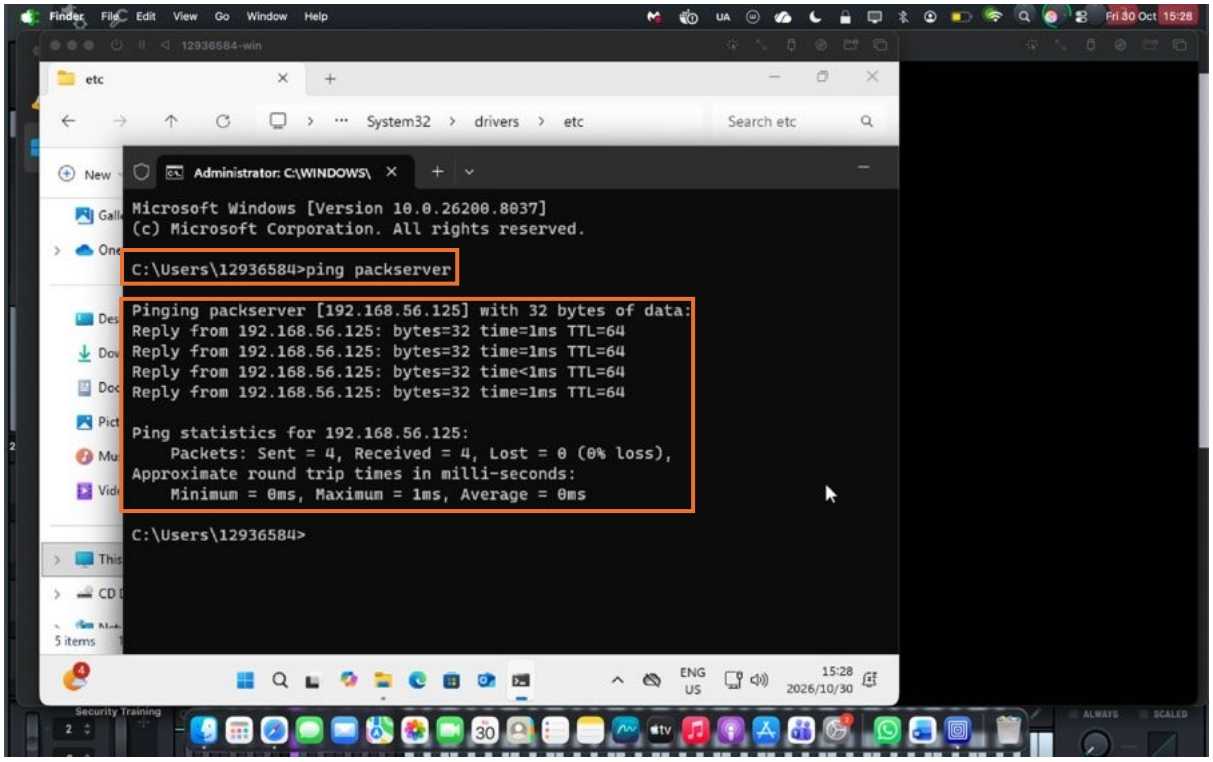
Configuring Ubuntu /etc/host with addition of Windows IP and naming.



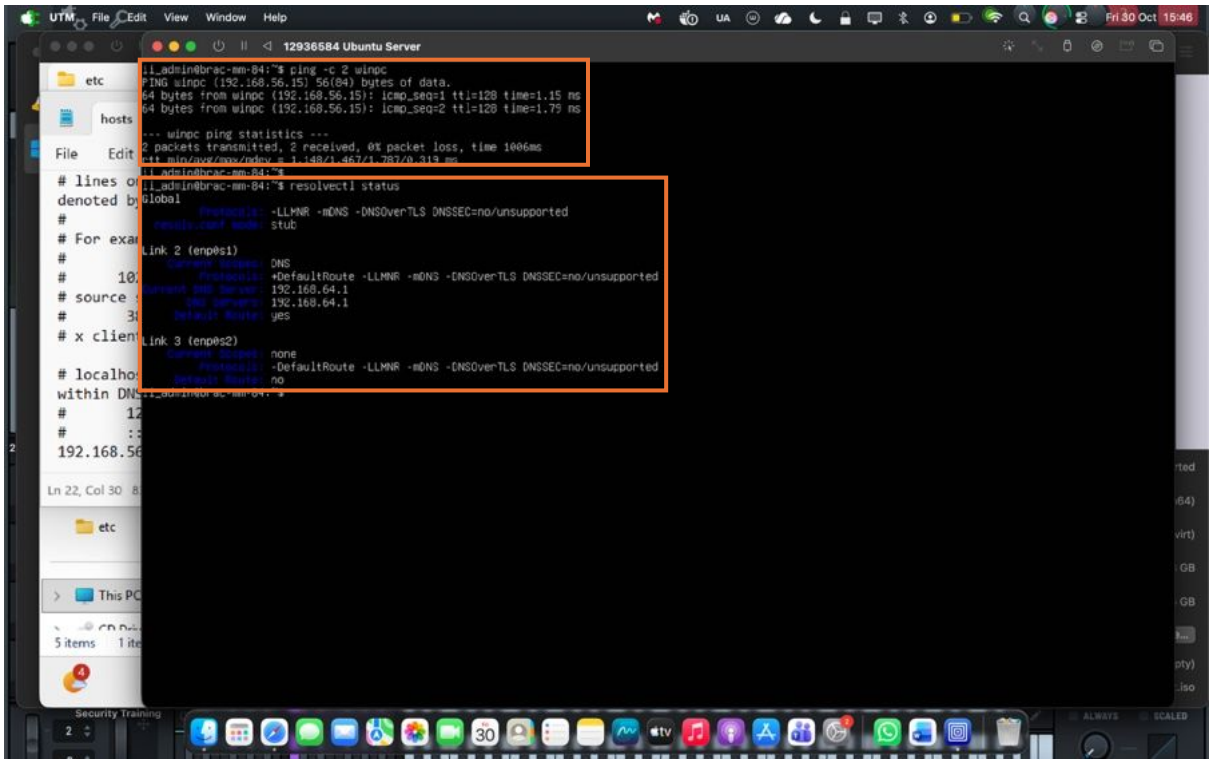
Configuring Windows /etc/host with addition of Ubuntu IP and naming.



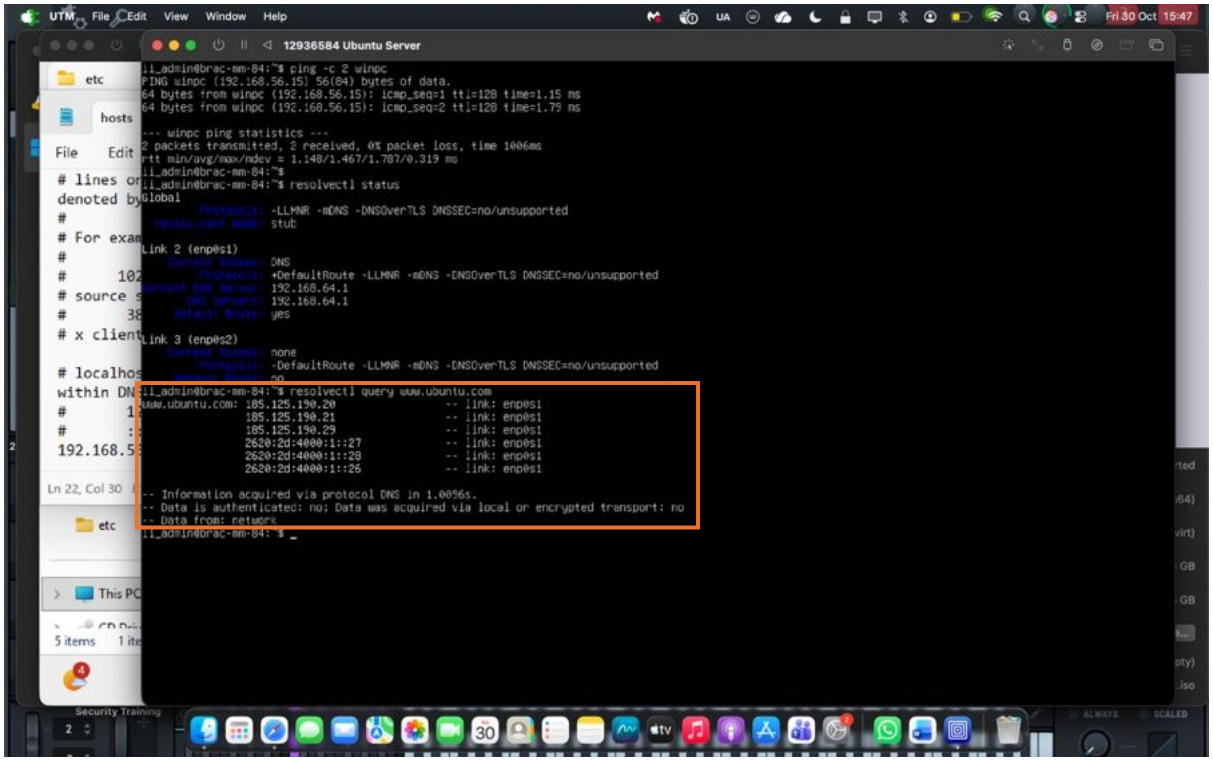
Pinging both devices to verify.



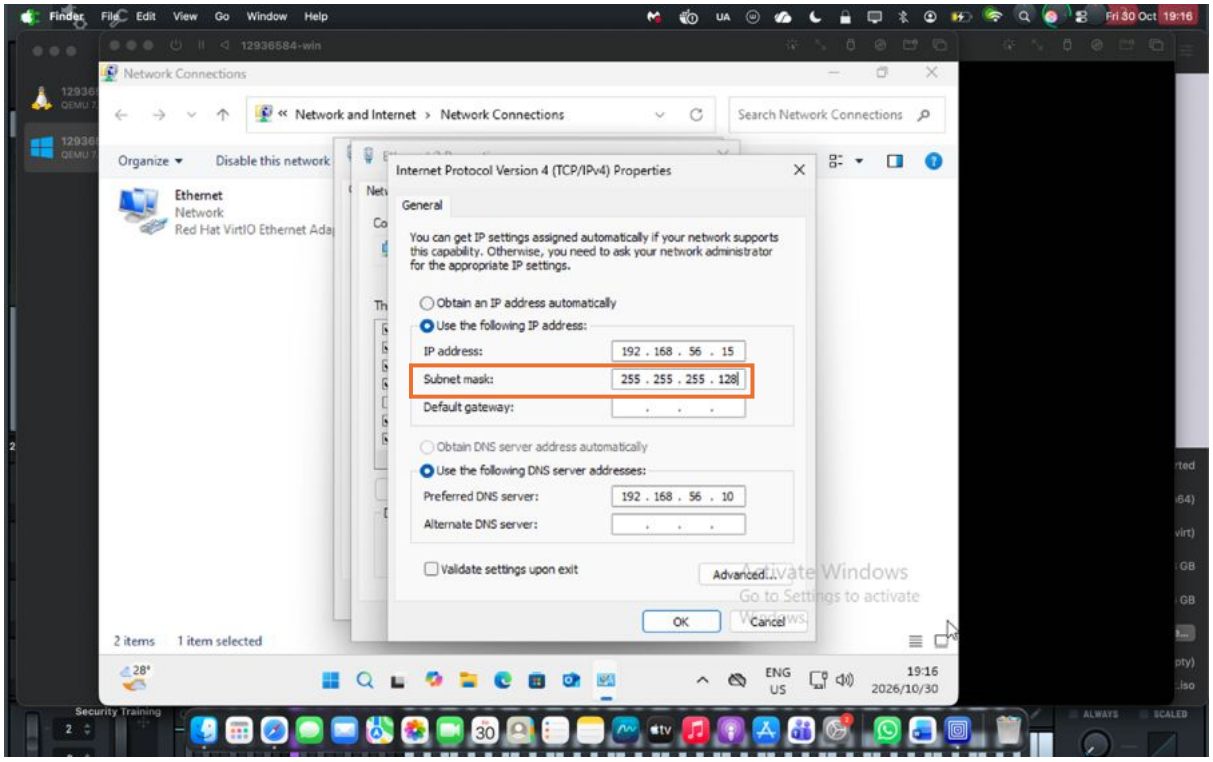
Pinging with ping packserver. (Windows resolving Ubuntu).



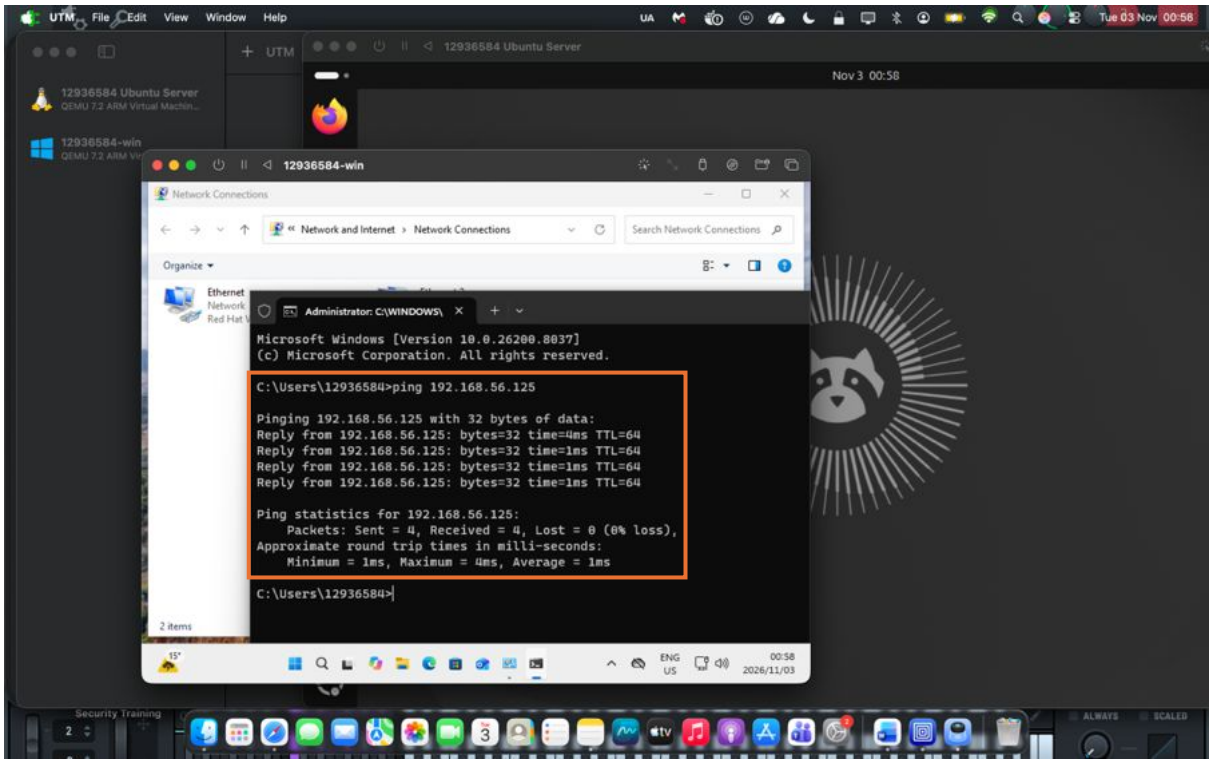
Pinging with ping -c 2 winpc. (Ubuntu resolving Windows).



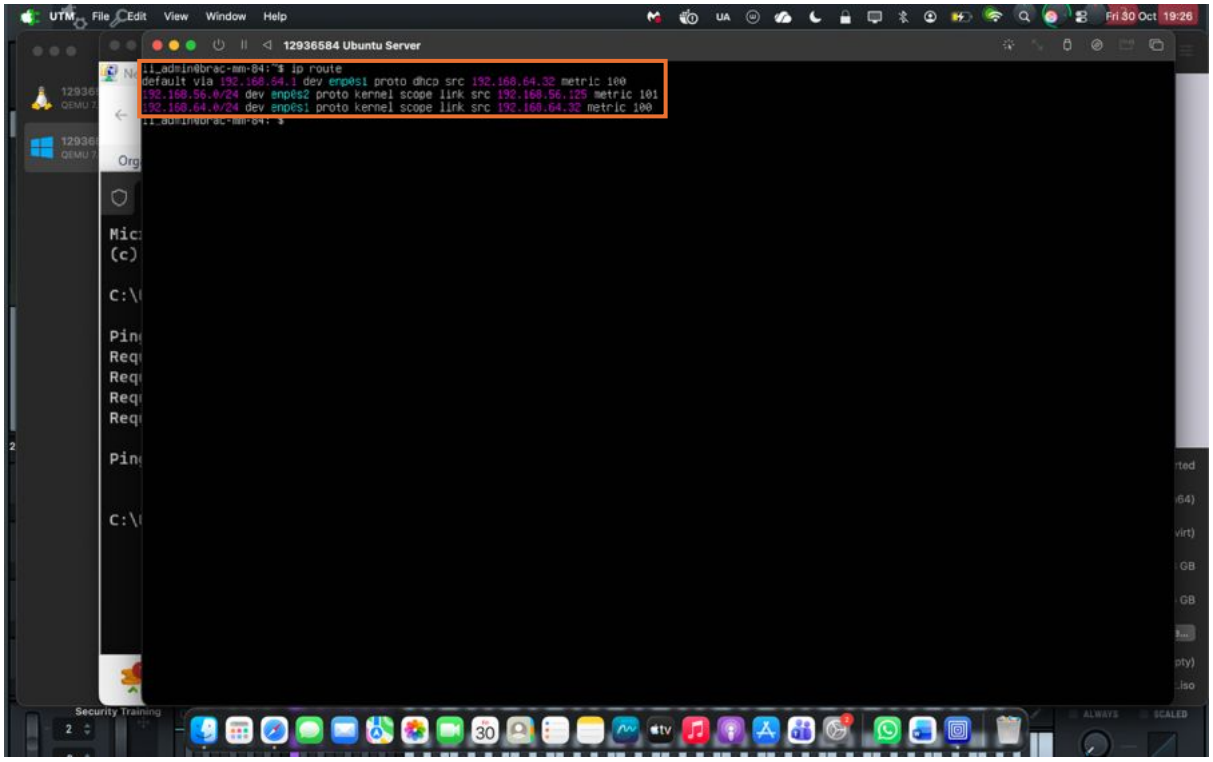
Ubuntu DNS Status proof and internet Query proof.



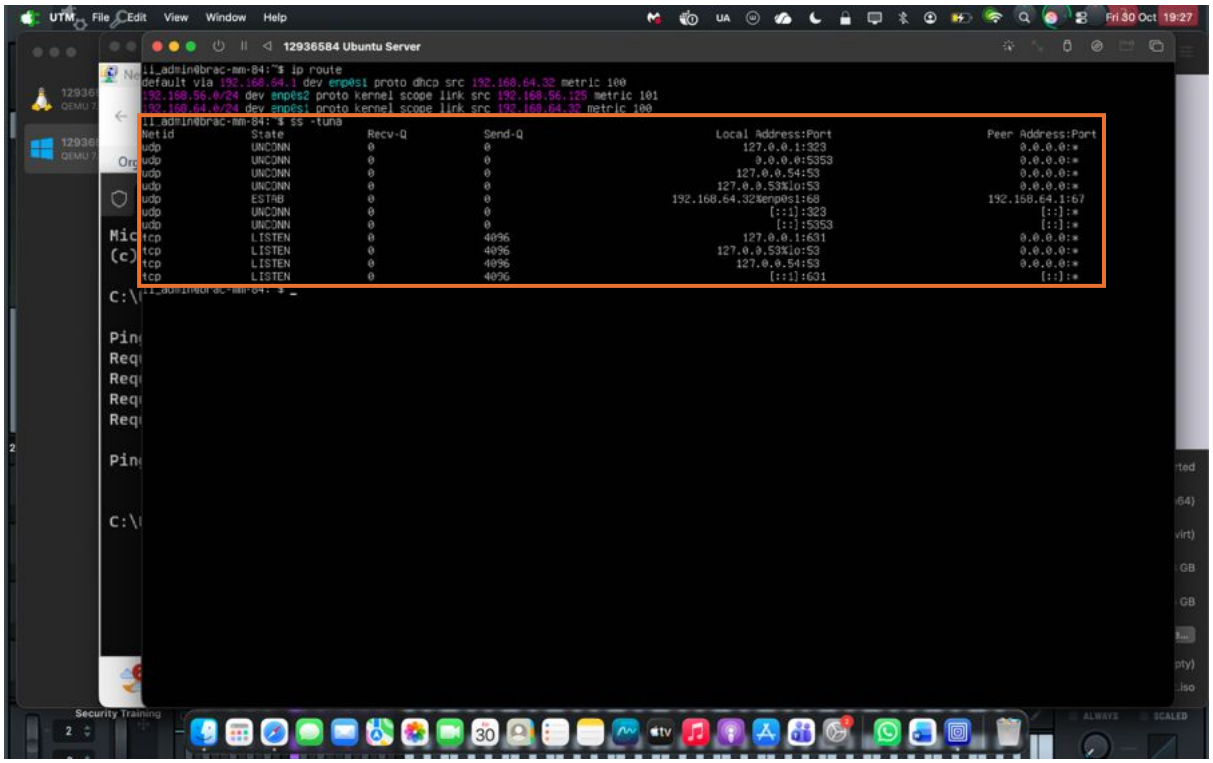
Reproducing issue on Windows.



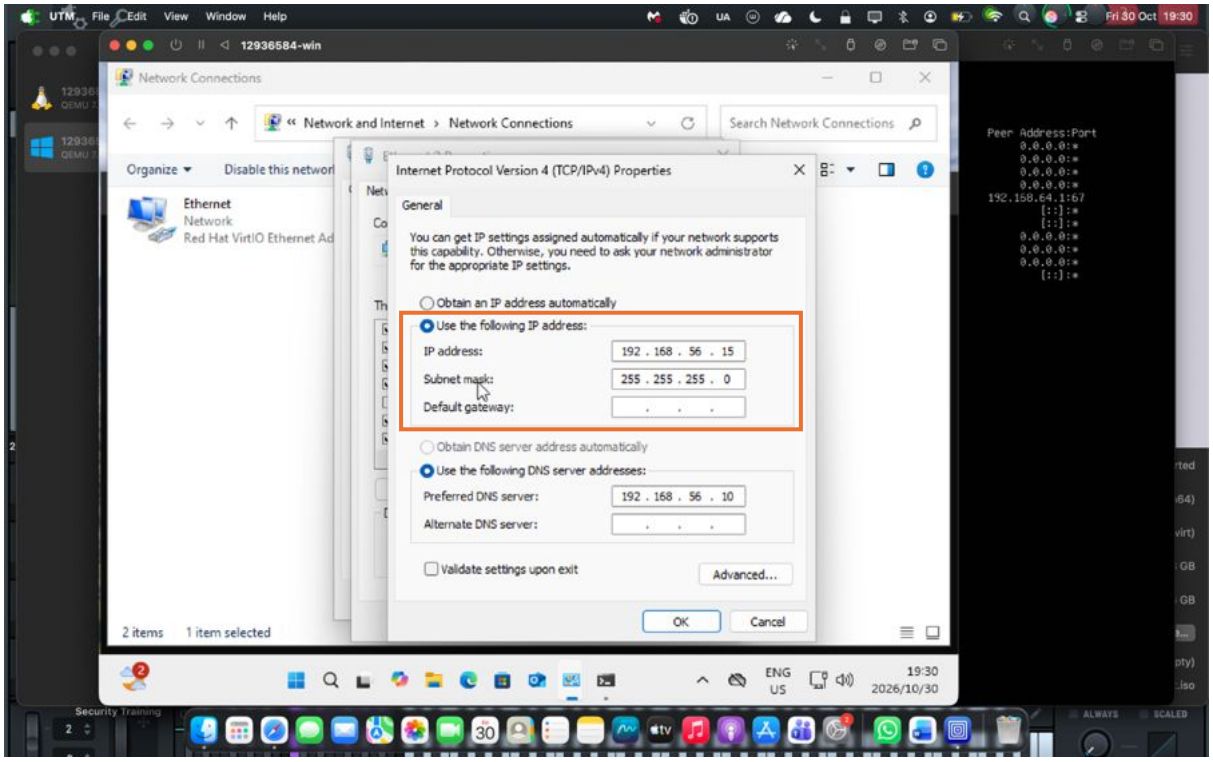
Pinging packserver (Ubuntu).



Linux diagnosis with ip route.



Linux diagnosis with ss -tuna.



Resolving and change back to /24 subnet mask.

4.3 My ping succeeded the two addresses because both VMs still fall within the same active half of the split subnet. With the maths of subnet 255.255.255.0 covering address from .1 to .254 split into 2 subnets. Subnet 1 from .1 too .126 and subnet 2 from .129 to .254.

4.4 Connectivity checklist

1 Verify host-only interface

- Ip -br addr show dev enp0s2
- Good Result: enp0s2 UP 192.168.56.[120+X]/24
- Bad Result: DOWN or Missing IP

2 Confirm local Routing priorities

- Ip route show
- Good Result: default via 10.0.2.2 dev enp0s2 proto dhcp src 10.10.2.15 metric 100 192.168.56.0/24 dev enp0s2 proto kernel scope link src 192.168.56.[120+X]
- Bad Result: exist default gateway on host-only breaking external internet access.

3 Test Local Connectivity and Host Mapping

- Ping -c winpc
- Good Result: PING winpc (192.168.56.[10+X]) 56(84) bytes of data,
  64 bytes from winpc (192.168.56.[10+X]): ICMP_seq=1 ttl=128 time=0.421
- Bad Result: if returns name or service not known.

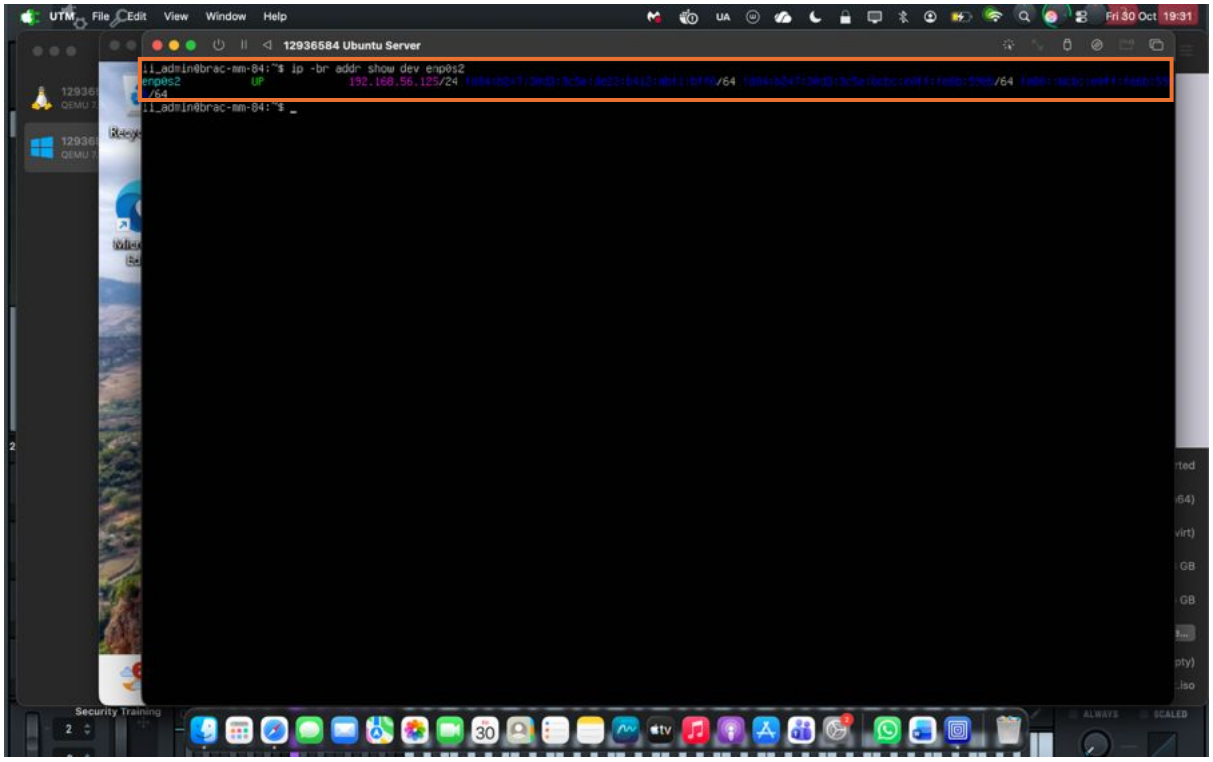
4 Audit Local Network Listening Socket

- ss -tln
- Good Result: State=Listen; Recv-Q=0; Send-Q Local=128; Address:Port=0.0.0.0:80; peer=; Address:Port=0.0.0.0* and Process
- Bad Results: no active services are bound to port 80 or exclusively bound to 127.0.0.1:80

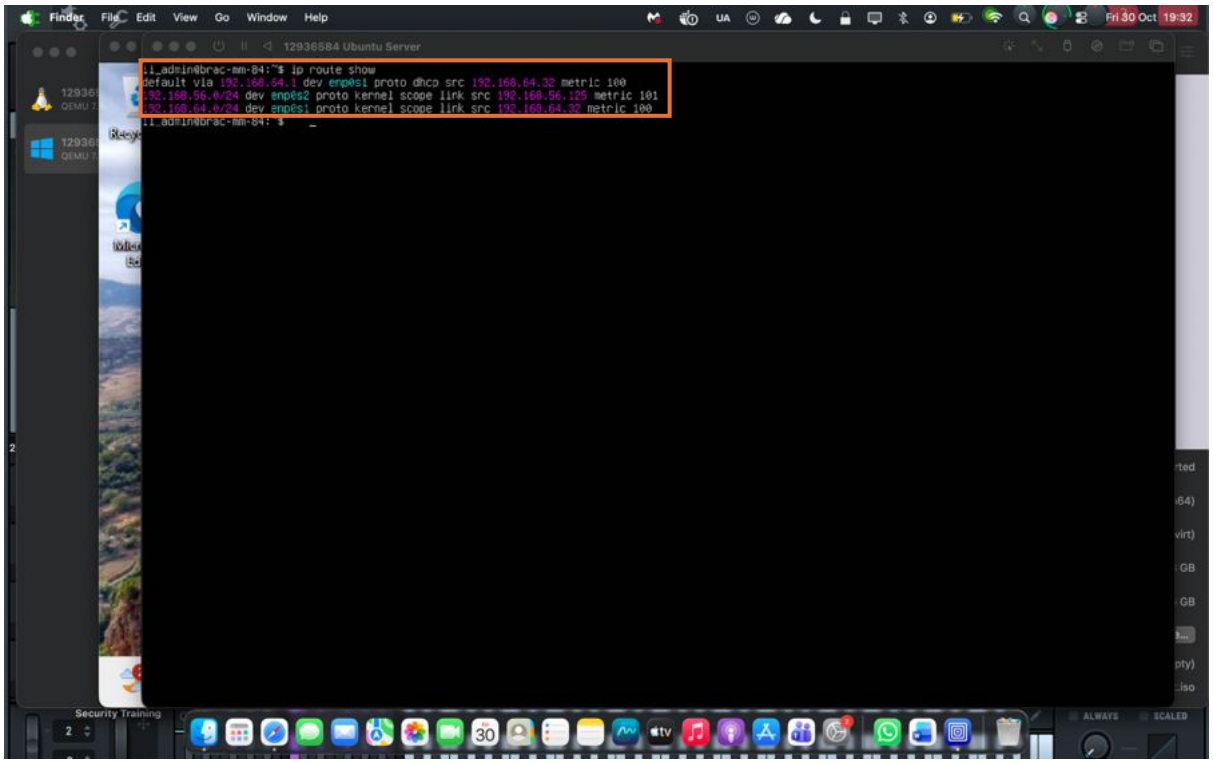
“Connection Refused” error when try to access packserver.

5 Verify Internet Name Resolution(DNS)

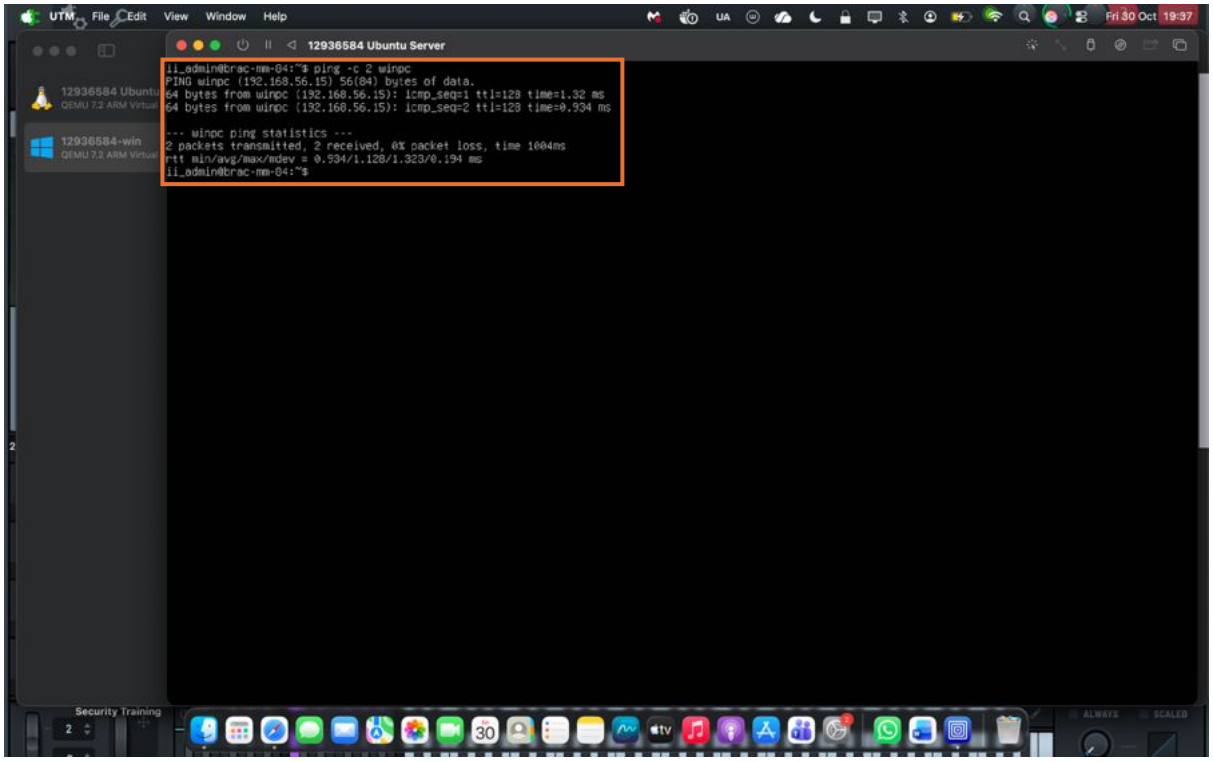
- Resolvectl query ://ubuntu.com
- Good Result: ://Ubuntu.com: 185.125.190.29 --Link:enp0s2 --information acquired via protocol DNS in 12ms.
- Bad Result: A timeout or failure error.


UTM File Edit View Window Help
Fri 30 Oct 19:31
12936584 Ubuntu Server
il_admin@brac-mm-84:~$ ip -br addr show dev enp6s2
enp6s2 UP 192.168.56.125/24
/64
il_admin@brac-mm-84:~$ _
Security Training

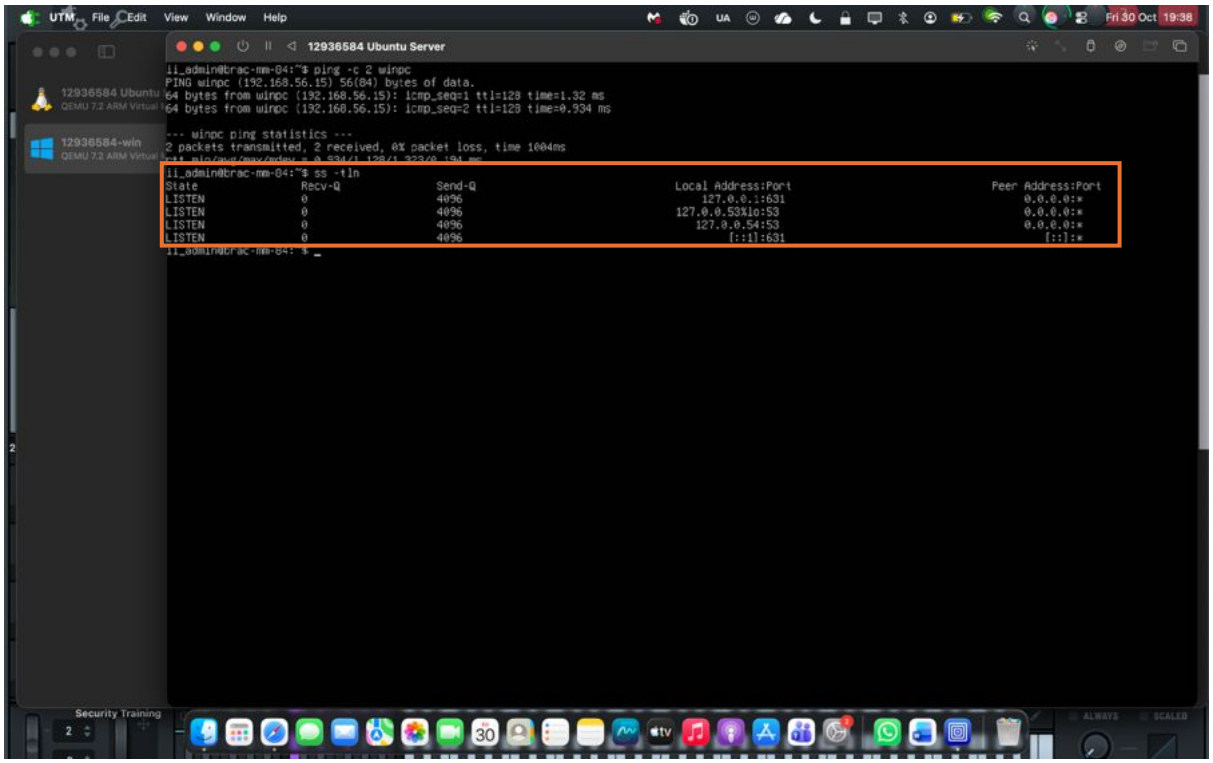
Step 1


Finder File Edit View Go Window Help
Fri 30 Oct 19:32
12936584 Ubuntu Server
il_admin@brac-mm-84:~$ ip route show
default via 192.168.64.1 dev enp6s1 proto dhcp src 192.168.64.32 metric 100
192.168.56.0/24 dev enp6s2 proto kernel scope link src 192.168.56.125 metric 101
192.168.64.0/24 dev enp6s1 proto kernel scope link src 192.168.64.32 metric 100
il_admin@brac-mm-84: $ _
Security Training

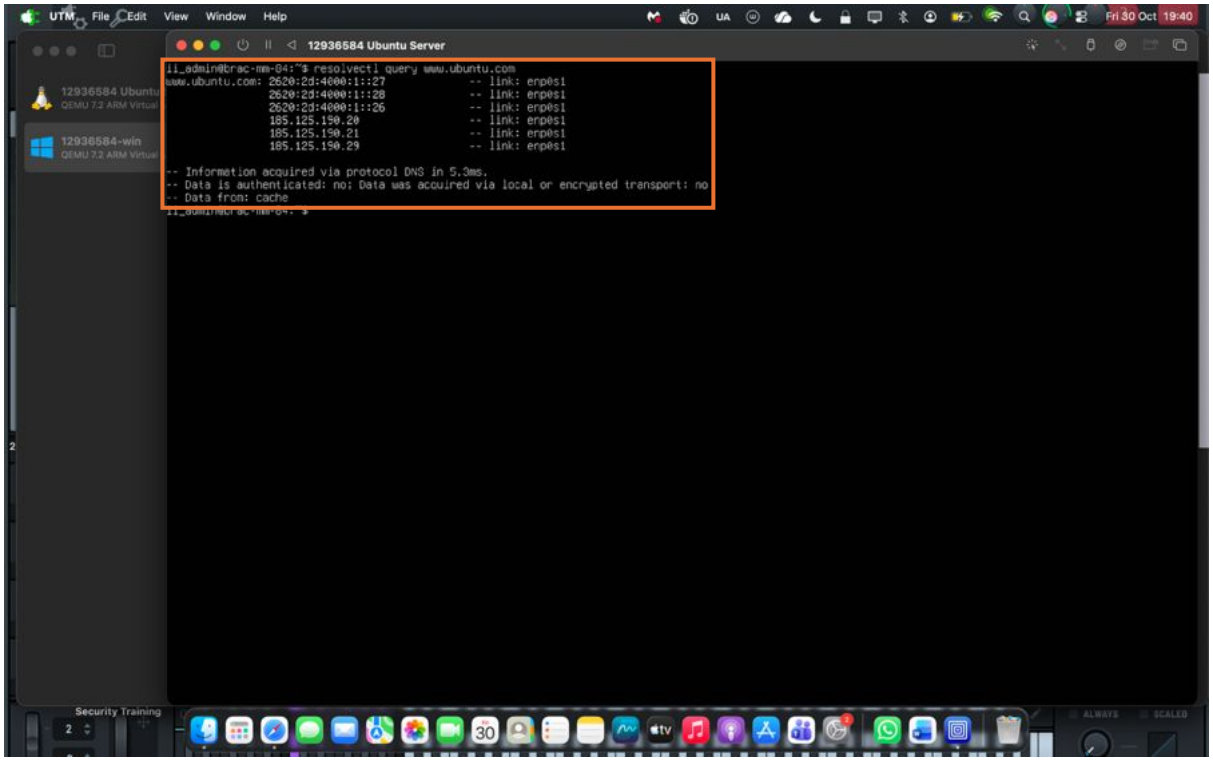
Step 2


UTM File Edit View Window Help
Fri 30 Oct 19:37
12936584 Ubuntu Server
ii_admin@trac-nm-04:~$ ping -c 2 winpc
PING winpc (192.168.56.15) 56(84) bytes of data.
64 bytes from winpc (192.168.56.15): icmp_seq=1 ttl=128 time=1.32 ms
64 bytes from winpc (192.168.56.15): icmp_seq=2 ttl=128 time=0.934 ms
--- winpc ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1004ms
rtt min/avg/max/mdev = 0.934/1.128/1.323/0.194 ms
ii_admin@trac-nm-04:~$
12936584 Ubuntu
QEMU 7.2 ARM Virtual
12936584-win
QEMU 7.2 ARM Virtual
Security Training

Step 3

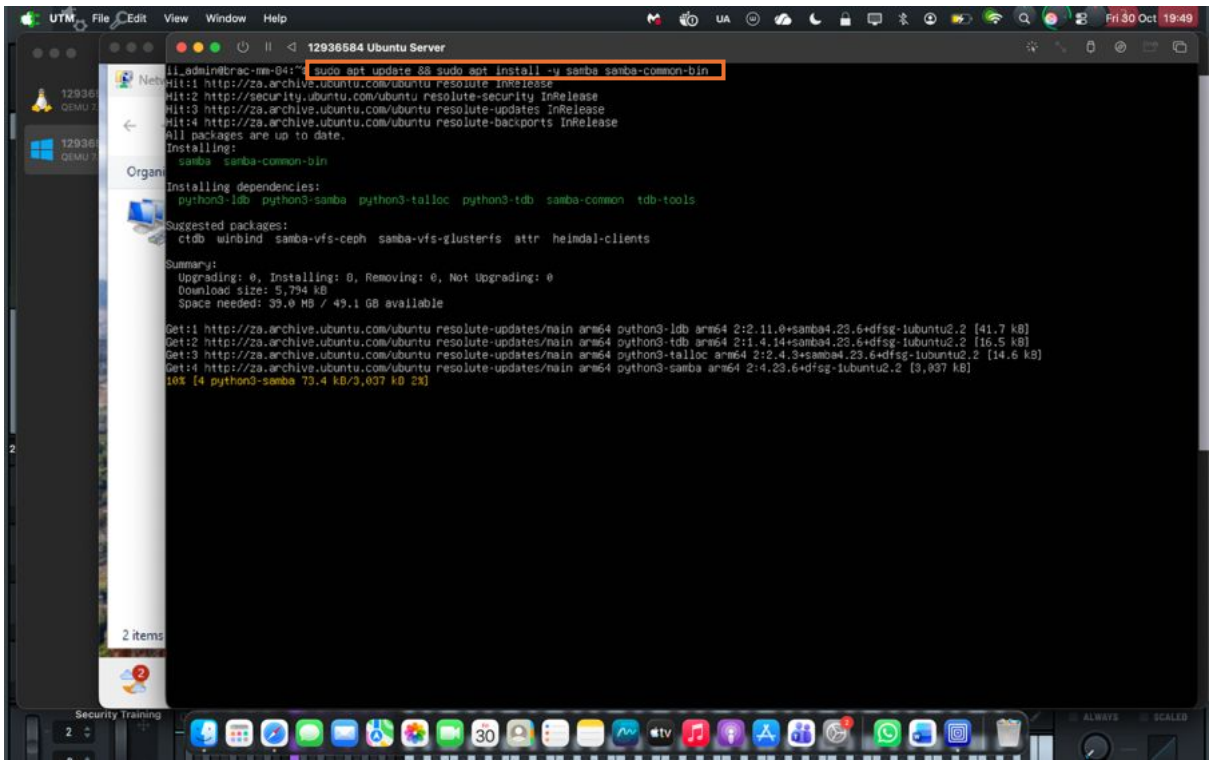

UTM File Edit View Window Help
Fri 30 Oct 19:38
12936584 Ubuntu Server
ii_admin@trac-nm-04:~$ ping -c 2 winpc
PING winpc (192.168.56.15) 56(84) bytes of data.
64 bytes from winpc (192.168.56.15): icmp_seq=1 ttl=128 time=1.32 ms
64 bytes from winpc (192.168.56.15): icmp_seq=2 ttl=128 time=0.934 ms
--- winpc ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1004ms
ii_admin@trac-nm-04:~$ ss -tln
State Recv-Q Send-Q Local Address:Port Peer Address:Port
LISTEN 0 4096 127.0.0.1:631 0.0.0.0:*
LISTEN 0 4096 127.0.0.53%lo:53 0.0.0.0:*
LISTEN 0 4096 127.0.0.54:53 0.0.0.0:*
LISTEN 0 4096 [::1]:631 [::]:*
ii_admin@trac-nm-04:~$
12936584 Ubuntu
QEMU 7.2 ARM Virtual
12936584-win
QEMU 7.2 ARM Virtual
Security Training

Step 4

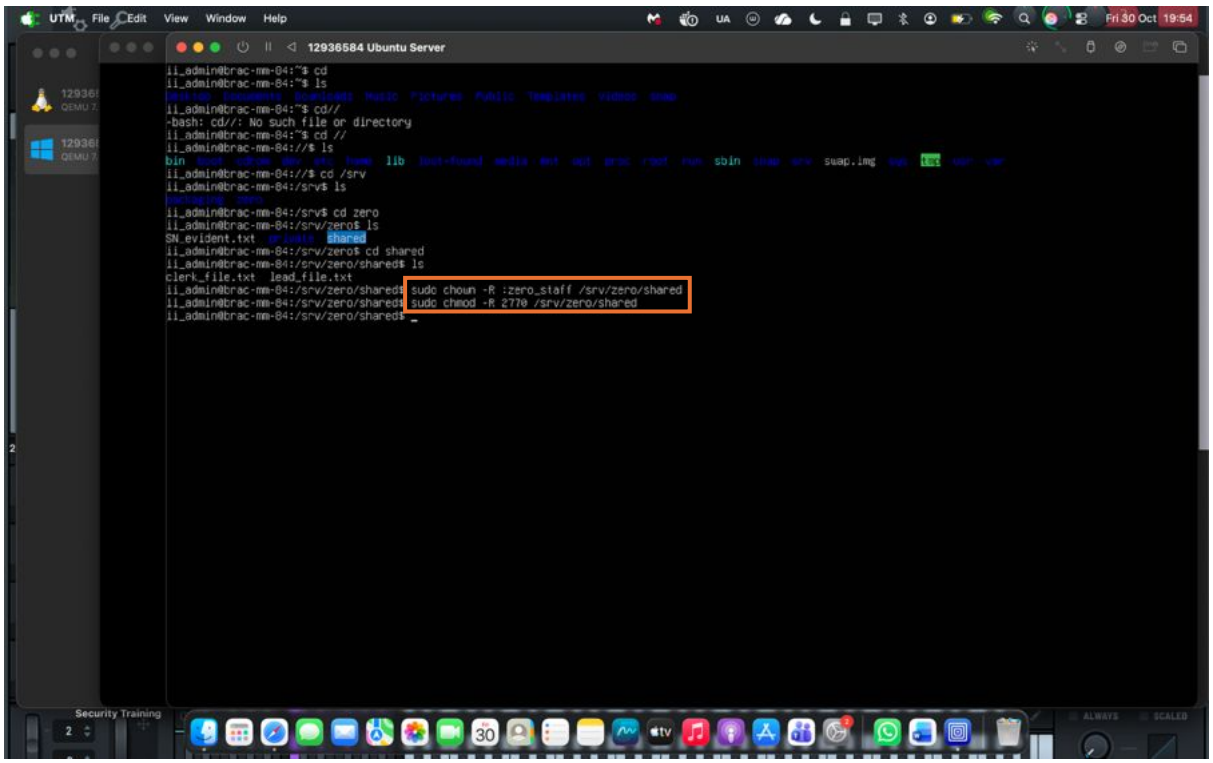

UTM File Edit View Window Help
Fri 30 Oct 19:40
12936584 Ubuntu Server
li_admin@brac-nm-04:~$ resolvectl query www.ubuntu.com
www.ubuntu.com: 2620:2d:4000:1::27 -- link: enp0s1
2620:2d:4000:1::28 -- link: enp0s1
2620:2d:4000:1::26 -- link: enp0s1
185.125.190.20 -- link: enp0s1
185.125.190.21 -- link: enp0s1
185.125.190.29 -- link: enp0s1
-- Information acquired via protocol DNS in 5.3ms.
-- Data is authenticated: no; Data was acquired via local or encrypted transport: no
-- Data from: cache
12936584 Ubuntu
QEMU 7.2 ARM Virtual
12936584-win
QEMU 7.2 ARM Virtual
Security Training

Step 5

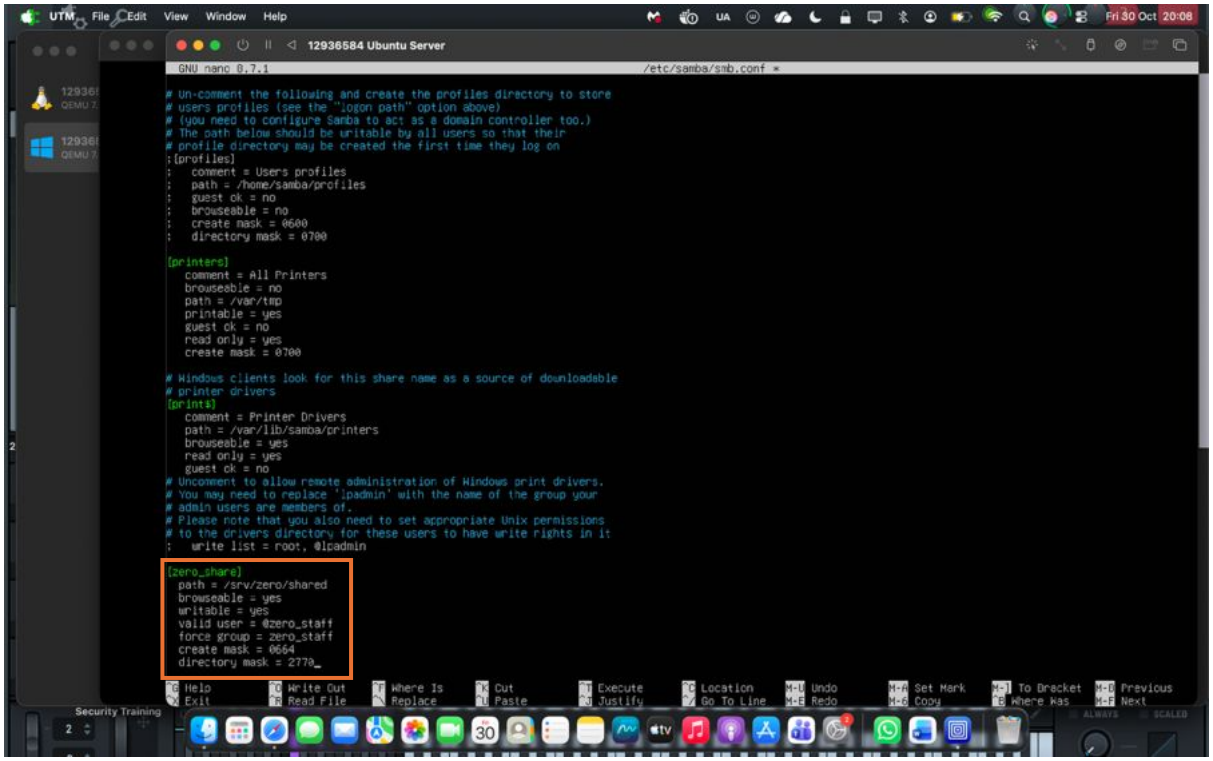
Question 5



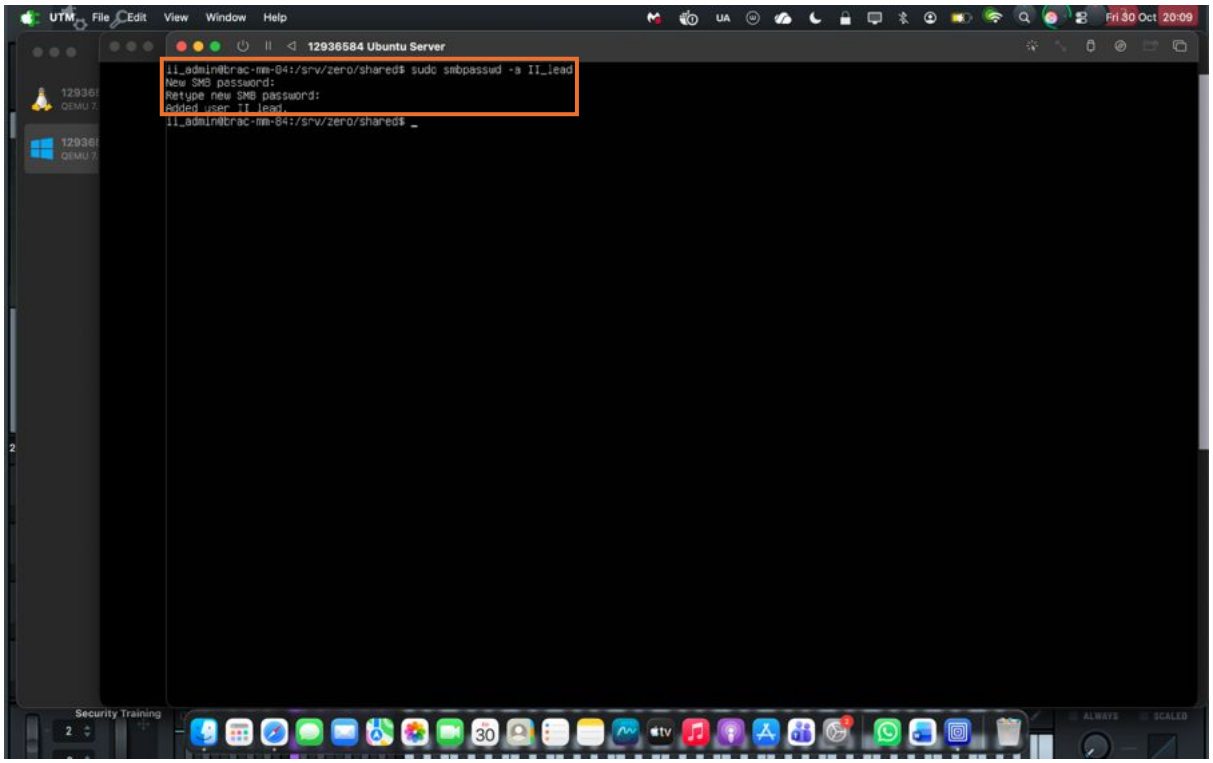
Updating package manager and installing samba.



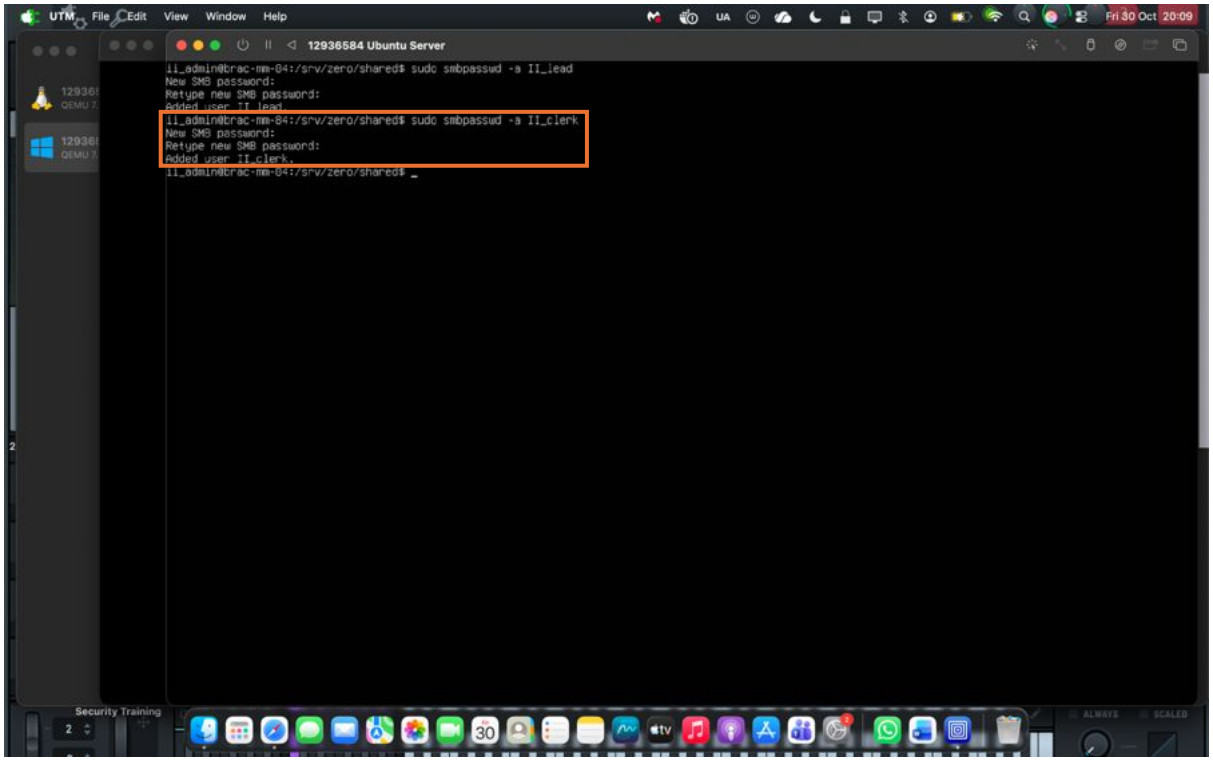
Configuring Shared Directory.



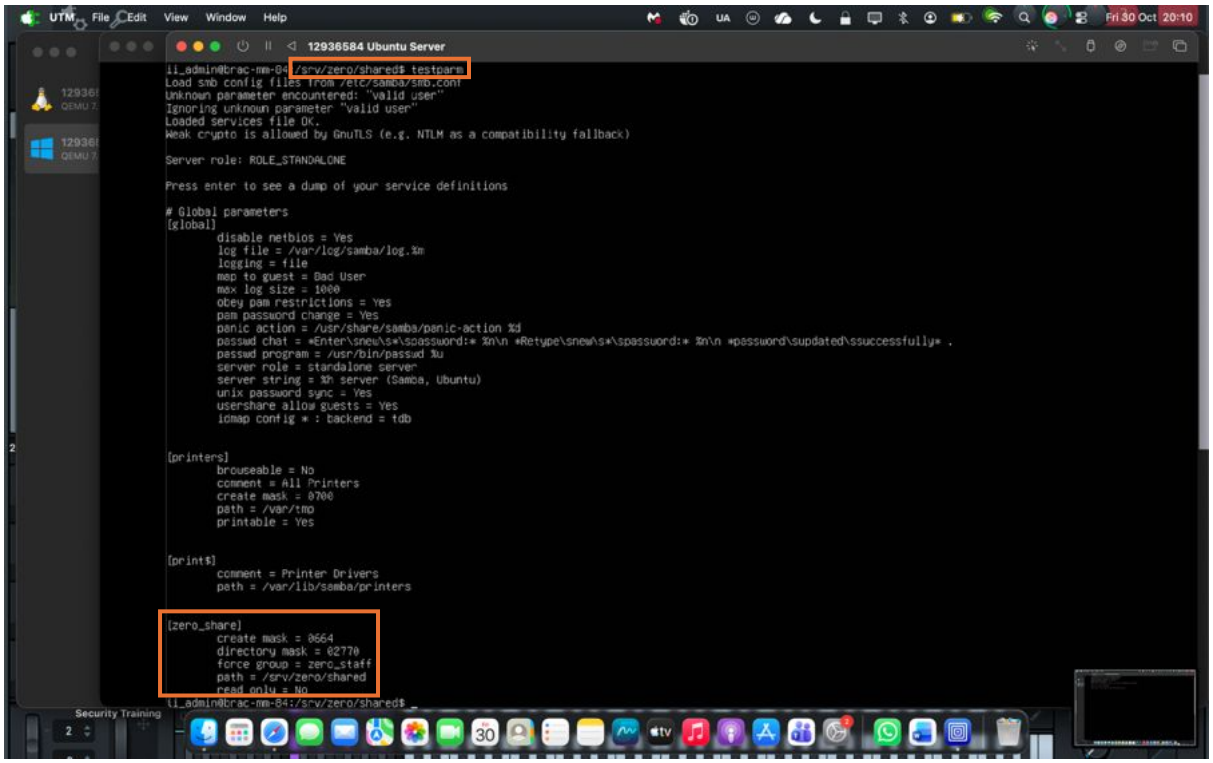
Configuring Samba Share [zero_share].



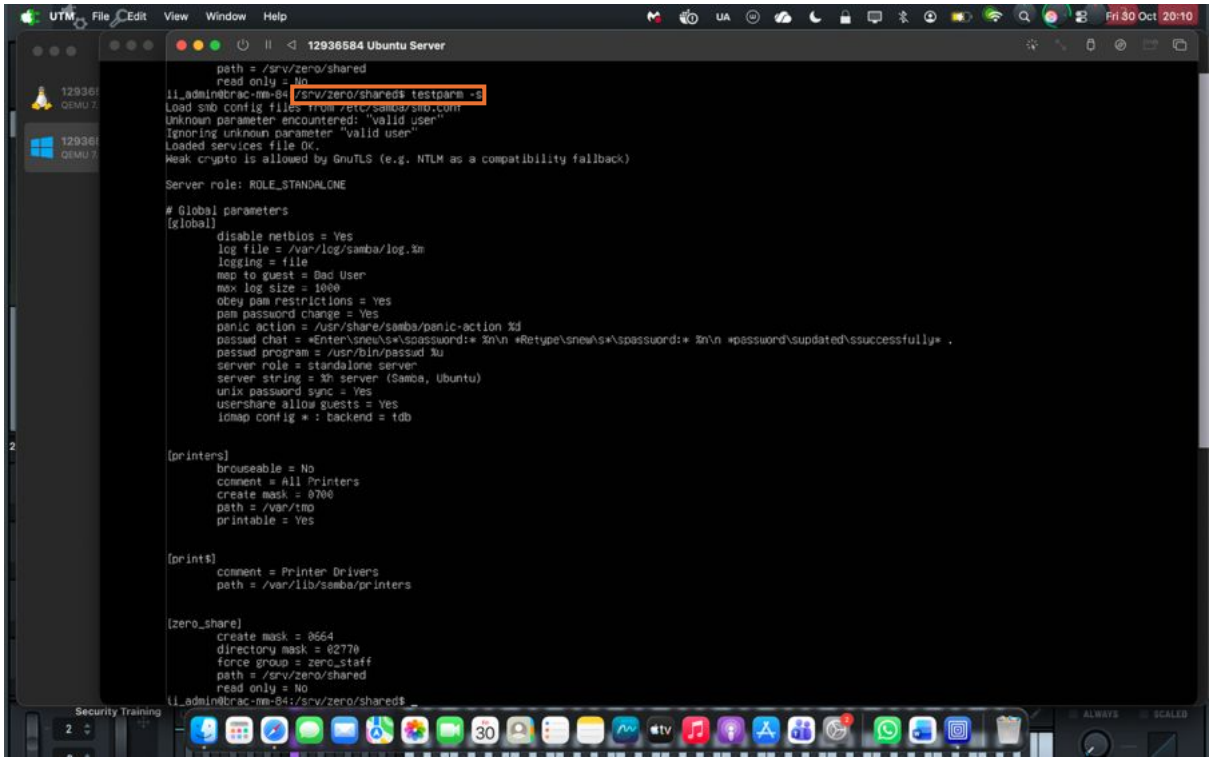
Adding Password for II_lead.



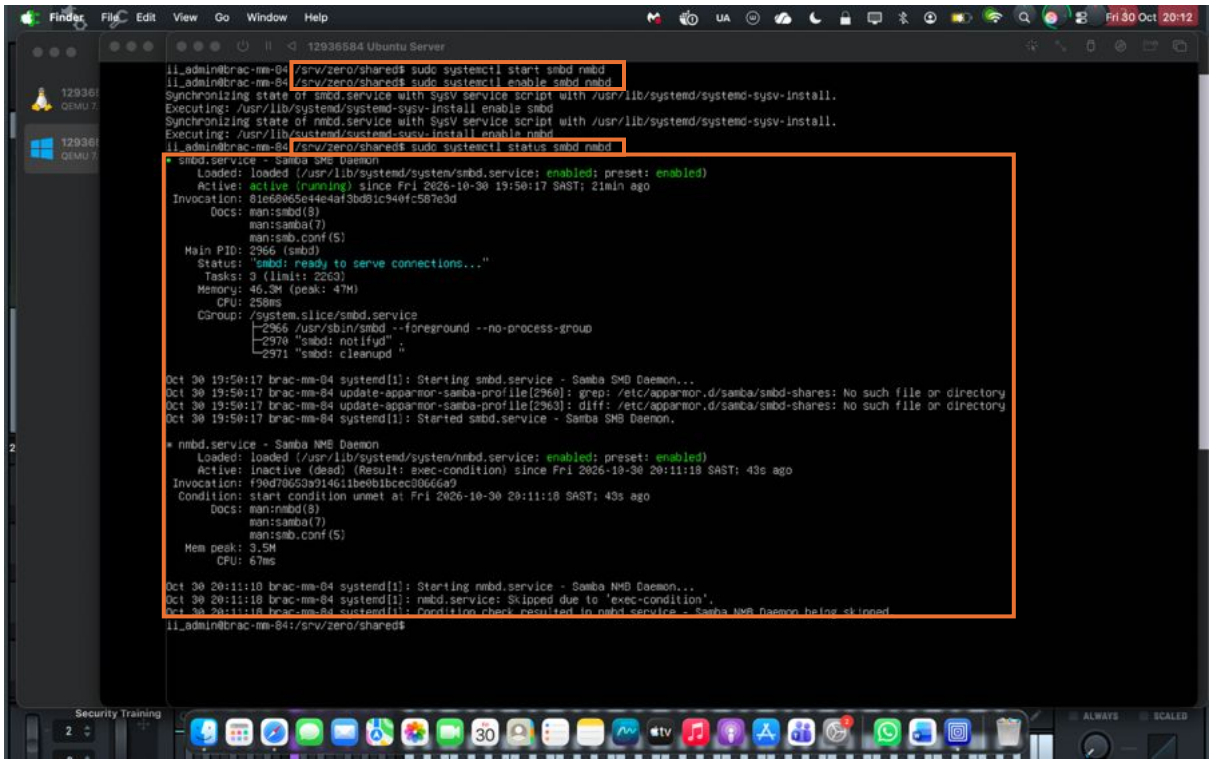
Adding Password for ll_clerk.



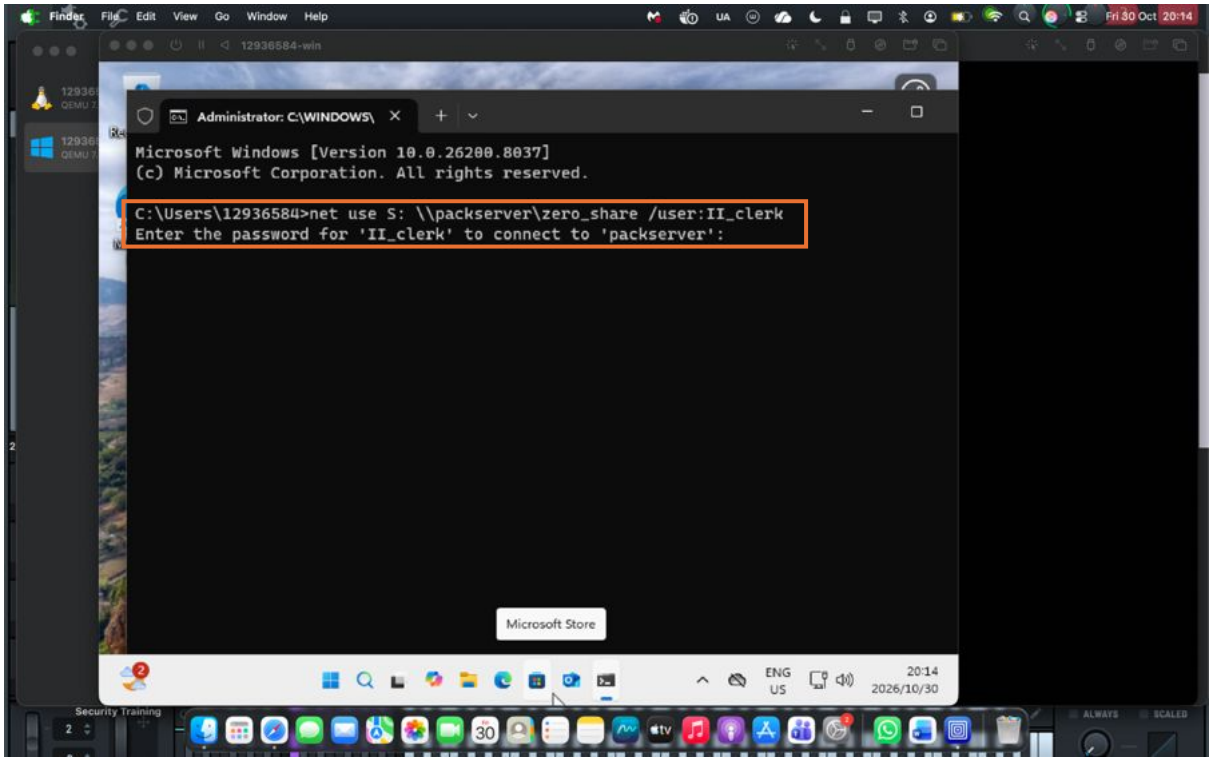
Validating Configuration with testparm.



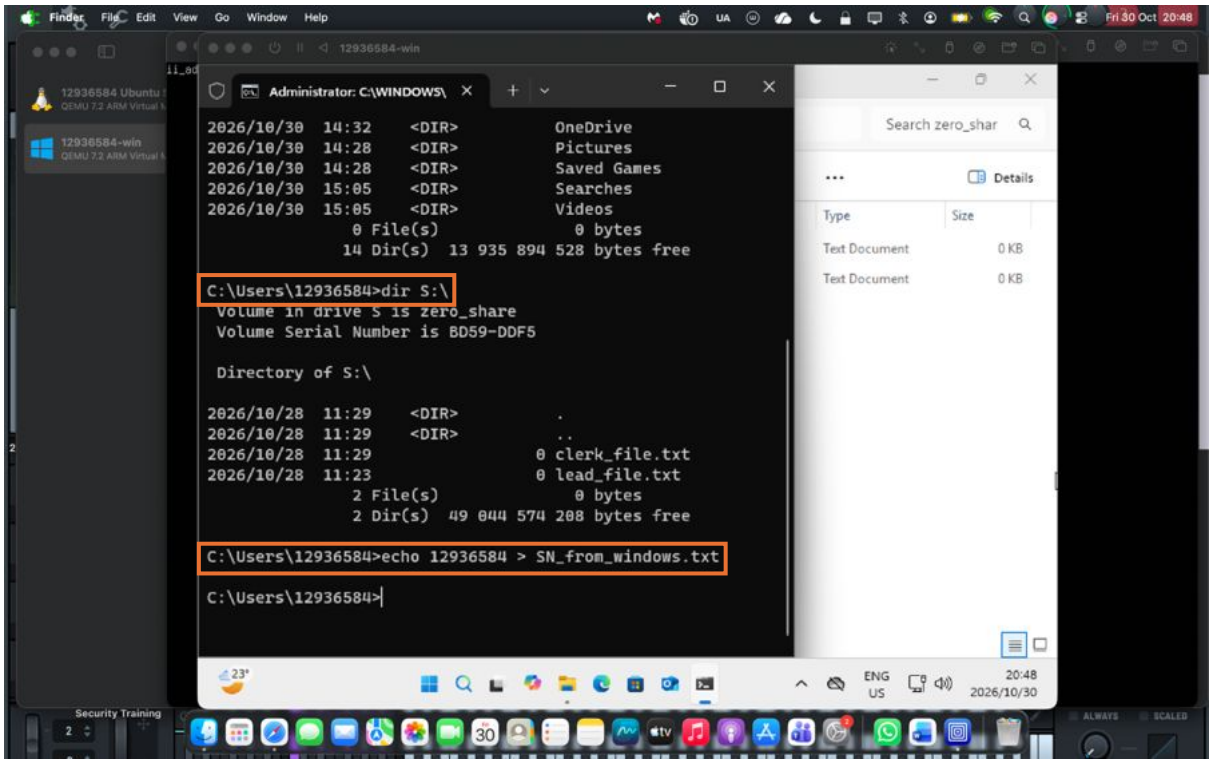
Validating Configuration with testparm.



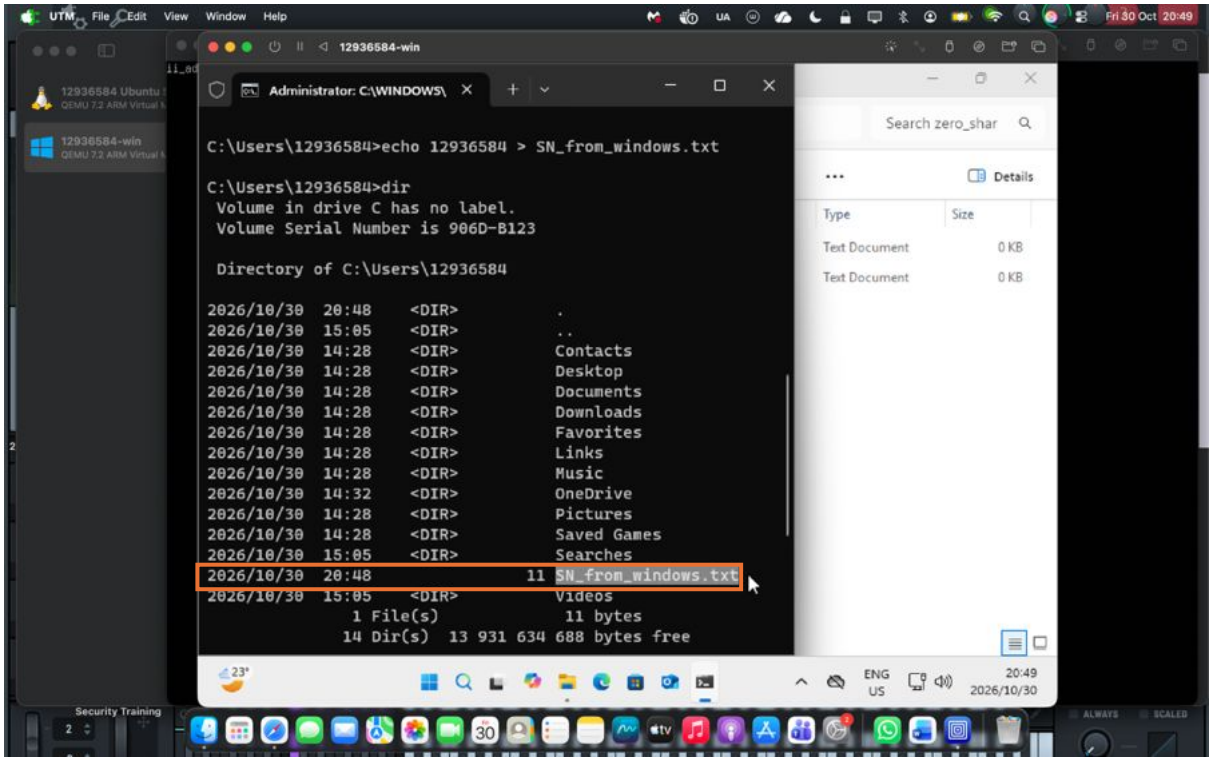
Enabling, Starting and viewing status of Samba service.



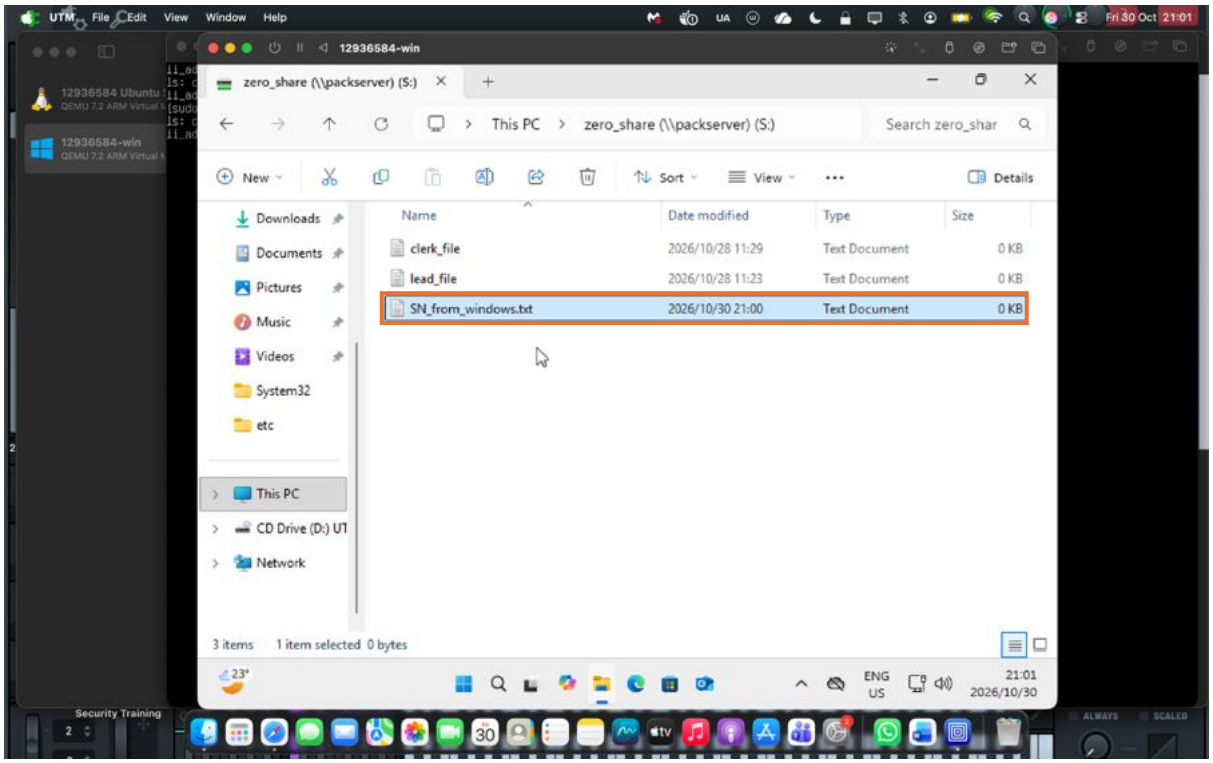
Mapping the Share with S: drive to user II_clerk on Windows.



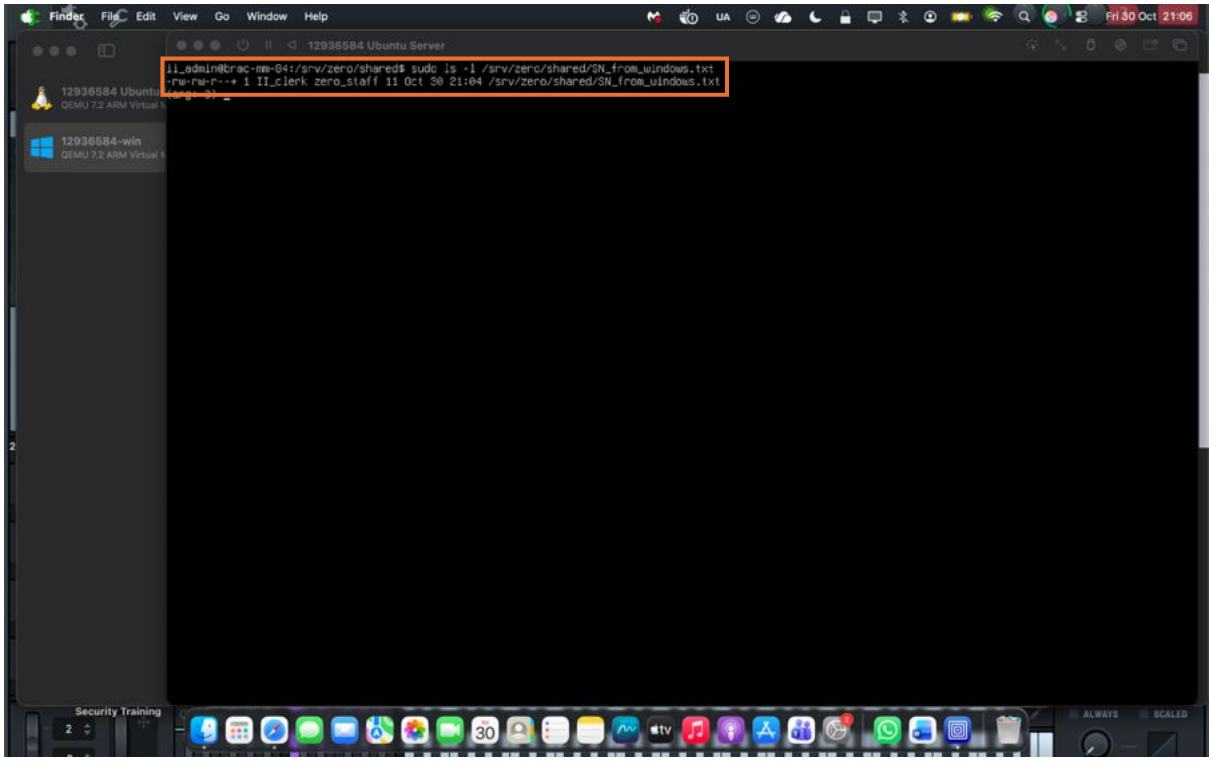
Navigating to S: drive and creating a file with my Student Number.



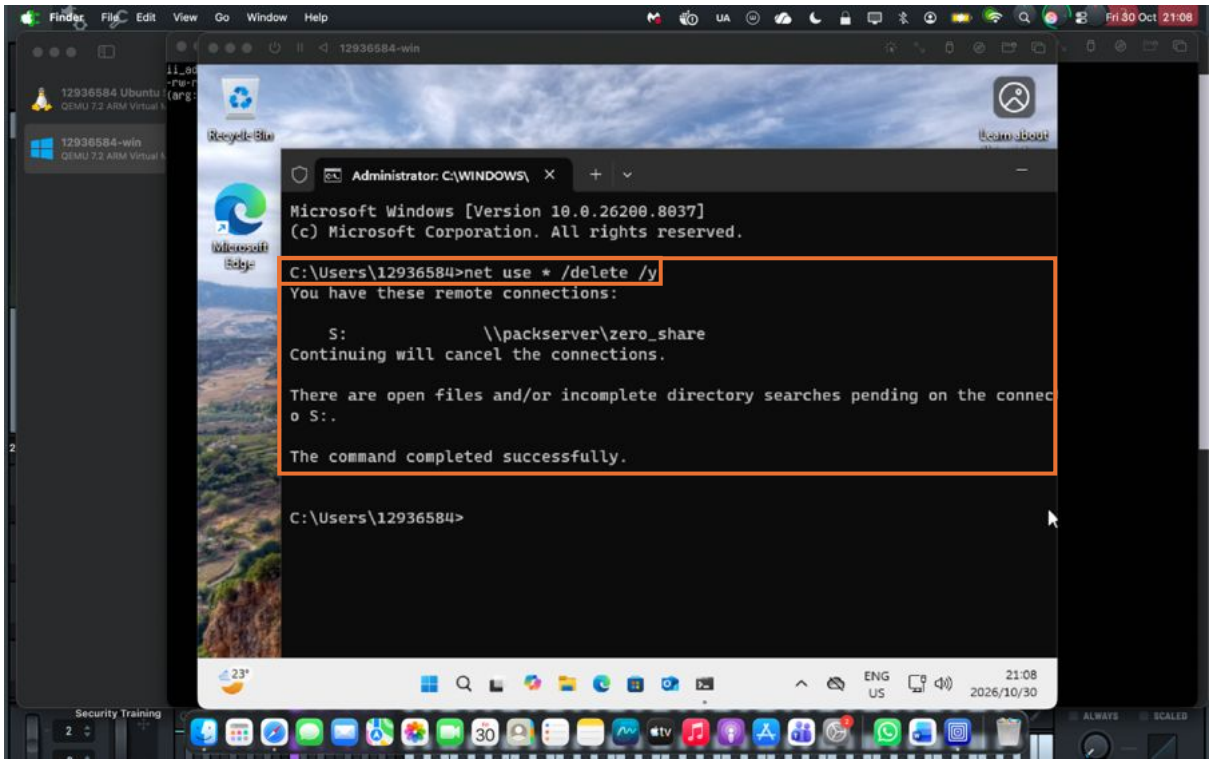
Verifying File.



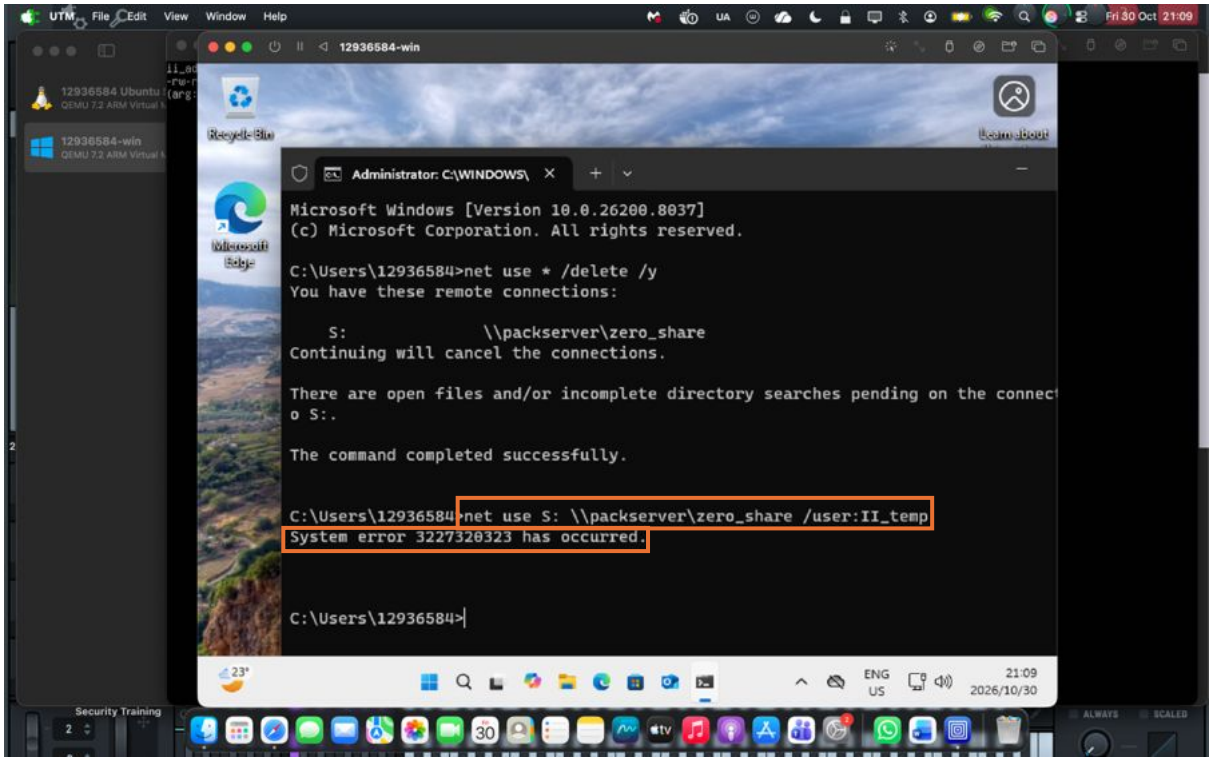
Verifying File.



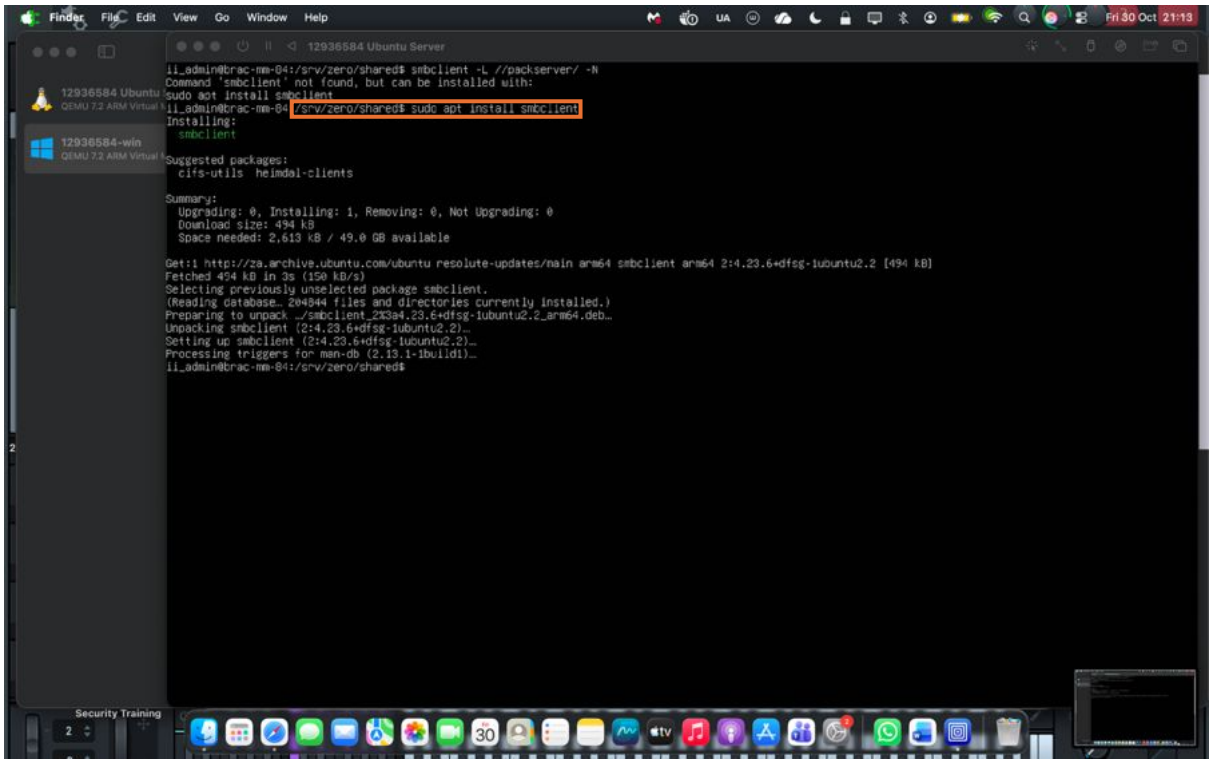
Inspecting file permissions and the setgid behaviour on Ubuntu.



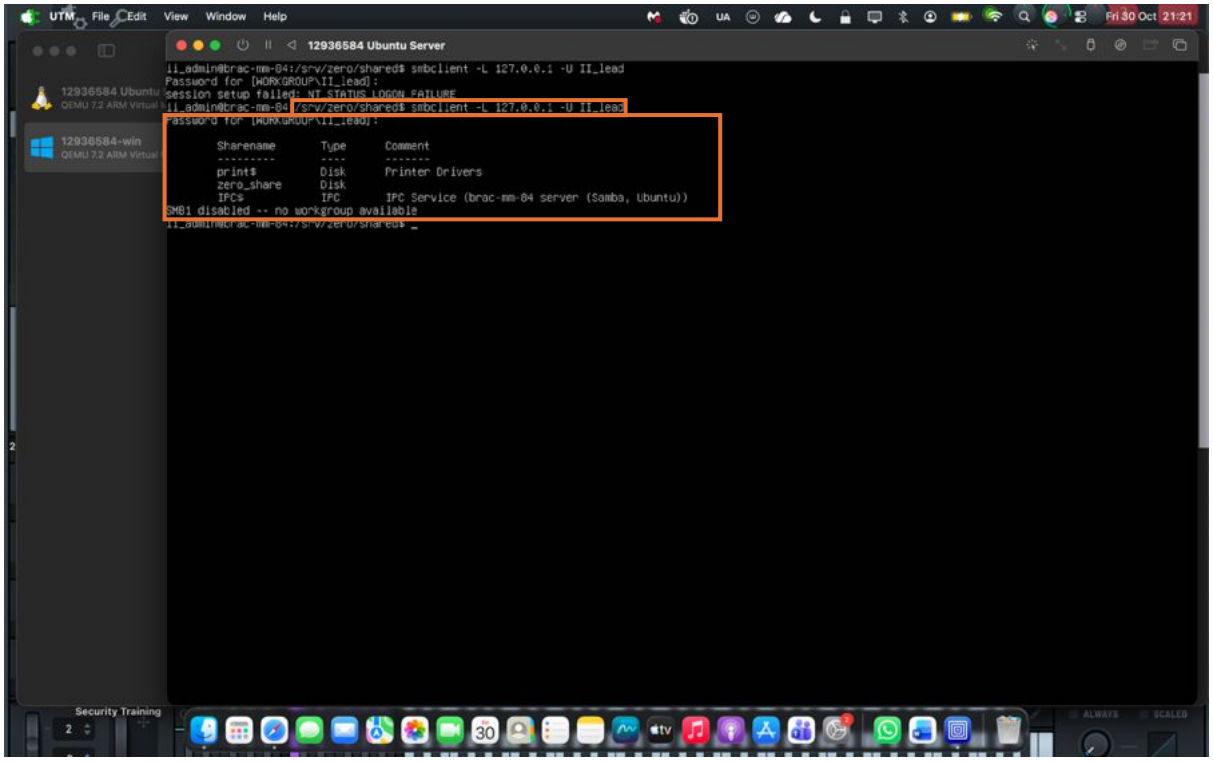
Clearing active connections.



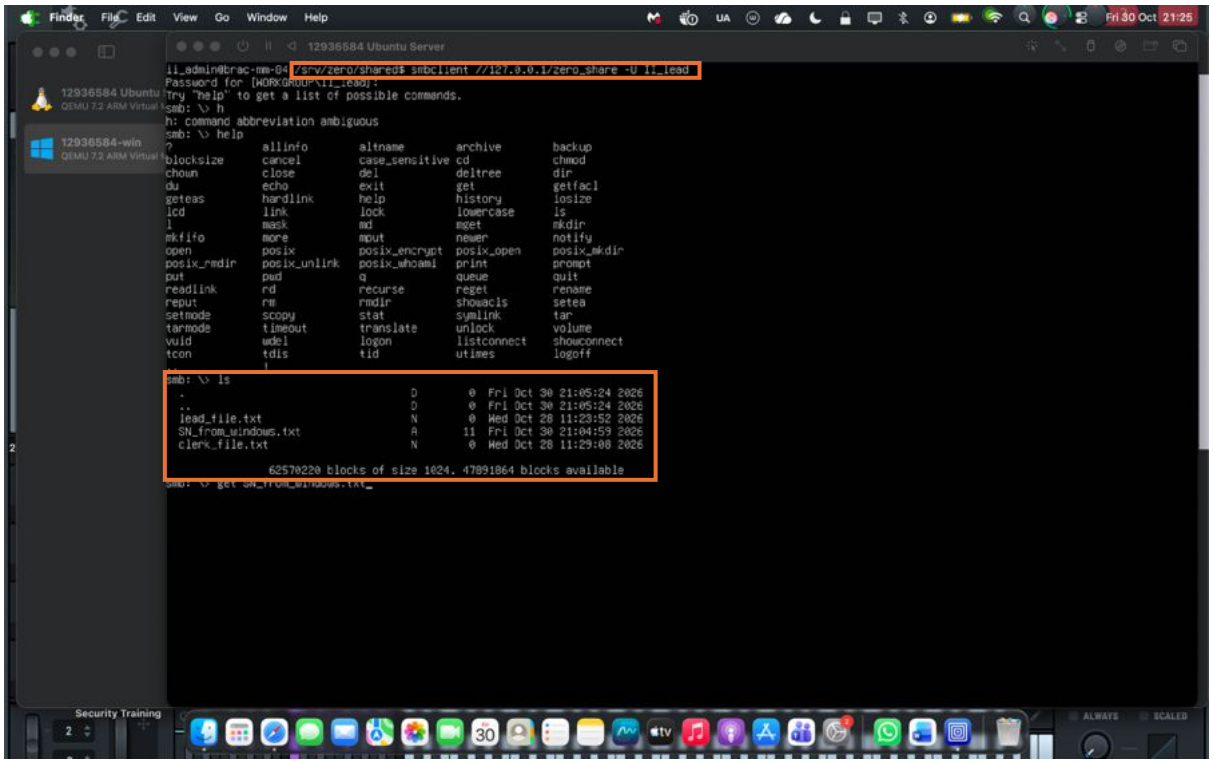
Proving Access is Refused for II_Temp by attempting to map as II_Temp on Windows.



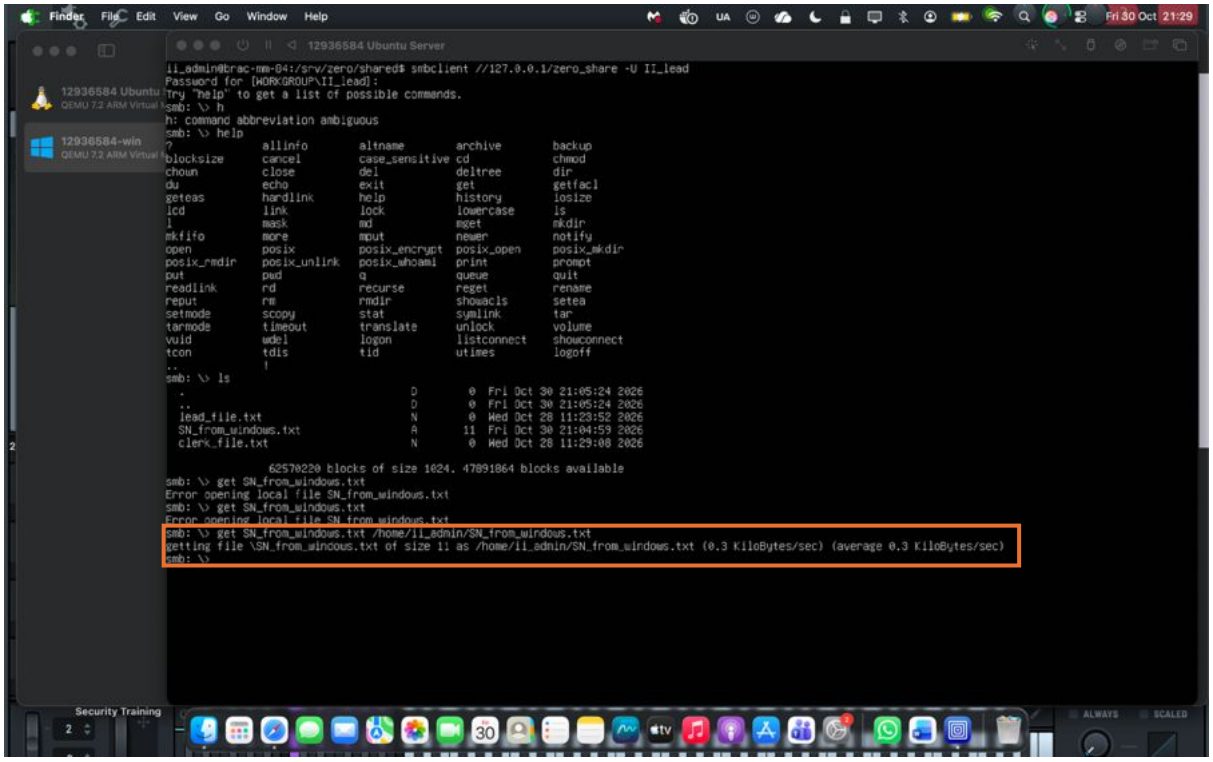
Installing smbclient on Ubuntu.



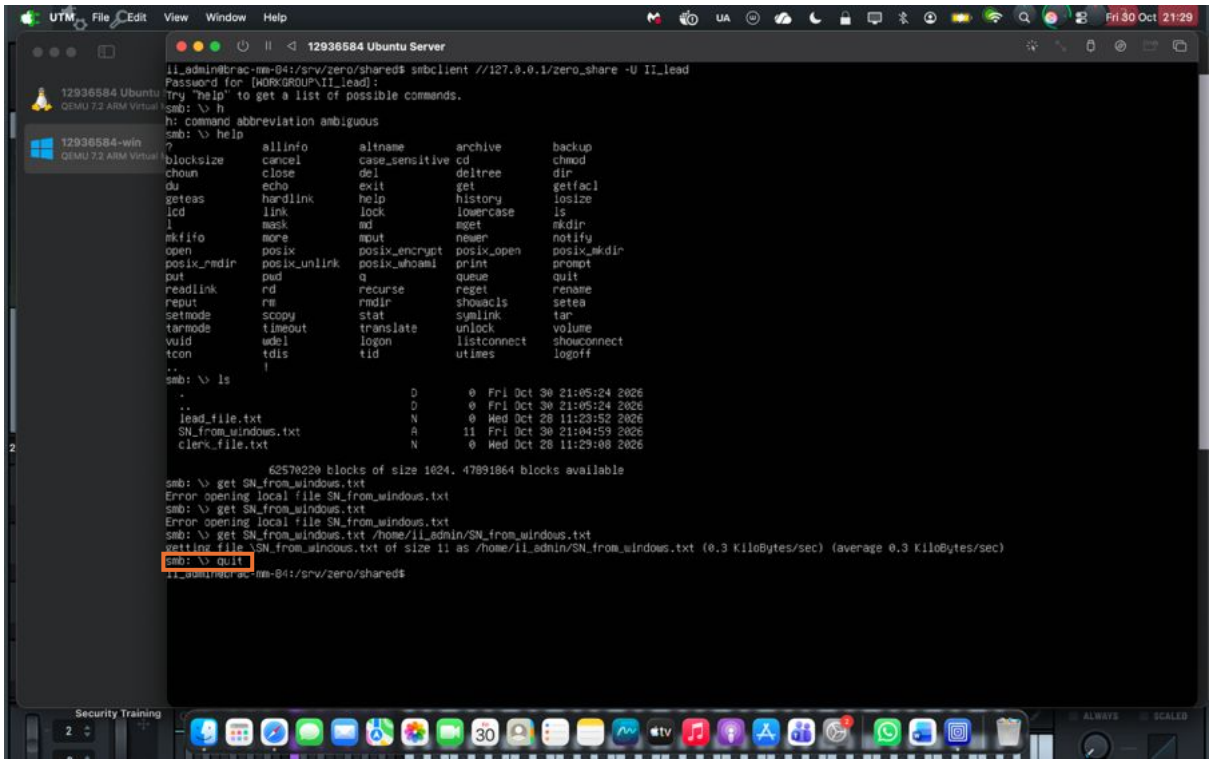
Testing with smbclient on Ubuntu.



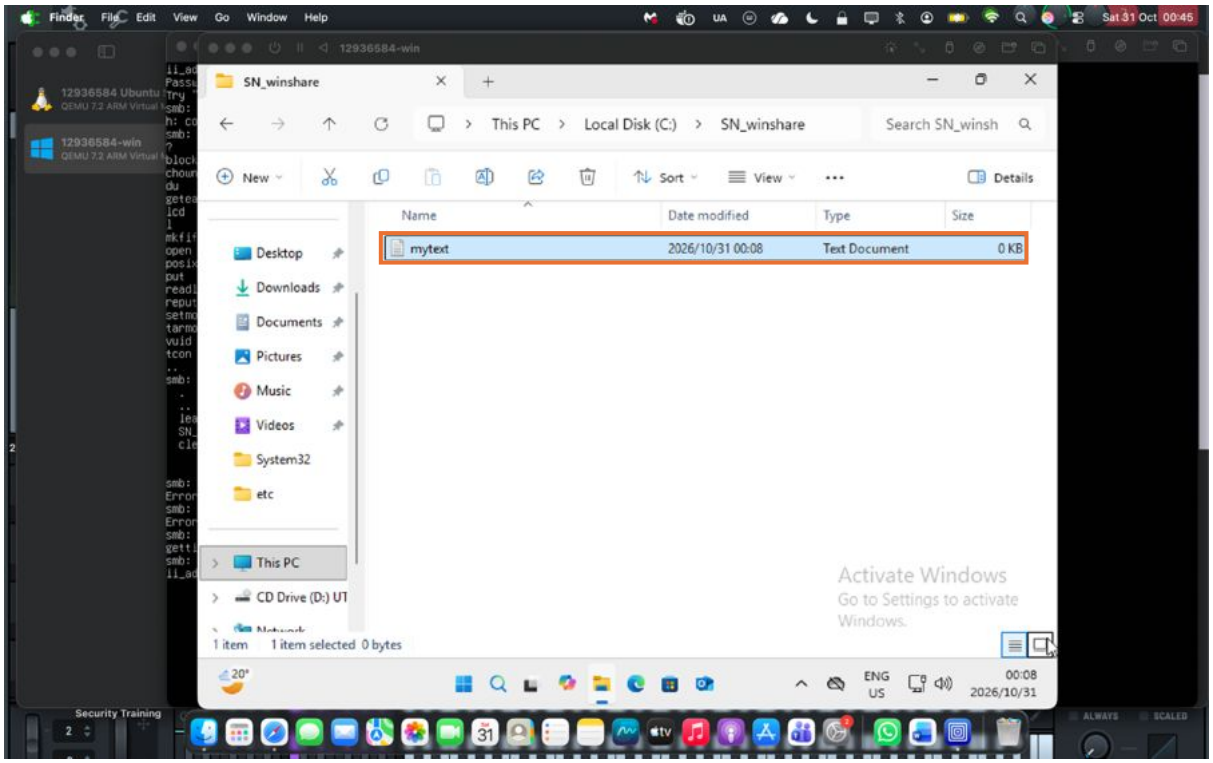
Connecting.



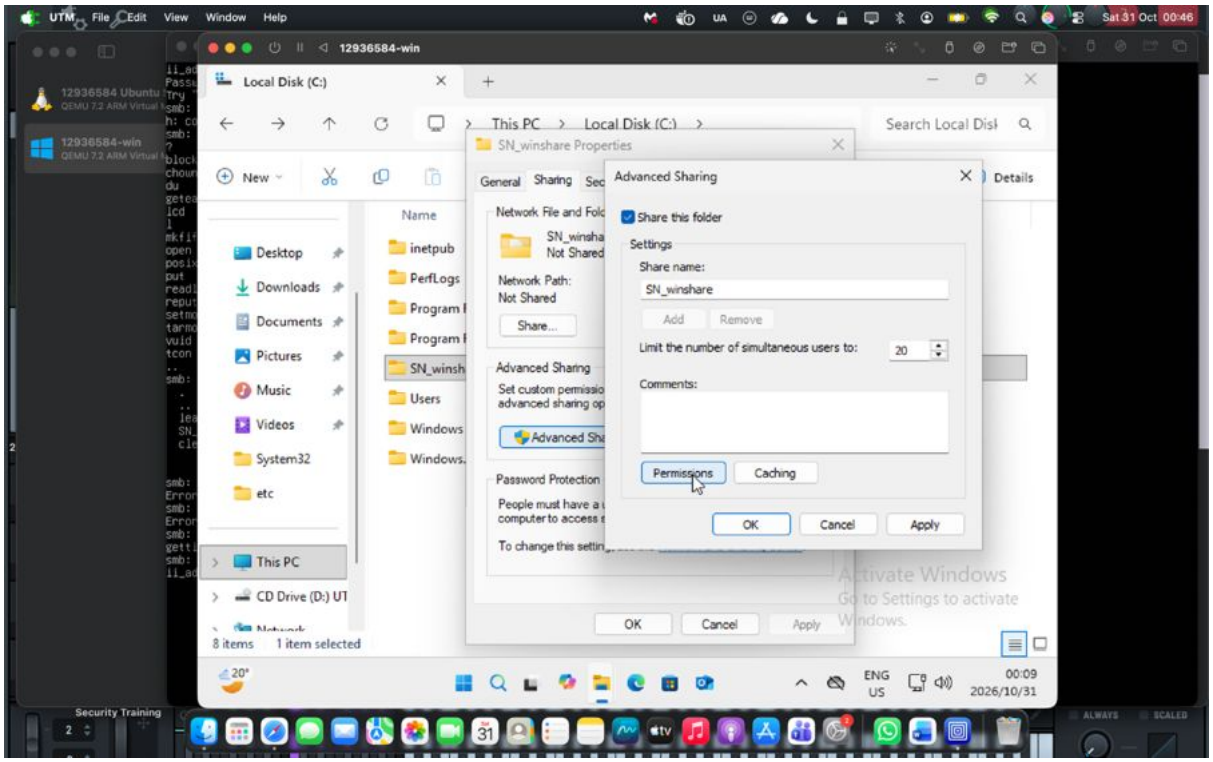
Downloading SN_from_windows.txt file with get.



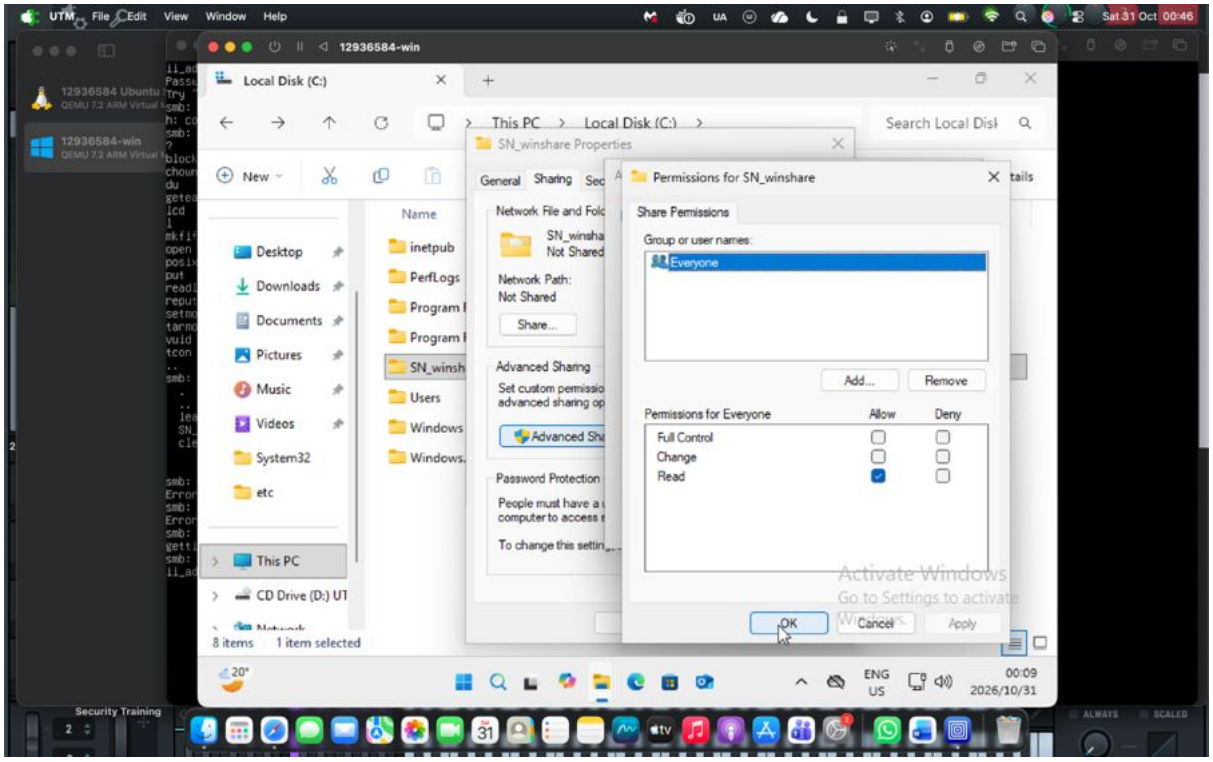
Quitting.



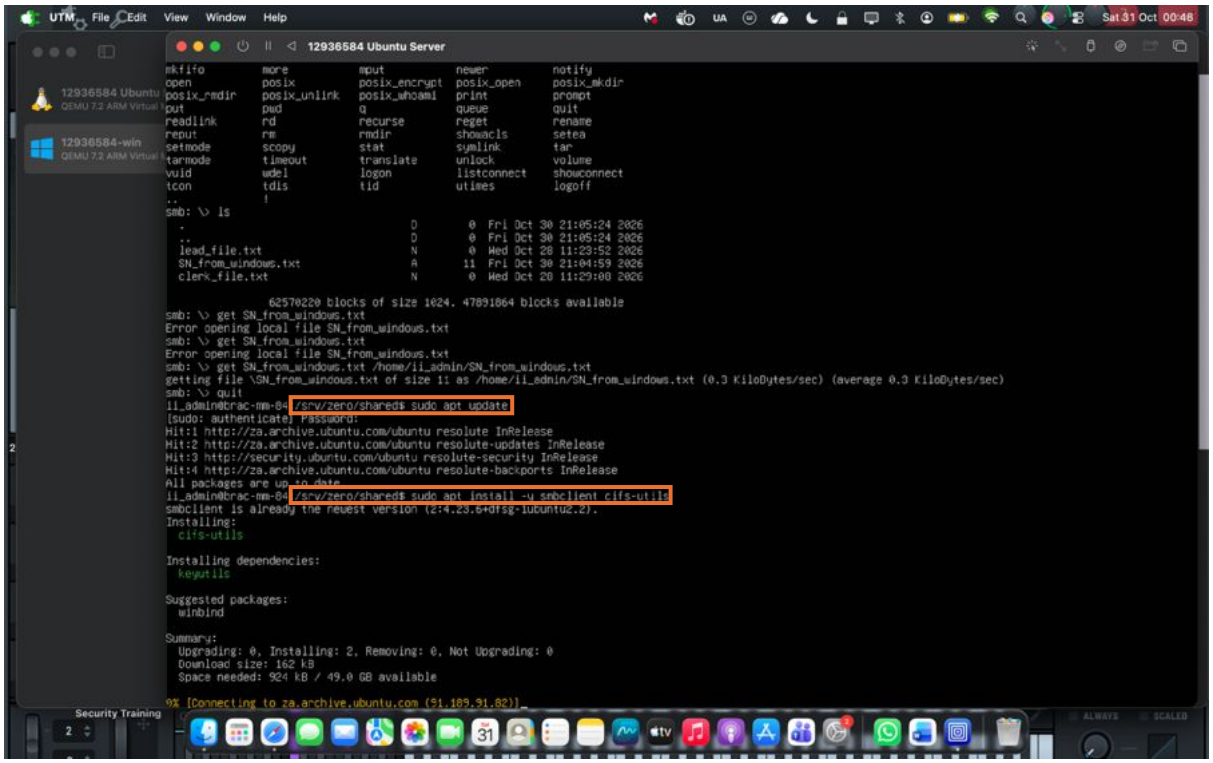
SN_winshare folder created with file named mytext.



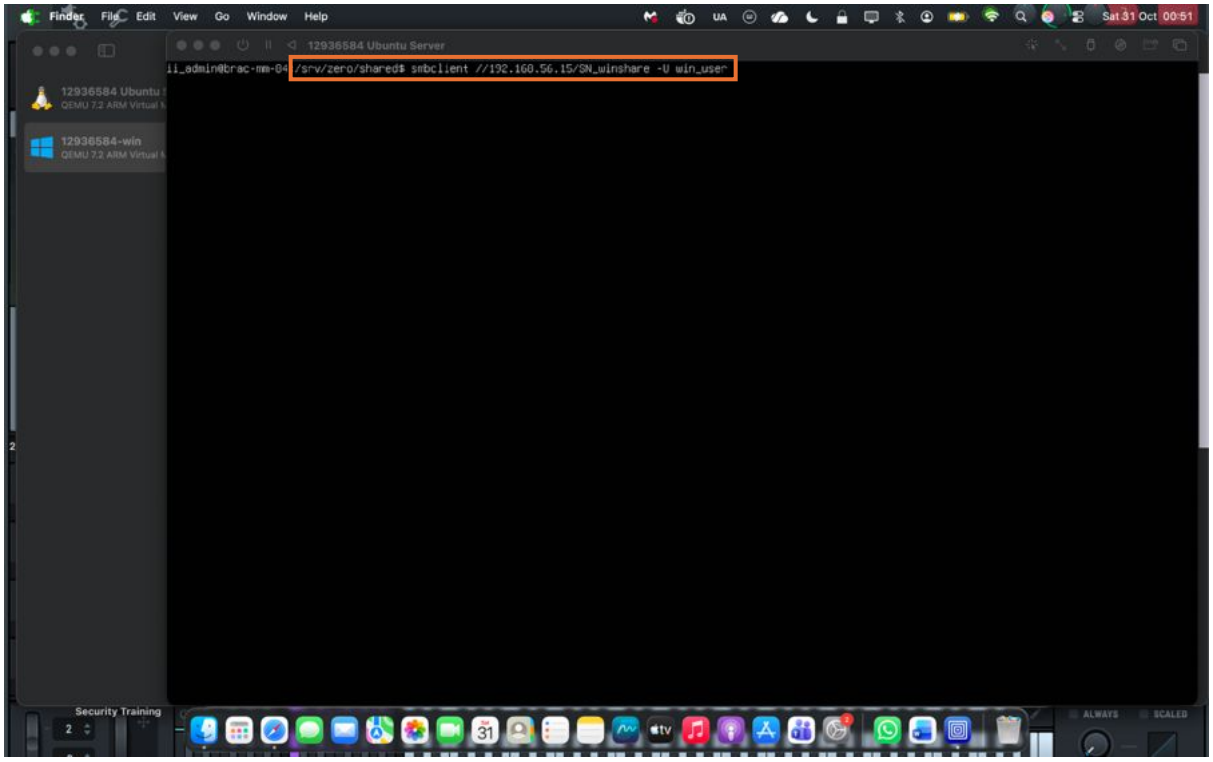
Creating a Share for this file.

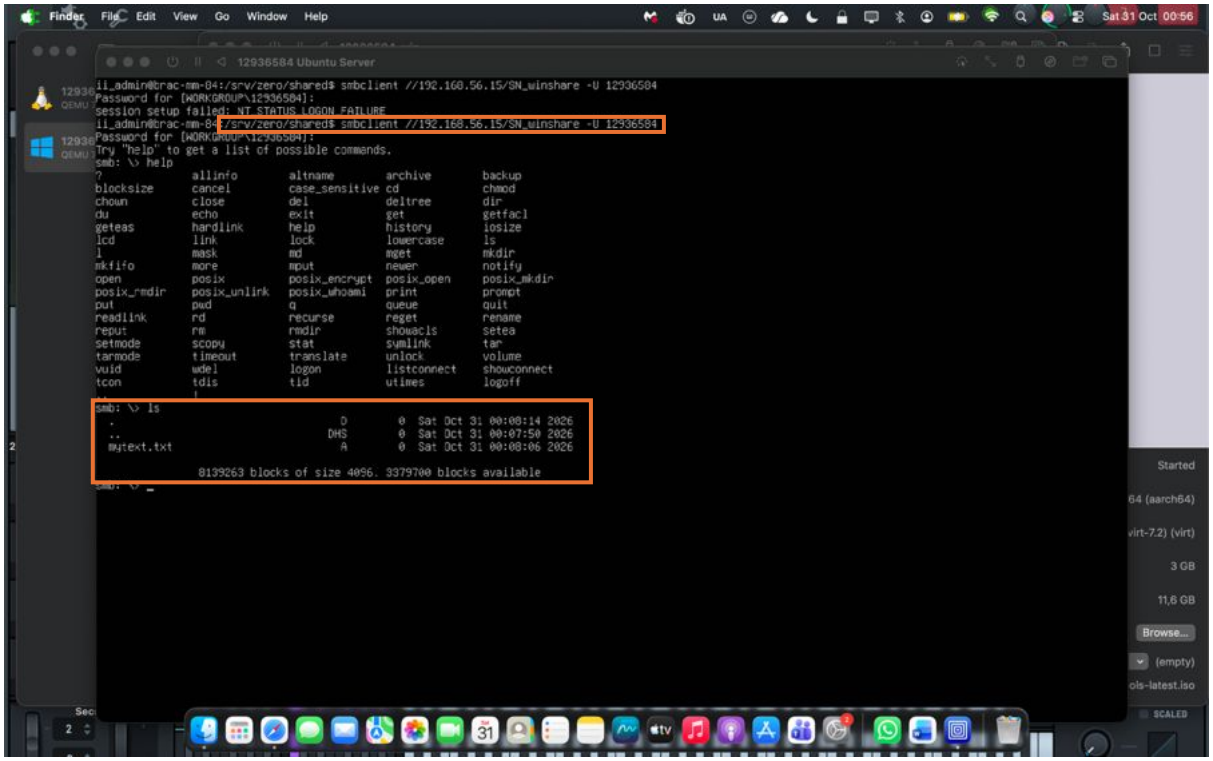


Sharing file to everyone.



Updating packaging index and installing smbclient cifs-utils (installing prerequisites) on Ubuntu.



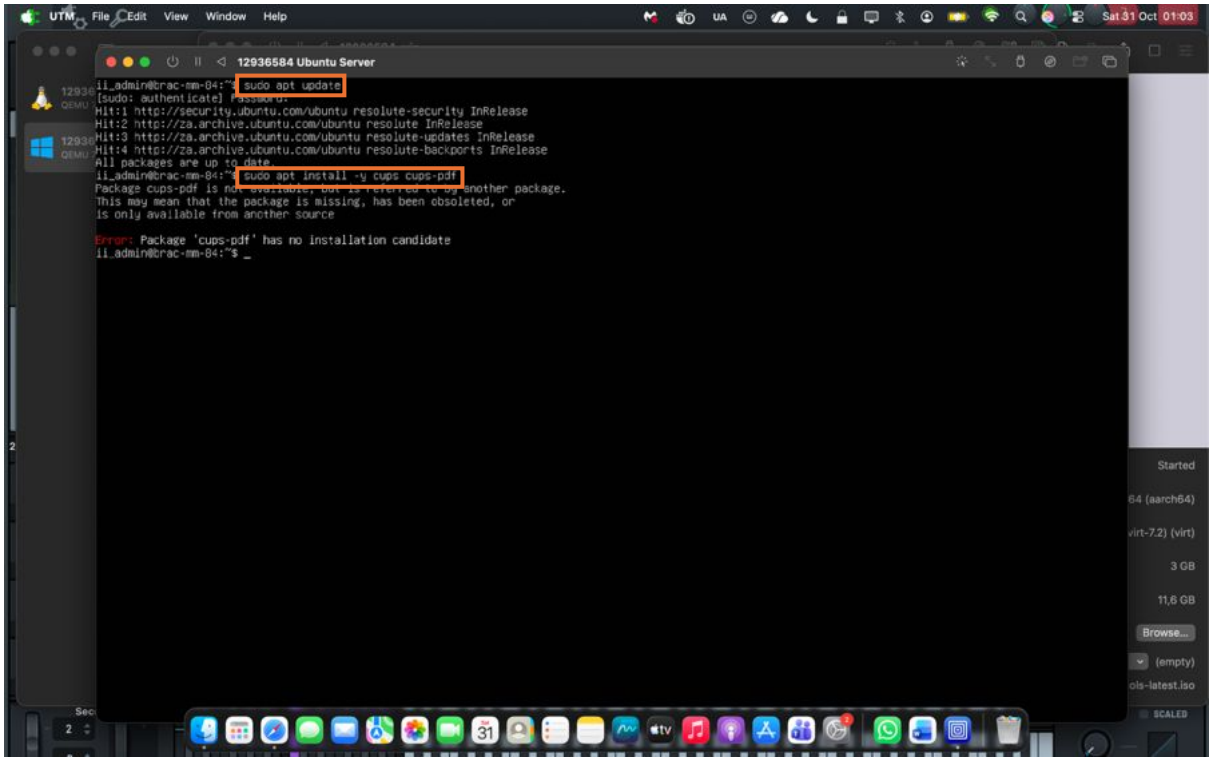


Accessing and viewing interactive file listing with smbclient.

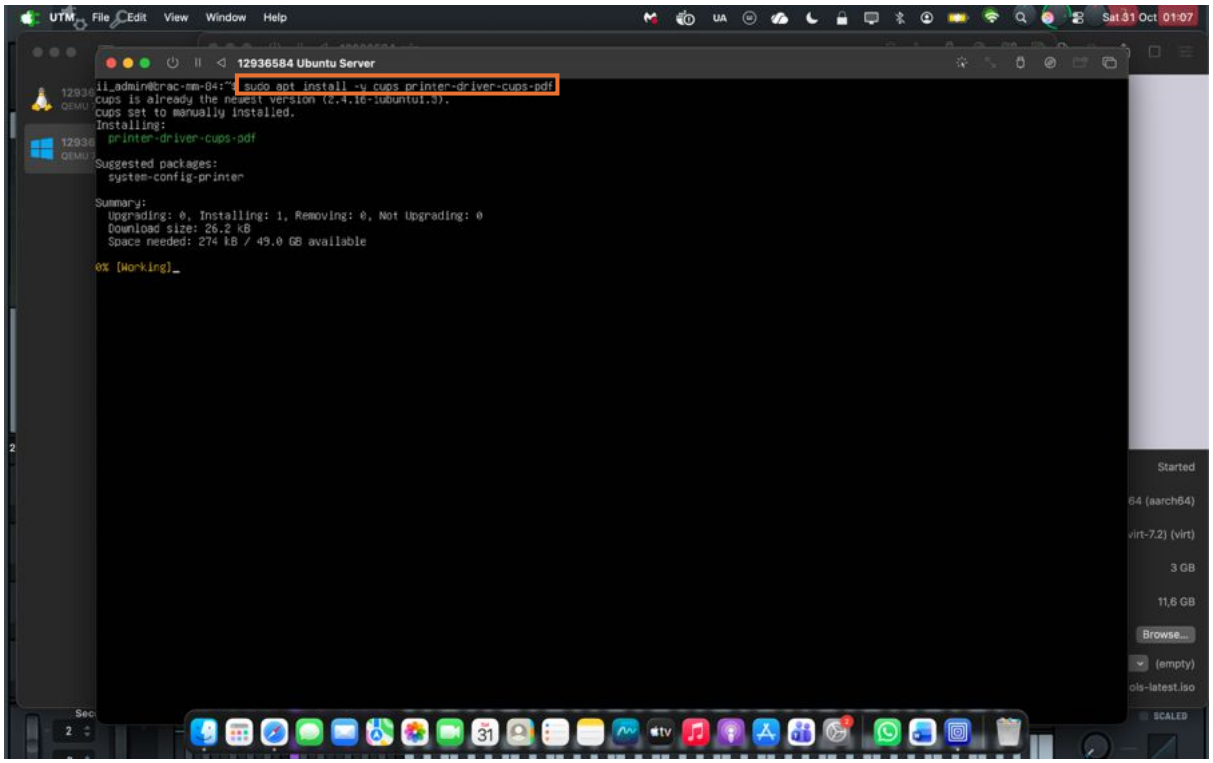
5.3

The difference between Linux Permissions and Samba Share permissions encountered are how permissions are mapped and inherited.

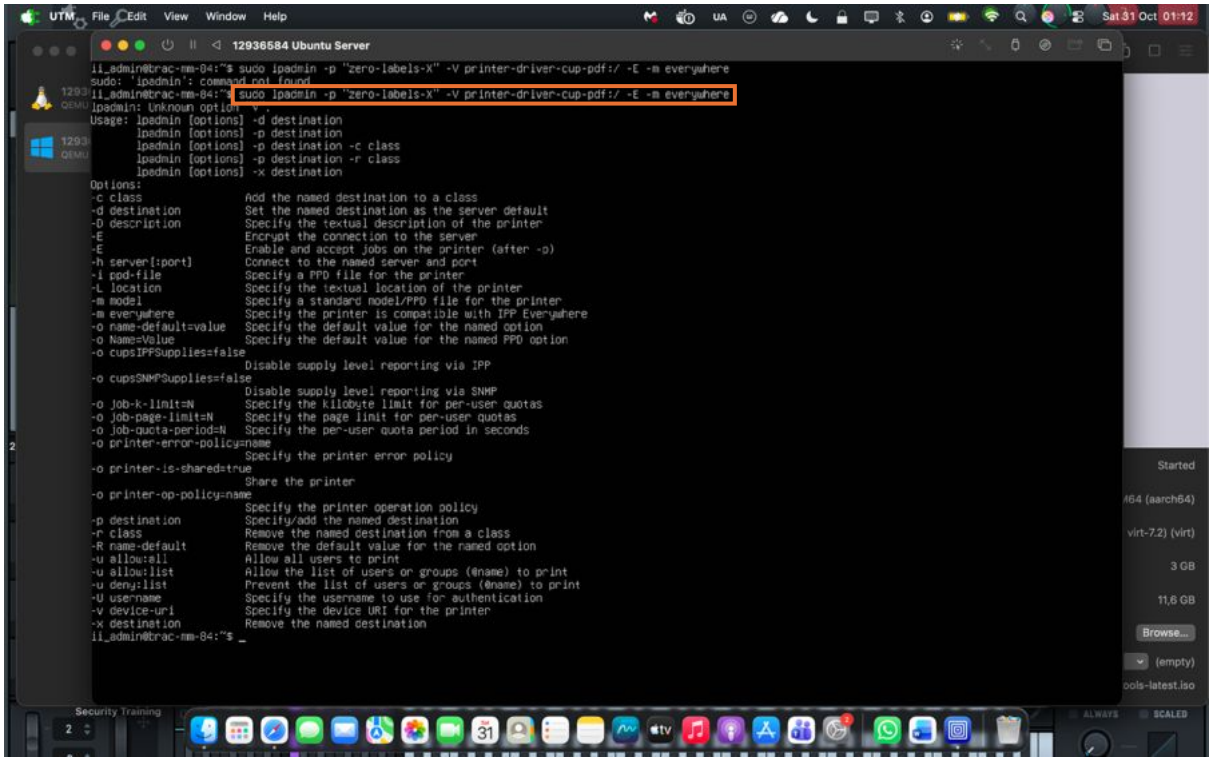
Samba share/NTFS permissions: where Windows manages file using Access Control Lists (ACL) and NTFS permissions where users and groups are identified by security identifiers (SIDs) with Windows not understanding native Linux Ownership (UID) or Linux Group (GID).
Linux Virtual File System (VFS) layer: where when using smbclient, you only just using an FTP like tool that passes commands through the network. Without it worrying about Local Linux file attribute. Although when using mount -t gifs, Linux assigns a local owner, group and permission bit mask (e.g. rwxr-xr-x) to file so Linux kernel can interact with them.



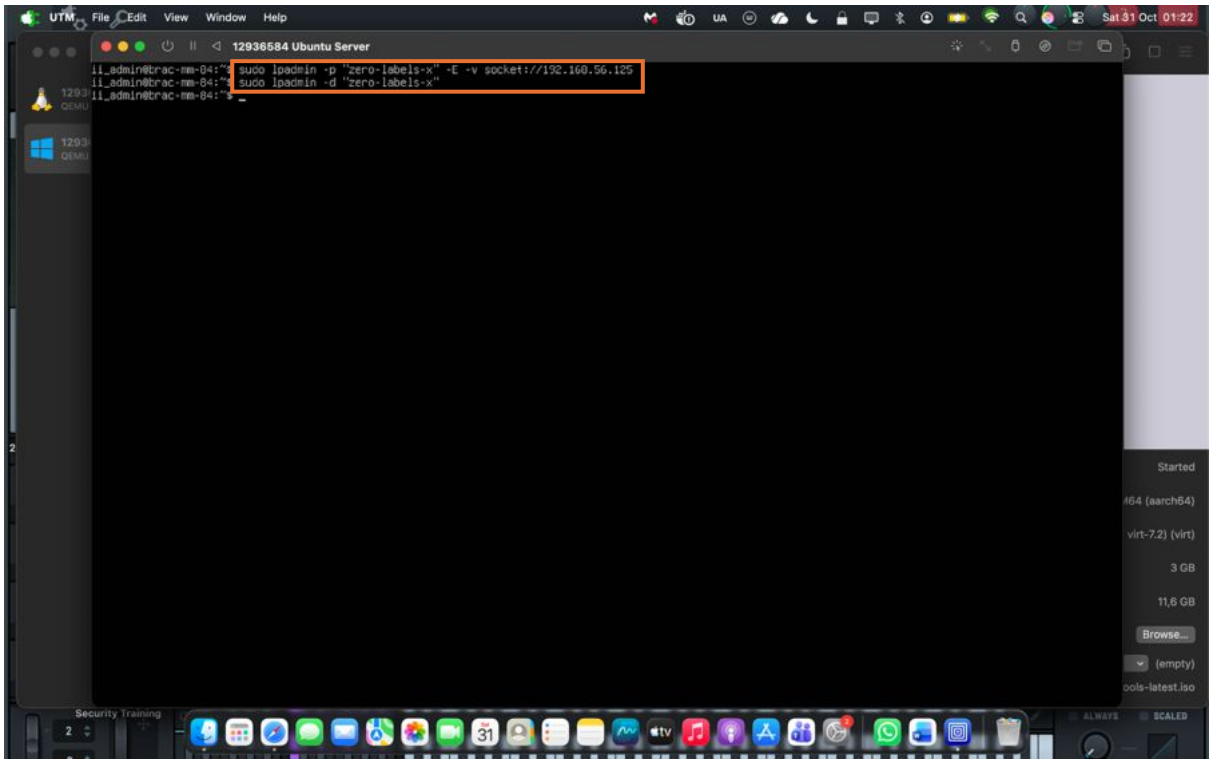
Updating packaging index.



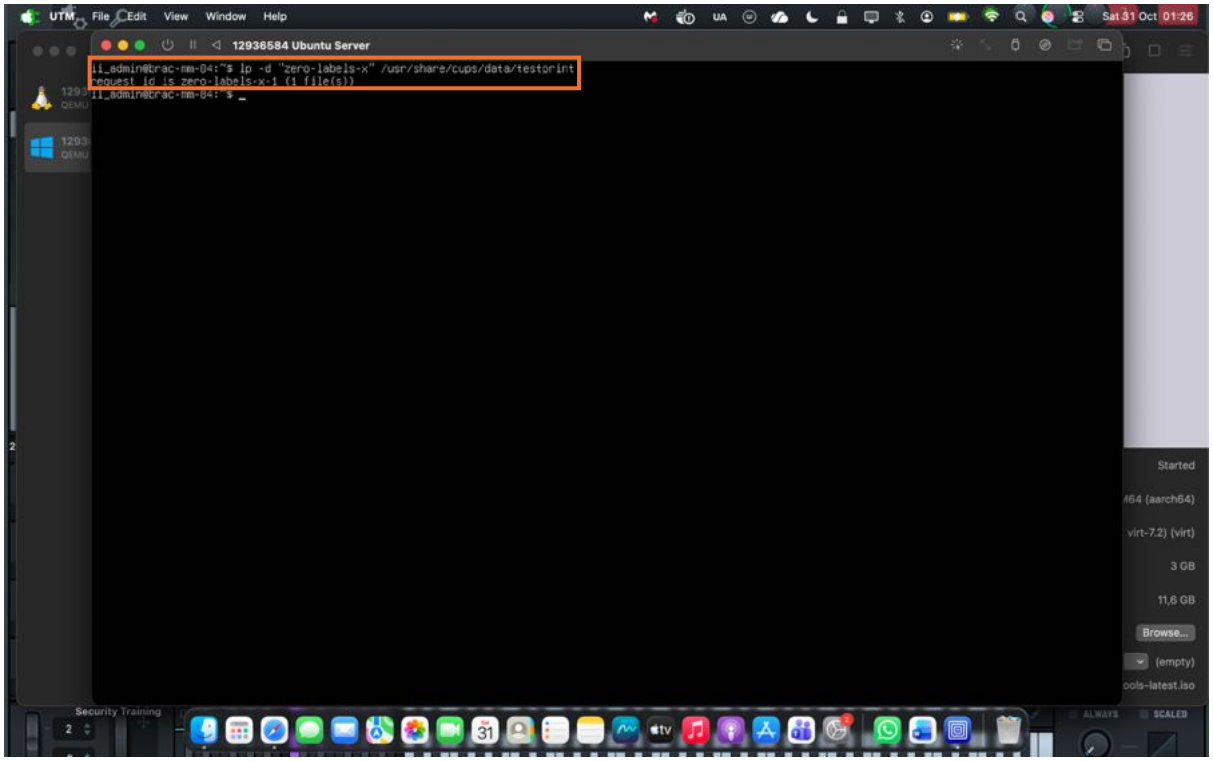
Installing CUPS and cups-pdf (printer-driver).



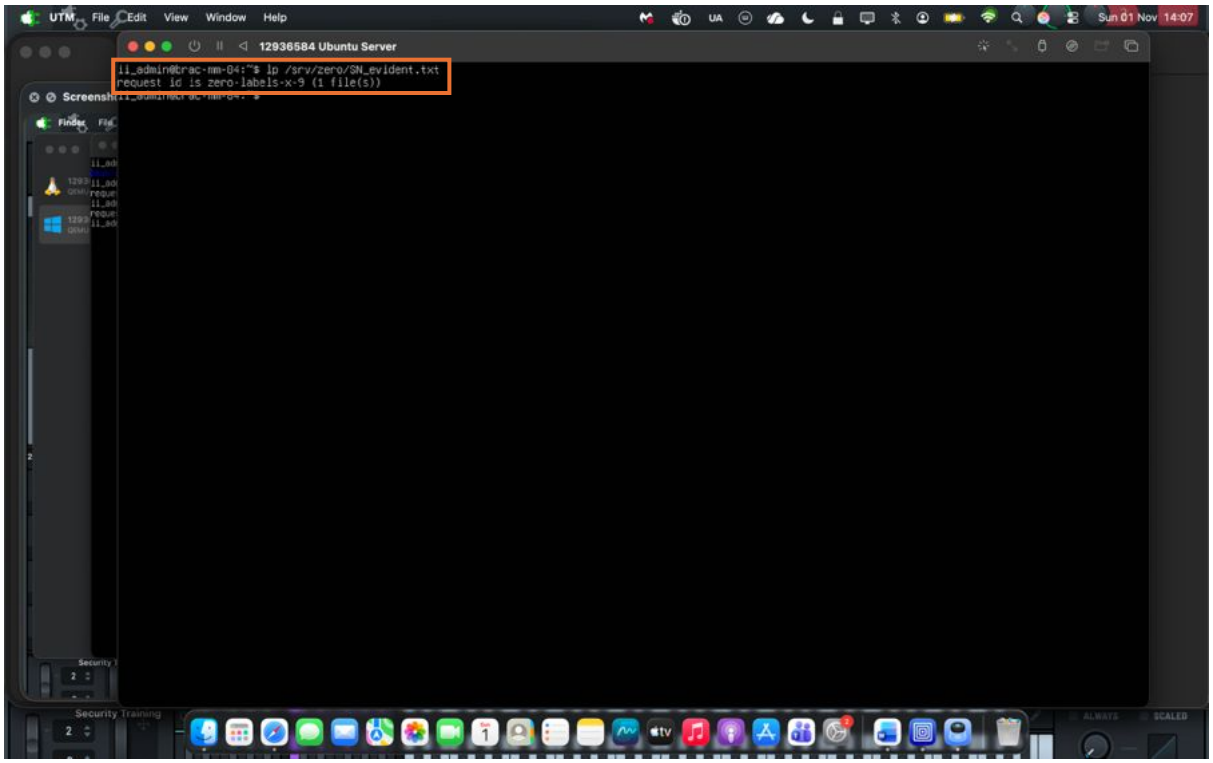
Adding printer queue using the cup-pdf backend.



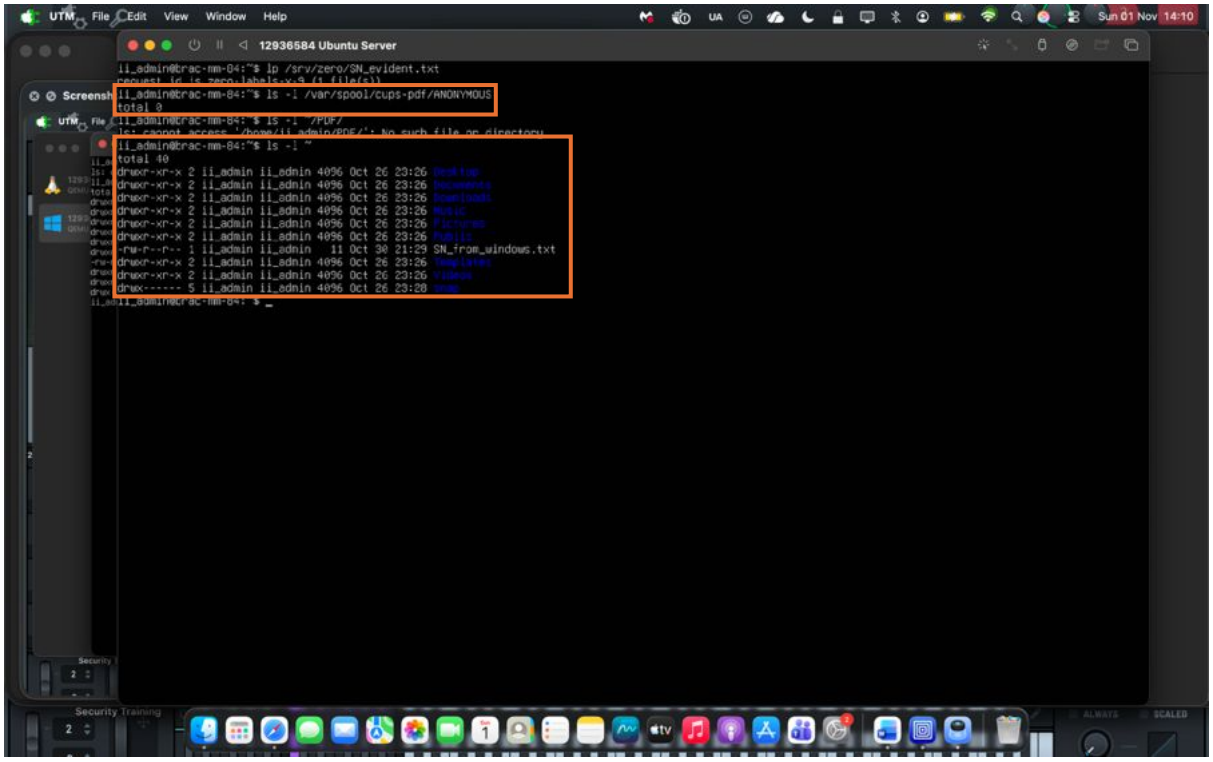
Setting the queue as the default printer for the system.



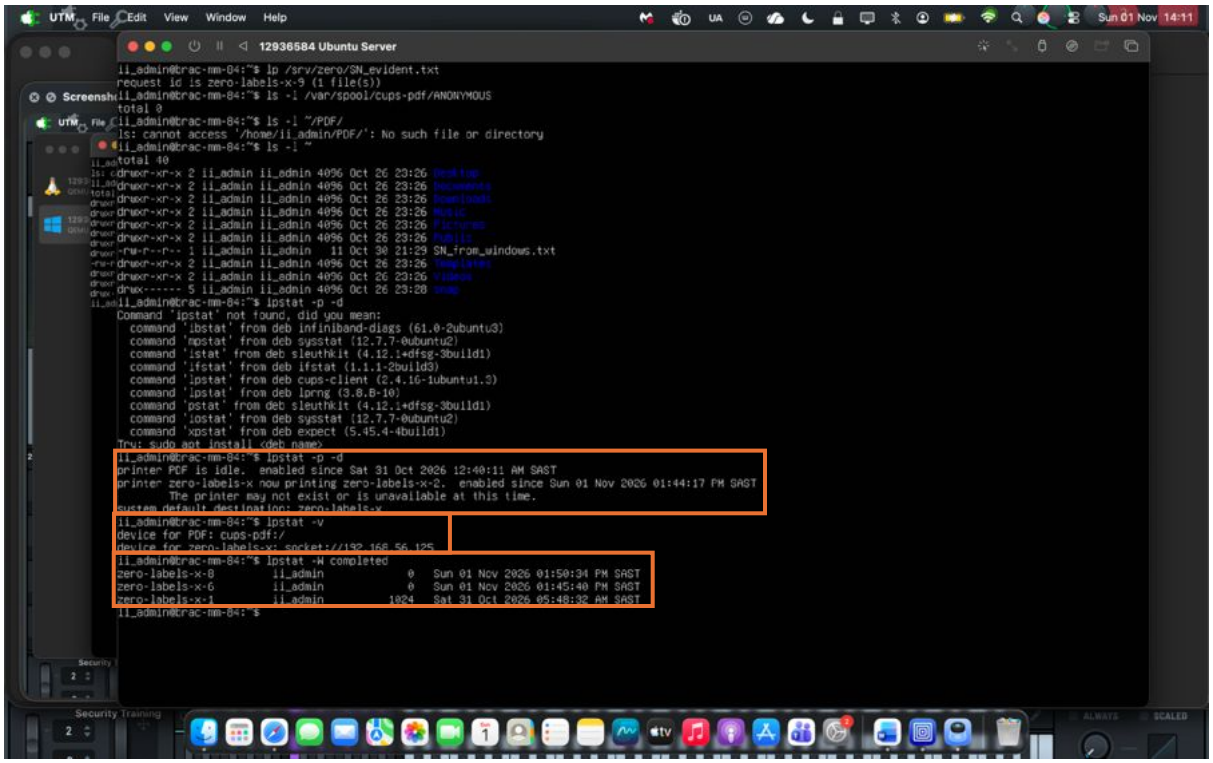
Printing standard CUPS test page.



Printing SN_evident.txt file.



Locating Results of PDF output for standard user.



Displaying Printer Status and Diagnostics (lpstat -p -d). Showing the backend device URL mapping (lpstat -v). Showing the queue's completed job history (lpstat -W completed).

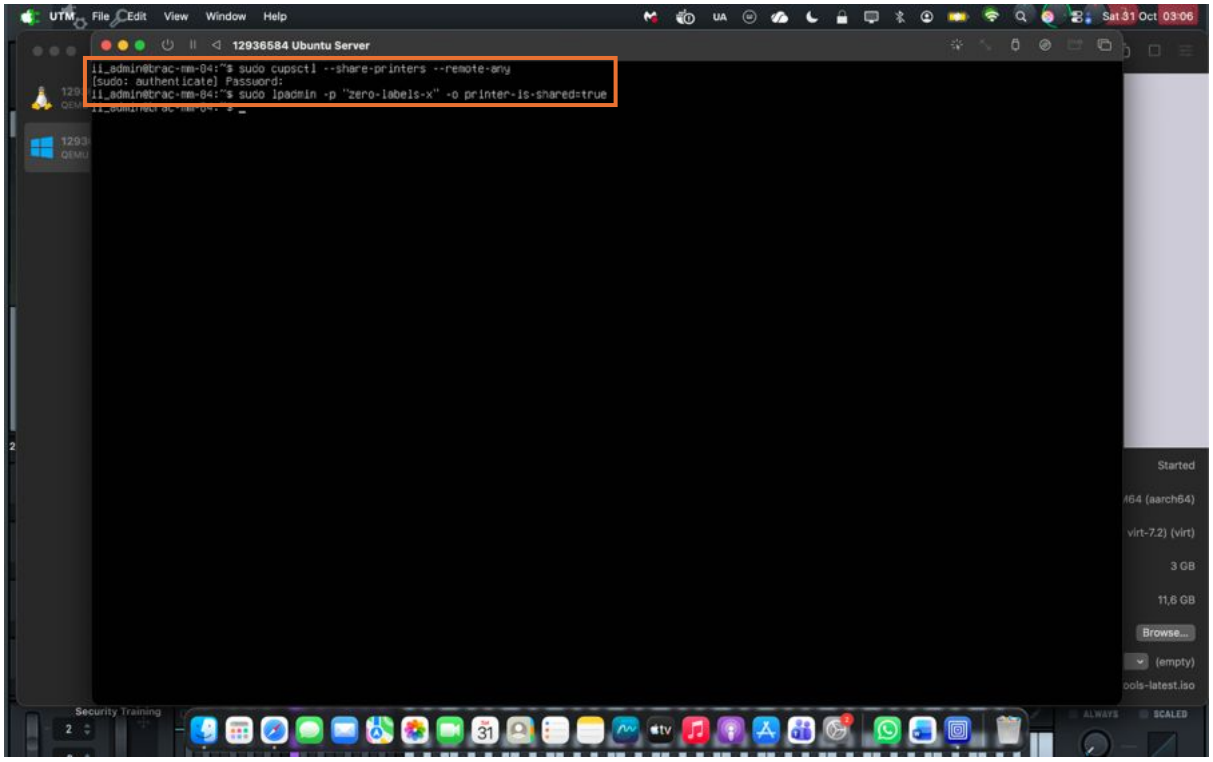
5.4

To replace the virtual CUPS-PDF setup with a physical network label printer like a Zebra, Brother or Dymo, you'll need to make two key changes.

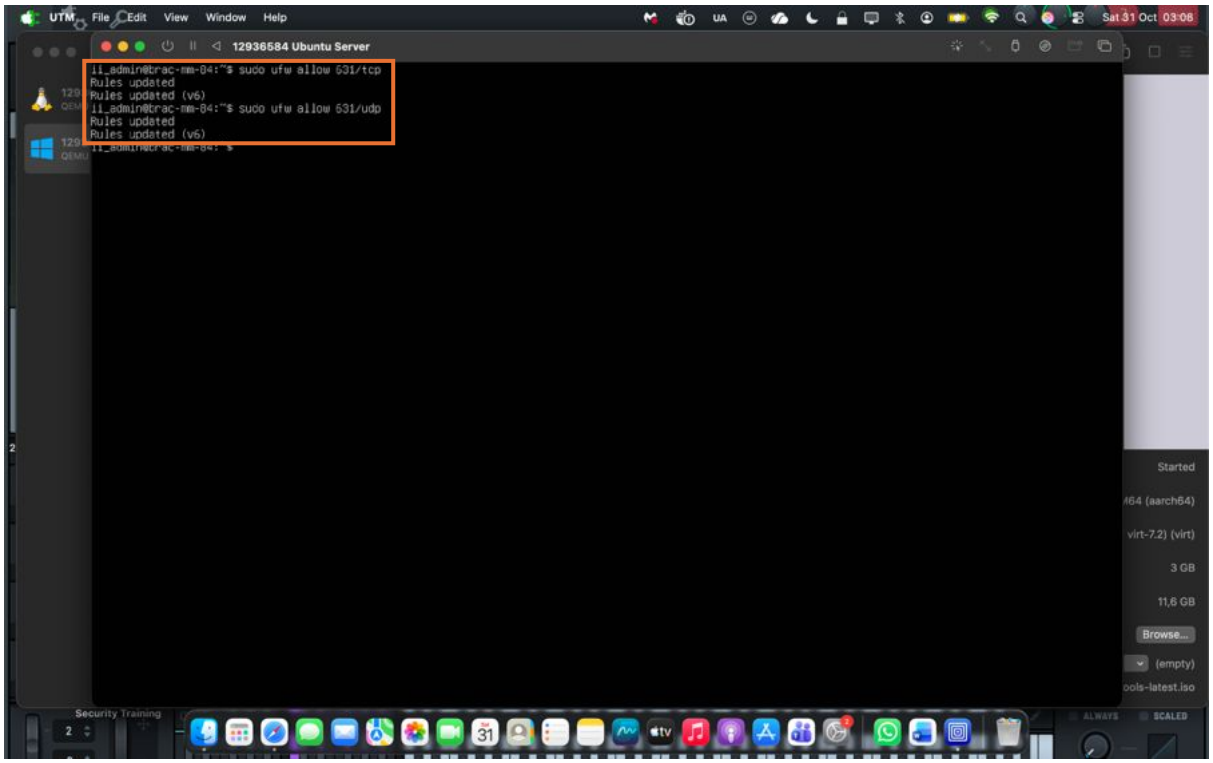
Firstly, the Device URI (-v flag) should be updated. Instead of pointing to cups-pdf:/, use a network protocol URI that matches the printer's interface. E.g.

- AppSocket/JetDirect: socket://<PRINTER_IP_ADDRESS>:9100 (most common for warehouse/office network printers)
- Internet Printing Protocol: ipp://<PRINTER_IP_ADDRESS>/ipp/print
- Line Printer Daemon: lpd://<PRINTER_IP_ADDRESS>/queue

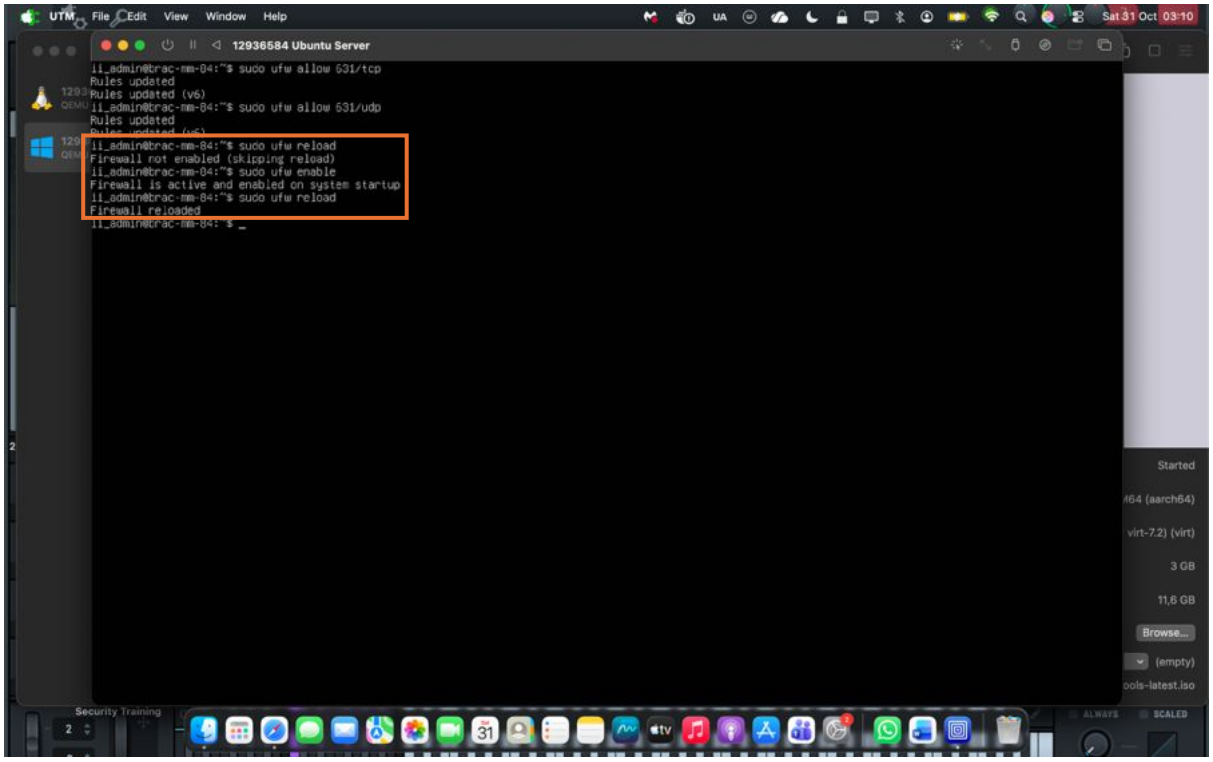
Secondly, the Printer Driver or PPD File (-m or -P flag) needs to be changed. Industrial label printers don't natively parse text/PDF format. Therefore, replace the generic PDF model selection with the specific make and model driver or its descriptive PostScript Printer Description (PPD) file to ensure the layout matches the precise media dimensions.



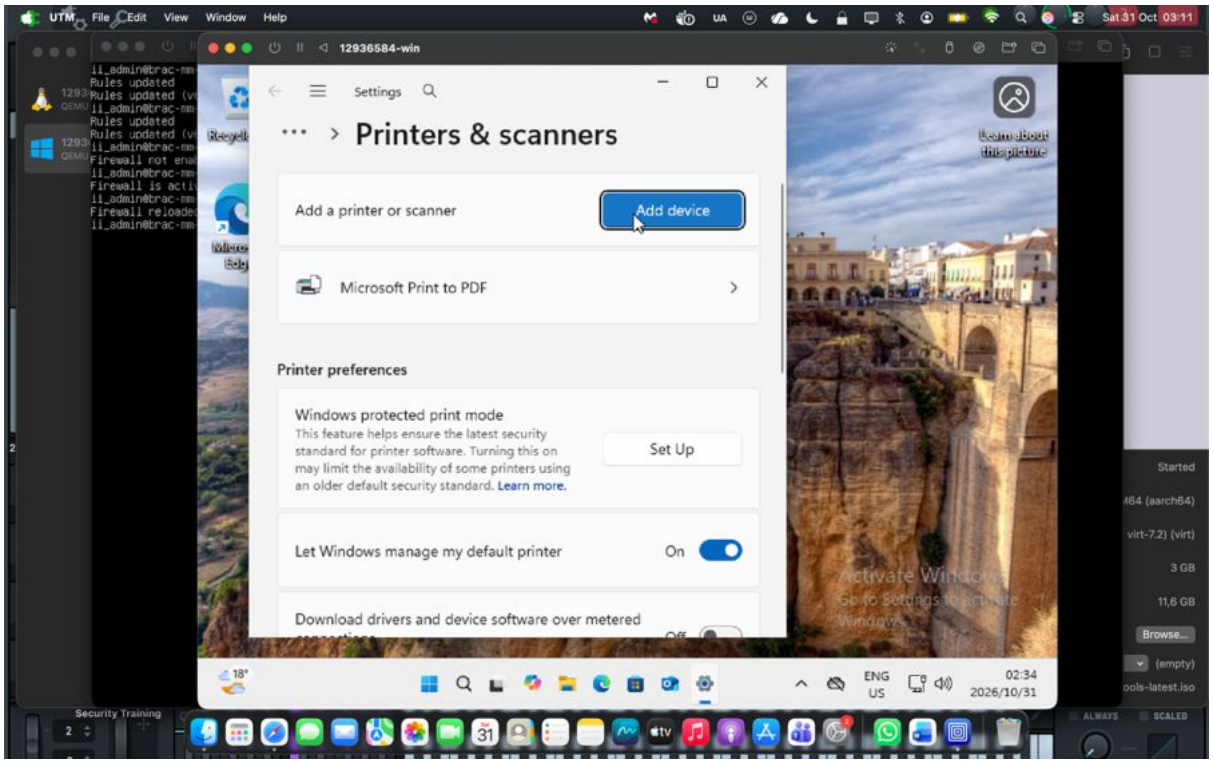
Enabling global sharing and remote administration/browsing. Explicitly ensuring the specific queue is marked as shared.



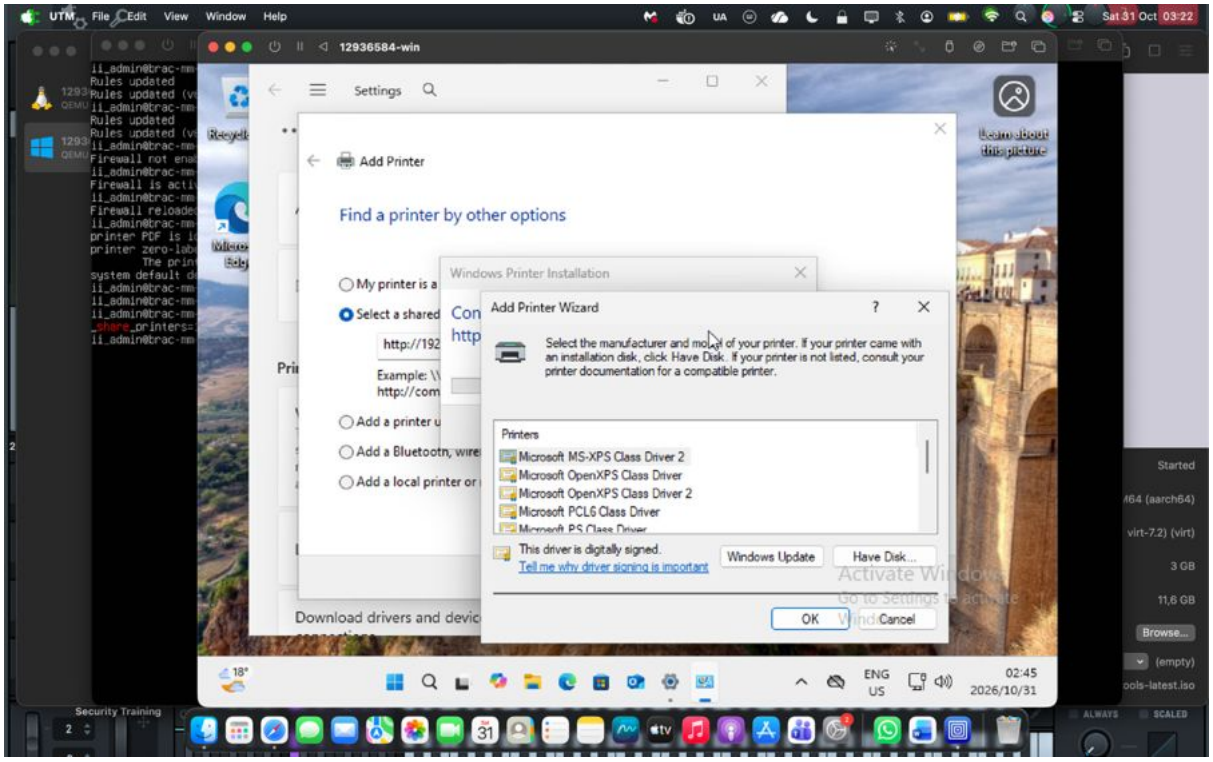
Allowing CUPS traffic from any network interface.



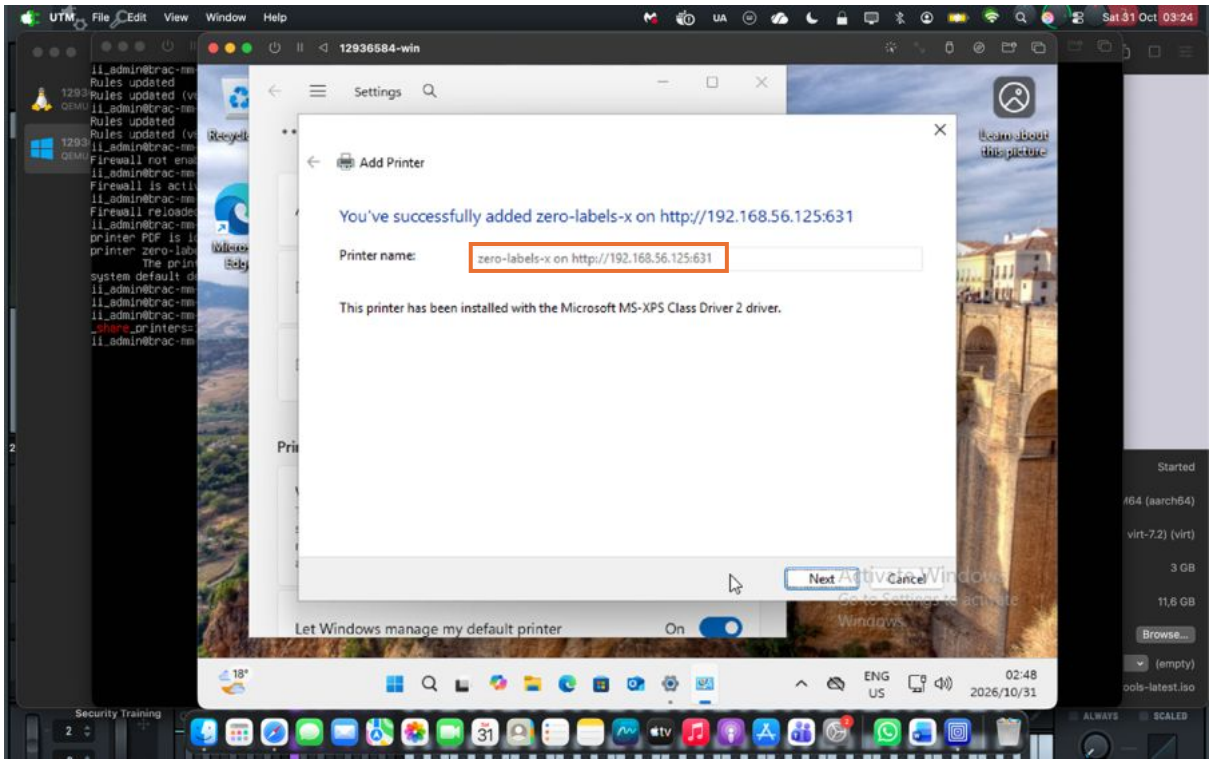
Enabling and reloading firewall to apply update.



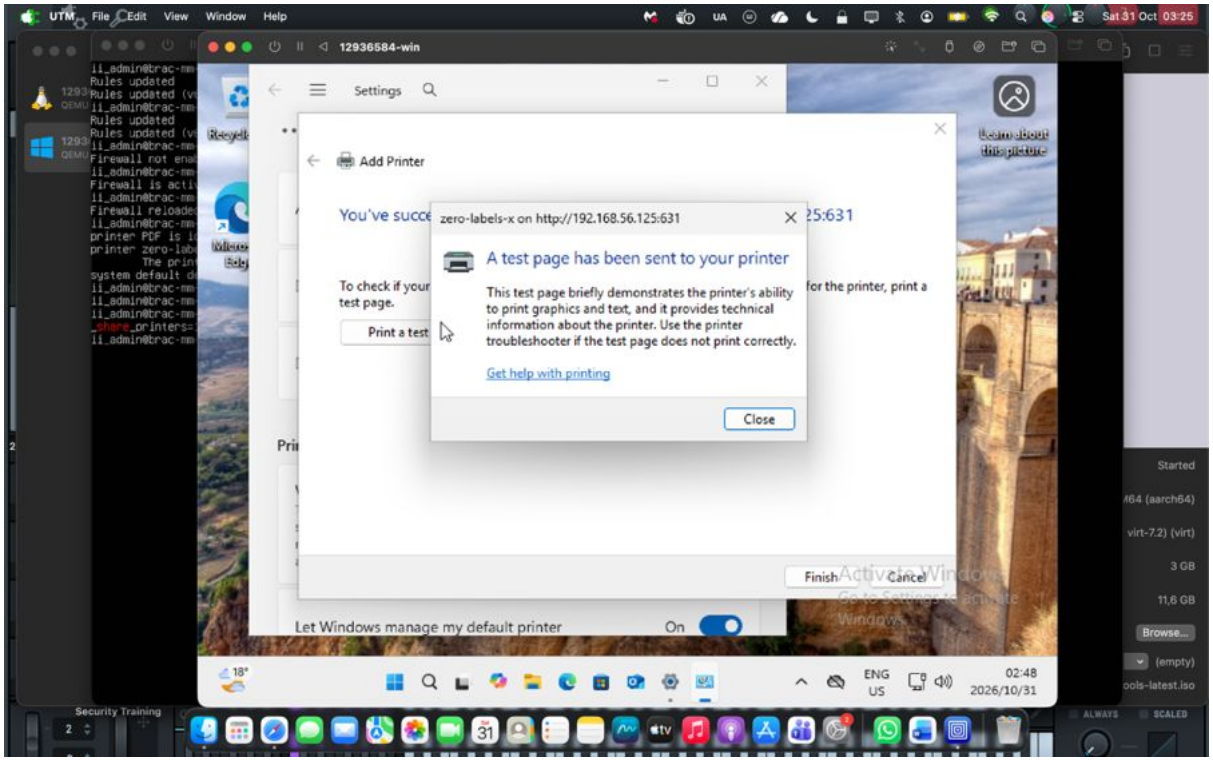
Adding a Printer.



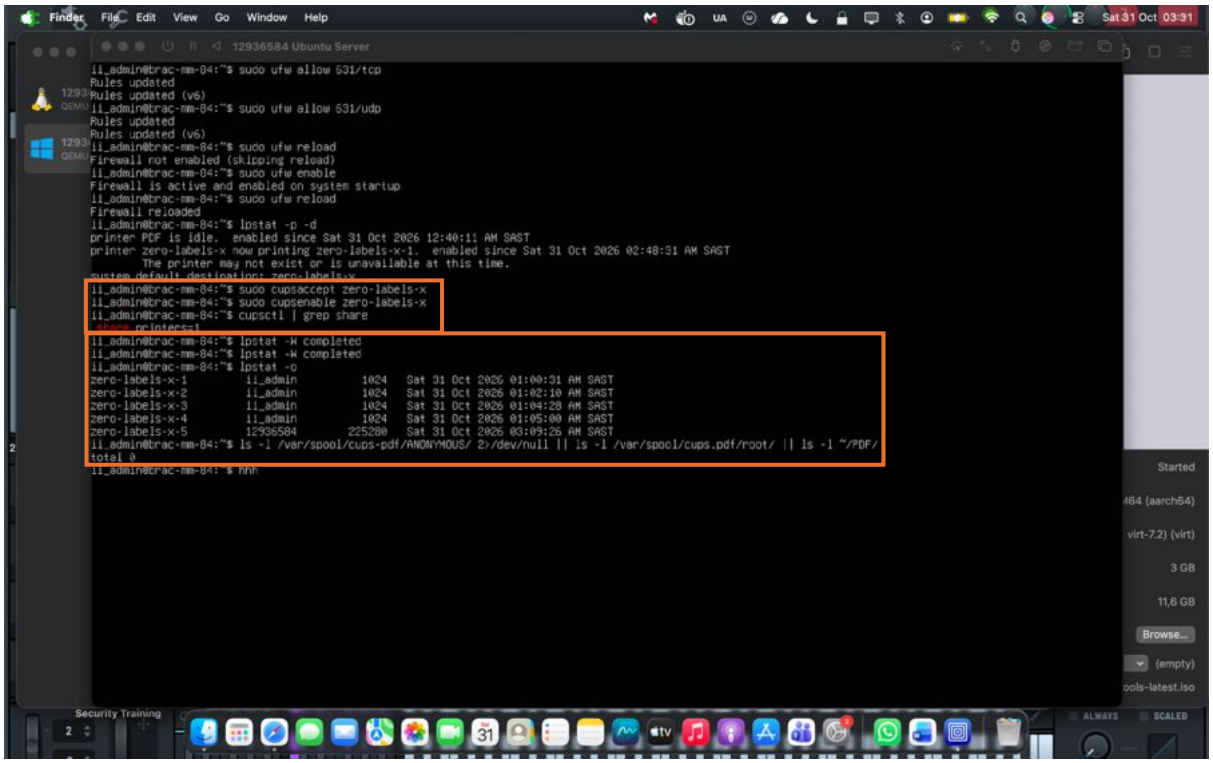
Adding a Printer.



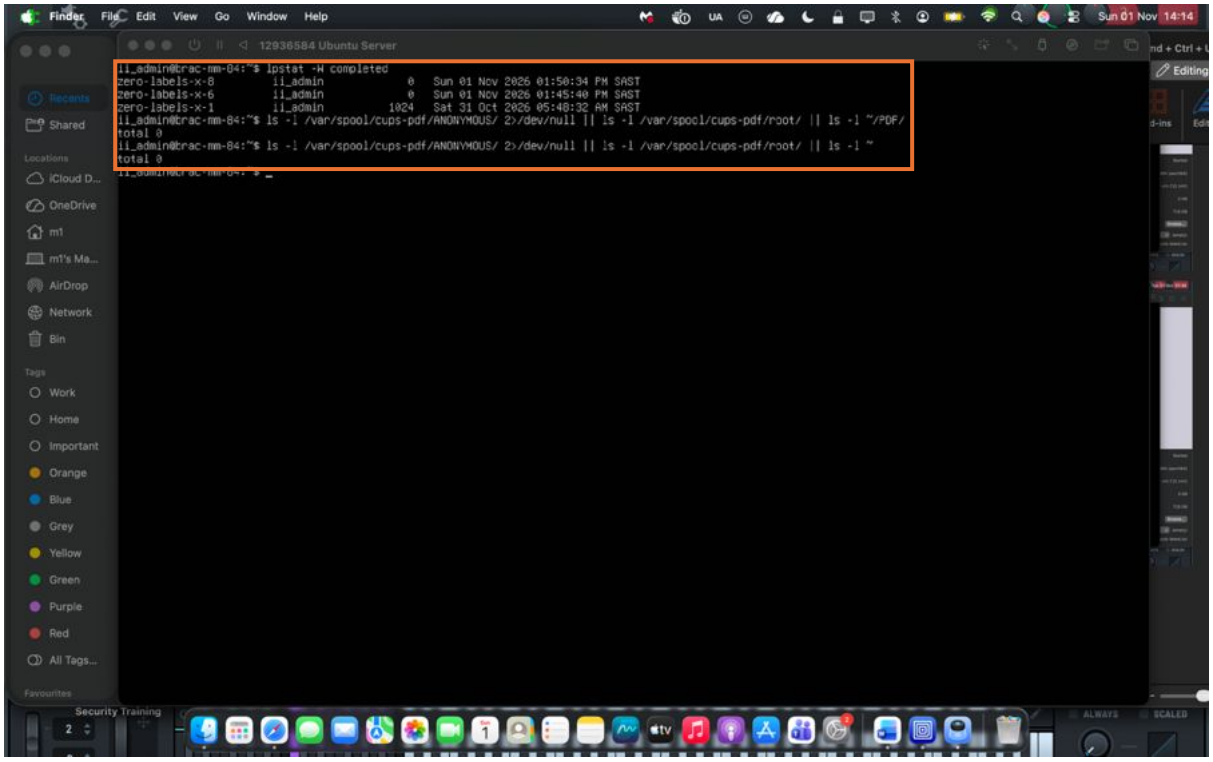
Successfully added a Printer.



Test page sent to Printer.



Checking completed job logs (lpstat -W -o completed).



Verifying the physical PDF was completed and list files to check for the newly generated document.

5.5

Print sharing works over the Internet Printing Protocol (IPP).

a networking standard based on HTTP that enables seamless communication between different operating systems. When you print from Windows the system wraps the document layout using standard PostScript metadata and sends it as an HTTP POST request to port 631 on the Linux server. The Linux CUPS scheduler then processes the document through its filter pipeline ultimately rendering it as a clean PDF using the cups-pdf backend.